# Vector Spaces (Chapter 1)

## $\mathbb{R}^n$ and $\mathbb{C}^n$ (Section 1A)

### Lists (Sub-Section 1A1)

**Def 1.8** (*list, length*)**:**

A list of length $n$ is the same as an $n$ tuple. We write a list of vectors of length $m$ $(v_1,\ldots,v_m)$ as $v_1,\ldots,v_m$ throughout this document without closing parentheses. Since lists are the same as tuples, two lists are only equal if they have the same length and the same elements in the same order.

**Def 1.11** ($\mathbb{F}^n$, *coordinate*)**:**

$\mathbb{F}^n := \left\{\left(x_1,\ldots,x_n\right) : x_k \in \mathbb{F},\ k \in \{1,\ldots,n\}\right\}$

## Definition of a Vector Space (Section 1B)

**Def 1.20** (*vector space*)**:**

A *"vector space"* is a set $V$ along with an addition on $V$ and a scalar multiplication on $V$ such that the following properties hold.

- **Commutativity:** $u+v=v+u \quad \forall u,v\in V$
- **Associativity:** $(u+v)+w=u+(v+w)$ and $(ab)v=a(bv) \quad \forall u,v,w\in V$ and $\forall a,b\in\mathbb{F}$
- **Additive identity:** $\exists 0\in V : v+0=v \quad \forall v\in V$.
  $\exists!$ is not a requirement, but a property of vector spaces. See 1.26.
- **Additive inverse:** $\forall v\in V, \quad \exists w\in V : v+w=0$
- **Multiplicative identity:** $1v=v \quad \forall v\in V$
- **Distributive properties:** $a(u+v)=au+av \quad \forall u,v\in V$ and
  $(a+b)v=av+bv \quad \forall a,b\in\mathbb{F}$ and $v\in V$.

**Def 1.24** (*notation* $\mathbb{F}^S$)**:**

If $S$ is a set, then $\mathbb{F}^S$ denotes the set of functions from $S$ to $\mathbb{F}$. Let $f,g\in\mathbb{F}^S$ and $\lambda\in\mathbb{F}$:

- The *"sum"* $f+g\in\mathbb{F}^S$ is defined as follows: $(f+g)(x):=f(x)+g(x) \quad \forall x\in S$.
- The *"product"* $\lambda f\in\mathbb{F}^S$ is defined as follows: $(\lambda f)(x):=\lambda f(x) \quad \forall x\in S$.

The vector space $\mathbb{F}^n :\equiv \mathbb{F}^{\{1,2,\ldots,n\}}$ is a special case, because each $(x_1,\ldots,x_n)\in\mathbb{F}^n$ can be thought of as a function $x$ from the set $\{1,2,\ldots,n\}$ to $\mathbb{F}$ by writing $x(k)$ instead of $x_k$ for the $k^{\text{th}}$ coordinate of $(x_1,\ldots,x_n)$.

**Thm 1.26** (*unique additive identity*)**:**

A vector space has a unique additive identity.

***Proof:*** Suppose $0$ and $0'$ are both additive identities. Then $0'=0'+0=0+0'=0$. So, $0'=0$. $\square$

**Thm 1.27** (*unique additive inverse*)**:**

Every element in a vector space has a unique additive inverse.

***Proof:*** Let $v\in V$. Suppose $u,w\in V$ are additive inverses of $v$. Using associativity and the definition of the additive inverse, we get

$$u=u+0=u+(v+w)=(u+v)+w=0+w=w.$$

Thus, $u=w$, as desired. $\square$

**Def 1.28:**

For $v, w \in V$, $-v$ denotes the additive inverse of $v$, and we define subtraction as $w - v := w + (-v)$.

**Def 1.29:**

For the rest of this summary, $V$ denotes a vector space over $\mathbb{F}$ most of the time.

**Thm 1.30** (*the number* 0 *times a vector*)**:**

$0v = 0 \quad \forall v \in V$.

***Proof:*** Let $v \in V$. Thus, $0v = (0+0)v = 0v + 0$, which implies $0 = 0v$. Here we have been using the definition of additive identity and the distributive property. After that, we subtract $0v$ on both sides of the equation. □

**Thm 1.31** (*a number times the vector* 0)**:**

$a\vec{0} = \vec{0} \quad \forall a \in \mathbb{F}$.

***Proof:*** For every $a \in \mathbb{F}$, we have $a\vec{0} = a(\vec{0} + \vec{0}) = a\vec{0} + a\vec{0}$. This implies $\vec{0} = a\vec{0}$. (Using the definition of additive identity ($\vec{0} + \vec{0} = \vec{0}$) and one of the distributive properties. After that, we subtract $\vec{0}a$ on both sides of the equation). □

**Thm 1.32** (*a number* $-1$ *times a vector*)**:**

$(-1)v = -v \quad \forall v \in V$.

***Proof:*** Let $v \in V$. Hence, $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$, using the distributive property and the definition of the multiplicative identity. This equation ($v + (-1)v = 0$) tells us that $(-1)v$ is the additive inverse of $v$, and therefore, $(-1)v = -v$. □

## Exercises about Definition of a Vector Space 1B

(1) Prove that $-(-v) = v \quad \forall v \in V$.

***Proof:*** Let $v \in V$. By definition, we know that $v + (-v) = 0$. We also know by definition, that the additive inverse of $(-v)$, which is unique by 1.26, is denoted by $-(-v)$. The very first equations shows that $-(-v) = v$, since $v + (-v) = -(-v) + (-v) = 0$. □

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(2) Suppose for $a \in \mathbb{F}$ and $v \in V$ we have $a \cdot v = 0$. Prove that $a = 0$ or $v = 0$.

***Proof:*** There are two cases for $a$. Either $a = 0$ or $a \neq 0$. If $a = 0$, we are done. If $a \neq 0$, then

$$av = 0 \iff \frac{1}{a}(av) = \frac{1}{a}0 \iff \left(\frac{1}{a}\right)v = 0 \iff v = 0.$$

So either $a = 0$ or $v = 0$. □

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(3) Suppose $v, w \in V$. Explain why there exists a unique $x \in V$ such that $v + 3x = w$.

***Solution:*** To solve this equation for $x$, we first need to subtract $v$ from $w$ and aftwewards devide by 3. The additive inverse of $v$ is unique. But also the result of multyplying a vector by a number is unique.

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(4) ***Solution:*** The only requirement for a vector space the empty set fails to satisfy is the existence of an additive idenity.

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(5) Show that in the definition of a vector space, the additive inverse condition can be replace with the condition that

$$0v = 0 \quad \forall v \in V.$$

***Proof:*** By the distributive properties, we get $\forall v \in V$ :

$$\begin{aligned} 0 = 0v &= (1 + (-1))v \\ &= 1v + (-1)v \\ &= v + (-1)v \end{aligned}$$

So, $\forall v \in V$ it holds that $0 = v + (-1)v$, meaning that for every $v$ we have an additive inverse, namely $(-1)v$. □

## Subspaces (Section 1C)

**Def 1.33** (*subspace*):
A subset $U$ of $V$ ($U \subseteq V$) is called a *"subspace"* of $V$ if $U$ is also a vector space with the same additive identity, addition, and scalar multiplication as on $V$.

**Thm 1.34** (*conditions for a subspace*):
A subset $U$ of $V$ is a subspace of $V \iff U$ satisfies the following three conditions:

- **Additive identity:** $0 \in U$
- **Closed under addition:** $u, w \in U \implies u + w \in U$
- **Closed under scalar multiplication:** $a \in \mathbb{F}$ and $u \in U \implies au \in U$

The additive identity condition above could be replaced by the condition that $U$ is nonempty (because then taking $u \in U$ and multiplying it by 0 would imply that $0 \in U$).

***Proof:*** We need to show the proposition in two directions.

$\Rightarrow$ ***direction:*** If $U$ is a subspace of $V$, then $U$ satisfies the three conditions above by the definition of a vector space.

$\Leftarrow$ ***direction:*** The first condition ensures that the additive identity 0 of $U$ is in $U$. The other two conditions make sure that addition and scalar multiplication make sense on $U$. Thanks to 1.32 and the third condition (closedness under scalar multiplication), every $u \in U$ has an additive inverse $-u = (-1)u$. The other parts of the definition of a vector space, such as associativity, commutativity, and the distributive properties, are automatically satisfied for $U$ because they hold on the larger space $V$. Thus, $U$ is a vector space and hence a subspace of $V$. $\square$

## Sum of Subspaces (Sub-Section 1C1)

**Def 1.36** (*sum of subspaces*):
If $V_1, \dots, V_m$ are subspaces of $V$, the *"sum"* of $V_1, \dots, V_m$, denoted by $V_1 + \cdots + V_m$, is the set of all possible sums of elements of $V_1, \dots, V_m$. More precisely,

$$V_1 + \cdots + V_m := \{v_1 + \cdots + v_m \mid v_1 \in V_1, \dots, v_m \in V_m\}.$$

**Thm 1.40** (*sum of subspaces is the smallest containing subspace*):
If $V_1, \dots, V_m$ are subspaces of $V$, then $V_1 + \cdots + V_m$ is the smallest subspace of $V$ containing $V_1, \dots, V_m$.

***Proof:*** We first prove that $W := \sum_{k=1}^m V_k$ is a subspace of $V$, and then show it is the smallest such subspace containing each $V_k$. Since $0 \in V_k$ for every $k \in \{1, \dots, m\}$, we have $0 = 0 + \cdots + 0 \in W$. If $u = \sum_{k=1}^m v_k \in W$ and $w = \sum_{k=1}^m v_k' \in W$ with $v_k, v_k' \in V_k$, then

$$u + w = \textstyle\sum_{k=1}^m (v_k + v_k') \in W, \quad \text{since each } V_k \text{ is closed under addition.}$$

For any $\lambda \in \mathbb{F}$,

$$\lambda u = \lambda \textstyle\sum_{k=1}^m v_k = \sum_{k=1}^m \underbrace{\lambda v_k}_{\in V_k} \in W, \quad \text{as each } V_k \text{ is closed under scalar multiplication.}$$

Thus, $W$ contains 0, is closed under addition and scalar multiplication, and hence, is a subspace of $V$ by 1.34. Now, we have to show the minimality of $W$ and that $W$ is contained in every subspace of $V$ containing $V_1, \dots, V_m$. The subspaces $V_1, \dots, V_m$ are all contained in $W$, i.e., $\forall k \in \{1, \dots, m\} : V_k \subseteq W$. This is because every $v \in V_k$ can be written as $v = 0 + \cdots + v + \cdots + 0$.

Conversely, every subspace of $V$ containing $V_1, \dots, V_m$ contains $W$, because $V$ is closed under addition. Thus, $W$ is the smallest subspace of $V$ containing $V_1, \ldots, V_m$. $\square$

## DIRECT SUMS (SUB-SECTION 1C2)

**DEF 1.41** (*direct sum*, $\oplus$)**:**

$V_1 + \cdots + V_m$ is called a *"direct sum"* if each element of $V_1 + \cdots + V_m$ can be written in only one way as a sum $v_1 + \cdots + v_m$, where each $v_k \in V_k$. In this case, we write

$$V_1 + \cdots + V_m = V_1 \oplus \cdots \oplus V_m$$

to denote that $V_1 + \cdots + V_m$ is a direct sum.

**EXAMPLE 1.42:**

Example: $\mathbb{F}^3 = \Big\{ \big(x, y, 0\big) \in \mathbb{F}^3 \mid x, y \in \mathbb{F} \Big\} \oplus \Big\{ \big(0, 0, z\big) \in \mathbb{F}^3 \mid z \in \mathbb{F} \Big\}$

**THM 1.45** (*condition for a direct sum*)**:**

$\sum_{k=1}^{m} V_k$ is a direct sum $\Big($i. e., $\sum_{k=1}^{m} V_k = \bigoplus_{k=1}^{m} V_k \Big) \iff$ the only way to express $0$ as $v_1 + \cdots + v_m$ with $v_k \in V_k$ is $v_1 = \cdots = v_m = 0$.

***PROOF:*** $\Rightarrow$ ***DIRECTION:*** First, suppose $\sum_{k=1}^{m} V_k$ is a direct sum. By uniqueness of decompositions, the only way to write $0 = v_1 + \cdots + v_m$ is $v_1 = \cdots = v_m = 0$.

$\Leftarrow$ ***DIRECTION:*** Second, suppose that the only way to write $0$ as a sum $w_1 + \cdots + w_m$, where each $w_k \in V_k$, is by taking each $w_k = 0$. Let $v \in \sum_{k=1}^{m} V_k$, with

$$v = v_1 + \cdots + v_m = u_1 + \cdots + u_m, \quad v_k, u_k \in V_k \text{ (for } k = 1, \ldots, m).$$

Subtracting gives

$$0 = \underbrace{(v_1 - u_1)}_{=0} + \cdots + \underbrace{(v_m - u_m)}_{=0}.$$

By hypothesis, each summand must be zero, so $v_1 = u_1, \ldots, v_m = u_m$. Hence, every element has a unique decomposition, so $\sum_{k=1}^{m} V_k$ is direct. $\square$

**THM 1.46** (*sum and intersection of two subspaces*)**:**

$U + W$ is a direct sum, i.e., $U + W = U \oplus W \iff U \cap W = \{0\}$.

***PROOF:*** Let $U$ and $W$ be subspaces of the same vector space.

$\Rightarrow$ ***DIRECTION:*** First, suppose $U + W$ is a direct sum. If $v \in U \cap W$, then $v, -v \in U$ and $v, -v \in W$. We also have that

$$0 = v + (-v), \text{ where } v \in U \text{ and } (-v) \in W.$$

Because $U + W$ is a direct sum and writing $0$ as $0 = 0 + (-0) = 0 + 0$ is unique, the equation above implies that $v = 0$. Therefore, $U \cap W = \{0\}$.

$\Leftarrow$ ***DIRECTION:*** Suppose $U \cap W = \{0\}$. Let $u \in U$ and $w \in W$ such that

$$0 = u + w. \tag{1.1}$$

By 1.45, we know that our goal is to show that this implies $u = w = 0$. The equation (1.1) above implies that $u = -w \in W$ and $w = -u \in U$ for all arbitrary choices of $U$ and $W$. Thus, $u, w \in U \cap W$. Hence, $u = w = 0$. Therefore, $U + W$ is a direct sum. $\square$

## EXERCISES ABOUT SUBSPACES 1C

*1* For each of the following subsets of $\mathbb{F}^3$, determine wether is is a subspace of $\mathbb{F}^3$.

**(a)** $\Big\{ \big(x_1, x_2, x_3\big) \in \mathbb{F} : x_1 + 2x_2 + 3x_3 = 0 \Big\}$

**(b)** $\left\{(x_1, x_2, x_3) \in \mathbb{F} : x_1 + 2x_2 + 3x_3 = 4\right\}$

**(c)** $\left\{(x_1, x_2, x_3) \in \mathbb{F} : x_1 x_2 x_3 = 0\right\}$

**(d)** $\left\{(x_1, x_2, x_3) \in \mathbb{F} : x_1 = 5x_3\right\}$

***Solution:***

**(a)** Let $E :\equiv \{(x_1, x_2, x_3) \in \mathbb{F} \mid x_1 + 2x_2 + 3x_3 = 0\}$, then $E$ is a subspace of $\mathbb{F}^3$ because it is a plane. We have the additive identity $0 \in \{(x_1, x_2, x_3) \in \mathbb{F} \mid x_1 + 2x_2 + 3x_3 = 0\}$. For $u, v \in E$ with their coordinates satisfying $u_1 + 2u_2 + 3u_3 = 0$ and $v_1 + 2v_2 + 3v_3 = 0$, we also have the condition that $(u_1 + v_1) + 2(u_2 + v_2) + 3(u_3 + v_3) = 0$. So $u + v \in E$. Hence $E$ is closed under addition. If $w \in E$ and $\lambda \in \mathbb{F}$, then if $u_1 + 2u_2 + 3u_3 = 0$, it also holds that $\lambda u_1 + 2\lambda u_2 + 3\lambda u_3 = 0$. Hence $\lambda u \in E$. So $E$ is closed under scalar multiplication. Therefore, $E$ is a subspace of $\mathbb{F}^3$ by 1.34.

**(b)** It is not a subspace because is does not contain 0. It is also not closed under addtition and scalar multiplication.

**(c)** Let $U :\equiv \{(x_1, x_2, x_3) \in \mathbb{F} \mid x_1 x_2 x_3 = 0\}$. Then $U$ is actually the union of the $xy$-, the $xz$- and the $yz$-plane, because its description tells us that at least one of its coordinates has to be zero. Thus we have

$$(1,1,0) + (0,1,1) = (1,2,1) \notin U.$$

Therefore, $U$ is not closed under addition. But it is closed under scalar multiplication and we also have $0 \in U$. Therefore, $U$ is not a subspace of $\mathbb{F}^3$.

**(d)** Let $U :\equiv \{(x_1, x_2, x_3) \in \mathbb{F} \mid x_1 = 5x_3\}$. Then $0 \in U$, so we have checked for additive identity. Let $u, v \in U$. Then $u + v = (5u_3 + 5v_3, u_2 + v_2, u_3 + v_3) = (5(u_3 + v_3), u_2 + v_2, u_3 + v_3) \in U$, so its closed under addition. Simmilarly for $\lambda \in \mathbb{F}$ and $w \in U$ we have $\lambda u = (\lambda 5u_3, \lambda u_2, \lambda u_3) = (5\lambda u_3, \lambda u_2, \lambda u_3) \in U$, so $U$ is a subspace of $\mathbb{F}^3$.

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2 Verify all assertions about subspaces in example 1.35.

**(a)** If $b \in \mathbb{F}$ then $U :\equiv \{(x_1, x_2, x_3, x_4) \in \mathbb{F} \mid x_3 = 5x_4 + b\}$ is a subspace of $\mathbb{F}^4 \iff b = 0$.

***Solution:*** If $v, u \in U$, then

$$\begin{aligned} v + u &= (v_1 + u_1, v_2 + u_2, v_3 + u_3, v_4 + u_4, v_5 + u_4) \\ &= (v_1 + u_1, v_2 + u_2, v_3 + u_3, 5v_4 + b + 5u_4 + b, v_5 + u_4) \\ &= (v_1 + u_1, v_2 + u_2, v_3 + u_3, 5(v_4 + u_4) + 2b, v_5 + u_4) \end{aligned}$$

Only if $b = 0$, it can hold that the third coordinate of $v + u$ is 5 times the fourth coordinate of $v + u$ plus an element of $\mathbb{F}$ acting as $b$. Also note that if $\lambda \in \mathbb{F}$ and $w \in U$, then $\lambda w = (\lambda w_1, \lambda w_2, \lambda u_4 + \lambda b, \lambda u_4)$. In this case, we have $\lambda w \in U$ if and only if $b = 0$.

**(b)** The set of continious real-valued functions on the intervall $[0,1]$ is a subspace of $\mathbb{R}^{[0,1]}$ (The set of real valued functions on the invervall $[0,1]$)

***Solution:*** The constant function 0 is continious. Adding two continious functions together yields a continious function. Multiplying a continious function with a scalar also gives us a continious function.

**(c)** The set of differentiable real-valued functions of $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.

***Solution:*** We check for all three properties.

**Additive identity:** $f(x) := 0 \quad \forall x \in \mathbb{R}$ is in the set of differentiable real-valued functions since $f'(x) = 0 \quad \forall x \in \mathbb{R}$ is also contained.

**Closededness under addition:** Let $h(x) := f(x) + g(x)$ for two differentiable real-valued functions $f$ and $g$. Then $h'(x) = \frac{d}{dx}(f(x)+g(x)) = f'(x)+g'(x)$, so this sum $h$ also belongs to set of differentiable real-valued functions.

**Closededness under multiplication:** Let $h(x) := cf(x)$. Then $h'(x) = \frac{d}{dx}(cf(x)) = cf'(x)$. So this product $h$ also belongs to the set of differentiable real-valued functions.

**(d)** The set $S$ of differentiable real-valued functions $f$ on the inverfall $(0,3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

***Solution:*** For $h(x) := 0$, we have $h'(x) = 0$. The condition $h'(2) = 0 = b$ only holds $\iff$ $b = 0$. Set $h(x) := f(x)+g(x)$, thus $h'(x) = f'(x)+g'(x)$. The condition $h'(2) = f'(2)+g'(2) = 2b = b$ is always true $\iff$ $b = 0$. For $h(x) := cf(x)$ we have $h'(x) = cf'(x)$. Now, $h'(2) = cf'(2) = cb = b$ can only hold for all constants $c \iff b = 0$.

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**3** Show that the set $S$ of differentiable real-valued functions $f$ on the invervall $(-4,4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbb{R}^{(-4,4)}$.

***Solution:*** We check for all three properties.

**Additive identity:** For $f(x) :\equiv 0, f'(x) = 0$ we have $f'(-1) = 0 = 3\cdot 0 = 3f(2)$, so $f'$ belongs so $S$ as well.

**Closed under addition:** For $h(x) :\equiv f(x)+g(x), h'(x) = f'(x)+g'(x)$ we got $h'(-1) = f'(-1) + g'(-1) = 3f(2)+3g(2) = 3(f(2)+g(2)) = 3h(2)$. Therefore, $h \in S$.

**Closed under scalar multiplication:** For $h(x) :\equiv cf(x), h'(x) = cf'(x)$ we got $h'(-1) = cf'(-1) = c3f(2) = 3cf(2) = 3h(x)$. So if $h' \in S$ whenever $h \in S$.

Hence $S$ is a subspace of $\mathbb{R}^{(-4,4)}$.

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**8** Prove or give a counterexample: If $U$ is a nonempty subset of $\mathbb{R}^2$ such that $U$ is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then $U$ is subspace of $\mathbb{R}^2$.

***Solution:*** This statement is wrong. Take $U :\equiv \{(x,y) \in \mathbb{R}^2 \mid (x,y) \in \mathbb{Z}^2\} \subset \mathbb{R}^2$. Then $U$ is closed under addtion since $\mathbb{Z}$ is closed under addition. Example: $(2,1)+(-2,1)+(1,1) = (1,1) \in U$. But if $\lambda \in \mathbb{R}$ such that $\lambda \notin \mathbb{Z}$ we have for $a, b \in \mathbb{Z}$:

$$\lambda(a,b) = (\lambda a, \lambda b) \notin U.$$

For example: $\pi(3,4) \notin U$. Therefore $U$ is not a subspace of $\mathbb{R}^2$.

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**12** Prove that the union $U \cup W$ of of two subspaces $U, W$ of $V$ is a subspace of $V \iff$ one of the subspaces is contained in the other. So $U \subseteq W$ or $W \subseteq U$.

***Proof:*** $\Leftarrow$ ***direction:*** Let $U$ and $W$ be two subspaces of $V$. If $U \subseteq W$ then $U \cup W = W \subseteq V$ and if $W \subseteq U$ then $U \cup W = U \subseteq V$.

$\Rightarrow$ ***direction:*** We can prove this either directly or by contradiction. In both proofs, let us assume that $U \cup W$ is a subspace of $V$, given that $U$ and $W$ are subspaces of $V$ as well.

$\Rightarrow$ ***direction proof by contradiction:*** Suppose $U \subsetneq W$ and $W \subsetneq U$. Note that this would imply that $U \neq \{0\}$ and $W \neq \{0\}$. Therefore, we have that

$$\exists u \in U : u \notin W \quad \text{and}$$
$$\exists w \in W : w \notin U.$$

Since $u, w \in U \cup W \implies u + w \in U \cup W$ we have that either $u + w \in U$ or $u + w \in W$ (or both).

- ***Case* $\mathbf{u+w \in U}$:** In this case, let $\widetilde{u} := u + w \in U$. Hence, $w = \widetilde{u} - u \in U$, a contradiction, because we had $w \notin U$

- ***Case*** $\mathbf{u}+\mathbf{w}\in\mathbf{W}$ : Let $\widetilde{w} := u+w \in W$. Hence $u = \widetilde{w} - u \in W$, again a contradiction, because we had $u \notin W$.

Therefore, the assumption that $U \subsetneq W$ and $W \subsetneq U$ can not hold. Hence, the opposite is true. By de Morgan's rule, it must hold that $U \subseteq W$ or $W \subseteq U$.

***Direct*** $\Rightarrow$ ***direction proof:*** We have 2 cases. $W \subseteq U$ or $W \subsetneq U$. If $W \subseteq U$, then the disjunction $W \subseteq U$ or $U \subseteq W$ is automatically true. So let us assume that $W \subsetneq U$ and prove that this implies that $U \subseteq W$. Let $u \in U$ and $w \in W \backslash U$. By the definition of union, it also holds that $u, w \in U \cup W$. Since $U \cup W$ is a subspace of $V$, we write

$$u + w \in U \cup W.$$

By the defintion of union, there are two possible cases. $u+w \in U$ or $u+w \in W$. The first case would imply

$$w = (u+w) + (-u) \in U,$$

which is impossible. Hence $u+w \in W$ and therefore,

$$u = (u+w) - w \in W$$

Since $u$ was arbitrary, $U \subseteq W$. □

23 Prove or give a counterexample: If $V_1, V_2, U$ are subspaces of $V$ such that $V = V_1 \oplus U$ and $V = V_2 \oplus U$, then $V_1 = V_2$.

***Solution:*** The statement is not true. One counterexample would be $V := \mathbb{R}^2$ and

$$U := \{(x_1, x_2) \mid x_2 = 0\},$$

where $U$ is the $x$-axis. Let $V_1$ and $V_2$ be defined as follows:

$$V_1 :\equiv \underbrace{\{(x_1,x_2) \in \mathbb{R}^2 \mid x_1 = 0\}}_{y\text{-axis}} \quad \text{and } V_2 :\equiv \underbrace{\{(x_1,x_2) \in \mathbb{R}^2 \mid x_2 = x_1\}}_{\text{straight line with slope 1}}$$

We can see that

$$V = \mathbb{R}^2 = V_1 \oplus U = V_2 \oplus U,$$

because each $(x,y) \in \mathbb{R}^2$ can be uniquely represented as

$$\underbrace{(a,0)}_{\in U} + \underbrace{(0,b)}_{\in V_1} \text{ or } \underbrace{(c,0)}_{\in U} + \underbrace{(k,k)}_{\in V_2} = (c+k,k) \text{ where } a,b,c,k \in \mathbb{R}$$

Therefore, the statement is not true.

# Finite Dimensional Vector Spaces (Chapter 2)

## Span and Linear Independence (Section 2A)

### Linear Combinations and Span (Sub-Section 2A1)

**Def 2.2** (*linear combination*):

$a_1v_1 + \cdots + a_mv_m$ is called a *"linear combination"* of a list of vectors $v_1, \ldots, v_m$ ($a_1, \ldots, a_m \in \mathbb{F}$).

**Def 2.4** (*span of a list of vectors*):

The *"span"* of a list of vectors $v_1, \ldots, v_m$ is defined as follows:

$$\mathrm{span}(v_1, \ldots, v_m) :\equiv \{a_1v_1 + \cdots + a_mv_m \mid a_1, \ldots, a_m \in \mathbb{F}\}$$

$$\mathrm{span}(\,) :\equiv \{0\} \quad \text{(The span of the empty list ( ) is defined to be } \{0\}\text{)}$$

The definition also implies that $\mathrm{span}(0) = \{0\}$, or written differently, $\mathrm{span}(\vec{0}) = \{\vec{0}\}$. Note that sometimes we write $\vec{0}$ and sometimes we write 0 throughout this document. Later on, 0 sometimes also means the zero-operator.

**Thm 2.6** (*the span of a list of vectors is a subspace*):

The span of a list of vectors in $V$ is the smallest subspace of $V$ containing all vectors in the list.

***Proof:*** Let $v_1, \ldots, v_m \in V$. We have

$$0 = 0v_1 + \cdots + 0v_m \in \mathrm{span}(v_1, \ldots, v_m),$$

and hence, the additive identity is in $\mathrm{span}(v_1, \ldots, v_m)$.

$\mathrm{span}(v_1, \ldots, v_m)$ is closed under addition, because for all $a$'s and $c$'s in $\mathbb{F}$, we have

$$\underbrace{(a_1v_1 + \cdots + a_mv_m)}_{\in \mathrm{span}(v_1, \ldots, v_m)} + \underbrace{(c_1v_1 + \cdots + c_mv_m)}_{\in \mathrm{span}(v_1, \ldots, v_m)} = (a_1 + c_1)v_1 + \cdots + (a_m + c_m)v_m \in \mathrm{span}(v_1, \ldots, v_m).$$

Furthermore, $\mathrm{span}(v_1, \ldots, v_m)$ is closed under scalar multiplication because for $\lambda \in \mathbb{R}$, we have:

$$\lambda \underbrace{(a_1v_1 + \cdots + a_mv_m)}_{\in \mathrm{span}(v_1, \ldots, v_m)} = \lambda a_1v_1 + \cdots + \lambda a_mv_m \in \mathrm{span}(v_1, \ldots, v_m).$$

Thus, $\mathrm{span}(v_1, \ldots, v_m)$ is a subspace of $V$ by 1.34. Every subspace of $V$ that contains each $v_k$ is closed under scalar multiplication and addition, and therefore, contains $\mathrm{span}(v_1, \ldots, v_m)$. Hence, $\mathrm{span}(v_1, \ldots, v_m)$ is the smallest subspace containing each $v_1, \ldots, v_m$. The fact that

$$v_j \in \mathrm{span}(v_1, \ldots, v_m) \quad (j = 1, \ldots, m)$$

is obvious. □

**Def 2.7** (*spanning a vector space*):

If $V = \mathrm{span}(v_1, \ldots, v_m)$, we say $v_1, \ldots, v_m$ *"spans"* $V$.

**Def 2.9** (*finite-dimensional vector space*):

Such a vector space is called finite-dimensional. We write

$$\dim V \neq \infty \text{ or } \dim V < \infty.$$

**Def 2.10** (*polynomial, $\mathcal{P}(\mathbb{F})$*):

A function $p : \mathbb{F} \to \mathbb{F}$ is called a *"polynomial"* with coefficients in $\mathbb{F}$ if there exist $a_0, \ldots, a_m \in \mathbb{F}$ such that

$$p(z) = a_0 + a_1z + a_2z^2 + \cdots + a_mz^m \quad \forall z \in \mathbb{F}.$$

We further denote

- $\mathcal{P}(\mathbb{F})$ as the set of all polynomials with coefficients in $\mathbb{F}$.
- Note that $\mathcal{P}(\mathbb{F})$ is a vector space over $\mathbb{F}$ and a subspace of $\mathbb{F}^{\mathbb{F}}$.

**Def 2.11** (*degree of a polynomial,* $\deg p$)**:**

The degree $m$ of a polynomial $p = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m$ is denoted with

$$\deg p := m.$$

We also define for the zero polynomial $p = 0$:

$$\deg p = \deg 0 := -\infty.$$

**Def 2.12** (*the set of all polynomials with degree m*)**:**

$\mathcal{P}_m(\mathbb{F}) = \operatorname{span}(1, z, \ldots, z^m)$ denotes the set of all polynomials with coefficients in $\mathbb{F}$ and degree at most $m$. One should actually use the lambda notation $\lambda z.z^m$ to speak of functions and write $\mathcal{P}_m(\mathbb{F}) = \operatorname{span}(\lambda z.1, \lambda z.z, \ldots, \lambda z.z^m)$, but this is not done most of the time. Note that

$$\dim \mathcal{P}_m(\mathbb{F}) = m + 1.$$

**Def 2.13** (*infinite-dimensional*)**:**

A vector space is called *"infinite-dimensional"* if it is not finite-dimensional.

## Linear Independence (Sub-Section 2A2)

**Def 2.15** (*linear independence*)**:**

There are 2 ways to think about *"linear independence"*. Let's first start with the motivation. A list of vectors $v_1, \ldots, v_m$ is called *"linearly independent"* if every combination of these vectors $w = b_1 v_1 + \cdots + b_m v_m \in \operatorname{span}(v_1, \ldots, v_m)$ has a unique choice of scalars $b_1, \ldots, b_m \in \mathbb{F}$. That means there does not exist another choice $b'_1, \ldots, b'_m \in \mathbb{F}$, with at least one $b_j \neq b'_j$, such that

$$b_1 v_1 + \cdots + b_m v_m = b'_1 v_1 + \cdots + b'_m v_m.$$

**The actual definition of linear independence:**

A list of vectors $v_1, \ldots, v_m$ is called linearly independentif the only way to combine them together and yield zero is if we choose every coefficient $\lambda_i$ to be zero:

$$\lambda_1 v_1 + \cdots + \lambda_m v_m = 0 \iff \lambda_1 = \cdots = \lambda_m = 0.$$

Another way to put it:

Let $w \in \operatorname{span}(v_1, \ldots, v_m)$ such that for $a_1, \ldots, a_m, c_1, \ldots, c_m \in \mathbb{F}$:

$$w = a_1 v_1 + \cdots + a_m v_m \text{ and}$$
$$w = c_1 v_1 + \cdots + c_m v_m$$
$$\iff$$
$$0 = \underbrace{(a_1 - c_1)}_{=0?} v_1 + \cdots + \underbrace{(a_m - c_m)}_{=0?} v_m.$$

If $a_1 = c_1, a_2 = c_2, \ldots, a_m = c_m$ is the only solution, the representation of $w$ as a linear combination of $v_1, \ldots, v_m$ is unique and the only way to add them up together equaling 0 is $0v_1 + \cdots + 0v_m = 0$. Both of these definitions are equivalent.

Note: The empty list () is also defined to be linearly independent.

**Def 2.16** (*linear dependence*)**:**

Otherwise, $v_1, \ldots, v_m$ is called *"linearly dependent"*. For example, let $V$ be a 2-dimensional vector space and let $u, v \in V$ be linearly dependent. Then we have $a, b$ such that $au + bv = 0$ with $a \neq 0$ and $b \neq 0$. Thus, we can write: $u = (-b/a)\, v$.

**Thm 2.19** (*linear dependence lemma*):

Suppose the list $v_1, \ldots, v_m \in V$ is linearly dependent. Then there is an index $k \in \{1, \ldots, m\}$ with

$$v_k \in \operatorname{span}(v_1, \ldots, v_{k-1}). \tag{2.1}$$

Furthermore, if $k$ satisfies the conditions above (2.1) and the $k^{\text{th}}$-term is removed from $v_1, \ldots, v_m$, then the span of the remaining list equals $\operatorname{span}(v_1, \ldots, v_m)$:

$$\operatorname{span}(v_1, \ldots, v_{k-1}, \cancel{v_k}, v_{k+1}, \ldots, v_m) = \operatorname{span}(v_1, \ldots, v_m).$$

***Proof:*** Because the list $v_1, \ldots, v_m$ is linearly dependent, we have a non-trivial linear combination

$$a_1 v_1 + \cdots + a_m v_m = 0,$$

where $a_1, \ldots, a_m \in \mathbb{F}$ and $a_j \neq 0$ and $a_k \neq 0$, for $j, k \in \{1, \ldots, m\}$ and $j \neq k$ (we know at least 2 $a$'s are not 0). Let $k$ be the largest element of $\{1, \ldots, m\}$ such that $a_k \neq 0$. Therefore,

$$v_k = -\frac{a_1}{a_k} v_1 - \cdots - \frac{a_{k-1}}{a_k} v_{k-1},$$

showing $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$. Now for the second claim, suppose $k$ is any element of $\{1, \ldots, m\}$ such that

$$v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$$

and write

$$v_k = b_1 v_1 + \cdots + b_{k-1} v_{k-1}, \quad \text{where } b_1, \ldots, b_{k-1} \in \mathbb{F}. \tag{2.2}$$

Suppose $u \in \operatorname{span}(v_1, \ldots, v_m)$. So,

$$u = c_1 v_1 + \cdots + c_k v_k + \cdots + c_m v_m, \quad \text{where } c_1, \ldots, c_m \in \mathbb{F}. \tag{2.3}$$

Now we can replace $v_k$ with the right side of (2.2) in equation (2.3):

$$\begin{aligned} u &= c_1 v_1 + \cdots + c_k \cdot (b_1 v_1 + \cdots + b_{k-1} v_{k-1}) + \cdots + c_m v_m \\ &= (c_1 + c_k b_1) \cdot v_1 + (c_2 + c_k b_2) \cdot v_2 + \cdots + (c_{k-1} + c_k b_{k-1}) \cdot v_{k-1} + c_{k+1} v_{k+1} + \cdots + c_m v_m, \end{aligned} \tag{2.4}$$

which shows that $u \in \operatorname{span}(v_1, \ldots, v_{k-1}, v_{k+1}, \ldots, v_m)$. Combining (2.3) and (2.4), we conclude that

$$\operatorname{span}(v_1, \ldots, v_{k-1}, \cancel{v_k}, v_{k+1}, \ldots, v_m) = \operatorname{span}(v_1, \ldots, v_m),$$

as desired. $\square$

**Thm 2.22** (*length of linearly independent list ≤ length of spanning list*):

In a finite-dimensionalvector space, the length of every linearly independent list of vectors is less than or equal to the length of every spanning list of vectors.

***Proof:*** Let $u_1, \ldots, u_m \in V$ be a linearly independent list and let $w_1, \ldots, w_n \in V$ such that

$$V = \operatorname{span}(w_1, \ldots, w_n).$$

We need to prove that $m \leq n$. We construct, in $m$ steps, a spanning list of size $n$ that incorporates $u_1, \ldots, u_m$. This will force $m \leq n$. Start with

$$B_0 := (w_1, \ldots, w_n), \quad V = \operatorname{span}(B_0).$$

***Step 1:*** Add $u_1$ to $B_0$, so set

$$B_1^* := (u_1, w_1, \ldots, w_n).$$

Since $u_1 \in \operatorname{span}(B_0)$, the enlarged list is dependent, so by the Linear Dependence Lemma 2.19, we can remove one of the original $w$'s, say $w_j$, obtaining

$$B_1 := B_1^* \setminus \{w_j\} = \left(u_1, w_1, \ldots, w_{j-1}, \cancel{w_j}, w_{j+1}, \ldots, w_n\right), \quad V = \operatorname{span}(B_1).$$

***Step k, for k=2, …, m:*** The list $B_{k-1}$ from the previous step $k-1$ spans $V$. So we have $u_k \in \operatorname{span}(B_{k-1})$, because the span never changed by our algorithm. Hence, if we add $u_k$ to the list, it becomes linearly dependent, since it has length $(n+1) = k + (n-(k-1))$. Let

$$B_k^* :\equiv \Big(u_1,\dots,u_{k-1},u_k,\underbrace{\widetilde{w}_1,\dots,\widetilde{w}_{n-(k-1)}}_{\text{remaining } w\text{'s}}\Big), \quad V = \operatorname{span}(B_k^*),$$

where $\widetilde{w}_1,\dots,\widetilde{w}_{n-(k-1)}$ are our remaining $n-(k-1)$ $w$'s at our current step $k$, which are the same as from the previous list $B_{k-1}$. We then remove another vector; since the $u$'s remain independent, that must be one of the $w$'s, say $\widetilde{w}_j$, obtaining

$$B_k := B_k^* \setminus \{\widetilde{w}_j\} = \Big(u_1,\dots,u_k,\widetilde{w}_1,\dots,\widetilde{w}_{j-1},\cancel{\widetilde{w}_j},\widetilde{w}_{j+1},\dots,\widetilde{w}_{n-(k-1)}\Big), \quad V = \operatorname{span}(B)_k.$$

After each step $k$, $|B_k| = n$ and $u_1,\dots,u_k \in B_k$.

**Conclusion:** After $m$ steps, $B_m$ is a spanning list of size $n$ containing the $m$ independent vectors; hence, $m \le n$. □

**Thm 2.25** (*finite-dimensional subspace*):

Every subspace of a finite-dimensional vector space is finite-dimensional.

***Proof:*** Suppose $\dim V < \infty$, and $U \subseteq V$. Let $m := \dim V$.

**Step 1:** If

$$U = \{0\},$$

we are done. Otherwise, pick any nonzero $u_1 \in U$.

***Step k, ($2 \le k \le$ m):*** If

$$U = \operatorname{span}(u_1,\dots,u_{k-1}),$$

then $U$ is finite-dimensionaland we are done.

If

$$U \neq \operatorname{span}(u_1,\dots,u_{k-1}),$$

then choose a vector $u_k \in U$ such that

$$u_k \notin \operatorname{span}(u_1,\dots,u_{k-1}).$$

After each step, as long as the process continues, we have constructed a list of vectors such that no vector in the list is in the span of the previous vectors. Hence, at each stage we maintain a linearly independent list (by the Linear Dependence Lemma (2.19)). No list of independent vectors in $V$ can exceed the length of any spanning list of $V$ (by 2.22). Thus, the process eventually terminates (after at most $m = \dim V$ steps), so $U$ has a finite basis. □

## Exercises about Span and Linear Independence 2A

*1* We want to find a list of four distinct vectors in $\mathbb{F}^3$ whose span equals

$$\Big\{\big(x,y,z\big) \in \mathbb{F}^3 \mid x+y+z=0\Big\}.$$

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*2* Claim: If $v_1,v_2,v_3,v_4$ spans $V$, then the list $v_1-v_2, v_2-v_3, v_3-v_4, v_4$ also spans $V$.

***Proof:*** Let $u \in V$ and $a_i, b_i \in \mathbb{F}$ such that

$$\begin{aligned} u &= a_1(v_1 - v_2) + a_2(v_2 - v_3) + a_3(v_3 - v_4) - a_4 v_4 \\ &= a_1 v_1 - a_1 v_2 + a_2 v_2 - a_2 v_3 + a_3 v_3 - a_3 v_4 + a_4 v_4 \\ &= \underbrace{a_1}_{b_1} v_1 + \underbrace{(a_2 - a_1)}_{b_2} v_2 + \underbrace{(a_3 - a_2)}_{b_3} v_3 + \underbrace{(a_4 - a_3)}_{b_4} v_4 \end{aligned}$$

$$\begin{aligned} \implies b_1 &= a_1 \\ b_2 &= a_2 - a_1 = a_2 - b_1 \iff a_2 = b_1 + b_2 \\ b_3 &= a_3 - a_2 = a_3 - b_1 - b_2 \iff a_3 = b_1 + b_2 + b_3 \\ b_4 &= a_4 - a_3 = a_4 - b_1 - b_2 - b_3 \iff a_4 = b_1 + b_2 + b_3 + b_4 \end{aligned}$$

So, if we have an arbitrary vector $u \in V$ such that

$$u = b_1 v_1 + b_2 v_2 + b_3 v_3 + b_4 v_4$$

it can be rewritten as a linear combination

$$u = b_1(v_1 - v_2) + (b_1 - b_2)(v_2 - v_3) + (b_1 + b_2 + b_3)(v_2 - v_4) + (b_1 + b_2 + b_3 + b_4)v_4$$

So, the list $v_1 - v_2, v_2 - v_3, v_3 - v_4, v_4$ also spans $V$. □

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**3** Claim: If $v_1, \ldots, v_m \in V$ and

$$w_k :\equiv v_1 + \cdots + v_k \quad \forall k \in \{1, \ldots, m\},$$

then $\mathsf{span}(v_1, \ldots, v_m) = \mathsf{span}(w_1, \ldots, w_m)$.

***Proof:*** Observe that

$$\begin{aligned} w_1 &= v_1 & v_1 &= w_1 \\ w_2 &= v_1 + v_2 & v_2 &= w_2 - v_1 = w_2 - w_1 \\ w_3 &= \boxed{v_1 + v_2 + v_3} & v_3 &= w_3 - v_1 - v_2 = w_3 - w_1 - w_2 + w_1 = w_3 - w_2 \\ w_4 &= v_1 + v_2 + v_3 + v_4 & v_4 &= w_4 - \boxed{(v_1 + v_2 + v_3)} = w_4 - w_3 \\ &\vdots & &\vdots \\ w_m &= v_1 + \cdots + v_m & v_m &= w_m - w_{m-1} \end{aligned}$$

***Step 1:*** If $u \in \mathsf{span}(v_1, \ldots, v_m)$ we have for some $a_1, \ldots, a_m \in \mathbb{F}$

$$\begin{aligned} u &= a_1 v_1 + \cdots + a_m v_m \\ &= a_1 w_1 + a_2(-w_1 + w_2) + a_3(-w_2 + w_3) + \cdots + a_m(-w_{m-1} + w_m) \\ &= (a_1 - a_2)w_1 + (a_2 - a_3)w_2 + \cdots + (a_{m-1} + a_m)w_{m-1} + a_m w_m \end{aligned}$$

Thus, $u \in \mathsf{span}(w_1, \ldots, w_m)$, and therefore,

$$\mathsf{span}(v_1, \ldots, v_m) \subseteq \mathsf{span}(w_1, \ldots, w_m). \tag{2.5}$$

***Step 2:*** If $u \in \mathsf{span}(w_1, \ldots, w_m)$ we have for some $a_1, \ldots, a_m \in \mathbb{F}$

$$\begin{aligned} u &= a_1 w_1 + \cdots + a_m w_m \\ &= a_1 v_1 + a_2(v_1 + v_2) + \cdots + a_m(v_1 + \cdots + v_m) \\ &= a_1 v_1 + (a_2 v_1 + a_2 v_2) + \cdots + (a_m v_1 + \cdots + a_m v_m) \\ &= (a_1 + \cdots + a_m)v_1 + (a_2 + \cdots + a_m)v_2 + \cdots + a_m v_m \end{aligned}$$

Thus, $u \in \mathsf{span}(v_1, \ldots, v_m)$ and therefore

$$\mathsf{span}(w_1, \ldots, w_m) \subseteq \mathsf{span}(v_1, \ldots, v_m). \tag{2.6}$$

(2.5) and (2.6) together imply that $\mathsf{span}(w_1, \ldots, w_m) = \mathsf{span}(v_1, \ldots, v_m)$. □

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**16** Explain why no list of four polynoials spans $\mathcal{P}_4(\mathbb{F})$.

***Solution:*** $\mathcal{P}_4(\mathbb{F})$ has 5 degrees of freedom and looks like this:

$$\mathcal{P}_4(\mathbb{F}) = \left\{a_0 + a_1 z + a_2 z^2 + a_3 z^3 + a_4 z^4 : a_0, \ldots, a4 \in \mathbb{F}\right\}.$$

If you span 4 polynomials, you can not get 5 degrees of freedom.

**20** Suppose $p_0, p_1, \ldots, p_m$ are polynomials in $\mathcal{P}_m(\mathbb{F})$ such that $p_k(2) = 0$ for $0 \le k \le m$. Prove that $p_0, p_1, \ldots, p_m$ is not linearly independent in $\mathcal{P}_m(\mathbb{F})$.

***Proof:*** Define an evaluation map $\varphi : \mathcal{P}_m(\mathbb{F}) \to \mathbb{F}$ such that

$$\varphi(p) := p(2) \quad \forall p \in \mathcal{P}(\mathbb{F}).$$

The polynomials that vanish at $z = 2$ lie in the subspace $\operatorname{null}(\varphi)$ of $\mathcal{P}_m(\mathbb{F})$. So, we have

$$\operatorname{null}(\varphi) \subseteq \mathcal{P}_m(\mathbb{F}).$$

Any polynomial $p \in \operatorname{null}(\varphi)$ can be written as

$$p(z) = (z-2)q(z),$$

where $q(z) \in \mathcal{P}_{m-1}(\mathbb{F})$. Therefore, $\dim \operatorname{null}(\varphi) = m$. Since $p_0, p_1, \ldots, p_m \in \operatorname{null}(\varphi)$, the given list is not linearly independent by 2.22, because it has length $m+1$. $\square$

## Bases (Section 2B)

**Def 2.26** (*basis*)**:**

A *"basis"* $\beta$ of $V$ is a list of vectors $\beta = v_1,\dots,v_n$ in $V$ that is linearly independent and spans $V$.

$$V = \mathrm{span}(v_1,\dots,v_n) = \mathrm{span}(\beta)$$

**Thm 2.28** (*condition for a basis*)**:**

$v_1,\dots,v_n$ is a basis of $V \iff$ every $v \in V$ can be written uniquely in the form

$$v = a_1v_1 + \cdots + a_nv_n, \quad \text{where } a_1,\dots,a_n \in \mathbb{F}.$$

Note that the actual definition of linear dependence has said nothing about uniqueness.

**Thm 2.30** (*every spanning list contains a basis*)**:**

Every spanning list $v_1,\dots,v_n$ of a vector space $V$ can be reduced to a basis of $V$. So, if $V = \mathrm{span}(v_1,\dots,v_n)$, then we can delete elements from that list, or leave the list unchanged, such that the list forms a basis.

***Proof:*** Suppose $V = \mathrm{span}(v_1,\dots,v_n)$. Let $B_0 :\equiv (v_1,\dots,v_n)$.

**Step 1:** If $v_1 = 0$, delete $v_1$ from $B_0$:

Therefore, let $B_1 :\equiv (v_2,\dots,v_n)$.

If $v_1 \neq 0$, then leave $B_0$ unchanged:

$B_1 :\equiv B_0 = (v_1,\dots,v_n)$.

**Step k:** Let $B_{k-1} = (\underbrace{v_1^*,\dots,v_{k-1}^*}_{\text{modified elements}},\underbrace{v_k,\dots,v_n}_{\text{original elements}})$ be our list from our previous steps, where $v_1^*,\dots,v_{k-1}^*$ are not necessarily the original elements $v_1,\dots,v_{k-1}$ from $B_0$. The $*$ indicates that those elements may have been modified in the previous steps.

If $v_k \in \mathrm{span}(v_1^*,\dots,v_{k-1}^*)$, then delete $v_k$ from list $B_{k-1}$ such that:

$B_k :\equiv (v_1^*,\dots,v_{k-1}^*,v_{k+1},\dots,v_n)$.

If $v_k \notin \mathrm{span}(v_1^*,\dots,v_{k-1}^*)$, leave the list unchanged:

$B_k :\equiv B_{k-1}$.

Stop the process after step $n$, getting a list $B_n$. The list $B_n$ spans $V$ because our original list $B_0$ spanned $V$, and we have discarded only vectors that were already in the span of the previous vectors. The process ensures that no vector in $B_n$ is in the span of the previous ones. Thus, $B_n$ is linearly independent. Hence, $B_n$ is a basis of $V$. □

**Thm 2.31** (*basis of finite-dimensional vector space*)**:**

Every finite-dimensional vector space has a basis.

***Proof:*** By definition, a finite-dimensional vector space has a spanning list. The previous result tells us that each spanning list can be reduced to a basis. □

**Thm 2.32:**

Every linearly independentlist of vectors in a finite-dimensional vector space can be extended to a basis of the vector space.

***Proof:*** Let $u_1,\dots,u_m \in V$ be linearly independent. Let $V = \mathrm{span}(w_1,\dots,w_n)$.

$\implies V = \mathrm{span}(u_1,\dots,u_m,w_1,\dots,w_n)$.

Applying the procedure of the proof of 2.30 to reduce this list to a basis of $V$ produces a basis consisting of vectors $u_1,\dots,u_m$ and some $w$'s. □

**Thm 2.33** (*every subspace of $V$ is part of a direct sum equal to $V$*)**:**
Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Then there is a subspace $W$ of $V$ such that $V = U \oplus W$.

Note: One may choose $W = V$ or $W = \{0\}$.

***Proof:*** Suppose $U$ is a subspace of a finite-dimensional vector space $V$. Therefore, by 2.25, $U$ is also finite-dimensional. Hence, there exist vectors $u_1, \dots, u_m \in U$ such that $u_1, \dots, u_m$ form a basis of $U$ by 2.31. We may extend $u_1, \dots, u_m$ to a basis $u_1, \dots, u_m, w_1, \dots, w_n$ of $V$, using 2.32. Set

$$W := \mathsf{span}(w_1, \dots, w_n).$$

By 1.46, to show that $V = U \oplus W$, it suffices to prove that $V = U + W$ and $U \cap W = \{0\}$.

***Proof of** $V = U + W$**:*** Let $v \in V$. Since $u_1, \dots, u_m, w_1, \dots, w_n$ is a basis of $V$, we may write

$$v = \underbrace{(a_1 u_1 + \cdots + a_m u_m)}_{u} + \underbrace{(b_1 w_1 + \cdots + b_n w_n)}_{w}$$

for some scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbb{F}$. With $u$ and $w$ defined as above, we have $v = u + w$, where $u \in U$ and $w \in W$. Hence, $v \in U + W$, and so $V \subseteq U + W$. Conversely, since $U, W \subseteq V$, we have $U + W \subseteq V$. Therefore, $V = U + W$.

***Proof of** $U \cap W = \{0\}$**:*** Let $v \in U \cap W$. Then, there exist scalars $a_1, \dots, a_m, b_1, \dots, b_n \in \mathbb{F}$ such that $v = a_1 u_1 + \cdots + a_m u_m$ and $v = b_1 w_1 + \cdots + b_n w_n$. Thus, $a_1 u_1 + \cdots + a_m u_m - b_1 w_1 - \cdots - b_n w_n = 0$. Linear independence of $u_1, \dots, u_m, w_1, \dots, w_n$ forces all $a_i = 0$ and $b_j = 0$. Therefore, $v$ must be zero, and $U \cap W = \{0\}$.

Hence, $V = U \oplus W$, as desired. □

## Exercises about Bases (Sub-Section 2B0)

6 Claim: If $v_1, v_2, v_3, v_4$ is a basis of $V$, then

$v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ is also a basis of $V$.

***Proof:*** ***Step 1:*** We first show linear independence. Let $a_1, a_2, a_3, a_4 \in \mathbb{F}$ such that

$$a_1(v_1 + v_2) + a_2(v_2 + v_3) + a_3(v_3 + v_4) + a_4 v_4 = 0.$$

This is the case if and only if

$$a_1 v_1 + a_1 v_2 + a_2 v_2 + a_2 v_3 + a_3 v_3 + a_3 v_4 + a_4 v_4 = a_1 v_1 + (a_1 + a_2) v_2 + (a_2 + a_3) v_3 + (a_3 + a_4) v_4 = 0$$

Since $v_1, \dots, v_4$ are linearly independent, this is only the case if $0 = a_1 = a_1 + a_2 = a_2 + a_3 = a_3 + a_4$ and therefore, since $a_1 = 0$, $\iff a_1 = \cdots = a_4 = 0$. So the list $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ is linearly independent as well.

***Step 2:*** Now we also have to show, that $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ spans $V$. Let $u \in V$. With our old basis, $u$ would have been written as

$$u = b_1 v_1 + \cdots + b_4 v_4. \tag{2.7}$$

So let $v \in V$ such that

$$\begin{aligned} v &= a_1(v_1 + v_2) + a_2(v_2 + v_3) + a_3(v_3 + v_4) + a_4 v_4 \\ &= a_1 v_1 + (a_1 + a_2) v_2 + (a_2 + a_3) v_3 + (a_3 + a_4) v_4 = 0 \end{aligned}$$

Comparing this with the equation of $u$ (2.7), we see that $v$ can be indeed any vector in $V$, since $v_1, v_2, v_3, v_4$ is a basis.

If we want to rewrite vector $u$ we would write $a_1 = b_1$. Next we would have $a_1 + a_2 = b_2$ and therefore $a_2 = b_2 - a_1 = b_2 - b_1$. Next we would have $a_2 + a_3 = b_3$ and thus $a_3 = b_3 - a_2 = b_3 - (b_2 - b_1)$.

Lastly $a_4 = b_4 - a_3 = b_4 - (b_3 - (b_2 - b_1))$. Putting this all together we have

$$\begin{aligned} u &= b_1 \cdot (v_1 + v_2) + (b_2 - b_1) \cdot (v_2 + v_3) + (b_3 - (b_2 - b_1)) \cdot (v_3 + v_4) + (b_4 - (b_3 - (b_2 - b_1))) \cdot v_4 \\ &= b_1(v_1 + v_2) + (b_2 - b_1)(v_2 + v_3) + (b_3 - b_2 + b_1)(v_3 + v_4) + (b_4 - b_3 + b_2 - b_1)v_4 \\ &= b_1 v_1 + b_2 v_2 + b_3 v_3 + b_4 v_4. \end{aligned}$$

which can be verified by comparing the coefficents of the $v$'s. For example, adding the two coefficients $(b_3 - b_2 + b_1)$ and $(b_4 - b_3 + b_2 - b_1)$ of $v_4$ together yields $b_4$. So every $u \in V$ can be expressed in either of the two lists $v_1, v_2, v_3, v_4$ or $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$. Since we have shown linear independence as well, $v_1 + v_2, v_2 + v_3, v_3 + v_4, v_4$ is a basis of $V$. □

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**7** Claim: If $v_1, v_2, v_3, v_4$ is a basis of $V$ and $U \subseteq V$, such that $v_1, v_2 \in U$ and $v_3 \notin U$ and $v_4 \notin U$, then $v_1, v_2$ is a basis of $U$.

***Proof:*** Any vector in $U$ can be represented as a unique linear combination of $v_1, v_2, v_3, v_4$ since $U \subseteq V$. Let $u \in U$ such that $u = a_1 v_1 + a_2 v_2 + a_3 v_3 + a_4 v_4$. In this case, $a_3$ must equal to 0, because otherwise, due to closedness under addition and multiplication, we could subtract $a_1 v_1 + a_2 v_2 + a_4 v_4$ from $u$ and multiply by $\frac{1}{a_3}$ to get $v_3$, which is not in $U$. The same goes for $a_4$, which must be 0 as well. So every vector in $U$ is of the form $u = a_1 v_1 + a_2 v_2$, since $v_1, v_2 \in U$ and $v_3, v_4 \notin U$. Since every basis has the same length and $v_1$ and $v_2$ are linearly independent, the list $v_1, v_2$ forms a basis of $U$. □

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**8** Suppose $v_1, \cdots, v_m$ is a list of vectors in $V$. For $k \in \{1, \ldots, m\}$, let

$$w_k = v_1 + \cdots + v_k$$

Show that $v_1, \ldots, v_m$ is a basis of $V$ if and only if $w_1, \ldots, w_m$ is a basis of $V$.

***Proof:*** Let us first observe the following two patterns

$$\begin{aligned} w_1 &= v_1 & v_1 &= w_1 \\ w_2 &= v_1 + v_2 & v_2 &= w_2 - w_1 \\ w_3 &= v_1 + v_2 + v_3 & v_3 &= w_3 - w_2 \\ w_4 &= v_1 + v_2 + v_3 + v_4 & v_4 &= w_4 - w_3 \\ &\vdots & &\vdots \\ w_n &= v_1 + \cdots + v_n & v_n &= w_n - w_{n-1} \end{aligned}$$

□

## Dimension (Section 2C)

**Thm 2.34** (*basis length*):
Any two bases of a finite-dimensional vector space have the same length. So, the basis length does not depend on the basis.

***Proof:*** Let $B_1$ and $B_2$ be 2 bases of $V$. Since both are linearly independent and both span $V$, we have $|B_1| \leq |B_2|$ and $|B_2| \leq |B_1|$ by using 2.22. Therefore, $|B_1| = |B_2|$. □

**Def 2.35** (*dimension*):
$\dim V :\equiv$ length of any basis of $V$.

**Thm 2.37** (*dimension of a subspace*):
If $V$ is finite-dimensionaland $U$ is a subspace of $V$, then

$$\dim U \leq \dim V.$$

***Proof:*** Since the length of every linearly independent list is smaller or equal to every spanning list in $V$ according to 2.22, we have the situation that the length of the basis of $U$ is less than or equal to the length of the basis of $V$. □

**Thm 2.38** (*linearly independent list of the right length*):
Every linearly independent list of vectors of length $\dim V$ is a basis.

***Proof:*** Let $\dim V = n$ and $v_1, \ldots, v_n$ be linearly independent in $V$. This list can be extended to a basis of $V$ by 2.32. However, every basis of $V$ has length $n$, so no elements are adjoined to $v_1, \ldots, v_n$ by such an extension process. □

**Thm 2.39** (*subspace of full dimension*):
If $U$ is a subspace of $V$ and $\dim U = \dim V \neq \infty$, then $U = V$.

***Proof:*** Let $u_1, \ldots, u_n$ be a basis of $U$. Thus, $n = \dim U$, and by the hypothesis, we also have $n = \dim V$. Thus, $u_1, \ldots, u_n$ is a linearly independent list of vectors in $V$ of length $\dim V$. So, by our previous theorem 2.38, we see that $u_1, \ldots, u_n$ is a basis of $V$. In particular, every vector in $V$ is a linear combination of $u_1, \ldots, u_n$. Thus, $U = V$. □

**Thm 2.42** (*spanning list of the right length*):
Every spanning list of vectors in $V$ ($\dim V \neq \infty$) with length $\dim V$ is a basis of $V$.

***Proof:*** Suppose $\dim V = n$ and $v_1, \ldots, v_n$ spans $V$. The list $v_1, \ldots, v_n$ can be reduced to a basis of $V$ by 2.30. But our list already has length $n$, so no elements will get deleted. Thus, $v_1, \ldots, v_n$ is a basis of $V$, as desired. □

**Thm 2.43** (*dimension of a sum*):
If $V_1$ and $V_2$ are subspaces of a finite-dimensional vector space $V$, then

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

***Proof:*** Let $r_1, \ldots, r_n \in V$ be a basis of $V_1 \cap V_2$, where $n \in \mathbb{N}_{\geq 0}$. This means the basis can also be the empty list, and $V_1 \cap V_2$ might equal $\{\vec{0}\}$. Note that the empty list is defined to be linearly independent, and the span of the empty list is defined to be equal to $\{\vec{0}\}$. By this definition, we have

$$\dim(V_1 \cap V_2) = n.$$

Since $r_1, \ldots, r_n$ is linearly independent in $V$ as well as in $V_1$ and $V_2$, this list can be extended to a basis of $V_1$ and $V_2$. So let

- $r_1, \ldots, r_n, u_1, \ldots, u_j$ be a basis of $V_1$, and
- $r_1, \ldots, r_n, w_1, \ldots, w_k$ be a basis of $V_2$ for $j, k \in \mathbb{N}_{\geq 0}$.

Then we have $\mathsf{dim}(V_1) = n + j$ and $\mathsf{dim}(V_2) = n + k$. Note that the extensions might be empty ($j, k \in \mathbb{N}_{\geq 0}$), meaning $V_1$ or $V_2$ could coincide with $V_1 \cap V_2$. If both subspaces equal $V_1 \cap V_2$, this would imply $V_1 = V_2$. The reader can easily verify that the formula still holds in this case: $\mathsf{dim}(V_2 + V_2) = \mathsf{dim}(V_2) = 2 \cdot \mathsf{dim}(V_2) - \mathsf{dim}(V_2 \cap V_2)$.

We will show that

$$r_1, \ldots, r_n, u_1, \ldots, u_j, w_1, \ldots, w_k \tag{2.8}$$

is a basis of $V_1 + V_2$. This will complete the proof because then we will have

$$\begin{aligned}\mathsf{dim}(V_1 + V_2) &= n + j + k \\ &= (n + j) + (n + k) - n \\ &= \mathsf{dim}(V_1) + \mathsf{dim}(V_2) - \mathsf{dim}(V_1 \cap V_2).\end{aligned}$$

The list (2.8) is contained in $V_1 + V_2$. The span of this list contains $V_1$ and $V_2$; therefore, it equals $V_1 + V_2$. Thus, to show that (2.8) is a basis of $V_1 + V_2$, we only need to show that it is linearly independent. Therefore, let $a_1, \ldots, a_n, b_1, \ldots, b_j, c_1, \ldots, c_k \in \mathbb{F}$, such that

$$0 = \underbrace{a_1 r_1 + \cdots + a_n r_n}_{\substack{\in V_1 \cap V_2 \\ \text{(common intersection)}}} + \underbrace{b_1 u_1 + \cdots + b_j u_j}_{\substack{\in V_1 \\ \text{(extension for } V_1\text{)}}} + \underbrace{c_1 w_1 + \cdots + c_k w_k}_{\substack{\in V_2 \\ \text{(extension for } V_2\text{)}}}. \tag{2.9}$$

We need to prove that all the $a$'s, $b$'s, and $c$'s equal 0. The equation above can be rewritten as

$$c_1 w_1 + \cdots + c_k w_k = -a_1 r_1 - \cdots - a_n r_n - b_1 u_1 - \cdots - b_j u_j,$$

which shows that $c_1 w_1 + \cdots + c_k w_k \in V_1$ because $r_1, \ldots, r_n, u_1, \ldots, u_j$ is a basis of $V_1$. But all the $w$'s are in $V_2$ as well, so this implies that $c_1 w_1 + \cdots + c_k w_k \in V_1 \cap V_2$. Because $r_1, \ldots, r_n$ is a basis of $V_1 \cap V_2$, we have

$$c_1 w_1 + \cdots + c_k w_k = d_1 r_1 + \cdots + d_n r_n, \text{ for some } d_1, \ldots, d_n \in \mathbb{F} \text{ (which are newly introduced here).}$$

But $r_1, \ldots, r_n, w_1, \ldots, w_k$ is linearly independent, because it is a basis of $V_2$, and the equation above can be rewritten as $0 = c_1 w_1 + \cdots + c_k w_k - d_1 r_1 - \cdots - d_n r_n$, so all the $c$'s and $d$'s must equal 0. Thus, (2.9) becomes the equation

$$a_1 r_1 + \cdots + a_n r_n + b_1 u_1 + \cdots + b_j u_j = 0.$$

Because $r_1, \ldots, r_n, u_1, \ldots, u_j$ is linearly independent, the equation above implies that all the $a$'s and $b$'s are 0. So we have shown that in equation (2.9) all coefficients ($a$, $b$, $c$) are 0. In symbols: $a_1 = 0, \ldots, a_n = 0, b_1 = 0, \ldots, b_j = 0, c_1 = 0, \ldots, c_k = 0$, completing the proof. □

## EXERCISES ABOUT DIMENSION 2C (SUB-SECTION 2C0)

**20** Explain why you might guess, motivated by analogy with the formula for the number of elements in the union of three finite sets, that if $V_1$, $V_2$, $V_3$ are subspaces of a finite-dimensional vector space $V$, then

$$\begin{aligned}\mathsf{dim}(V_1 + V_2 + V_3) = \mathsf{dim}(V_1) &+ \mathsf{dim}(V_2) + \mathsf{dim}(V_3) \\ &- \mathsf{dim}(V_1 \cap V_2) - \mathsf{dim}(V_1 \cap V_3) - \mathsf{dim}(V_2 \cap V_3) \\ &+ \mathsf{dim}(V_1 \cap V_2 \cap V_3).\end{aligned}$$

Then either prove the formula above or give a counterexample.

***SOLUTION:*** The formula is **wrong**. Consider the following subspaces of $\mathbb{R}^2$:

- $V_1 := \{(x,0) \mid x \in \mathbb{R}\}$ (the $x$-axis)
- $V_2 := \{(x,x) \mid x \in \mathbb{R}\}$ (the line $y = x$)
- $V_3 := \{(0,y) \mid y \in \mathbb{R}\}$ (the $y$-axis)

The right-hand side of the formula gives: $1+1+1-0-0-0+0 = 3$, whereas the left-hand side is $\dim(V_1+V_2+V_3) = 2$. So the formula **does not hold** for this example.

However, **it does hold** in every case where one of the subspaces is contained in another; hence, without loss of generality, assume that $V_1 \subseteq V_3$. For this situation, we have the following lemma.

***Lemma:*** If $V_1, V_2, V_3$ are finite-dimensional subspaces of a vector space, and $V_1 \subseteq V_3$, then

$$(V_1+V_2)\cap V_3 = (V_1\cap V_3)+(V_2\cap V_3)$$

***Proof:*** If $v \in (V_1\cap V_3)+(V_2\cap V_3)$, then we can write $v = u+w$, where $u \in V_1\cap V_3$ and $w \in V_2\cap V_3$. Because $V_3$ is closed under addition and both $u$ and $w$ are in $V_3$, so is $v$. Hence, $v \in V_3$ and $v = u+w$, such that $u \in V_1$ and $w \in V_2$. Therefore, $v \in (V_1+V_2)\cap V_3$. We conclude:

$$(V_1\cap V_3)+(V_2\cap V_3) \subseteq (V_1+V_2)\cap V_3 \tag{2.10}$$

Conversely, if $v \in (V_1+V_2)\cap V_3$, then we can write $v = u+w$, where $u \in V_1$, $w \in V_2$, and $v \in V_3$. Since $V_1 \subseteq V_3$ (by our assumption), we can rewrite

$$w = \underbrace{v}_{\in V_3} - \underbrace{u}_{\in V_3}.$$

Since both $u$ and $v$ belong to $V_3$, so does $w$. Therefore, $v = u+w$ such that $u \in V_1\cap V_3$ and $w \in V_2\cap V_3$. Hence, $v \in (V_1\cap V_3)+(V_2\cap V_3)$. We conclude:

$$(V_1+V_2)\cap V_3 \subseteq (V_1\cap V_3)+(V_2\cap V_3) \tag{2.11}$$

Inclusion (2.10) and (2.11) together yield

$$(V_1+V_2)\cap V_3 = (V_1\cap V_3)+(V_2\cap V_3)$$

for $V_1 \subseteq V_3$. □

Using the standard dimension formula for the sum of two subspaces (2.43), we obtain

$$\begin{aligned}\dim(V_1+V_2+V_3) &= \dim((V_1+V_2)+V_3)\\ &= \dim(V_1+V_2)+\dim(V_3)-\dim((V_1+V_2)\cap V_3)\\ &= \dim(V_1)+\dim(V_2)+\dim(V_3)\\ &\quad -\dim(V_1\cap V_2)-\dim((V_1+V_2)\cap V_3)\end{aligned} \tag{2.12}$$

Using the assumption that $V_1 \subseteq V_3$, we calculate:

$$\begin{aligned}\dim((V_1+V_2)\cap V_3) &= \dim((V_1\cap V_3)+(V_2\cap V_3))\\ &= \dim(V_1\cap V_3)+\dim(V_2\cap V_3)-\underbrace{\dim((V_1\cap V_3)\cap(V_2\cap V_3))}_{=\dim((V_1\cap V_2\cap V_3))}\end{aligned}$$

If we plug in this result in the first formula (2.12), we have

$$\begin{aligned}\dim(V_1+V_2+V_3) &= \dim(V_1)+\dim(V_2)+\dim(V_3)-\dim(V_1\cap V_2)-\dim((V_1+V_2)\cap V_3)\\ &= \dim(V_1)+\dim(V_2)+\dim(V_3)-\dim(V_1\cap V_2)\\ &\quad -(\dim(V_1\cap V_3)+\dim(V_2\cap V_3)-\dim((V_1\cap V_2\cap V_3)))\\ &= \dim(V_1)+\dim(V_2)+\dim(V_3)\\ &\quad -\dim(V_1\cap V_2)-\dim(V_1\cap V_3)-\dim(V_2\cap V_3)\\ &\quad +\dim(V_1\cap V_2\cap V_3).\end{aligned}$$

So, under certain conditions (i.e. $V_1 \subseteq V_3$), the formula is **true**.

**21** Show that for a finite-dimensional vector space $V$ with subspaces $V_1$, $V_2$, $V_3$ the following formula holds:

$$\begin{aligned}\dim(V_1 + V_2 + V_3) = {} & \dim(V_1) + \dim(V_2) + \dim(V_3) \\ & - \frac{\dim(V_1 \cap V_2) + \dim(V_1 \cap V_3) + \dim(V_2 \cap V_3)}{3} \\ & - \frac{\dim\big((V_1 + V_2) \cap V_3\big) + \dim\big((V_1 + V_3) \cap V_2\big) + \dim\big((V_2 + V_3) \cap V_1\big)}{3}\end{aligned}$$

***Solution:*** Recall that for any two subspaces $U$ and $W$ of a finite-dimensional vector space we have

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

So we can use associativity and commutativity of the sum $V_1 + V_2 + V_3$. In particular:

$$\begin{aligned}V_1 + V_2 + V_3 &= (V_1 + V_2) + V_3 \\ &= V_1 + (V_2 + V_3) \\ &= (V_1 + V_3) + V_2\end{aligned}$$

$$\begin{aligned}\dim((V_1 + V_2) + V_3) &= \dim(V_1 + V_2) + \dim(V_3) - \dim((V_1 + V_2) \cap V_3) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3) \\ \dim(V_1 + (V_2 + V_3)) &= \dim(V_2 + V_3) + \dim(V_1) - \dim((V_2 + V_3) \cap V_1) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3) - \dim((V_2 + V_3) \cap V_1) \\ \dim((V_1 + V_3) + V_2) &= \dim(V_1 + V_3) + \dim(V_2) - \dim((V_1 + V_3) \cap V_2) \\ &= \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_3) - \dim((V_1 + V_3) \cap V_2)\end{aligned}$$

Therefore, adding the three identities yields

$$\begin{aligned}3\dim(V_1 + V_2 + V_3) = {} & \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_2) - \dim((V_1 + V_2) \cap V_3) \\ & + \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_2 \cap V_3) - \dim((V_2 + V_3) \cap V_1) \\ & + \dim V_1 + \dim V_2 + \dim V_3 - \dim(V_1 \cap V_3) - \dim((V_1 + V_3) \cap V_2) \\ = {} & 3\dim V_1 + 3\dim V_2 + 3\dim V_3 \\ & - \dim(V_1 \cap V_2) - \dim(V_2 \cap V_3) - \dim(V_1 \cap V_3) \\ & - \dim((V_2 + V_3) \cap V_1) - \dim((V_1 + V_3) \cap V_2) - \dim((V_2 + V_3) \cap V_1)\end{aligned}$$

Dividing both sides by 3 gives the formula in question.

# Linear Maps (Chapter 3)

## Vector Space of Linear Maps (Section 3A)

### Definition and Examples of Linear Maps (Sub-Section 3A1)

**Def 3.1** (*linear map*):

A linear mapfrom $V$ to $W$ is a function $T : V \to W$ with the following properties:

- **additivity:** $T(u+v) = Tu + Tv \qquad \forall u, v \in V$
- **homogeneity:** $T(\lambda v) = \lambda(Tv) \qquad \forall \lambda \in \mathbb{F},\ \forall v \in V$

**Def 3.2** (*notation $\mathcal{L}(V,W)$, $\mathcal{L}(V)$*):

The set of all linear maps from $V$ to $W$ is denoted by $\mathcal{L}(V,W)$. For the set of all linear maps from $V$ to itself, we use $\mathcal{L}(V) :\equiv \mathcal{L}(V,V)$.

**Thm 3.4** (*linear map lemma*):

Suppose $v_1,\ldots,v_n$ is a basis of $V$ and $w_1,\ldots,w_n \in W$, where the $w$'s do not have any specific properties. Then there exists a unique linear map $T \in \mathcal{L}(V,W)$ such that

$$Tv_k = w_k \quad \forall k \in \{1,\ldots,n\}.$$

It is of the form

$$T(c_1v_1 + \cdots + c_nv_n) = c_1w_1 + \cdots + c_nw_n.$$

***Proof:*** Since the list $v_1,\ldots,v_n$ is a basis of $V$, every $v \in V$ has a unique expansion

$$v = c_1v_1 + \cdots + c_nv_n,$$

where $c_1,\ldots,c_n \in \mathbb{F}$. Define $T : V \to W$ by

$$T(c_1v_1 + \cdots + c_nv_n) := c_1w_1 + \cdots + c_nw_n.$$

By uniqueness of the basis expansion, this rule defines a well-defined function $T\colon V \to W$ (otherwise a single $v$ could receive different images). In particular, choosing $v = v_k$ (so $c_k = 1$, $c_i = 0$ for $i \neq k$) gives

$$T(v_k) = w_k.$$

Next, we verify that $T$ is linear. Let $u = a_1v_1 + \cdots + a_nv_n \in V$ and $v = b_1v_1 + \cdots + b_nv_n \in V$, and let $\lambda \in \mathbb{F}$. Then,

$$\begin{aligned} T(u+v) &= T((a_1+b_1)v_1 + \cdots + (a_n+b_n)v_n) \\ &= (a_1+b_1)w_1 + \cdots + (a_n+b_n)w_n \\ &= (a_1w_1 + \cdots + a_nw_n) + (b_1w_1 + \cdots + b_nw_n) \\ &= T(u) + T(v), \end{aligned}$$

and similarly,

$$\begin{aligned} T(\lambda v) &= T(\lambda b_1v_1 + \cdots + \lambda b_nv_n) \\ &= \lambda b_1w_1 + \cdots + \lambda b_nw_n \\ &= \lambda(b_1w_1 + \cdots + b_nw_n) \\ &= \lambda T(v). \end{aligned}$$

Hence, $T$ is linear, so $T \in \mathcal{L}(V,W)$.

Finally, to prove uniqueness, suppose $S \in \mathcal{L}(V,W)$ is any linear map such that $S(v_k) = w_k$ for each $k = 1, \dots, n$. Then, for any $v = c_1v_1 + \cdots + c_nv_n \in V$, we have

$$\begin{aligned} S(v) &= c_1S(v_1) + \cdots + c_nS(v_n) \\ &= c_1w_1 + \cdots + c_nw_n \\ &= T(v), \end{aligned}$$

so $S = T$. Therefore, the unique linear map sending each $v_k$ to $w_k$ (for $k \in \{1, \dots, n\}$) is $T$. □

**DEF 3.5** (*addition and scalar multiplication on $\mathcal{L}(V, W)$*):
Suppose $S, T \in \mathcal{L}(V, W)$ and $\lambda \in \mathbb{F}$.
The sum $S + T \in \mathcal{L}(V, W)$ and the product $\lambda T \in \mathcal{L}(V, W)$ are defined by:

- $(S + T)(v) := Sv + Tv \quad \forall v \in V$
- $(\lambda T)(v) := \lambda(Tv) \quad \forall v \in V$

## ALGEBRAIC OPERATIONS ON $\mathcal{L}(V, W)$ (SUB-SECTION 3A2)

**THM 3.6** ($\mathcal{L}(V, W)$):
With these operations of addition and scalar multiplication as defined above, $\mathcal{L}(V, W)$ is a vector space.

**DEF 3.7**:
If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, then the *"product"* $ST \in \mathcal{L}(U, W)$ is defined by

$(ST)(u) := S(Tu) \quad \forall u \in U.$

Thus, $ST$ is just the usual composition $S \circ T$ of two functions. But with linear maps, we have distributive and associative properties (see next result), so the notation seems natural.

**THM 3.8** (*algebraic properties of products of linear maps*):
With these definitions, we have:

- **ASSOCIATIVITY:** $(T_1T_2)T_3 = T_1(T_2T_3)$, whenever $T_3$ maps into the domain of $T_2$ and $T_2$ maps into the domain of $T_1$.
- **IDENTITY:** $TI = IT = T$ for $T \in \mathcal{L}(V, W)$ (The first $I$ is the identity operator on $V$ and the second $I$ is the identity operator on $W$. We could also write $TI_V = I_WT$).
- **DISTRIBUTIVE PROPERTIES:** For $T, T_1, T_2 \in \mathcal{L}(U, V)$ and $S, S_1, S_2 \in \mathcal{L}(V, W)$:
  $(S_1 + S_2)T = S_1T + S_2T$ and $S(T_1 + T_2) = ST_1 + ST_2$.
- **NON-COMMUTATIVE:** $ST \neq TS$ in general.

**THM 3.10** (*linear maps take* $0$ *to* $0$):
$T \in \mathcal{L}(V, W) \implies T(0) = 0.$

***PROOF:*** $T(0) = T(0 + 0) = T(0) + T(0)$. Now subtract $T(0)$ on both sides. □

## EXERCISES ABOUT VECTOR SPACE OF LINEAR MAPS 3A

1. Show that every linear map from a one-dimensional vector space to itself is a multiplication by some scalar. More precisely, prove that if $\dim V = 1$ and $T \in \mathcal{L}(V)$, then there exists $\lambda in \mathbb{F}$ such that $Tv = \lambda v$ for all $v \in V$.

   ***PROOF:*** If $T = 0$, take $\lambda = 0$ and there's nothing more to show. Otherwise, because $T \neq 0$ there exists $v_1 \in V$ with $T(v_1) \neq 0$. Since $\dim V = 1$, $\{v_1\}$ is a basis of $V$, so every $v \in V$ can be written uniquely as $v = a_1v_1$, $a_1 \in \mathbb{F}$. Because $T(v_1)$ lies in the one-dimensional space $V = \operatorname{span}(v_1)$, there is a unique scalar $\lambda \in \mathbb{F}$ such that

$$T(v_1) = \lambda v_1.$$

Then for any $v = a_1 v_1$,

$$T(v) = T(a_1 v_1) = a_1 T(v_1) = a_1(\lambda v_1) = \lambda(a_1 v_1) = \lambda v$$

Hence, $T$ acts by multiplication by $\lambda$ on all of $V$, as desired. $\square$

**12** Suppose $U$ is a subspace of $V$ with $U \neq V$. Let $S \in \mathcal{L}(U, W)$ and $S \neq 0$, which means that $Su \neq 0$ for some $u \in U$. Define $T : V \to W$ by

$$Tv :\equiv \begin{cases} Sv & \text{if } v \in U, \\ 0 & \text{if } v \in V \text{ and } v \notin U \quad (v \in V \backslash U) \end{cases}$$

Prove that $T$ is not a linear map on $V$.

***Proof:*** Let $u \in U$ and $v \in V$ with $v \notin U$ (i.e. $v \in V \backslash U$). Consider the sum $u + v$. We claim that $u + v \notin U$, because if $u + v \in U$ then by closure of $U$, we would have $(u + v) - u = v \in U$, which contradicts $v \notin U$. Therefore, it holds that

$$T(\underbrace{u + v}_{\notin U}) = 0.$$

Whereas

$$Tu + Tv = Su + 0.$$

We know that $Su \neq 0$ for some $u \in U$. This demonstrates that $T$ is not a linear map because it fails to satisfy additivity. The failure of $T$ to be a linear map arises because $T$ is defined differently on $U$ and $V \backslash U$, breaking the additivity required for linearity. $\square$

**13** Suppose $V$ is finite-dimensional. Prove that every linear map on a subspace of $V$ can be extended to a linear map on $V$. In other words, show that if $U$ is a subspace of $V$ and $S \in \mathcal{L}(U, W)$, then $\exists T \in \mathcal{L}(V, W)$ such that

$$Tu = Su, \ \ \forall u \in U.$$

***Proof:*** Let $U$ be a subspace of a finite-dimensional vector space $V$, and let $S \in \mathcal{L}(U, W)$. Choose a subspace $X \subseteq V$ with $V = U \oplus X$. Let $u_1, \ldots, u_m$ be a basis of $U$ and let $x_{m+1}, \ldots, x_n$ be a basis of $X$. This makes

$$u_1, \ldots, u_m, x_{m+1}, \ldots, x_n$$

a basis of $V$. Every $v \in V$ has a unique representation

$$v = c_1 u_1 + \cdots + c_m u_m + c_{m+1} x_{m+1} + \cdots + c_n x_n.$$

Define $T \in \mathcal{L}(V, W)$ as follows:

$$T(v) := c_1 S u_1 + \cdots + c_m S u_m.$$

Clearly, $Tu = Su$ for all $u \in U$. It remains only to check linearity. By defining $T(u_i) = S(u_i)$ for $1 \leq i \leq m$, and $T(x_j) = 0$ for $m + 1 \leq j \leq n$, we already ensured that $T$ is extended linearly on every basis vector of $V$. For illustration, we perform one more check of linearity. Let $v, v' \in V, \lambda, \lambda' \in \mathbb{F}$, and $a_i, b_i$ be scalars in $\mathbb{F}$ (for $i = 1, \ldots, n$) such that $v = \sum_{i=1}^{m} a_i u_i + \sum_{i=m+1}^{n} a_i x_i$ and

$v' = \sum_{i=1}^{m} b_i u_i + \sum_{i=m+1}^{n} b_i x_i$. We calculate

$$\begin{aligned} T\left(\lambda v + \lambda' v'\right) &= T\left(\lambda\left(\sum_{i=1}^{m} a_i u_i + \sum_{i=m+1}^{n} a_i x_i\right) + \lambda'\left(\sum_{i=1}^{m} b_i u_i + \sum_{i=m+1}^{n} b_i x_i\right)\right) \\ &= \lambda\left(\sum_{i=1}^{m} T\left(a_i u_i\right) + \sum_{i=m+1}^{n} \underbrace{T\left(a_i x_i\right)}_{=0}\right) + \lambda'\left(\sum_{i=1}^{m} T\left(b_i u_i\right) + \sum_{i=m+1}^{n} \underbrace{T\left(b_i x_i\right)}_{=0}\right) \\ &= \lambda\left(\sum_{i=1}^{m} S\left(a_i u_i\right)\right) + \lambda'\left(\sum_{i=1}^{m} S\left(b_i u_i\right)\right) \\ &= \lambda\left(\sum_{i=1}^{m} S\left(a_i u_i\right) + \sum_{i=m+1}^{n} T\left(a_i x_i\right)\right) + \lambda'\left(\sum_{i=1}^{m} S\left(b_i u_i\right) + \sum_{i=m+1}^{n} T\left(b_i x_i\right)\right) \\ &= \lambda T(v) + \lambda' T(v'). \end{aligned}$$

Hence, $T \in \mathcal{L}(V, W)$ is linear and extends $S$, as required. □

**15** Suppose $v_1, \ldots, v_m$ is a linearly dependent list of vectors in $V$. Suppose also that $W \neq \{0\}$. Prove that there exists $w_1, \ldots, w_m$ in $W$ such that no $T \in \mathcal{L}(V, W)$ satisfies

$$T v_k = w_k \quad (\text{for } k = 1, \ldots, m).$$

***Solution:*** Let $v_1, \ldots, v_m \in V$ be a linearly dependent list and let $T \in \mathcal{L}(V, W)$ be a linear map satisfying $T v_j = w_j$ (for $j = 1, \ldots, m$) for some $w_1, \ldots, w_m \in W$. Since $v_1, \ldots, v_m$ is a linearly dependent list, there exists a $k \in \{1, \ldots, m\}$ such that $v_k \in \operatorname{span}(v_1, \ldots, \cancel{v_k}, \ldots, v_m)$ (by 2.19). Hence, we write

$$v_k = \sum_{\substack{j=1 \\ j \neq k}}^{m} c_j v_j, \quad \text{where } c_j \in \mathbb{F}\, (j = 1, \ldots, m).$$

If we apply $T$, we get

$$T v_k = \sum_{\substack{j=1 \\ j \neq k}}^{m} T(c_j v_j) \iff w_k \quad = \sum_{\substack{j=1 \\ j \neq k}}^{m} c_j w_j.$$

This means that $w_1, \ldots, w_m$ must be a linearly dependent list as well and satisfy the equation above. Since $W \neq \{0\}$, choose a nonzero $w \in W$. If we now put instead $w_k := w$ and $w_j := 0$ (for $j \neq k$), every possible $T \in \mathcal{L}(V, W)$ would violate the linear dependence enforced by the list $v_1, \ldots, v_m$ with the same arguments as before (i.e., the forced relation $w_k = \sum_{j \neq k} c_j w_j$ becomes $w = 0$, a contradiction). With this choice of the $w_j$, no $T \in \mathcal{L}(V, W)$ can satisfy $T(v_j) = w_j$ for all $j \in \{1, \ldots, m\}$.

# Null Spaces and Ranges (Section 3B)

## Null Space and Injectivity (Sub-Section 3B1)

**Def 3.11** (*null space*, $\operatorname{null} T$):

$\operatorname{null} T :\equiv \ker T :\equiv \{v \in V \mid Tv = 0\} \subseteq V$ for $T \in \mathcal{L}(V, W)$.

**Thm 3.13** (*null space is a subspace*):

For $T = \mathcal{L}(V, W)$, $\operatorname{null} T$ is a subspace of $V$.

***Proof:*** We have that $0 \in \operatorname{null} T$, because $T(0) = 0$. Suppose $u, v \in \operatorname{null} T$ and $\lambda \in \mathbb{F}$. Then $T(u+v) = Tu + Tv = 0 + 0 = 0$. Hence, $u + v \in \operatorname{null} T$. Similarly, $T(\lambda u) = \lambda Tu = \lambda 0 = 0$, so $\lambda u \in \operatorname{null} T$. □

**Def 3.14** (*injectivity*):

A function $T : V \to W$ is called *"injective"* if

$$Tu = Tv \implies u = v \quad \forall u, v \in V.$$

Or analogously:

$$u \neq v \implies Tu \neq Tv \quad \forall u, v \in V.$$

**Thm 3.15** (*injectivity and null space property*):

Let $T \in \mathcal{L}(V, W)$. Then

$$T \text{ is injective} \iff \operatorname{null} T = \{0\}.$$

***Proof:*** $\Rightarrow$ ***direction:*** First, assume that $T$ is injective. Since $\operatorname{null} T$ is a subspace of $V$ (by 3.13), we have $\{0\} \subseteq \operatorname{null} T$. To prove that $\operatorname{null} T \subseteq \{0\}$, suppose $v \in \operatorname{null} T$. Consequently,

$$T(v) = 0.$$

Moreover, since linear maps send 0 to 0 (by 3.10), we have

$$T(0) = 0.$$

Since $T$ is injective, both equations imply that $v = 0$. Thus, we can conclude that $\operatorname{null} T = \{0\}$, as desired.

$\Leftarrow$ ***direction:*** Second, assume $\operatorname{null} T = \{0\}$. Let $u, v \in V$ such that $Tu = Tv$. Then

$$0 = Tu - Tv = T(u - v).$$

Thus, $u - v \in \operatorname{null} T = \{0\}$, so $u - v = 0$, which gives $u = v$. Therefore, $T$ is injective. □

## Range and Surjectivity (Sub-Section 3B2)

**Def 3.16** (*range*):

$\operatorname{range} T := \{Tv \mid v \in V\} \subseteq W$ for $T \in \mathcal{L}(V, W)$.

**Thm 3.18** (*the range is a subspace*):

If $T \in \mathcal{L}(V, W)$, then $\operatorname{range} T$ is a subspace of $W$.

***Proof:*** Let $T \in \mathcal{L}(V, W)$. By 3.10, $T(0) = 0$, so $0 \in \operatorname{range} T$. Thus, the additive identity 0 is part of $\operatorname{range} T$. If $w_1, w_2 \in \operatorname{range} T$, then there exist some $v_1, v_2 \in V$ such that $Tv_1 = w_1$ and $Tv_2 = w_2$, for which we have $w_1 + w_2 = Tv_1 + Tv_2 = T(v_1 + v_2) \in \operatorname{range} T$. Hence, $\operatorname{range} T$ is closed under addition. If $w \in \operatorname{range} T$ and $\lambda \in \mathbb{F}$, then there exists a $v \in V$ such that $w = Tv$. For $w$ it holds that $\lambda w = \lambda Tv = T(\lambda v) \in \operatorname{range} T$. Therefore, $\operatorname{range} T$ is closed under scalar multiplication. By 1.34, $\operatorname{range} T$ is a subspace of $W$, since we have checked for all 3 conditions. □

**Def 3.19** (*surjectivity*):
If range $T = W$, then $T : V \to W$ is called *"surjective"* or *"onto"*.

## Fundamental Theorem of Linear Maps (Sub-Section 3B3)

**Thm 3.21** (*Fundamental Theorem of Linear Maps or "Rank Nullity Theorem"*):
For $T \in \mathcal{L}(V, W)$, where $V$ is finite-dimensional, we have that range$(T)$ is also finite-dimensionaland

$$\dim V = \underbrace{\dim \operatorname{null} T}_{\text{nullity}} + \underbrace{\dim \operatorname{range} T}_{\text{rank}}.$$

Note that $T \in \mathcal{L}(V, W)$ is not a typo.

***Proof:*** Let $u_1, \dots, u_m$ be a basis of null $T$ with $\dim \operatorname{null} T = m$. Using 2.32, this basis, which forms a linearly independent list inside $V$, can be extended to a basis of $V$, which looks like this:

$u_1, \dots, u_m, v_1, \dots, v_n$ (we have been using the fact that $V$ is finite-dimensional.)

Thus, $\dim V = m + n$. To complete the proof, we need to show that range $T$ is finite-dimensional and $\dim \operatorname{range} T = n$. We will do this by proving that $Tv_1, \dots, Tv_n$ is a basis of range $T$.

Let $v \in V$ such that for $a_1, \dots, a_m \in \mathbb{F}$ and $b_1, \dots, b_n \in \mathbb{F}$:

$$v = (a_1 u_1 + \cdots + a_m u_m) + (b_1 v_1 + \cdots + b_n v_n).$$

If we apply $T$, we get $Tv = b_1 Tv_1 + \cdots + b_n Tv_n$. Since we can do this for every $v$, this implies that range $T = \operatorname{span}(Tv_1, \dots, Tv_n)$ and that range $T$ is finite-dimensional.

To show that $Tv_1, \dots, Tv_n$ is linearly independent, suppose $c_1, \dots, c_n \in \mathbb{F}$ such that

$$c_1 Tv_1 + \cdots + c_n Tv_n = 0.$$

Then $T(c_1 v_1 + \cdots + c_n v_n) = 0$. Hence, $c_1 v_1 + \cdots + c_n v_n \in \operatorname{null} T$. Because $u_1, \dots, u_m$ spans null $T$, we can write

$$c_1 v_1 + \cdots + c_n v_n = d_1 u_1 + \cdots + d_m u_m \quad \text{or}$$

$$0 = c_1 v_1 + \cdots + c_n v_n - d_1 u_1 - \cdots - d_m u_m,$$

where the $d$'s are in $\mathbb{F}$. This last equation implies that all the $c$'s and $d$'s are 0 because the $u$'s and $v$'s form a basis of $V$ and are therefore linearly independent. Thus, $Tv_1, \dots, Tv_n$ is linearly independent (because all $c$'s are 0) and hence is a basis of range $T$, as desired. □

**Thm 3.22** (*linear map to a lower-dimensional space is not injective*):
Suppose $V$ and $W$ are finite-dimensional vector spaces such that $\dim V > \dim W$. Then no linear map from $V$ to $W$ is injective.

***Proof:*** Let $T \in \mathcal{L}(V, W)$ such that $\dim V > \dim W$. Using 3.21, we have

$$\begin{aligned} \dim \operatorname{null} T &= \dim V - \dim \operatorname{range} T \\ &\geq \dim V - \dim W \\ &> 0. \end{aligned}$$

The first inequality follows from 2.37, because range $T$ is a subspace of $W$. The second inequality holds because we assumed $\dim V > \dim W$. Now, the equation tells us that $\dim \operatorname{null} T > 0$, implying null $T$ is not equal to $\{0\}$ (meaning that null $T$ contains vectors other than 0). Thus, $T$ is not injective by 3.15. □

**Thm 3.24** (*linear map to a higher-dimensional space is not surjective*):
Suppose $V$ and $W$ are finite-dimensional vector spaces such that $\dim V < \dim W$. Then no linear map

from $V$ to $W$ is surjective.

***Proof:*** Let $T \in \mathcal{L}(V, W)$ such that $\dim V < \dim W$. Then

$$\begin{aligned}\dim \operatorname{range} T &= \dim V - \dim \operatorname{null} T\\ &\leq \dim V\\ &< \dim W.\end{aligned}$$

Thus, we have $\dim \operatorname{range} T < \dim W$. So $\operatorname{range} T \neq W$. Hence, $T$ is not surjective by 3.19. □

**Thm 3.26** (*homogeneous system of linear equations*):
A homogeneous system of linear equations with more variables than equations has nonzero solutions.

***Proof:*** *"Homogeneous"* in this context means that the constant term on the right side of each equation below is 0. Fix positive integers $m$ and $n$, and let $A_{j,k} \in \mathbb{F}$ for $j = 1, \dots, m$ and $k = 1, \dots, n$. Consider:

$$\begin{aligned}A_{1,1}x_1 + A_{1,2}x_2 + \cdots + A_{1,n}x_n &= \sum_{k=1}^{n} A_{1,k}\, x_k = 0,\\ A_{2,1}x_1 + A_{2,2}x_2 + \cdots + A_{2,n}x_n &= \sum_{k=1}^{n} A_{2,k}\, x_k = 0,\\ &\vdots\\ A_{m,1}x_1 + A_{m,2}x_2 + \cdots + A_{m,n}x_n &= \sum_{k=1}^{n} A_{m,k}\, x_k = 0.\end{aligned}$$

Clearly, $x_1 = \cdots = x_n = 0$ is a solution. Define $T : \mathbb{F}^n \to \mathbb{F}^m$ by

$$T(x_1, \dots, x_n) := \left( \sum_{k=1}^{n} A_{1,k}\, x_k, \sum_{k=1}^{n} A_{2,k}\, x_k, \dots, \sum_{k=1}^{n} A_{m,k}\, x_k \right).$$

Thus, we want to know if $\operatorname{null} T$ is strictly bigger than $\{0\}$, which is equivalent to $T$ not being injective (by 3.15). By 3.22, this is the case if $\dim \mathbb{F}^n > \dim \mathbb{F}^m$, so if $n > m$. This is the case if we have more variables than equations. □

**Thm 3.28** (*system of linear equations with more equations than variables*):
An inhomogeneous system of linear equations with more equations than variables has no solutions for some choice of constant terms.

***Proof:*** We want to know whether a system of linear equations has no solutions for some choice of constant terms on the right-hand side. Fix positive integers $m$ and $n$, and let $A_{j,k} \in \mathbb{F}$ for $j = 1, \dots, m$ and $k = 1, \dots, n$. For $c_1, \dots, c_m \in \mathbb{F}$, consider:

$$\begin{aligned}A_{1,1}x_1 + A_{1,2}x_2 + \cdots + A_{1,n}x_n &= \sum_{k=1}^{n} A_{1,k}\, x_k = c_1,\\ &\vdots\\ A_{m,1}x_1 + A_{m,2}x_2 + \cdots + A_{m,n}x_n &= \sum_{k=1}^{n} A_{m,k}\, x_k = c_m.\end{aligned} \tag{3.1}$$

Define $T : \mathbb{F}^n \to \mathbb{F}^m$ by

$$T(x_1, \dots, x_n) := \left( \sum_{k=1}^{n} A_{1,k}\, x_k, \sum_{k=1}^{n} A_{2,k}\, x_k, \dots, \sum_{k=1}^{n} A_{m,k}\, x_k \right).$$

Therefore, we can rewrite (3.1) as $T(x_1, \dots, x_n) = (c_1, \dots, c_m)$. Hence, we want to know if $\operatorname{range} T = \mathbb{F}^m$ or $\operatorname{range} T \neq \mathbb{F}^m$. What condition ensures that $T$ is not surjective? By 3.24, this is the case if $\dim \mathbb{F}^n < \dim \mathbb{F}^m$ (i.e., if $n < m$). This means if we have more equations than variables in a system of linear equations, then there are no solutions for some constant terms. □

## Exercises about Null Spaces and Ranges 3B

(1) Exercise 1

(2) Suppose $S, T \in \mathcal{L}(V)$ are such that $\operatorname{range} S \subseteq \operatorname{null} T$. Prove that $(ST)^2 = 0$.

***Proof:*** For any $v \in V$, $(ST)^2(v) = (ST)(ST)(v) = ST(ST(v))$. Since $T(v) \in V$, we have $ST(v) = S(T(v)) \in \operatorname{range} S \subseteq \operatorname{null} T$, so $T(S(T(v))) = 0$. Thus, we compute

$$(ST)^2(v) = S(T(S(T(v)))) = S(0) = 0.$$

Therefore, $(ST)^2$ sends every $v$ to 0, i.e. $(ST)^2 = 0$. □

(8) Suppose $T \in \mathcal{L}(V, W)$ is injective and $v_1, \dots, v_n$ is linearly independent in $V$. Prove that $Tv_1, \dots, Tv_n$ is linearly independent in $W$.

***Proof:*** Let $a_1, \dots, a_n \in \mathbb{F}$ such that $a_1 Tv_1 + \cdots + a_n Tv_n = 0$. By the linearity of $T$, we can rewrite this equation as $T(a_1 v_1 + \cdots + a_n v_n) = 0$. Since $T$ is injective, and $T(0) = 0$ by 3.10, $T(u) = 0$ if and only if $u = 0$. Thus, $a_1 v_1 + \cdots + a_n v_n = 0$. Because $v_1, \dots, v_n$ are linearly independent vectors in $V$, this forces $a_1 = 0, \dots, a_n = 0$. Hence, $Tv_1, \dots, Tv_n$ is a linearly independent list of vectors in $W$. □

(9) Suppose $v_1, \dots, v_n$ spans $V$ and $T \in \mathcal{L}(V, W)$. Show that $Tv_1, \dots, Tv_n$ spans $\operatorname{range} T$.

***Proof:*** Let $v = a_1 v_1 + \cdots + a_n v_n \in V$, where $a_1, \dots, a_n \in \mathbb{F}$. If we apply $T$, we get

$$Tv = a_1 Tv_1 + \cdots + a_n Tv_n \in \operatorname{range} T.$$

Thus, every element of $Tv \in \operatorname{range} T$ can be written as a linear combination of $Tv_1, \dots, Tv_n$. Hence,

$$\operatorname{range} T \subseteq \operatorname{span}(Tv_1, \dots, Tv_n).$$

Conversely, $Tv_1, \dots, Tv_n \in \operatorname{range} T$ and $\operatorname{range} T$ is a subspace of $V$ which is closed under addition and multiplication. Therefore, $\operatorname{span}(Tv_1, \dots, Tv_n) \subseteq \operatorname{range} T$ and hence, $\operatorname{range} T = \operatorname{span}(Tv_1, \dots, Tv_n)$. □

(12) Suppose $T$ is a linear map from $\mathbb{F}^4$ to $\mathbb{F}^2$ such that

$$\operatorname{null} T :\equiv \{(x_1, x_2, x_3, x_4) \in \mathbb{F} \mid x_1 = 5x_2 \text{and} x_3 = 7x_4\}.$$

Prove that $T$ is surjective.

***Proof:*** To prove that $T$ is surjective, we have to find the dimension of $\operatorname{null} T$. We can rewrite:

$$\begin{aligned}\operatorname{null} T &= \{(5a, a, 7b, b) \mid a \in \mathbb{F} \text{ and } b \in \mathbb{F}\}\\ &= \{t_1(5,1,0,0) + t_2(0,0,7,1) \mid t_1 \in \mathbb{F} \text{ and } t_2 \in \mathbb{F}\}.\end{aligned}$$

Which shows that $\operatorname{null} T$ is a 2-dimensional plane in $\mathbb{F}^4$, since $(5,1,0,0)$ and $(0,0,7,1)$ are obviously linearly independent. According to the rank-nullity theorem, we have $\dim \mathbb{F}^4 = \dim \operatorname{null} T + \dim \operatorname{range} T$, and thus we derive that $\dim \operatorname{range} T = 2$. Theorem 2.39 states that if a subspace of a vector space has the same dimension as the vector space itself, then the subspace equals the whole space. In this case, it follows that $\operatorname{range} T = \mathbb{F}^2$. Therefore, $T$ is surjective. □

(13) Suppose $U$ is a three-dimensional subspace of $\mathbb{R}^8$ and $T$ is a linear map from $\mathbb{R}^8$ to $\mathbb{R}^5$ such that

$$\operatorname{null} T = U.$$

Prove that $T$ is surjective.

***PROOF:*** According to the rank-nullity theorem we have

$$\dim \mathbb{R}^8 = \dim \operatorname{null} T + \dim \operatorname{range} T,$$

or

$$8 = 3 + \dim \operatorname{range} T.$$

Thus, $\dim \operatorname{range} T = 5$. We also know that $\operatorname{range} T \subseteq \mathbb{R}^5$. Since a subspace of full dimension equals the whole space itself, according to 2.39, we conclude:

$$\operatorname{range} T = \mathbb{R}^5.$$

Hence, $T$ is surjective. □

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**14** Prove that there does not exist a linear map from $\mathbb{F}^5$ to $\mathbb{F}^2$ whose null space equals

$$S := \{(x_1, x_2, x_3, x_4, x_5) \in \mathbb{F}^5 : x_1 = 3x_2 \ \text{ and } \ x_3 = x_4 = x_5\}.$$

***PROOF:*** We can rewrite $S$ as

$$\begin{aligned} S &= \{(3a, a, b, b, b) : a \in \mathbb{F} \ \text{ and } \ b \in \mathbb{F}\} \\ &= \{t_1(3,1,0,0,0) + t_2(0,0,1,1,1) : t_1 \in \mathbb{F} \ \text{ and } \ t_2 \in \mathbb{F}\}. \end{aligned}$$

Clearly, $\dim S = 2$. Using the rank-nullity theorem 3.21, we have for every $T \in \mathcal{L}(\mathbb{F}^5, \mathbb{F}^2)$:

$$\dim \mathbb{F}^5 = \dim \mathbb{F}^2 + \dim \operatorname{null} T$$

Hence, $\dim \operatorname{null} T = 3$. Therefore, $\operatorname{null} T$ could never equal $S$. □

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**15** Suppose there exists a linear map $T \in \mathcal{L}(V)$ on $V$ such that its null space $\operatorname{null}(T)$ and range $\operatorname{range}(T)$ are both finite-dimensional. Prove that $V$ is finite-dimensional.

***PROOF:*** According to the rank-nullity theorem, we have: $\dim V = \dim \operatorname{range}(T) + \dim \operatorname{null}(T)$. Since both $\dim \operatorname{range} T$ and $\dim \operatorname{null} T$ are finte by assumption, their sum is also finite. Hence, $\dim V$ is finite, and V is finite-dimensional. □

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**16** Suppose $V$ and $W$ are both finite-dimensional. Prove that there exists an injective linear map $T \in \mathcal{L}(V, W) \iff \dim V \leq \dim W$.

***PROOF:*** $\Rightarrow$ ***DIRECTION:*** If there exists an injective linear map $T \in \mathcal{L}(V, W)$, then $\operatorname{null}(T) = \{0\}$, so $\dim \operatorname{null}(T) = 0$. By the rank-nullity theorem:

$$\dim V = \dim \operatorname{range}(T) \leq \dim W,$$

because $\operatorname{range}(T) \subseteq W$.

$\Leftarrow$ ***DIRECTION:*** Let $\dim V \leq \dim W$. For any $T \in \mathcal{L}(V, W)$, we know

$$\dim \operatorname{null}(T) = \dim V - \dim \operatorname{range}(T) \geq \dim V - \dim W.$$

Since $\dim V \leq \dim W$, it follows that $\dim \operatorname{null}(T) \geq 0$, which is consistent with the existence of an injective map $T \in \mathcal{L}(V, W)$ where $\operatorname{null}(T) = \{0\}$.

***EXPLICIT CONSTRUCTION OF*** $T$: Suppose $\dim V \leq \dim W$. Let $v_1, \ldots, v_n$ be a basis for $V$ (where $n = \dim V$), and let $w_1, \ldots, w_m$ be a basis of $W$ (where $m = \dim W$). Define $T$ by $T(v_i) = w_i$ for $1 \leq i \leq n$. Then $T$ is injective, as $\operatorname{null}(T) = \{0\}$. □

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**17** Suppose $V$ and $W$ are both finite-dimensional. Prove that there exists a surjective linear map from $V$ onto $W \iff \dim V \geq \dim W$.

***PROOF:*** $\Rightarrow$ ***DIRECTION:*** Suppose $\exists T \in \mathcal{L}(V, W)$ such that $T$ is surjective. Then $\operatorname{range}(T) = W$, so by the rank-nullity theorem:

$$\begin{aligned}\dim V &= \dim \operatorname{range}(T) + \dim \operatorname{null}(T)\\ &= \dim W + \dim \operatorname{null}(T).\\ &\geq \dim W.\end{aligned}$$

$\Leftarrow$ ***direction:*** Suppose $\dim V \geq \dim W$. For any $T \in \mathcal{L}(V, W)$, the rank-nullity theorem implies:

$$\dim V = \dim \operatorname{range}(T) + \dim \operatorname{null}(T).$$

Since $\dim V \geq \dim W$, we can select $T$ such that $\operatorname{range}(T) = W$, ensuring surjectivity.

***Explicit Construction of $T$:***

Suppose $\dim V \geq \dim W$. Let $v_1, \dots, v_n$ be a basis of $V$ (where $n = \dim V$) and let $w_1, \dots, w_m$ be a basis of $W$ (where $m = \dim W$). Define $T$ on $V$ by setting $T(v_i) = w_i$ for $1 \leq i \leq m$, and $T(v_i) = 0$ for $m+1 \leq i \leq n$. Note that we have $\dim V \geq \dim W$, which is the same as $n \geq m$. So if $n = m$, then we do not set any $T(v_i)$ to be 0 and $\operatorname{null} T = \{0\}$. If $T$ is constructed like this, $T$ is surjective, as $\operatorname{range}(T) = W$. To show that, let $w \in W$ such that $w = a_1 w_1 + \cdots + a_m w_m$. We see that: $T(a_1 v_1 + \cdots + a_m v_m) = a_1 w_1 + \cdots a_m w_m = w$. Thus, for every $w \in W$, we have found a preimage $v \in V$ such that $T(v) = w$, proving that $T$ is surjective. $\square$

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**18** Suppose $V$ and $W$ are finite-dimensional, and that $U$ is a subspace of $V$. Prove that there exists a $T \in \mathcal{L}(V, W)$ such that $\operatorname{null} T = U \iff \dim U \geq \dim V - \dim W$.

***Proof:*** $\Rightarrow$ ***direction:*** Suppose $T \in \mathcal{L}(V, W)$ satisfies $\operatorname{null} T = U$. By the rank-nullity theorem:

$$\begin{aligned}\dim V &= \dim \operatorname{range}(T) + \dim \operatorname{null}(T)\\ &= \dim \operatorname{range}(T) + \dim U.\end{aligned}$$

Since $\dim \operatorname{range}(T) \leq \dim W$, it follows that:

$$\dim V \leq \dim W + \dim U \qquad \text{or equivalently,}$$

$$\dim U \geq \dim V - \dim W.$$

$\Leftarrow$ ***direction:*** Suppose $\dim U \geq \dim V - \dim W$. By the rank-nullity theorem, for any $T \in \mathcal{L}(V, W)$:

$$\dim V = \dim \operatorname{range}(T) + \dim \operatorname{null}(T).$$

Setting $\operatorname{null}(T) :\equiv U$, we have

$$\dim \operatorname{range}(T) = \dim V - \dim U.$$

Since $\dim V - \dim U \leq \dim W$, it follows that $\operatorname{range}(T) \subseteq W$, so $T \in \mathcal{L}(V, W)$ exists. Now we want to show how such a $T$ could look like.

***Explicit Construction of $T$:*** Suppose $\dim U \geq \dim V - \dim W$. Let $u_1, \dots, u_k$ be a basis for $U$ (where $k = \dim U$), and extend it to a basis $u_1, \dots, u_k, v_1, \dots, v_m$ for $V$ (where $m = \dim V - \dim U$). Let $w_1, \dots, w_m$ be a basis for $W$. Define $T \in \mathcal{L}(V, W)$ by setting $T(u_i) = 0$ for $1 \leq i \leq k$, and $T(v_i) = w_i$ for $1 \leq i \leq m$. Then $\operatorname{null}(T) = U$ and $\operatorname{range}(T) = W$. $\square$

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**19** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is injective $\iff \exists S \in \mathcal{L}(W, V)$ such that $ST$ is the identity operator on $V$.

***Proof:*** $\Rightarrow$ ***direction:*** Let $T \in \mathcal{L}(V, W)$ be injective. Let $v_1, \dots, v_m$ be a basis of $V$. We know from exercise 10 that $Tv_1, \dots, Tv_m$ spans $\operatorname{range} T$. We also know from exercise 9 that $Tv_1, \dots, Tv_m$ is linearly independent in $W$ and thus is a basis of $\operatorname{range} T$. Let $n :\equiv \dim W$. This basis can be extended to a basis of

$$Tv_1, \dots, Tv_m, w_{m+1}, \dots, w_n \tag{3.2}$$

of $W$, given $m < n$ ($\dim V < \dim W$). Therefore, every $w \in W$ can be written as follows:

$$w = a_1 Tv_1 + \cdots + a_m Tv_m + a_{m+1}w_{m+1} + \cdots + a_n w_n,$$

for some unique scalars $a_1, \ldots, a_m, a_{m+1}, \ldots, a_n \in \mathbb{F}$. Given the basis ((**1**)) of $W$, we can define $S \in \mathcal{L}(W, V)$ for any $w \in W$, letting

$$S(w) = S(a_1 Tv_1 + \cdots + a_m Tv_m + a_{m+1}w_{m+1} + \cdots + a_n w_n) :\equiv a_1 v_1 + \cdots + a_m v_m$$

Now it is easy to see that for any $v \in V$:

$$\begin{aligned} ST(v) &= ST(a_1 v_1 + \cdots + a_m v_m) \\ &= S(a_1 Tv_1 + \cdots + a_m Tv_m) \\ &= a_1 v_1 + \cdots + a_m v_m = v. \end{aligned}$$

Thus, $ST$ is the identity operator on $V$.

$\Leftarrow$ ***direction:*** Let $ST$ be the identity operator on $V$. That means: $\forall v \in V : S(T)(v) = v$. Suppose $u \in V$ satisfies $T(u) = 0$. Then we have $S(T(u)) = S(0) = 0$. Since $ST = I$, it follows that $u = 0$. We conclude that $\operatorname{null} T = \{0\}$, hence $T$ is injective. □

**20** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V, W)$. Prove that $T$ is surjective $\iff \exists S \in \mathcal{L}(W, V)$ such that $TS$ is the identity operator on $W$.

***Proof:*** $\Rightarrow$ ***direction:*** Let $T \in \mathcal{L}(V, W)$ such that $T$ is surjective. Hence, $\operatorname{range} T = W$. Let $w_1, \ldots, w_m$ be a basis of $W$ and, hence, of $\operatorname{range} T$ as well. Since $T$ is surjective, we have for every $j \in \{1, \ldots, m\}$ a preimage $v_j \in V$ of $w_j$ such that

$$w_j = T(v_j).$$

Based on this list $v_1, \ldots, v_j$ which has no other special properties, we can define $S \in \mathcal{L}(W, V)$ for each $w \in W$ as follows:

$$S(w) = S(a_1 w_1 + \cdots a_m w_m) := a_1 v_1 + \cdots + a_n v_n.$$

Now we have $\forall w \in W$:

$$TS(w) = a_1 TS(w_1) + \cdots + a_n TS(w_n) = a_1 w_1 + \cdots + a_n w_n = w, \tag{3.3}$$

which makes $TS$ the identity operator on $W$. A different way of calculating $TS$ would be $TS(w) = T(a_1 v_1 + \cdots + a_n v_n) = a_1 Tv_1 + \cdots + a_n Tv_n = a_1 w_1 + \cdots + a_n w_n = w$. In equation ((**1**)) we were using $S(w_j) = v_j$ for $1 \le j \le m$.

$\Leftarrow$ ***direction:*** Let $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, V)$ such that $TS$ is the identity operator on $W$. Let $w \in W$ and $v \in V$ such that $v = S(w)$. Now we apply $T$ on both sides and get $T(v) = T(S(w)) = w$. So for any $w \in W$ we found a preimage $v \in V$ such that $T(v) = w$, which makes $T$ surjective. □

**21** Suppose $V$ is finite-dimensional, $U$ is a subspace of $V$ and $T \in \mathcal{L}(V, W)$. Prove that

$$X :\equiv \{v \in V \mid Tv \in U\}$$

is a subspace of $V$ and

$$\dim X = \dim\{v \in V \mid Tv \in U\} = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T).$$

***Proof:*** Let $v_1, v_2 \in X$. Thus $v_1, v_2 \in V$ as well as $T(v_1 + v_2) = Tv_1 + Tv_2 \in U$, because $U$ is closed under addition. $\implies v_1 + v_2 \in X$. So $X$ is closed under addition as well. Similarly, $\lambda v_1 \in X$ for $\lambda \in \mathbb{F}$. So $X$ is closed under scalar multiplication. We also have the additive identity $0 \in X$ because $T0 = 0 \in U$. Therefore, $X$ is a subspace of $V$.

The formula we need to prove reminds us of the ranke nulllity theorem. We want to show that:

$$\begin{aligned} \dim X &= \dim \operatorname{null} T_X + \dim \operatorname{range} T_X \\ &= \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T) \end{aligned} \tag{3.4}$$

The first line of the equation above is just the rank-nullity theorem of the restriction of $T$ on $X$. For the second line in formula ((1)), we have $\operatorname{null} T_X = T$ and $\operatorname{range} T_X = (U \cap \operatorname{range} T)$. Now $X$ is the set of all vectors in $V$ such that $Tv$ is in $U$. Because $0 \in U$, all the vectors $v \in V$ sucht that $Tv = 0$ are contained in $X$. In symbols: $\operatorname{null} T \subseteq X$. Hence, $\operatorname{null} T|_X = \operatorname{null} T$.

Now we want to show that $\operatorname{range} T|_X \subseteq (U \cap \operatorname{range} T)$ and $(U \cap \operatorname{range} T) \subseteq \operatorname{range} T|_X$, which implies

$$\operatorname{range} T|_X = (U \cap \operatorname{range} T).$$

If $u \in (U \cap \operatorname{range} T)$, then $\exists v \in V$ such that $u = Tv$ and $u \in U$. So $v \in X$ because $Tv \in U$. Hence $u \in \operatorname{range} T|_X$ and

$$(U \cap \operatorname{range} T) \subseteq \operatorname{range} T|_X.$$

Conversely, let $u \in \operatorname{range} T|_X$. According to the definition of $X$, there exists a $w \in V$ such that $Tw \in U$. For this $w$ we have $u = T|_X(w)$. Since $T|_X$ ist just $T$ on a restricted domain $X \subseteq V$, we have $u \in U$ and $u \in \operatorname{range} T$, hence $u \in (U \cap \operatorname{range} T)$. We conclude:

$$\operatorname{range} T|_X \subseteq (U \cap \operatorname{range} T).$$

Therefore, $\operatorname{range} T|_X = (U \cap \operatorname{range} T)$, which validates equation ((1)) □

22 **(a)** Suppose $U$, $V$ and $W$ are finite-dimensional vector spaces and $S \in \mathcal{L}(V,W)$ and $T \in \mathcal{L}(U,V)$. Prove that $\dim \operatorname{null} ST \leq \dim \operatorname{null} S + \dim \operatorname{null} T$.

***PROOF:*** Note that $ST \in \mathcal{L}(U,W)$ and $\operatorname{null} T \subseteq \operatorname{null} ST$ (this implies $\dim \operatorname{null} T \leq \dim \operatorname{null} ST$). According to the rank-nullity theorem, the following two formulas hold:

$$\dim U = \dim \operatorname{range} T \quad + \dim \operatorname{null} T$$
$$\dim U = \dim \operatorname{range} ST + \dim \operatorname{null} ST$$

Equating the two formulas, we obtain:

$$\dim \operatorname{null} ST = \dim \operatorname{range} T - \dim \operatorname{range} ST + \dim \operatorname{null} T \tag{3.5}$$

So, what is $\dim \operatorname{range} T - \dim \operatorname{range} ST$? It equals the dimension of the null space of the restriction map $S|_{\operatorname{range} T}$. If we can show that $\dim \operatorname{range} T - \dim \operatorname{range} ST \leq \dim \operatorname{null} S$, the proof is complete. Consider the restriction map

$$S|_{\operatorname{range} T} \colon \operatorname{range} T \to W.$$

Notice two things. First, for every $x \in U$ we have $S(T(x)) = (S|_{\operatorname{range} T})(T(x))$; hence,

$$\operatorname{range} S|_{\operatorname{range} T} = \operatorname{range} ST. \tag{3.6}$$

Second, $\operatorname{null}(S|_{\operatorname{range} T}) = \{y \in \operatorname{range} T \mid S(y) = 0\}$. Clearly, every such $y$ is contained in $\operatorname{null} S$, so

$$\operatorname{null} S|_{\operatorname{range} T} \subseteq \operatorname{null} S. \tag{3.7}$$

Now we can apply the Rank-Nullity Theorem to $S|_{\operatorname{range} T}$ and use ((**a**)) and ((**a**)) consecutively. Thus,

$$\begin{aligned} \dim \operatorname{range} T &= \dim \operatorname{null} S|_{\operatorname{range} T} + \dim \operatorname{range} S|_{\operatorname{range} T} \\ &= \dim \operatorname{null} S|_{\operatorname{range} T} + \dim \operatorname{range} ST \\ &\leq \dim \operatorname{null} S + \dim \operatorname{range} ST. \end{aligned}$$

Rearranging, we obtain:

$$\dim \operatorname{range} T - \dim \operatorname{range} ST \leq \dim \operatorname{null} S.$$

Therefore, ((**a**)) becomes

$$\dim \operatorname{null} ST \leq \dim \operatorname{null} S + \dim \operatorname{null} T.$$

Note: If $\dim \operatorname{range} T = \dim \operatorname{range} ST$, ((**a**)) becomes $\dim \operatorname{null} ST = \dim \operatorname{null} T$. ☐

**(b)** Give an example of $S, T \in \mathcal{L}(\mathbb{F}^5)$ with $ST = 0$ and $\dim \operatorname{range} TS = 2$.

***Solution:*** Let $S, T \in \mathcal{L}(\mathbb{F}^5)$ such that for all $x \in \mathbb{F}^5$:

$$S(x) :\equiv (0,0,0,x_4,x_5) \quad \text{and} \quad T(x) :\equiv (x_1 + x_4, x_2 + x_5, 0,0,0)$$

If we compute $ST$ and $TS$ for $x \in \mathbb{F}^5$ we get

$$ST(x) = S((x_1 + x_4, x_2 + x_5, 0,0,0)) = (0,0,0,0,0) = 0$$
$$TS(x) = T((0,0,0,x_4,x_5)) = (0 + x_4, 0 + x_5, 0,0,0) = (x_4, x_5, 0,0,0)$$

Hence, $\dim \operatorname{range} TS = 2$.

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(24) Suppose $\dim V = 5$ and $S, T \in \mathcal{L}(V)$ are such that $ST = 0$. Prove that $\dim \operatorname{range} TS \leq 2$.

***Proof:*** According to the rank-nullity theorem, we have

$$5 = \dim \operatorname{range} T + \dim \operatorname{null} T.$$
$$5 = \dim \operatorname{range} S + \dim \operatorname{null} S$$

From exercise 3B.22, we know

$$\begin{aligned} \dim \operatorname{null} ST &\leq \dim \operatorname{null} S + \dim \operatorname{null} T, \\ 5 &\leq \dim \operatorname{null} S + \dim \operatorname{null} T. \end{aligned} \tag{3.8}$$

From exercise 3B.23, we know

$$\begin{aligned} \dim \operatorname{range} TS &\leq \min\{\dim \operatorname{range} S, \dim \operatorname{range} T\} \\ &= \min\{5 - \dim \operatorname{null} S, 5 - \dim \operatorname{null} T\} \end{aligned} \tag{3.9}$$

Equation ((1)) and the equation ((1)) above tell us, that $\operatorname{range} TS$ is at its maximum, if either $\dim \operatorname{null} S$ or $\dim \operatorname{null} T$ is 2 and the other one is 3. Hence $\dim \operatorname{range} TS = \min\{2,1\} = 2$. ☐

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(25) Suppose $W$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} S \subseteq \operatorname{null} T \iff$ there exists a $E \in \mathcal{L}(W)$ such that $T = ES$.

***Proof:*** $\Rightarrow$ ***direction:*** Let $S, T \in \mathcal{L}(V, W)$ such that $\operatorname{null} S \subseteq \operatorname{null} T$. (i.e. $\dim \operatorname{null} S \leq \dim \operatorname{null} T$). Let $U$ be a subspace of $V$ such that $V = \operatorname{null} S \oplus U$. This implies that

$$S|_U \colon U \to \operatorname{range} S$$

is injective, because $\operatorname{null} S|_U = \{0\}$. Using the rank nullity-theorem and the fact that $\dim \operatorname{null} S|_U = 0$, we can conclude that:

$$\dim U = \dim \operatorname{range} S|_U = \dim \operatorname{range} S.$$

The equation above also made use of the fact that $\operatorname{range} S|_U = \operatorname{range} S$. This is because any vector $v \in V$ can be written as $v = r + u$, where $r \in \operatorname{null} S$ and $u \in U$. For this $v$ we have:

$$S(v) = S(r) + S(u) = S(u) = S|_U(u) = S|_U(r) + S|_U(u) = S|_U(v).$$

Thus, $S$ and $S|_U$ have the same range, $\operatorname{range} S|_U = \operatorname{range} S$.

Now if $u_1, \ldots, u_n$ is a basis of $U$, then $Su_1, \ldots, Su_n$ is a basis of $\operatorname{range} S \subseteq W$. Therefore, we can extend $Su_1, \ldots, Su_n$ to a basis

$$Su_1, \ldots, Su_n, w_{n+1}, \ldots, w_m$$

of $W$. Note that the extension might be empty and no $w$'s are added. If $n = m$ the basis of $W$ just consists of $Su_1, \ldots, Su_n$ and $\operatorname{range} S = W$. So we have $n \leq m$ for $n = \dim \operatorname{span}(S)$ and $m = \dim W$. Now we define $E : W \to W$ for $w \in W$, letting

$$E(w) = E(a_1 Su_1 + \cdots + a_n Su_n + a_{n+1} w_{n+1} + \cdots + a_m w_m) :\equiv a_1 Tu_1 + \cdots + a_n Tu_n.$$

If we decompose $v \in V = \operatorname{null} S \oplus U$ again as $v = r + u$ where $r \in \operatorname{null} S$ and $u \in U$, then

$$ES(v) = ES(r + u) = ES(u) = E(a_1 Su_1 + \cdots + a_n Su_n) = a_1 Tu_1 + \cdots + a_n Tu_n.$$

Since $\operatorname{null} S \subseteq \operatorname{null} T$, and we take the same $v = r + u$, we have $T(r) = 0$. Hence:

$$T(v) = T(r) + T(u) = T(u) = a_1 T u_1 + \cdots + a_n T u_n.$$

Thus we conclude: $ES = T$.

$\Leftarrow$ **direction:** Assume $W$ is finite-dimensional and we have $S, T \in \mathcal{L}(V, W)$ and $E \in \mathcal{L}(W)$ such that $T = ES$. Now assume that $\operatorname{null} S$ is not a subset of $\operatorname{null} T$. Then there exists a $v \in V$ such that $S(v) = 0$ but $T(v) \neq 0$. Because of 3.10 it must be that $E(0) = 0$, and thus $ES(v) = 0$. But we assumed that $T = ES$. Now we have $ES \neq T$ (because $T(v) \neq 0$). The assumption that $\operatorname{null} S$ is not a subset of $\operatorname{null} T$ has lead to a contradiction. Therefore, $\operatorname{null} S \subseteq \operatorname{null} T$, as desired. $\square$

**26** Suppose that $V$ is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{range} S \subseteq \operatorname{range} T \iff$ there exists $E \in \mathcal{L}(V)$ such that $S = TE$.

***Proof:*** $\Rightarrow$ **direction:** Let $V$ be finite-dimensional and $S, T \in \mathcal{L}(V, W)$ such that $\operatorname{range} S \subseteq \operatorname{range} T$. Let $Q$ be a subspace of $V$ such that $V = \operatorname{null} S \oplus Q$. Let $q_1, \ldots, q_m$ be a basis of $Q$. We can extend this list to a basis

$$q_1, \ldots, q_m, r_{m+1}, \ldots, r_n$$

of $V$ with $\dim V = n$. Note that $r_{m+1}, \ldots, r_n$ is a basis of the null space of $S$, i.e $\operatorname{null} S$. The restriction $S|_Q$ is injective because $\operatorname{null} S|_Q = \{0\}$. This makes

$$Sq_1, \ldots, Sq_m$$

a basis of the range of $S$, i.e $\operatorname{range} S$. From $\operatorname{range} S \subseteq \operatorname{range} T$, it follows that $\forall j \in \{1, \ldots, m\}$:

$$\exists u_j \in V \text{ such that } T(u_j) = S(q_j).$$

So we have a list of distinct preimage vectors $u_1, \ldots, u_m$. The linear map lemma 3.4 tells us, we can construct $E : V \to V$ as follows:

$$\begin{aligned} E(q_k) &= u_k \quad \forall k \in \{1, \ldots, m\} \\ E(r_k) &= 0 \quad \forall k \in \{m+1, \ldots, n\} \end{aligned} \tag{3.10}$$

This lemma guarantees the existence (and uniqueness) of a linear map on $V$ that sends the basis vectors $q$'s and $r$'s to the specified vectors $u$'s and $0$'s respectively. This ensures $E$ is well defined. For a $v \in V$, given as $v = (c_1 q_1 + \cdots c_m q_m) + (c_{m+1} r_{m+1} + \cdots + c_n r_n)$ with its coordinates and its basis, the linear map $E$ looks like this:

$$E(v) :\equiv c_1 u_1 + \cdots + c_m u_m$$

One can verify that this coincides with the definition (1) above, using the fact that $E$ is linear. Now what does $TE$ look like? It computes as follows:

$$\begin{aligned} T(E(v)) &= TE\big((c_1 q_1 + \cdots + c_m q_m) + (c_{m+1} r_{m+1} + \cdots + c_n r_n)\big) \\ &= T(c_1 u_1 + \cdots + c_m u_m) = c_1 T(u_1) + \cdots + c_m T(u_m) \\ &= c_1 S(q_1) + \cdots + c_m S(q_m). \end{aligned}$$

Compare this with $S$:

$$\begin{aligned} S(v) &= S\big((c_1 q_1 + \cdots + c_m q_m) + (c_{m+1} r_{m+1} + \cdots + c_n r_n)\big) = S(c_1 q_1 + \cdots + c_m q_m) \\ &= c_1 S(q_1) + \cdots + c_m S(q_m). \end{aligned}$$

Therefore, $S = TE$.

$\Leftarrow$ **direction:** Suppose there exists $E \in \mathcal{L}(V)$ such that $S = TE$. Then for any $v \in V$, $S(v) = T(E(v))$ implies that $S(v) \in \operatorname{range} T$. Hence, $\operatorname{range} S \subseteq \operatorname{range} T$. $\square$

**27** Suppose $P \in \mathcal{L}(V)$ and $P^2 = P$. Prove that $V = \operatorname{null} P \oplus \operatorname{range} P$.

***Proof:*** Suppose $U$ is a subspace of a finite-dimensional vector space $V$ such that $V = \operatorname{null} P \oplus U$.

We want to show that $U = \operatorname{range} P$. We do this by showing $\operatorname{range} P \subseteq U$ and $U \subseteq \operatorname{range} P$.

***Step 1:*** Let $v \in \operatorname{range} P$. We can write $v = r + u$ such that $r \in \operatorname{null} P$ and $u \in U$. Thus, we have

$$P(P(v)) = P(v) \in \operatorname{range} P,$$

and $P(v) = P(r) + P(u) = P(u)$. The formula $P^2 = P$ tells us that $P$ acts as the identity operator on $\operatorname{range} P$. So we can just write: $v = u$; hence, $v \in U$. Therefore,

$$\operatorname{range} P \subseteq U.$$

***Step 2:*** Conversely, let $u \in U$. Notice that we can write

$$u = \underbrace{(u - P(u))}_{\in \operatorname{null} P} + P(u). \tag{3.11}$$

$u - P(u)$ lies in $\operatorname{null} P$, because

$$P(u - P(u)) = P(u) - P(P(u)) = P(u) - P(u) = 0.$$

The decomposition of any vector in $U$ (or $V$) as an element of $\operatorname{null} P$ plus an element of $U$ is unique. Thus, the two decompositions of $u \in U$, $u = 0 + u$ and (***1***) must coincide. This forces $u - P(u) = 0$ and $P(u) = u$. Therefore, we conclude that $u = P(u)$, which means that $u \in \operatorname{range} P$. It follows that

$$U \subseteq \operatorname{range} P$$

Therefore, $U = \operatorname{range} P$.

***Alternative proof:*** For any $v \in V$, let $u := P(v)$ and $w := v - P(v)$. Then clearly, $v = w + u$. By definition, $u \in \operatorname{range} P$. If we apply $P$ to $w$, we have

$$P(w) = P(v - P(v)) = P(v) - P(P(v)) = P(v) - P(v) = 0.$$

Thus, $w \in \operatorname{null} P$.

Now we have to verify the uniqueness of the sum. Suppose a vector $u$ lies both in $\operatorname{null} P$ and $\operatorname{range} P$. Then, $u = P(v)$ for some $v \in V$ and also $P(u) = 0$. But since $P(u) = u$, we have $u = 0$. Therefore, we conclude that the intersection is trivial:

$$\operatorname{null} P \cap \operatorname{range} P = \{0\}.$$

This implies,

$$V = \operatorname{null} P \oplus \operatorname{range} P.$$

$\square$

# MATRICES (SECTION 3C)

## REPRESENTING A LINEAR MAP BY A MATRIX (SUB-SECTION 3C1)

**DEF 3.31** (*matrix of a linear map, $\mathcal{M}(T)$*):

Let $T \in \mathcal{L}(V,W)$, let $v_1,\dots,v_n$ be a basis of $V$, and let $w_1,\dots,w_m$ be a basis of $W$. The *"matrix of T"* with respect to these bases is the $m$-by-$n$ matrix $\mathcal{M}(T)$ whose entries $A_{j,k}$ are defined by

$$Tv_k = A_{1,k}w_1 + \cdots + A_{m,k}w_m = \sum_{j=1}^{m} A_{j,k}w_j.$$

If the bases are not clear from the context, one writes $\mathcal{M}(T,(v_1,\dots,v_n),(w_1,\dots,w_m))$. One remembers how to construct $\mathcal{M}(T)$ if one writes the bases on the left-hand side and on top of the matrix. It looks as follows:

$$\mathcal{M}(T) :\equiv \begin{array}{c} \\ w_1 \\ \vdots \\ w_m \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_k & \cdots & v_n \end{array} \\ \begin{pmatrix} & & A_{1,k} & & \\ & & \vdots & & \\ & & A_{m,k} & & \end{pmatrix} \end{array}$$

**DEF** (*matrix notation for operators (see later)*):

For $T \in \mathcal{L}(V)$, we use the notation

$$\mathcal{M}(T,(v_1,\dots,v_n)) := \mathcal{M}(T,(v_1,\dots,v_n),(v_1,\dots,v_n)).$$

## ADDITION AND SCALAR MULTIPLICATION OF MATRICES (SUB-SECTION 3C2)

**THM 3.35** (*matrix of the sum of linear maps*):

$\mathcal{M}(S+T) = \mathcal{M}(S) + \mathcal{M}(T)$ for $S,T \in \mathcal{L}(V,W)$.

**THM 3.38** (*matrix of a scalar times a linear map*):

$\mathcal{M}(\lambda T) = \lambda\mathcal{M}(T)$ for $\lambda \in \mathbb{F}$ and $T \in \mathcal{L}(V,W)$.

**DEF 3.39** (*notation $\mathbb{F}^{m,n}$*):

$\mathbb{F}^{m,n}$ denotes the set of all $m$-by-$n$ matrices with entries in $\mathbb{F}$, where $m,n \in \mathbb{N}$.

**THM 3.40** (*dimension of $\mathbb{F}^{m,n}$*):

$\mathbb{F}^{m,n}$ is a vector space with $\dim \mathbb{F}^{m,n} = mn$.

***PROOF:*** It is easy to verify that $\mathbb{F}^{m,n}$ is a vector space. The list of length $mn$ that consists of distinct $m$-by-$n$ matrices that have 0 in all entries except a 1 in one entry is a basis of $\mathbb{F}^{m,n}$. Hence, $\dim \mathbb{F}^{m,n} = mn$. □

## MATRIX MULTIPLICATION (SUB-SECTION 3C3)

Let $v_1,\dots,v_n$ be a basis of $V$, $w_1,\dots,w_m$ be a basis of $W$, and $u_1,\dots,u_p$ be a basis of $U$. Consider the linear maps $T: U \to V$ and $S: V \to W$. The composition $ST$ is a linear map from $U$ to $W$. Is it the case that

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)\,?$$

Later on, matrix multiplication will be defined accordingly such that this question has a positive answer. For now, let $A := \mathcal{M}(S) \in \mathbb{F}^{m,n}$ and $B := \mathcal{M}(T) \in \mathbb{F}^{n,p}$. The matrices of $S$ and $T$ look as follows:

$$A := \mathcal{M}(S) = \begin{array}{cc} & \begin{matrix} v_1 & \cdots & v_r & \cdots & v_n \end{matrix} \\ \begin{matrix} w_1 \\ \vdots \\ w_j \\ \vdots \\ w_m \end{matrix} & \begin{pmatrix} & & A_{1,r} & & \\ & & \vdots & & \\ & & A_{j,r} & & \\ & & \vdots & & \\ & & A_{m,r} & & \end{pmatrix} \end{array}, \quad B := \mathcal{M}(T) = \begin{array}{cc} & \begin{matrix} u_1 & \cdots & u_k & \cdots & u_p \end{matrix} \\ \begin{matrix} v_1 \\ \vdots \\ v_r \\ \vdots \\ v_n \end{matrix} & \begin{pmatrix} & & B_{1,k} & & \\ & & \vdots & & \\ & & B_{r,k} & & \\ & & \vdots & & \\ & & B_{n,k} & & \end{pmatrix} \end{array}$$

For $1 \leq k \leq p$, we have:

$$(ST)u_k = S(Tu_k) = S\left(\sum_{r=1}^{n} B_{r,k} v_r\right) = \sum_{r=1}^{n} B_{r,k} S v_r = \sum_{r=1}^{n} B_{r,k} \sum_{j=1}^{m} A_{j,r} w_j = \sum_{j=1}^{m} \left(\sum_{r=1}^{n} A_{j,r} B_{r,k}\right) w_j$$

Thus, $\mathcal{M}(ST)$ is the $m$-by-$p$ matrix whose entry in row $j$, column $k$ equals $\sum_{r=1}^{n} A_{j,r} B_{r,k}$:

$$\mathcal{M}(ST) = \begin{array}{cc} & \begin{matrix} u_1 & \cdots & u_k & \cdots & u_p \end{matrix} \\ \begin{matrix} w_1 \\ \vdots \\ w_j \\ \vdots \\ w_m \end{matrix} & \begin{pmatrix} & & \sum_{r=1}^{n} A_{1,r} B_{r,k} & & \\ & & \vdots & & \\ & & \sum_{r=1}^{n} A_{j,r} B_{r,k} & & \\ & & \vdots & & \\ & & \sum_{r=1}^{n} A_{m,r} B_{r,k} & & \end{pmatrix} \end{array}$$

**Def 3.41** (*matrix multiplication*)**:**

Let $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$. Then $AB \in \mathbb{F}^{m,p}$ is the matrix whose entry in row $j$, column $k$, is given by the equation

$$(AB)_{j,k} :\equiv \sum_{r=1}^{n} A_{j,r} B_{r,k}.$$

The octave code for calculating the entry $j,k$ of the result of $A,B$ with the *"sum-scheme"* like above looks as follows:

```
function retval = standard_matrix_mult (A,B)
  m = rows(A);
  n = columns(A);
  p = columns(B);
  retval=zeros(m,p)
  if (columns(A) != rows(B))
    error('columns of A and rows of B dont match');
  end
  for (j=1:m)
    for (k=1:p)
      for (r=1:n)
        retval(j,k)=retval(j,k)+A(j,r)*B(r,k);
      end
    end
  end
end
```

**Thm 3.43** (*matrix of product of linear maps*)**:**

If $T \in \mathcal{L}(U,V)$ and $S \in \mathcal{L}(V,W)$, then

$$\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T).$$

***Proof:*** See at the beginning of the sub-subsection. □

**Def 3.44** (*notation: $A_{j,\bullet}, A_{\bullet,k}$*):

Let $A \in \mathbb{F}^{m,n}$. Then:

- If $1 \le j \le m$, then $A_{j,\bullet}$ denotes the 1-by-$n$ matrix consisting of row $j$ of $A$.
- If $1 \le k \le n$, then $A_{\bullet,k}$ denotes the $m$-by-1 matrix consisting of column $k$ of $A$.

**Thm 3.46** (*entry of matrix product equals row times column*):

If $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$, then we have for $1 \le j \le m$ and $1 \le k \le p$:

$$(AB)_{j,k} = A_{j,\bullet} B_{\bullet,k}.$$

The octave code for multiplying rows and columns directly at entry $j,k$ looks as follows:

```
function retval = row_wise_matrix_mult (A,B)
  m = rows(A);
  n = columns(A);
  p = columns(B);
  retval=zeros(m,p)
  if (n != rows(B))
    error('columns of A and rows of B dont match');
  end
  for (j=1:m)
    for (k=1:p)
      retval(j,k)=A(j,:)*B(:,k);
    end
  end
end
```

**Thm 3.48** (*column of matrix product equals matrix times column*):

If $A \in \mathbb{F}^{m,n}$ and $B \in \mathbb{F}^{n,p}$, then we have for $1 \le k \le p$:

$$(AB)_{\bullet,k} = AB_{\bullet,k}.$$

In other words, column $k$ of $AB$ equals $A$ times column $k$ of $B$. The octave code looks as follows:

```
function retval = matrix_times_columns_matrix_mult (A,B)
  m = rows(A);
  p = columns(B);
  retval=zeros(m,p)
  if (columns(A) != rows(B))
    error('columns of A and rows of B dont match');
  end
  for (k=1:p)
    retval(:,k)=A*B(:,k);
  end
end
```

**Thm 3.50** (*linear combination of columns*):

Suppose $A \in \mathbb{F}^{m,n}$ and $b = (b_1, \dots, b_n)^t \in \mathbb{F}^{n,1}$. Then

$$Ab = b_1 A_{\bullet,1} + \dots + b_n A_{\bullet,n}.$$

See 3.54 for the definition of $(^t)$.

**Thm 3.51** (*matrix multiplication as linear comb. of columns or rows*):

Let $C \in \mathbb{F}^{m,c}$ and $R \in \mathbb{F}^{c,n}$.

**Column-wise:** If $k \in \{1, \dots, n\}$, then column $k$ of $CR \in \mathbb{F}^{m,n}$ is a linear combination of the columns of

$C$, with the coefficients of this linear combination coming from column $k$ of $R$ $[R_{\bullet,k}]$.

$$CR = \left( \underbrace{\left( \begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,1} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,1} \right)}_{\text{1st column}} \dots \underbrace{\boxed{\begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,k} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,k}}}_{k\text{th column}} \dots \underbrace{\left( \begin{pmatrix} C_{1,1} \\ \vdots \\ C_{m,1} \end{pmatrix} R_{1,n} + \cdots + \begin{pmatrix} C_{1,c} \\ \vdots \\ C_{m,c} \end{pmatrix} R_{c,n} \right)}_{\text{last column}} \right)$$

The octave-code for column-wise multiplication is as follows.

```
function retval = column_wise_matrix_mult (C,R)
  m = rows(C);
  c = columns(C);
  n = columns (R);
  retval=zeros(m,n);
  if (c != rows(R))
    error('columns of C and rows of R dont match');
  end
  for (k=1:n)
    for (i=1:c)
      retval(:,k)=retval(:,k)+C(:,i)*R(i,k);
    end
  end
end
```

**Row-wise:** If $j \in \{1,\dots,m\}$, then row $j$ of $CR$ is a linear combination of the rows of $R$, with the coefficients of this linear combination coming from row $j$ of $C$ $[C_{j,\bullet}]$:

$$CR = \begin{pmatrix} C_{1,1}(R_{1,1},\dots,R_{1,n}) + C_{1,2}(R_{2,1},\dots,R_{2,n}) + \cdots + C_{1,c}(R_{c,1},\dots,R_{c,n}) \\ C_{2,1}(R_{1,1},\dots,R_{1,n}) + C_{2,2}(R_{2,1},\dots,R_{2,n}) + \cdots + C_{2,c}(R_{c,1},\dots,R_{c,n}) \\ \vdots \\ C_{j,1}(R_{1,1},\dots,R_{1,n}) + C_{j,2}(R_{2,1},\dots,R_{2,n}) + \cdots + C_{j,c}(R_{c,1},\dots,R_{c,n}) \\ \vdots \\ C_{m,1}(R_{1,1},\dots,R_{1,n}) + C_{m,2}(R_{2,1},\dots,R_{2,n}) + \cdots + C_{m,c}(R_{c,1},\dots,R_{c,n}) \end{pmatrix}$$

The octave-code for row-wise multiplication is as follows.

```
function retval = row_wise_matrix_mult (C,R)
  m = rows(C);
  c = columns(C);
  n = columns (R);
  retval=zeros(m,n);
  if (c != rows(R))
    error('columns of C and rows of R dont match');
  end
  for (j=1:m)
    for (i=1:c)
      retval(j,:) = retval(j,:) + C(j,i) * R(i,:);
    end
  end
end
```

## Column-Row Factorization and Rank of a Matrix (Sub-Section 3C4)

**Def 3.52** (*column rank, row rank*):

For $A \in \mathbb{F}^{m,n}$, the "*column rank*" of $A$ is the dimension of the span of the columns of $A$ in $\mathbb{F}^{m,1}$. The "*row rank*" of $A$ is the dimension of the span of the rows of $A$ in $\mathbb{F}^{1,n}$.

**Def 3.54** (*transpose, $A^t$*):

The "*transpose*" of a matrix $A \in \mathbb{F}^{m,n}$, denoted by $A^t \in \mathbb{F}^{n,m}$, is obtained by interchanging rows and columns. Its entries are given by the equation

$$\left(A^t\right)_{k,j} = A_{j,k} \quad \forall k \in \{1,\ldots,m\}, j \in \{1,\ldots,n\}.$$

**Thm 3.56** (*column-row factorization*):

Suppose $A \in \mathbb{F}^{m,n}$ with column rank $c \geq 1$. Then there exist $C \in \mathbb{F}^{m,c}$ and $R \in \mathbb{F}^{c,n}$ such that

$$A = CR.$$

**Thm 3.57** (*column rank and row rank*):

If $A \in \mathbb{F}^{m,n}$, then column rank of $A$ = row rank of $A$. This leads to the next definition.

**Def 3.58** (*rank*):

The "*rank*" of a matrix $A \in \mathbb{F}^{m,n}$ is the column rank of $A$.

## Exercises about Matrices 3C

# INVERTIBILITY AND ISOMORPHISMS (SECTION 3D)

## INVERTIBLE LINEAR MAPS (SUB-SECTION 3D1)

**DEF 3.59** (*invertible, inverse*):

$S \in \mathcal{L}(W,V)$ with $ST = I_V$ and $TS = I_W$ is called the "*inverse*" of the "*invertible*" linear map $T \in \mathcal{L}(V,W)$.

**THM 3.60** (*inverse is unique*):

An invertible linear map has a unique inverse.

***PROOF:*** Suppose $T \in \mathcal{L}(V,W)$ is invertible and $S_1$ and $S_2$ are inverses of $T$. Hence,

$$S_1 = S_1 I = S_1(TS_2) = (S_1 T)S_2 = IS_2 = S_2.$$

Therefore, $S_1 = S_2$. □

**DEF 3.61** (*notation $T^{-1}$*):

Since the inverse is unique, the inverse is denoted by $T^{-1} \in \mathcal{L}(W,V)$ for $T \in \mathcal{L}(V,W)$. We have

$$T^{-1}T = I_V \text{ and } TT^{-1} = I_W.$$

**THM 3.62** (*invertibility, injectivity and surjectivity*):

A linear map is invertible $\iff$ it is injective and surjective.

**THM 3.65** (*injectivity is equivalent to surjectivity*):

For $T \in \mathcal{L}(V,W)$, if $V$ and $W$ are finite-dimensional, injectivity is equivalent to surjectivity:

$$T \text{ is invertible} \iff T \text{ is injective} \iff T \text{ is surjective.}$$

Equivalently:

$$T \text{ is not invertible} \iff T \text{ is not injective} \iff T \text{ is not surjective.}$$

***PROOF:*** The rank-nullity theorem (3.21) or fundamental theorem of linear maps states that

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T. \tag{3.12}$$

If $T$ is injective, $\dim \operatorname{null} T = 0$, and therefore

$$\dim \operatorname{range} T = \dim V = \dim W,$$

which means $T$ is surjective ($\operatorname{range} T = \dim W$). This is because every linearly independent list of vectors of length $\dim W$ is a basis by 2.38. And that's how dimension is defined. (TODO)

If $T$ is surjective, we have $\dim \operatorname{range} T = \dim W$ to begin with, and therefore (3D1) becomes

$$\dim \operatorname{null} T = \dim V - \dim \operatorname{range} T = \dim V - \dim W = 0,$$

which makes $T$ also injective. Since injectivity and surjectivity together imply invertibility, this ends the proof. □

**THM 3.68** (*$ST = I \iff TS = I$ on vector spaces of the same dimension*):

Suppose $V$ and $W$ are finite-dimensional vector spaces of the same dimension. Let $S \in \mathcal{L}(W,V)$ and $T \in \mathcal{L}(V,W)$. Then $ST = I \iff TS = I$.

***PROOF:*** $\Rightarrow$ **DIRECTION:** Let $S \in \mathcal{L}(W,V)$ and $T \in \mathcal{L}(V,W)$ such that $ST = I$. If $v \in V$ and $Tv = 0$, then

$$v = Iv = (ST)v = S(Tv) = S(0) = 0.$$

Thus, $\operatorname{null} T = \{0\}$, and therefore, $T$ is injective by 3.15. Because $V$ and $W$ have the same dimension, $T$ is invertible by 3.65. Now, multiply both sides of the equation $ST = I$ by $T^{-1}$ to conclude:

$$S = T^{-1}.$$

Hence, $TS = TT^{-1} = I$, as desired.

$\Leftarrow$ ***direction:*** Right above, we have shown that $ST = I$ implies that $TS = I$. To prove the implication in the other direction, simply reverse the roles of $S$ and $T$ and $V$ and $W$, showing that $TS = I$ implies that $ST = I$. □

## Isomorphic Vector Spaces (Sub-Section 3D2)

**Def 3.69:**

An isomorphism is an invertible linear map. Two vector spaces are called isomorphic if there is an isomorphism from one vector space to the other, $T : V \to W$.

**Thm 3.70** (*dimension shows whether vector spaces are isomorphic*)**:**

Two finite-dimensionalvector spaces $U, V$ over $\mathbb{F}$ are isomorphic $\iff$ they have the same dimension (i.e., $\dim U = \dim V$).

***Proof:*** exercise

$\Rightarrow$ ***direction:*** rank-nullity theorem

$\Leftarrow$ ***direction:*** surjectivity and injectivity. □

**Thm 3.71** ($\mathcal{L}(V, W)$ *and* $\mathbb{F}^{m,n}$ *are isomorphic*)**:**

Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$. Then $\mathcal{M}$ is an isomorphism between $\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$.

***Proof:*** exercise (injectivity and surjectivity) □

**Thm 3.72** ($\dim \mathcal{L}(V, W) = (\dim V) \cdot (\dim W)$)**:**

Suppose $V, W$ are finite-dimensional. Then $\mathcal{L}(V, W)$ is finite-dimensional, and

$$\dim \mathcal{L}(V, W) = (\dim V) \cdot (\dim W).$$

***Proof:*** Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $w_1, \ldots, w_m$ is a basis of $W$. By 3.71, there is an isomorphism between $\mathcal{L}(V, W)$ and $\mathbb{F}^{m,n}$, and thus, they have the same dimension. Now, 3.40 tells us that $\dim \mathbb{F}^{m,n} = mn$; therefore,

$$\dim \mathcal{L}(V, W) = \dim \mathbb{F}^{m,n} = m \cdot n = (\dim V) \cdot (\dim W),$$

as desired. □

## Linear Maps Thought of as Matrix Multiplication (Sub-Section 3D3)

**Def 3.73** (*matrix of a vector,* $\mathcal{M}(u)$)**:**

Let $u \in V$ and let $v_1, \ldots, v_m$ be a basis of $V$ such that

$$u = b_1 v_1 + \cdots + b_m v_m. \text{ We define } \mathcal{M}(u) := \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}.$$

The vector depends on the basis, but it is not included in the notation.

**Thm 3.75** ($\mathcal{M}(T)_{\bullet,k} = \mathcal{M}(Tv_k)$)**:**

Suppose $T \in \mathcal{L}(V, W)$, $v_1, \ldots, v_n$ is a basis of $V$, and $w_1, \ldots, w_m$ is a basis of $W$. Let $1 \le k \le n$. Then the

$k^{\text{th}}$ column of $\mathcal{M}(T)$, which is denoted by $\mathcal{M}(T)_{\bullet,k}$, equals $\mathcal{M}(Tv_k)$:

$$\mathcal{M}(T)_{\bullet,k} = \mathcal{M}(Tv_k) \quad \forall k \in \{1, \dots, n\}.$$

**Thm 3.76** (*linear maps act like matrix multiplication*):

Suppose $T \in \mathcal{L}(V, W)$ and $v \in V$. Suppose $v_1, \dots, v_n$ is a basis of $V$ and $w_1, \dots, w_m$ is a basis of $W$. Then

$$\mathcal{M}(Tv) = \mathcal{M}(T) \cdot \mathcal{M}(v).$$

Or using different notation:

$$\mathcal{M}(Tv, (v_1, \dots, v_n), (w_1, \dots, w_m)) = \mathcal{M}(T, (v_1, \dots, v_n), (w_1, \dots, w_m)) \cdot \mathcal{M}(v).$$

**Thm 3.78** (*dimension of* $\operatorname{range} T$ *and column rank of* $\mathcal{M}(T)$):

Suppose $\dim V < \infty$ and $\dim W < \infty$. Then, for $T \in \mathcal{L}(V, W)$:

$$\dim \operatorname{range} T = \text{column rank of } \mathcal{M}(T).$$

## Change of basis (Sub-Section 3D4)

**Def 3.80** (*invertible, inverse,* $A^{-1}$):

A square matrix $A$ is called invertible if there is some square matrix $B$ of the same size such that

$$AB = BA = I.$$

We call $B$ the inverse of $A$, denoted by $A^{-1} :\equiv B$. Rules:

- $(A^{-1})^{-1} = A$
- $(AC)^{-1} = C^{-1}A^{-1}$ (Because $(AC)(C^{-1}A^{-1}) = I$ and $(C^{-1}A^{-1})(AC) = I$)

**Thm 3.81** (*matrix of product of linear maps*):

Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$. If $u_1, \dots, u_m$ is a basis of $U$, $v_1, \dots, v_n$ is a basis of $V$, and $w_1, \dots, w_p$ is a basis of $W$. Note that $\dim U = m$, $\dim V = n$, $\dim W = p$. Then we have:

$$\mathcal{M}(ST, (u_1, \dots, u_m), (w_1, \dots, w_p)) =$$
$$\mathcal{M}(S, (v_1, \dots, v_n), (w_1, \dots, w_p)) \cdot \mathcal{M}(T, (u_1, \dots, u_m), (v_1, \dots, v_n)).$$

Or using different notation:

$$\mathcal{M}(ST) = \mathcal{M}(S) \cdot \mathcal{M}(T).$$

***Proof:*** This is just another way of writing 3.43. □

**Thm 3.82** (*matrix of identity operator with respect to two bases*):

Suppose that $u_1, \dots, u_m$ and $v_1, \dots, v_m$ are bases of $V$. Then the matrices

$$\mathcal{M}\Big(I, (u_1, \dots, u_m), (v_1, \dots, v_m)\Big) \qquad \text{and} \qquad \mathcal{M}\Big(I, (v_1, \dots, v_m), (u_1, \dots, u_m)\Big)$$

are invertible, and each is the inverse of the other.

***Proof:*** Careful application of 3.81 yields

$$\mathcal{M}\Big(I_V\Big) = I_n = \mathcal{M}\Big(I_V I_V\Big) = \mathcal{M}\Big(I_V, (v_1, \dots, v_m), (u_1, \dots, u_m)\Big) \cdot \mathcal{M}\Big(I_V, (u_1, \dots, u_m), (v_1, \dots, v_m)\Big).$$

Now, interchange the roles of the $u$'s and the $v$'s, getting

$$\mathcal{M}\Big(I_V\Big) = I_n = \mathcal{M}\Big(I_V I_V\Big) = \mathcal{M}\Big(I_V, (u_1, \dots, u_m), (v_1, \dots, v_m)\Big) \cdot \mathcal{M}\Big(I_V, (v_1, \dots, v_m), (u_1, \dots, u_m)\Big).$$

So, the product of both matrices equals $I_n$, which is the identity in $\mathbb{F}^{n,n}$. □

**Thm 3.84** (*change-of-basis formula*):

Let $T \in \mathcal{L}(V)$. Suppose $u_1, \ldots, u_m$ and $v_1, \ldots, v_m$ are bases of $V$. Let

$$A := \mathcal{M}\Big(T, (u_1, \ldots, u_m)\Big),$$
$$B := \mathcal{M}\Big(T, (v_1, \ldots, v_m)\Big), \text{ and}$$
$$C := \mathcal{M}\Big(I, (u_1, \ldots, u_m), (v_1, \ldots, v_m)\Big).$$

Then

$$A = C^{-1}BC.$$

**Thm 3.86:**

If $v_1, \ldots, v_m$ is a basis of $V$ and $T \in \mathcal{L}V$ is invertible, then

$$\mathcal{M}(T^{-1}) = (\mathcal{M}(T))^{-1},$$

where both matrices are with respect to the basis $v_1, \ldots, v_m$, that is,

$$\mathcal{M}(T^{-1}, (v_1, \ldots, v_m)) = \mathcal{M}(T, (v_1, \ldots, v_m))^{-1}.$$

## Exercises about Invertibility and Isomorphisms 3D

(1) Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that $T^{-1}$ is invertible and $(T^{-1})^{-1} = T$.

***Proof:*** $T^{-1}$ is the unique element in $\mathcal{L}(W, V)$ such that

$$T^{-1}T = I_V \text{ and } TT^{-1} = I_W.$$

But this makes $T$ the unique element in $\mathcal{L}(V, W)$ such that

$$T(T^{-1}) = I_W \text{ and } T^{-1}T = I_V.$$

This makes $T$ the inverse of $T^{-1}$. Since the inverse is unique by THM 3.60, we conclude that $(T^{-1}) = T$. □

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(2) Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

***Proof:*** Let $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ be both invertible linear maps. Thus:

$$(ST)^{-1}(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SS^{-1} = I_W$$
$$(T^{-1}S^{-1})(ST)^{-1} = T^{-1}(S^{-1}S)T = T^{-1}T = I_U$$

Therefore, $(ST)^{-1} = T^{-1}S^{-1}$. □

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(3) Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent:

**(a)** $T$ is invertible.

**(b)** $Tv_1, \ldots, Tv_n$ is a basis of $V$ for every basis $v_1, \ldots, v_n$ of $V$.

**(c)** $Tv_1, \ldots, Tv_n$ is a basis of $V$ for some basis $v_1, \ldots, v_n$ of $V$.

***Proof:*** For all proof-steps, let $T \in \mathcal{L}(V)$ and $V$ be a finite-dimensional vector space. In each step of the proof, we show one equivalence listed above.

***Step 1:*** Clearly, (b) implies (c).

***Step 2:*** We start by showing that (c) implies (a). Actually, the proof that (b) implies (a) would look the same. Suppose $Tv_1, \ldots, Tv_n$ is a basis of $V$ for some basis $v_1, \ldots, v_n$. Since $\dim V = n$, we have $\operatorname{range} T = V$. It follows that $Tv_1, \ldots, Tv_n$ is a basis of $\operatorname{range} T$. So $T$ is a surjective map on $V$ by the definition 3.19 of surjectivity. Using theorem 3.65, we conclude that $T$ is invertible. It states that for a linear map from a vector space to another one of the same dimension, invertibility, injectivity and surjectivity are equivalent.

***Step 3:*** Now we want to show that (c) implies (b). So let $Tv_1, \ldots, Tv_n$ be a basis of $V$ for some basis of $v_1, \ldots, v_n$ of $V$. Let $u_1, \ldots, u_n$ be another basis of $V$. We already know that (c) $\Longrightarrow$

(a), so $T$ is injective. This means, that $Tu_1, \dots, Tu_n$ is a basis of $V$ as well. Since $u_1, \dots, u_n$ was arbitrary, we have shown that (c) $\implies$ (b).

***Step 4:*** Lastly, we want to show that (a) implies (c). Suppose $T \in \mathcal{L}(V)$ such that $T$ is invertible. Let $v_1, \dots, v_n$ be a basis of $V$. Let $a_1, \dots, a_n \in \mathbb{F}$ such that

$$0 = a_1 Tv_1 + \cdots a_n Tv_n$$

Hence,

$$\begin{aligned} T^{-1}(0) = T^{-1}(a_1 Tv_1 + \cdots a_n Tv_n) &\iff \\ 0 = a_1 T^{-1}Tv_1 + \cdots + a_n T^{-1}Tv_n &\iff \\ 0 = a_1 v_1 + \cdots a_n v_n. \end{aligned}$$

Therefore, $0 = a_1 = \cdots = a_n$, because all the $v$'s are linearly independent. We conclude that $Tv_1, \cdots, Tv_n$ is a linearly independent list of length $n = \dim V$ and hence a basis of $V$. Thus, (a) implies (c).

Overall, we have shown that (b) $\implies$ (c), (c) $\implies$ (b), (c) $\implies$ (a), and (a) $\implies$ (c). So all three statements are equivalent. $\square$

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5 Suppose $V$ is finite-dimensional, $U$ is a subspace of $V$, and $S \in \mathcal{L}(U, V)$. Prove that there exists an invertible linear map $T \in \mathcal{L}(U, V)$ such that $Tu = Su \;\; \forall u \in U \iff S$ is injective.

***Proof:*** Suppose $V$ is finite-dimensional, $U$ is a subspace of $V$ and $S \in \mathcal{L}(U, V)$.

$\Rightarrow$ ***direction:*** Suppose for every $u \in U$ we have that $Tu = Su$. Let $u_1, u_2 \in U$ such that $S(u_1) = S(u_2)$. This implies that

$$\begin{aligned} &T(u_1) = T(u_2) \\ &T^{-1}T(u_1) = T^{-1}T(u_2) \\ &u_1 = u_2 \end{aligned}$$

Hence, $S$ is injective.

$\Leftarrow$ ***direction:*** Suppose $S$ is injective. Let $u_1, \dots, u_m$ be a basis of $U$ and $Su_1, \dots, Su_m$ be a basis of range $S$. We extend

$$u_1, \dots, u_m, v_{m+1}, \dots, v_m \tag{3.13}$$

to a basis of $V$, and

$$Su_1, \dots, Su_m, r_{m+1}, \dots, r_n \tag{3.14}$$

to a basis of $V$ as well. We define $T \in \mathcal{L}(V)$ for each $v \in V$ as follows:

$$\begin{aligned} T(v) &= T(c_1 u_1 + \cdots + c_m u_m + c_{m+1} v_{m+1} + \cdots + c_n v_n) \\ &:= c_1 Su_1 + \cdots + c_m Su_m + c_{m+1} r_{m+1} + \cdots + c_n r_n. \end{aligned}$$

Clearly, $\forall u \in U : T(u) = S(u)$. Now we have to show that $T$ is injective and surjective. If $v_1, v_2 \in V$ such that

$$y_1 = T(v_1) = T(v_2) = y_2$$

All the $c$ coefficents of $y_1$ and $y_2$, which are defined over the basis (1) match the $c$ coefficients of $v_1$ and $v_2$, which are defined over the basis (1). So these representations are unique and therefore $v_1 = v_2$. We conclude that $T$ is injective. To prove that $T$ is surjective, we have to show that for any given $y \in V$ there is a $v \in V$ such that $T(v) = y$. So let

$$y := c_1 Su_1 + \cdots + c_m Su_m + c_{m+1} r_{m+1} + \cdots + c_n r_n.$$

Then there is exactly one $v \in V$ such that $T(v) = y$, namely

$$v := c_1 u_1 + \cdots + c_m u_m + c_{m+1} v_{m+1} + \cdots + c_n v_n.$$

For this choice of $y$ and $v$, we have $T(v) = y$. Hence, $T$ is injective and surjective. This makes $T$

invertible. $\square$

# Products and Quotients of Vector Spaces (Section 3E)

## Products of Vector Spaces (Sub-Section 3E1)

**Def 3.87** (*product of vector spaces*):

The product, addition, and scalar multiplication of a list of vector spaces $V_1, \dots, V_m$ are defined as follows:

$$\begin{aligned} V_1 \times \cdots \times V_m &:\equiv \{(v_1, \dots, v_m) \mid v_1 \in V_1, \dots, v_m \in V_m\} \\ (u_1, \dots, u_m) + (v_1, \dots, v_m) &:\equiv (u_1 + v_1, \dots, u_m + v_m) \\ \lambda(v_1, \dots, v_m) &:\equiv (\lambda v_1, \dots, \lambda v_m) \end{aligned}$$

**Thm 3.89** (*product of vector space is a vector space*):

$V_1 \times \cdots \times V_m$ together with addition and scalar multiplication is a vector space over $\mathbb{F}$.

**Thm 3.92** (*dimension of a product is the sum of dimensions*):

Suppose $V_1, \dots, V_m$ are finite-dimensional vector spaces. Then $V_1 \times \cdots \times V_m$ is finite-dimensional, and

$$\dim(V_1 \times \cdots \times V_m) = \dim V_1 + \cdots + \dim V_m.$$

***Proof:*** Choose a basis $v_{k_1}, \dots, v_{k_{\dim V_k}}$ for each $V_k$. For each basis vector of each $V_k$, consider the element of $V_1 \times \cdots \times V_m$ that equals the basis vector in the $k$-th slot and 0 in the other slots:

$$\underbrace{\begin{pmatrix} v_{1_1} \\ 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} v_{1_2} \\ 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} v_{1_{\dim V_1}} \\ 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \end{pmatrix}}_{\text{constructed from basis vectors of } V_1}, \dots, \underbrace{\begin{pmatrix} 0 \\ \vdots \\ v_{k_1} \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \vdots \\ v_{k_2} \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ \vdots \\ v_{k_{\dim V_k}} \\ 0 \\ \vdots \\ 0 \end{pmatrix}}_{\text{constructed from basis vectors of } V_k}, \dots, \underbrace{\begin{pmatrix} 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \\ v_{m_1} \end{pmatrix}, \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \\ v_{m_2} \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \\ v_{m_{\dim V_m}} \end{pmatrix}}_{\text{constructed from basis vectors of } V_m}$$

The list of all such vectors is linearly independent and spans $V_1 \times \cdots \times V_m$. Thus, it is a basis of $V_1 \times \cdots \times V_m$, and the length of this basis is $(\dim V_1 + \cdots + \dim V_m)$. □

In the next result, the map $\Gamma$ is surjective by the definition of $V_1 + \cdots + V_m$ because of closedness. Thus, the last word in the result below could be changed from *"injective"* to *"invertible"*.

**Thm 3.93** (*products and direct sums*):

Let $\Gamma : V_1 \times \cdots \times V_m \to V_1 + \cdots + V_m$ such that

$$\Gamma(v_1, \dots, v_m) = v_1 + \cdots + v_m \quad \forall v_1 \in V_1, \dots, v_m \in V_m.$$

Then

$$V_1 + \cdots + V_m \text{ is a direct sum} \iff \Gamma \text{ is injective/invertible.}$$

***Proof:*** By 3.15, $\Gamma$ is injective $\iff v_1 = \cdots = v_m = 0$ [Stated correctly, $\Gamma$ is injective $\iff \operatorname{null}\Gamma = \{\vec{0}\}$]. This means the only way to write 0 as a sum $v_1 + \cdots + v_m$, where each $v_k \in V_k$, is by taking each $v_k = 0$. Thus, 1.45 shows that $\Gamma$ is injective $\iff V_1 + \cdots + V_m$ is a direct sum, as desired. □

**Thm 3.94** (*a sum is a direct sum if and only if the dimensions add up*):

Suppose $V$ is finite-dimensional and $V_1, \dots, V_m$ are subspaces of $V$. Then

$$V_1 + \cdots + V_m \text{ is a direct sum (i.e., } \sum_{i=1}^{m} V_i = \bigoplus_{i=1}^{m} V_i) \iff \dim(V_1 + \cdots + V_m) = \dim V_1 + \cdots + \dim V_m.$$

***Proof:*** The map $\Gamma : V_1 \times \cdots \times V_m \to V_1 + \cdots + V_m$ in 3.93 is surjective. Thus, by the rank-nullity theorem

3.21, $\Gamma$ is injective $\iff \operatorname{dim}\operatorname{null}\Gamma = 0$, and therefore

$$\underbrace{\operatorname{dim}(V_1 \times \cdots \times V_m)}_{\text{domain}} = \underbrace{\operatorname{dim}\operatorname{null}\Gamma}_{=0} + \underbrace{\operatorname{dim}(V_1 + \cdots + V_m)}_{\text{codomain} \subseteq V}$$

$$\operatorname{dim}(V_1 + \cdots + V_m) = \operatorname{dim}(V_1 \times \cdots \times V_m).$$

We can use 3.92 to conclude that

$$\operatorname{dim}(V_1 \times \cdots \times V_m) = \operatorname{dim} V_1 + \cdots + \operatorname{dim} V_m \text{ and therefore}$$

$$\operatorname{dim}(V_1 + \cdots + V_m) = \operatorname{dim} V_1 + \cdots + \operatorname{dim} V_m.$$

Since $\Gamma$ is injective, 3.93 tells us that $V_1 + \cdots + V_m$ is a direct sum. So, we have that

$$\operatorname{dim}(V_1 + \cdots + V_m) = \operatorname{dim} V_1 + \cdots + \operatorname{dim} V_m$$

if and only if $V_1 + \cdots + V_m$ is a direct sum.

***Special case for m=2:*** For the special case where $m = 2$, we can use 2.37, where

$$\operatorname{dim}(V_1 + V_2) = \operatorname{dim} V_1 + \operatorname{dim} V_2 - \operatorname{dim}(V_1 \cap V_2),$$

and we get $\operatorname{dim}(V_1 \cap V_2) = 0 \iff V_1 + V_2 = V_1 \oplus V_2$ by 2.43. $\square$

## Quotient Spaces (Sub-Section 3E2)

**Def 3.95** (*notation $v + U$*)**:**

$v + U :\equiv \{v + u \mid u \in U\}$ for $v \in V$ and $U \subseteq V$.

**Def 3.97** (*translate*)**:**

The set $v + U$ is said to be a "*translate*" of $U$, where $v \in V$ and $U \subseteq V$.

**Def 3.99** (*quotient space, $V/U$*)**:**

Suppose $U$ is a subspace of $V$. Then the "*quotient space*" $V/U$ is the set of all translates of $U$. Thus,

$$V/U :\equiv \{v + U \mid v \in V\}.$$

**Example 3.100:**

If $U = \{(x, 2x) \in \mathbb{R}^2 \mid x \in \mathbb{R}\} \implies \mathbb{R}^2/U$ is the set of all lines with slope 2.

If $U$ is a plane in $\mathbb{R}^3 \implies \mathbb{R}^3/U$ is the set of all planes parallel to $U$.

**Thm 3.101** (*two translates of a subspace are equal or disjoint*)**:**

If $U$ is a subspace of $V$ and $v, w \in V$ (not a typo!), then

$$v - w \in U \iff v + U = w + U \iff (v + U) \cap (w + U) \neq \varnothing.$$

***Proof:*** ***Step 1:*** First, suppose $v - w \in U$. If $u \in U$, then

$$v + u = (w - w + v + u) = w + \Big(\underbrace{(v - w)}_{\in U} + u\Big), \text{ thus } v + u \in w + U.$$

Similarly, because $v - w \in U$, we also have $w - v \in U$, and thus,

$$w + u = (v - v + w + u) = v + \Big(\underbrace{(w - v)}_{\in U} + u\Big), \text{ hence } w + u \in v + U.$$

We have shown that $\forall u \in U : v + u \in w + U$, which is equivalent to

$$v + U \subseteq w + U$$

and $\forall u \in U : w + u \in v + U$, which is equivalent to

$$w + U \subseteq v + U.$$

This implies $v + U = w + U$, given that $v - w \in U$, which proves the first equivalence.

***Step 2:*** The equation $v + U = w + U$ implies that $(v + U) \cap (w + U) \neq \varnothing$.

***STEP 3:*** Now, suppose $(v+U)\cap(w+U)\neq\varnothing$. Thus, there exist $u_1, u_2 \in U$ such that

$$v + u_1 = w + u_2,$$

Therefore, $v - w = u_2 - u_1$. Hence, $v - w \in U$. □

**DEF 3.102** (*addition and scalar multiplication on $V/U$*)**:**
Suppose $U$ is a subspace of $V$. Then *"addition"* and *"scalar multiplication"* are defined on $V/U$ by

$$(v+U)+(w+U) :\equiv (v+w)+U$$
$$\lambda(v+U) :\equiv (\lambda v)+U$$

$\forall v, w \in V$ and $\forall \lambda \in \mathbb{F}$.

**THM 3.103** (*quotient space is a vector space*)**:**
$V/U$ is a vector space with additive identity $0+U$, which is equal to $U$, and the additive inverse $(-v)+U$.

**DEF 3.104** (*quotient map, $\pi$*)**:**
Suppose $U \subseteq V$. The *"quotient map"* $\pi : V \to V/U$ is the linear map defined by

$$\pi(v) = v + U \quad \forall v \in V.$$

**THM 3.105:**
Suppose $\dim V < \infty$ and $U \subseteq V$.

$$\implies \dim V/U = \dim V - \dim U.$$

**DEF 3.106** (*notation $\widetilde{T}$*)**:**
Suppose $T \in \mathcal{L}(V, W)$. Define $\widetilde{T} : V/(\operatorname{null} T) \to W$ by

$$\widetilde{T}(v + \operatorname{null} T) = Tv.$$

**THM 3.107** (*null space and range of $\widetilde{T}$*)**:**
Suppose $T \in \mathcal{L}(V, W)$. Then:

**(a)** $\widetilde{T}\circ\pi = T$, where $\pi$ is the quotient map $V$ onto $V/(\operatorname{null} T)$;
**(b)** $\widetilde{T}$ is injective;
**(c)** $\operatorname{range}\widetilde{T} = \operatorname{range} T$;
**(d)** $V/(\operatorname{null} T)$ and $\operatorname{range} T$ are isomorphic vector spaces.

## EXERCISES ABOUT NULL SPACES AND RANGES 3E

*6* Suppose that $v, x \in V$ and that $U, W$ are subspaces of $V$ such that $v + U = x + W$. Prove that $U = W$. ***PROOF:*** Since $v \in v + U$ and $v + U = x + W$, it follows that $v \in x + W$. Hence, there is $w_0 \in W$ with $v = x + w_0$, so $(x - v) = -w_0 \in W$. Now,

$$U = (-v) + (v + U) = (-v) + (x + W) = (x - v) + W = W,$$

where the last equality uses $x - v \in W$ and the fact that translating a subspace by one of its own elements leaves it unchanged (i.e., $(x-v)+W = W$). Hence, $U = W$. □

*10* Suppose $A_1 = v + U_1$ and $A_2 = w + U_2$ for some $v, w \in V$ and some subspaces $U_1, U_2$ of $V$. Prove that the intersection $A_1 \cap A_2$ is either a translate of some subspace of $V$ or is the empty set. ***PROOF:*** By the definition of intersection, we get

$$A_1 \cap A_2 = \{x \in V : x = v + u_1 \text{ and } x = w + u_2,\ u_1 \in U_1,\ u_2 \in U_2\}.$$

The condition $v + u_1 = w + u_2$ is equivalent to

$$v - w = u_2 - u_1, u_1 \in U_1, u_2 \in U_2 \quad \Longleftrightarrow \quad v - w \in U_1 + U_2.$$

If $v - w \notin U_1 + U_2$, there is no solution for $u_1$ and $u_2$ and $A_1 \cap A_2 = \varnothing$. If a solution exists, put

$$\begin{aligned} r^0 &:= v + u_1^0 \\ &:= w + u_2^0, \end{aligned}$$

where $r^0 \in A_1 \cap A_2$ and for appropriate $u_1^0 \in U_1, u_2^0 \in U_2$ (i.e., $v - w = u_2^0 - u_1^0$). Define

$$S := \{r^0 + u : u \in U_1 \cap U_2\}.$$

We want to show that $A_1 \cap A_2 = S$. Therefore, take any $x \in A_1 \cap A_2$, so that $x = v + u_1$ or $x = w + u_2$ with $u_1 \in U_1, u_2 \in U_2$. Now, it follows that

$$\begin{aligned} x - r^0 &= (v + u_1) - (v + u_1^0) = u_1 - u_1^0 \in U_1, \text{ and} \\ x - r^0 &= (w + u_2) - (w + u_2^0) = u_2 - u_2^0 \in U_2, \end{aligned}$$

and therefore, $x - r^0 \in U_1 \cap U_2$. Since $x = r^0 + (x - r^0)$, it is immediate that

$$x \in r^0 + (U_1 \cap U_2).$$

Hence, $A_1 \cap A_2 \subseteq S$. Conversely, pick any $u \in U_1 \cap U_2$, so that $r^0 + u \in S$. Then

$$\begin{aligned} r^0 + u &= (v + u_1^0) + u = v + (u_1^0 + u) \in v + U_1 = A_1, \text{ and} \\ r^0 + u &= (w + u_2^0) + u = w + (u_2^0 + u) \in w + U_2 = A_2. \end{aligned}$$

Thus, $S \subseteq A_1 \cap A_2$. Therefore, we have shown that

$$A_1 \cap A_2 = \begin{cases} S, & \text{if } v - w \in U_1 + U_2, \\ \varnothing, & \text{otherwise.} \end{cases}$$

$\square$

# Duality (Section 3F)

## Dual Space and Dual Map (Sub-Section 3F1)

**Def 3.108:**

A *"linear functional"* $\phi$ is an element of $\mathcal{L}(V,\mathbb{F})$. So, $\phi \in \mathcal{L}(V,\mathbb{F})$.

**Examples:**

$$\begin{aligned}
&\phi : \mathbb{R}^3 \to \mathbb{R}, && \phi(x,y,z) && \mapsto 4x - 5y - 2z \\
&\phi : \mathbb{F}^n \to \mathbb{F}, && \phi(x_1,\dots,x_n) && \mapsto c_1x_1 + \cdots + c_nx_n \\
&\phi : \mathcal{P}(\mathbb{R}) \to \mathbb{R}, && \phi(p) && \mapsto 3p''(5) + 7p(4) \\
&\phi : \mathcal{P}(\mathbb{R}) \to \mathbb{R}, && \phi(p) && \mapsto \int_0^1 p(x)dx
\end{aligned}$$

**Thm 3.110:**

The dual space of $V$, denoted by $V'$ or $V^*$, is the vector space of all linear functionals on $V$, i.e.,

$$V' := \mathcal{L}(V,\mathbb{F}).$$

**Thm 3.111** ($\dim V' = \dim V$)**:**

Suppose $V$ is finite-dimensional. Then $V'$ is also finite-dimensional, and

$$\dim V' = \dim V.$$

***Proof:*** By 3.72, we have $\dim V' = \dim \mathcal{L}(V,\mathbb{F}) = (\dim V)\cdot(\dim \mathbb{F}) = \dim V$. $\square$

**Def 3.112:**

If $v_1,\dots,v_m$ is a basis of $V$, then the *"dual basis"* of $v_1,\dots,v_m$ is the list $\varphi_1,\dots,\varphi_m \in V'$, where each $\varphi_j$ is the linear functional such that

$$\varphi_j(v_k) = \delta_{j,k} = \begin{cases} 1, & \text{if } k = j \\ 0, & \text{if } k \neq j. \end{cases}$$

Note that this is not the definition of each $\varphi_j$ but a property of each $\varphi_j$.

**Thm 3.114:**

Let $v_1,\dots,v_m$ be a basis of $V$. $\varphi_1,\dots,\varphi_m$ is called *"a dual basis"* of $v_1,\dots,v_m$ because, for all $v \in V$:

$$v = \varphi_1(v)\cdot v_1 + \cdots + \varphi_m(v)\cdot v_m.$$

***Proof:*** Let $v \in V$ such that $v = c_1v_1 + \cdots + c_mv_m$. Applying $\varphi_j$ on both sides gives $\varphi_j(v) = c_j$ for $j = 1,\dots,m$. Hence, $v = \varphi_1(v)v_1 + \cdots + \varphi_m(v)v_m$. $\square$

**Thm 3.116** (*basis of the dual space*)**:**

The dual basis of a basis of $V$ is a basis of the dual space $V'$.

***Proof:*** Suppose $v_1,\dots,v_n$ is a basis of $V$. Let $\varphi_1,\dots,\varphi_n$ denote its dual basis. Suppose

$$a_1\varphi_1 + \cdots + a_n\varphi_n = 0 \text{ for some } a_1,\dots,a_n \in \mathbb{F}.$$

Which means that we get a function in $\mathcal{L}(V,\mathbb{F})$ that always returns 0. In other words, $(a_1\varphi_1 + \cdots + a_n\varphi_n)(v) = 0$ for all $v \in V$. This implies

$$(a_1\varphi_1 + \cdots + a_n\varphi_n)(v_k) = a_k \quad \forall k \in \{1,\dots,n\}.$$

Thus, $a_1 = \cdots = a_n = 0$. Hence, the list $\varphi_1,\dots,\varphi_n$ is linearly independent. Since the list is a linearly independent list in $V'$ whose length equals $\dim V'$ by 3.111, we conclude that $\varphi_1,\dots,\varphi_n$ is a basis of $V'$ by 2.38. $\square$

**Def 3.118** (*dual map, $T'$*)**:**

The *"dual map"* $T'$ of $T \in \mathcal{L}(V,W)$ is the linear map $T' \in \mathcal{L}\big(W',V'\big) = \mathcal{L}\big(\mathcal{L}(W,\mathbb{F}),\mathcal{L}(V,\mathbb{F})\big)$ defined as follows:

$$T'(\varphi) := \varphi \circ T \quad \text{for every } \varphi \in W' = \mathcal{L}(W,\mathbb{F}).$$

Therefore, we have that $T'(\varphi) \in V' = \mathcal{L}(V,\mathbb{F})$. Hence, $T' \in \mathcal{L}\big(W',V'\big)$ is indeed a linear map from $W'$ to $V'$, since:

- $\varphi,\psi \in W' \implies T'(\varphi+\psi) = (\varphi+\psi)\circ T = \varphi\circ T + \psi\circ T = T'(\varphi) + T'(\psi)$
- $\lambda \in \mathbb{F}, \varphi \in W' \implies T'(\lambda\varphi) = (\lambda\varphi)\circ T = \lambda(\varphi\circ T) = \lambda T'(\varphi)$.

**Thm 3.120** (*algebraic properties of dual maps*)**:**

$T \in \mathcal{L}(V,W) \implies$

**(a)** $(S+T)' = S' + T' \quad \forall S \in \mathcal{L}(V,W)$

**(b)** $(\lambda T)' = \lambda T' \qquad \forall \lambda \in \mathbb{F}$

**(c)** $(ST)' = T'S' \quad \forall S \in \mathcal{L}(W,U)$

## Dual Space and Range of Dual of Linear Map (Sub-Section 3F2)

**Def 3.121** (*annihilator, $U^0$*)**:**

For $U \subseteq V$, the *"annihilator"* of $U$, denoted by $U^0$, is defined by

$$U^0 = \{\varphi \in V' \mid \varphi(u) = 0 \quad \forall u \in U\}.$$

**Thm 3.124** (*the annihilator is a subspace*)**:**

If $U \subseteq V$, then $U^0$ is a subspace of $V'$.

**Thm 3.125** (*dimension of the annihilator*)**:**

If $U \subseteq V$ and $\dim V < \infty$, then

$$\dim U^0 = \dim V - \dim U.$$

**Thm 3.127** (*condition for the annihilator to equal* $\{0\}$ *or the whole space*)**:**

If $\dim V < \infty$ and $U \subseteq V$, then:

**(a)** $U^0 = \{0\} \iff U = V$

**(b)** $U^0 = V' \iff U = \{0\}$

**Thm 3.128** (*the null space of $T'$*)**:**

If $\dim V < \infty$, $\dim W < \infty$, and $T \in \mathcal{L}(V,W)$, then:

**(a)** $\operatorname{null} T' = (\operatorname{range} T)^0$

**(b)** $\dim \operatorname{null} T' = \dim \operatorname{null} T + \dim W - \dim V$

**Thm 3.129** (*$T$ surjective, $T'$ injective*)**:**

If $\dim V < \infty$, $\dim W < \infty$, and $T \in \mathcal{L}(V,W)$, then:

$$T \text{ is surjective} \iff T' \text{ is injective.}$$

**Thm 3.130** (*the range of $T'$*)**:**

If $\dim V < \infty$, $\dim W < \infty$, and $T \in \mathcal{L}(V,W)$, then:

**(a)** $\dim \operatorname{range} T' = \dim \operatorname{range} T$

**(b)** $\operatorname{range} T' = (\operatorname{null} T)^0$

**Thm 3.131** (*$T$ injective, $T'$ surjective*)**:**

If $\dim V < \infty$, $\dim W < \infty$, and $T \in \mathcal{L}(V,W)$, then:

$T$ is injective $\iff$ $T'$ is surjective.

## Matrix of Dual of Linear Map (Sub-Section 3F3)

**Thm 3.132:**

Suppose $V, W$ are finite-dimensional and $T \in \mathcal{L}(V,W)$.

$\implies \mathcal{M}(T') = (\mathcal{M}(T))^\top$.

**Thm 3.133:**

If $A \in \mathbb{F}^{m,n}$, then the column rank of $A$ equals the row rank of $A$.

## Exercises about Duality 3F

(1) Explain why each linear functional is surjective or the zero map.

***Solution:*** $\varphi \in V' = \mathcal{L}(V,\mathbb{F})$ implies that for every $\lambda \in \mathbb{F}$, and every $v \in V$,

$$\varphi(\lambda v) = \lambda\varphi(v).$$

If $\varphi$ is not the zero map, then $\varphi$ evaluated at $v \in V$ or $\lambda v \in V$ is a scalar in $\mathbb{F}$ for some $v \in V$ for which it holds that $\varphi(v) \neq 0$. In this case, we have $\varphi(v) \in \mathbb{F}$ and $\varphi(\lambda v) = \lambda\varphi(v) \in \mathbb{F}$. Hence, $\operatorname{range}\varphi = \mathbb{F}$, since $\lambda \in \mathbb{F}$ was arbitrary.

---

(3) Suppose $V$ is finite-dimensional and $v \in V$ is nonzero. Prove that there exists $\varphi \in V'$ such that $\varphi(v) = 1$.

***Solution:*** Let $v_1, \ldots, v_n$ be a basis of $V$. Let $v \in V$ be nonzero such that $v = a_1v_1 + \cdots + a_nv_n$, where $a_i \in \mathbb{F}$. Since $v$ is nonzero, there exists a $k$ such that $a_k \neq 0$. Define $\varphi : V \to \mathbb{R}$ by

$$\varphi(u) = \frac{c_k}{a_k},$$

whenever $u = c_1v_1 + \cdots + c_nv_n \in V$. Note that this definition corresponds to the use of the linear map lemma (3.4), where we define $\varphi$ on each basis vector as follows:

$$\varphi(v_1) = 0,\ \ldots,\ \varphi(v_{k-1}) = 0,\ \varphi(v_k) = \frac{1}{a_k},\ \varphi(v_{k+1}) = 0,\ \ldots,\ \varphi(v_n) = 0.$$

Hence, $\varphi(a_1v_1 + \cdots + a_nv_n) = \frac{a_k}{a_k} = 1$. It is easy to check that $\varphi$ is a linear map.

---

(5) Suppose $V$ is finite dimensional and $U$ is a subspace to $V$, where $U \neq V$. Prove that there exists a $\varphi \in V'$ such that $\varphi(u) = 0$ for every $u \in U$ but $\varphi \neq 0$.

***Solution:*** Choose $W \subset V$ such that $V = U \oplus W$. Hence, there is a basis

$$u_1, \ldots, u_n, w_1, \ldots, w_m$$

of $V$ such that the $u$'s and $w$'s form a basis of $U$ and $W$ respectively. We define $\varphi \in V'$ such that for $v \in V$, written as

$$v = a_1u_1 + \cdots + a_nu_n + b_1w_1 + \cdots + b_mw_m,$$

we have

$$\varphi(v) = b_1 + \cdots + b_m.$$

It is easy to see that $\varphi(u) = 0$ vor every $u \in U$ (since $U \cap W = \{0\}$, by 1.46) and that $\varphi$ is linear.

---

(10) Suppose $m$ is positive integer.

**(a)** Show that $1, x-5, \ldots, (x-5)^m$ is a basis of $\mathcal{P}_m(\mathbb{R})$.

**(b)** What is the dual basis of the basis in (a)?

***Solution:*** We first start the exercise by solving it for $\mathcal{P}_4(\mathbb{R})$ straight forward before solving it more abstract and generic for $\mathcal{P}_m(\mathbb{R})$. Recall that $e_0 = x^0, e_1 = x^1, \dots, e_4 = x^4$ is the standard basis of $\mathcal{P}_4(\mathbb{R})$. We want to show that $v_0 = 1, v_1 = x-5, \dots, v_4 = (x-5)^4$ is a basis as well. If we do some rewiting and expansion of the formulas, we get

$$\begin{aligned}
v_0 &= (x^1 - 5x^0)^0 = 1x^0\ , \\
v_1 &= (x^1 - 5x^0)^1 = -5x^0 + x^1\ , \\
v_2 &= (x^1 - 5x^0)^2 = 25x^0 - 10x^1 + x^2\ , \\
v_3 &= (x^1 - 5x^0)^3 = -125x^0 + 75x^1 - 15x^2 + x^3\ , \\
v_4 &= (x^1 - 5x^0)^4 = 625x^0 - 500x^1 + 150x^2 - 20x^3 + x^4\ .
\end{aligned}$$

Now, for a given $p \in \mathcal{P}_4(\mathbb{R})$ so that $p(x) = b_0x^0 + b_1x^1 + \cdots + b_4x^4$ ($b_j \in \mathbb{F}$) we want to show that there exists $a_i \in \mathbb{F}$ such that $\sum_{j=0}^4 b_je_j = \sum_{j=0}^4 a_jv_j$. So let's expand

$$\begin{aligned}
\textstyle\sum_{j=0}^4 a_jv_j = a_0(1x^0) &+ a_1(-5x^0 + x^1) + a_2(25x^0 - 10x^1 + x^3) \\
&+ a_3(-125x^0 + 75x^1 - 15x^2 + x^3) + a_4(625x^0 - 500x^1 + 150x^2 + x^4) \\
&= (a_0 - 5a_1 + 25a_2 - 125a_3 + 625a_4) + (a_1 - 10a_2 + 75a_3 - 500a_4)x \\
&\quad + (a_2 - 15a_3 + 150a_4)x^2 + (a_3 - 20a_4)x^3 + a_4x^4 \\
&= b_0x^0 + b_1x^1 + \cdots + b_4x^4 = \textstyle\sum_{j=0}^4 b_je_j.
\end{aligned}$$

We can solve this for $a_j$ by back substitution:

$a_4 = b_4,$

$a_3 = b_3 + 20a_4,$

$a_2 = b_2 + 15a_3 - 150a_4,$

$a_1 = b_1 + 10a_2 + 75a_3 - 500a_4,$ and finally

$a_0 = b_0 + 5a_1 - 25a_2 + 125a_3 - 625a_4.$

Hence, $v_1, \dots, v_4$ is indeed a basis of $\mathcal{P}_4(\mathbb{R})$. Now, we want to find the dual basis $\psi_0, \psi_1, \dots, \psi_4$ of $v_1, \dots, v_4$. Recall that the condition

$$\phi_j(v_k) = \delta_{j,k} = \begin{cases} 1, & j = k \\ 0, & j \neq k \end{cases}$$

must hold, and we thus want to define each $\phi_j$ on the standard basis $e_0, \dots, e_4$. So let's start with $\psi_0$, where $\psi_0(v_0) = 1$ and zero on all other $v$'s.

$$\begin{aligned}
&\psi_0(v_0) = 1\psi_0(x^0) = 1\ . \implies \psi_0(x^0) = 1. \\
&\psi_0(v_1) = -5\psi_0(x^0) + \psi_0(x^1) = 0. \implies \psi_0(x^1) = 5\ . \\
&\psi_0(v_2) = 25psi_0(x^0) - 10\psi_0(x^1) + \psi_0(x^2) = 0. \implies \phi(x^2) = 25\ . \\
&\psi_0(v_3) = -125\psi_0(x^0) + 75\psi_0(x^1) - 15\psi_0(x^2) + \psi_0(x^3) = 0. \implies \psi_0(x^3) = 125. \\
&\psi_0(v_4) = 625\psi_0(x^0) - 500\psi_0(x^1) + 150\psi_0(x^2) - 20\psi_0(x^3) + \psi_0(x^4) = 0. \implies \psi_0(x^4) = 625.
\end{aligned}$$

Now, the same for $\psi_1$:

$$\begin{aligned}
&\psi_1(v_0) = 1\psi_1(x^0) = 0\ . \implies \psi_0(x^0) = 0. \\
&\psi_1(v_1) = -5\psi_1(x^0) + \psi_1(x^1) = 1. \implies \psi_1(x^1) = 1\ . \\
&\psi_1(v_2) = 25psi_0(x^0) - 10\psi_1(x^1) + \psi_1(x^2) = 0. \implies \psi_1(x^2) = 10\ . \\
&\psi_1(v_3) = -125\psi_1(x^0) + 75\psi_1(x^1) - 15\psi_1(x^2) + \psi_1(x^3) = 0. \implies \psi_1(x^3) = 75. \\
&\psi_1(v_4) = 625\psi_1(x^0) - 500\psi_1(x^1) + 150\psi_1(x^2) - 20\psi_1(x^3) + \psi_1(x^4) = 0. \implies \psi_1(x^4) = 500.
\end{aligned}$$

Now, $\psi_2$:

$\psi_2(v_0) = 1\psi_2(x^0) = 0\,. \implies \psi_2(x^0) = 0.$

$\psi_2(v_1) = -5\psi_2(x^0) + \psi_2(x^1) = 0. \implies \psi_2(x^1) = 0\,.$

$\psi_2(v_2) = 25\psi_2(x^0) - 10\psi_2(x^1) + \psi_2(x^2) = 1. \implies \psi_2(x^2) = 1\,.$

$\psi_2(v_3) = -125\psi_2(x^0) + 75\psi_2(x^1) - 15\psi_2(x^2) + \psi_2(x^3) = 0. \implies \psi_2(x^3) = 15.$

$\psi_2(v_4) = 625\psi_2(x^0) - 500\psi_2(x^1) + 150\psi_2(x^2) - 20\psi_2(x^3) + \psi_2(x^4) = 0. \implies \psi_2(x^4) = 150.$

For $\psi_3$ we get $\psi_3(x^0) = \psi_3(x_1) = \psi_3(x^2) = 0$ and $\psi_3(x^3) = 1$. Now what is left to calculate is

$$\psi_3(v_4) = 625\psi_2(x^0) - 500\psi_2(x^1) + 150\psi_2(x^2) - 20\psi_2(x^3) + \psi_2(x^4) = 0.$$

Hence, $\psi_4(v_4) = 20$. For $\psi_4$ we get $\psi_4(x^0) = \psi_4(x_1) = \cdots = \psi_4(x^3) = 0$, and $\psi_4(v_4) = 1$, and thus, $\psi_4(x^4) = 1$. Now, let's take a more generic approach making extensive use of the binomic formula

$$(a+b)^n = \sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i}, \quad (a, b \in \mathbb{R}, n \in \mathbb{N}),$$

and Pascal's identity

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \quad (n, k \in \mathbb{N}).$$

We start by rewriting once again

$$\begin{aligned}
v_0 &= (x^1 - 5x^0)^0 = \binom{0}{0}(-5)^0 e_0 = e^0\,, \\
v_1 &= (x^1 - 5x^0)^1 = \binom{1}{0}(-5)^1 e_0 + \binom{1}{1}(-5)^0 e_1\,, \\
v_2 &= (x^1 - 5x^0)^2 = \binom{2}{0}(-5)^2 e_0 + \binom{2}{1}(-5)^1 e_1 + \binom{2}{2}(-5)^0 e_2\,, \\
&\vdots \\
v_m &= (x^1 - 5x^0)^m = \binom{m}{0}(-5)^m e_0 + \binom{m}{1}(-5)^{m-1} e_1 + \cdots + \binom{m}{m}(-5)^0 e_m = \sum_{i=0}^{m} \binom{m}{i}(-5)^{m-i} e_i.
\end{aligned}$$

Now, $\psi_j(e_k) = \binom{k}{j} 5^{k-j}$, taylor polynomials. (why?).

ChatGTP: Now, why is $\psi_j(e_k) = \binom{k}{j} \cdot 5^{k-j}$? We want to define the dual basis $\{\psi_j\}_{j=0}^m$ of $\{v_k\}_{k=0}^m$, meaning we require

$$\psi_j((x-5)^k) = \delta_{j,k}.$$

Using the binomial theorem, we can expand each $(x-5)^k$ as

$$(x-5)^k = \sum_{i=0}^{k} \binom{k}{i}(-5)^{k-i} e_i.$$

To satisfy the condition $\psi_j((x-5)^k) = \delta_{j,k}$, it is enough to define $\psi_j$ on the standard basis $\{e_k\}$ by

$$\psi_j(e_k) = \binom{k}{j} \cdot 5^{k-j}.$$

This works because $\psi_j$ corresponds to applying the $j$-th derivative at $x = 5$, divided by $j!$, to polynomials:

$$\psi_j(p(x)) = \frac{1}{j!} p^{(j)}(5),$$

and

$$\frac{1}{j!} \cdot \frac{d^j}{dx^j} x^k \Big|_{x=5} = \binom{k}{j} \cdot 5^{k-j}.$$

Thus, $\{\psi_j\}_{j=0}^m$ is indeed the dual basis of $\{v_k\}_{k=0}^m$.

---

**11** Suppose $v_1, \ldots, v_n$ is a basis of $V$ and $\varphi_1, \ldots, \varphi_n$ is the corresponding dual basis of $V'$. Suppose

$\psi \in V'$. Prove that

$$\psi = \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n. \tag{3.15}$$

***Solution:*** For a given $\psi \in V'$, let $\Gamma \in V'$ defined by

$$\Gamma := \psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n$$

juste like in (1). We want to show that $\Gamma = \psi$. Observe that $\Gamma$ is a linear functional, since it is the linear combination of some linear functionals $\varphi_1, \ldots, \varphi_n$ (which are the dual basis of $V$). In this case, $\psi(v_1), \ldots, \psi(v_n)$ are the scalars of this sum, which do not change for a given $\psi$ and a given basis of $V$. If we now apply $\Gamma$ to each $v_k$ (for $k = 1, \ldots, n$), we get

$$\begin{aligned}\Gamma(v_k) &= (\psi(v_1)\varphi_1 + \cdots + \psi(v_n)\varphi_n)(v_k)\\ &= \psi(v_1)\varphi_1(v_k) + \cdots + \psi(v_k)\varphi_n(v_k) + \cdots + \psi(v_n)\varphi_n(v_k)\\ &= \psi(v_1)\cdot 0 + \cdots + \psi(v_k)\cdot 1 + \cdots + \psi(v_k)\cdot 0\\ &= \psi(v_k).\end{aligned}$$

Hence, $\Gamma$ and $\psi$ agree on every basis vector $v_k$. Therefore, $\Gamma = \psi$.

**14** Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T(x,y,z) := (4x + 5y + 6z, 7x + 8y + 9z).$$

Suppose $\varphi_1, \varphi_2$ denotes the dual basis of the standard basis of $\mathbb{R}^2$ and $\psi_1, \psi_2, \psi_3$ denotes the dual basis of the standard basis of $\mathbb{R}^3$.

**(a)** Describe the linear functionals $T'(\varphi_1)$ and $T'(\varphi_2)$.

**(b)** Write $T'(\varphi_1)$ and $T'(\varphi_2)$ as a linear combination of $\psi_1, \psi_2, \psi_3$.

***Solution:*** Let's start by getting comfortable about the notation. Let $e_1, e_2, e_3$ denote the standard basis of $\mathbb{R}^3$ and $e_1, e_2$ denote the standard basis of $\mathbb{R}^2$. In this case, $T(x,y,z) = (4x + 5y + 6z, 7x + 8y + 9z)$ actually means

$$T(xe_1 + ye_2 + ze_3) = (4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2$$

Thus, we have $Te_1 = 4e_1 + 7e_2$, $Te_2 = 5e_1 + 8e_2$, and $Te_3 = 6e_1 + 9e_2$. The Matrix $\mathcal{M}(T)$ of $T$ would look like this:

$$\mathcal{M}(T) = \begin{array}{c c} & \begin{array}{ccc} e_1 & e_2 & e_3 \end{array} \\ \begin{array}{c} e_1 \\ e_2 \end{array} & \begin{pmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \end{array}$$

Now, $T'(\varphi_1) = \varphi_1 \circ T$ and $T'(\varphi_2) = \varphi_2 \circ T$. Hence,

$$T'(\varphi_1)(xe_1 + ye_2 + ze_3) = \varphi_1((4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2) = 4x + 5y + 6z\,, \text{ and}$$
$$T'(\varphi_2)(xe_1 + ye_2 + ze_3) = \varphi_2((4x + 5y + 6z)e_1 + (7x + 8y + 9z)e_2) = 7x + 8y + 9z,$$

which answers question (a). For question (b), we use the result (1) from question 11 and write

$$T'(\varphi_1) = \sum_{k=1}^{3} T'(\varphi_1)(e_k)\psi_k = 4\psi_1 + 5\psi_2 + 6\psi_3\,, \text{ and}$$
$$T'(\varphi_2) = \sum_{k=1}^{3} T'(\varphi_2)(e_k)\psi_k = 7\psi_1 + 8\psi_2 + 9\psi_3$$

***Variation of Qustion 14:*** The question becomes more interesting if we use another basis for $\mathbb{R}^2$ and $\mathbb{R}^3$, say $v_1 = (1,1), v_2 = (2,1)$ for $\mathbb{R}^2$ and $w_1 = (2,2,0), w_2 = (4,2,0), w_3 = (0,0,1)$ for $\mathbb{R}^3$. In this case, let $\phi_1, \phi_2$ denote the dual basis of $v_1, v_2$ and $\psi_1, \psi_2, \psi_3$ denote the dual basis of $w_1, w_2, w_3$. Now, we have to calculate $\phi$ and $\psi$ first. Per definition, we have

$$\phi(v) = \phi(a_1 v_1 + a_2 v_2) = a_1\phi_1(v_1) + a_2\phi_2(v_2).$$

And we must assure that $\phi_1(v_1) = 1, \phi(v_2) = 0$ and $\phi_2(v_2) = 0$ and $\phi_2(v_2) = 1$. Using the facts that $v_1 = e_1 + e_2$ and $v_2 = 2e_1 + e_2$, we get

$$\left| \begin{array}{r} 1\phi_1(e_1) + 1\phi_1(e_2) = 1 \\ 2\phi_1(e_1) + 1\phi_1(e_2) = 0 \end{array} \right| \implies \left| \begin{array}{r} 1\phi_1(e_1) + 1\phi_1(e_2) = 1 \\ -2\phi_1(e_2) = -2 \end{array} \right|,$$

and hence, $\phi_1(e_1) = -1$ and $\phi_1(e_2) = 2$. Likewise,

$$\left| \begin{array}{r} 1\phi_2(e_1) + 1\phi_2(e_2) = 0 \\ 2\phi_2(e_1) + 1\phi_2(e_2) = 1 \end{array} \right| \implies \left| \begin{array}{r} 1\phi_2(e_1) + 1\phi_2(e_2) = 0 \\ -2\phi_2(e_2) = 1 \end{array} \right|.$$

Therefore, $\phi_2(e_1) = 1$ and $\phi_2(e_2) = -1$. To write $\phi_1$ and $\phi_2$ as members of $\mathbb{R}^2 \to \mathbb{R}$, we have $\phi_1(x,y) = -x + 2y$ and $\phi_2(x,y) = x - y$. Now,

$$T'(\phi_1) = \phi_1 \circ T = -(4x + 5y + 6z) + 2(7x + 8y + 9z) = 10x + 11y + 12z\,, \text{ and}$$
$$T'(\phi_2) = \phi_2 \circ T = 4x + 5y + 6z - (7x + 8y + 9z) = -3x - 3y - 3z = -3(x + y + z).$$

With the same approach, we can calculate the dual basis for $w_1, w_2, w_3$. We get $\psi_1(x,y,z) = -\frac{1}{2}x + y$, $\psi_2(x,y,z) = \frac{1}{2}x - \frac{1}{2}y$ and $\psi_3(x,y,z) = z$. To answer question (b), we use the theorem (**1**) from question 11 and write

$$\begin{aligned} T'(\varphi_1) = \sum_{k=1}^{3} T'(\varphi_1)(w_k)\psi_k &= (20 + 22 + 0)\psi_1 + (40 + 22 + 0)\psi_2 + (0 + 0 + 12)\psi_3 \\ &= 42\psi_1 + 62\psi_2 + 12\psi_3\,, \text{ and} \end{aligned}$$
$$T'(\varphi_2) = \sum_{k=1}^{3} T'(\varphi_2)(w_k)\psi_k = -12\psi_1 - 18\psi_2 - 3\psi_3.$$

---

**16** Suppose $W$ is finite-dimensional and $T \in \mathcal{L}(V,W)$. Prove that

$$T' = 0 \iff T = 0.$$

***Proof:*** $\Rightarrow$ ***direction:*** Assume $T' = 0$. By definition, $T' = \varphi \circ T = 0$ for every $\varphi \in W'$. If $W = \{0\}$, then necessarily $T = 0$, so $T' = 0$ holds trivially. For this reason, assume for now that $W \neq 0$ and let $w_1, \dots, w_n$ be a basis of $W$. Suppose for the sake of contradiction that $T \neq 0$. Hence, there exists a nonzero $v \in V$ and a nonzero $w \in W$ such that $w = T(v)$. Express $w$ as $w = a_1w_1 + \cdots + a_nw_n$. Since $w \neq 0$, pick $k$ such that $a_k \neq 0$. Define $\varphi \in W'$ by

$$\varphi(b_1w_1 + \cdots + b_nw_n) = \frac{b_k}{a_k},$$

where $b_k$ is the $k$-th coordinate of $\varphi$'s input. Hence, $\varphi(w) = 1$. But then

$$T'(\varphi)(v) = \varphi(T(v)) = \varphi(w) = 1 \neq 0.$$

Thus, $T'(\varphi) \neq 0$, contradicting $T' = 0$. Hence, $T = 0$.

$\Leftarrow$ ***direction:*** If $T = 0$, then for each $\varphi \in W'$ and $T' \in \mathcal{L}\big(W', V'\big)$ we have

$$T'(\varphi)(v) = \varphi(T(v)) = \varphi(0) = 0.$$

Hence, $T'(\varphi)$ is always sent to a map in $W'$ that is the zero map. □

---

**16** Suppose $V$ and $W$ are finite-dimensional and $T \in \mathcal{L}(V,W)$. Prove that

$$T \text{ is invertible} \iff T' \in \mathcal{L}(W', V') \text{ is invertible.}$$

***Proof:*** $\Rightarrow$ ***direction:*** Let $T \in \mathcal{L}(V,W)$ be invertible and $T^{-1} \in \mathcal{L}(W,V)$ be the inverse of $T$. We want first want to show that $\left(T^{-1}\right)'$ is the inverse of $T'$, that is

$$\left(T^{-1}\right)' = \left(T'\right)^{-1}.$$

For $\left(T^{-1}\right)' \in \mathcal{L}\big(V', W'\big)$ to be an inverse of $T' \in \mathcal{L}\big(W', V'\big)$, it must hold that $\left(T^{-1}\right)' T' = I_{W'}$ and

$T'\left(T^{-1}\right)' = I_{V'}$. Note that $I_W \neq I_{W'}$ and $I_V \neq I_{V'}$. Therefore, let's compute for $\psi \in V'$:

$$\left(\left(T^{-1}\right)' \circ T'\right)(\psi) = \left(T^{-1}\right)'\left(\psi \circ T^{-1}\right) = \left(\psi \circ T\right) \circ T^{-1} = \psi \circ \left(TT^{-1}\right) = \psi.$$

Thus, $T'\left(T^{-1}\right)' = I_{V'}$. The other composition works by the same argument. Let $\psi \in W'$. So,

$$\left(T' \circ \left(T^{-1}\right)'\right)(\varphi) = T'\left(\varphi \circ T^{-1}\right) = \left(\varphi \circ T^{-1}\right) \circ T = \varphi \circ \left(T^{-1}T\right) = \varphi,$$

for that $\left(T^{-1}\right)' T' = I_{W'}$. Hence, $T'$ is invertible with $\left(T^{-1}\right)' = \left(T'\right)^{-1}$.

$\Leftarrow$ ***direction:*** If $T'$ is invertible, then it is surjective (and also injective) by 3.65. Take any nonzero $v \in V$ and $\varphi \in V'$ with $\varphi(v) \neq 0$. Surjectivity of $T'$ yields a $\psi \in W'$ with $T'(\psi) = \varphi$, or $\varphi = \psi \circ T$. Therefore,

$$0 \neq \varphi(v) = \psi(Tv) = (\psi \circ T)(v).$$

This implies that $Tv \neq 0$, and thus, $\operatorname{null} T = \{0\}$ (since $v$ was arbitrary and nonzero). This forces $T$ to be injective. Because $T'$ is invertible, we have $\dim W' = \dim V'$, and hence, $\dim W = \dim V$ (see 3.111). An injective linear map between finite-dimensional vector spaces of the same dimension is surjective, so $T$ is bijectiive, and therefore, invertible. $\square$

# Polynomials (Chapter 4)

$\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$.

## Zeros of polynomials (Section 4A)

$p : \mathbb{F} \to \mathbb{F}$, $p(z) = a_0 + a_1 z + \cdots + a_n z^m \ \forall z \in \mathbb{F}$.

**Def 4.5** (*zero of a polynomial*):
$\lambda \in \mathbb{F}$ is called a *"zero"* (or *"root"*) of a polynomial $p$ if $p(\lambda) = 0$.

**Thm 4.6** (*each zero of a polynomial corresponds to a degree-one factor*):
$\forall p \in \mathcal{P}_m(\mathbb{F}) : \ \exists \lambda \in \mathbb{F}$ such that

$$p(\lambda) = 0 \iff \exists q \in \mathcal{P}_{m-1} : p(z) = (z - \lambda)q(z), \quad \forall z \in \mathbb{F}.$$

***Proof:*** Let $p \in \mathcal{P}_m(\mathbb{F})$ for $m \in \mathbb{N}$.

$\Leftarrow$ ***direction:*** Suppose there is $q \in \mathcal{P}(\mathbb{F})$ such that $p(z) = (z - \lambda)q(z) \quad \forall z \in \mathbb{F}$. Then

$$p(\lambda) = (\lambda - \lambda)q(\lambda) = 0, \text{ as desired.}$$

$\Rightarrow$ ***direction:*** Now, suppose $p(\lambda) = 0$. Let $a_0, a_1, \ldots, a_m \in \mathbb{F}$ such that

$$p(z) :\equiv a_0 + a_1 z + \cdots a_m z^m.$$

Thus,

$$p(z) = p(z) - p(\lambda) = a_1(z - \lambda) + a_2(z^2 - \lambda^2) + \cdots + a_m(z^m - \lambda^m) \quad \forall z \in \mathbb{F}.$$

Using the binomic formula $z^k - \lambda^k = (z - \lambda)\sum_{j=1}^{k} \lambda^{j-1} z^{k-j} \quad \forall k \in \{1, \ldots, m\}$, we get

$$p(z) = (z - \lambda)\underbrace{\left(a_1 + a_2(z + \lambda) + a_3(z^2 + \lambda z + \lambda^2) + a_4(z^3 + \lambda z^2 + \lambda^2 z + \lambda^3) + \cdots + a_m \sum_{j=1}^{m} \lambda^{j-1} z^{m-j}\right)}_{\text{a polynomial with degree } m-1}.$$

Thus, $p$ equals $z - \lambda$ times some polynomial of degree $m - 1$, as desired. □

In case you are wondering where the $a_0$ went by subtracting $p(\lambda)$ from $p(z)$, it is hidden in the roots!
For example, $p \in \mathcal{P}_1(\mathbb{F}) : p = a_0 + a_1 z = \left(z - \left(-\frac{a_0}{a_1}\right)\right)(a_1)$, where $\left(-\frac{a_0}{a_1}\right)$ is the only root!

**Thm 4.8** (*degree m implies at most m roots / zeros*):
A polynomial $p \in \mathcal{P}_m(\mathbb{F})$ of degree $m$ has at most $m$ zeros in $\mathbb{F}$.

***Proof:*** For $m = 1$, $a_0 + a_1 z$ has only one zero, which is $-\frac{a_0}{a_1}$.

Now, for $m > 1$, let us assume the desired result holds for $m - 1$.

If $p \in \mathcal{P}_m(\mathbb{F})$ has no zeros, we are done.

If $p$ has a zero $\lambda \in \mathbb{F}$, by 4.6, there is a $q \in \mathcal{P}_{m-1}(\mathbb{F})$ such that

$$p(z) = (z - \lambda)q(z).$$

Since $q$ has at most $m - 1$ zeros, $p$ has at most $m$ zeros. □

## Division Algorithm for Polynomials (Section 4B)

**Thm 4.9**:
Suppose $p, s \in \mathcal{P}(\mathbb{F})$, $s \neq 0$. Then $\exists q, r \in \mathcal{P}(\mathbb{F})$ such that

$$p = sq + r \quad \text{and} \quad \deg r < \deg s.$$

***Proof:*** Let $n :\equiv \deg p$ and $m :\equiv \deg s$.

If $n < m$, then take $q = 0$ and $r = p$.

Thus, now we assume $n \geq m$. The list

$$1, z, \ldots, z^{m-1}, s, zs, \ldots, z^{n-m}s \tag{4.1}$$

is linearly independent because each polynomial in this list has a different degree.

The list has length $1 + (m-1) + 1 + (n-m) = n+1$, which equals $\dim \mathcal{P}_n(\mathbb{F})$. Hence, (4B) is a basis of $\mathcal{P}_n(\mathbb{F})$ by 2.38. Therefore, we have $a_0, a_1, \ldots, a_{m-1} \in \mathbb{F}$ and $b_0, b_1, \ldots, b_{n-m} \in \mathbb{F}$ such that

$$\begin{aligned} p &= a_0 + a_1 z + a_2 z^2 + \cdots + a_{m-1} z^{m-1} + b_0 s + b_1 zs + b_2 z^2 s + \cdots + b_{n-m} z^{n-m} s \\ &= \underbrace{\left(a_0 + a_1 z + a_2 z^2 + \cdots + a_{m-1} z^{m-1}\right)}_{r} + \underbrace{(s)}_{s} \underbrace{\left(b_0 + b_1 z + b_2 z^2 + \cdots + b_{n-m} z^{n-m}\right)}_{q} \\ &= r + sq = sq + r, \quad \text{with } \deg r = m-1 \leq \deg s = m, \end{aligned}$$

as desired.

The uniqueness of $q, r \in \mathcal{P}(\mathbb{F})$ follows from the uniqueness of the $a$'s and $b$'s, because (4B) is a basis of $\mathcal{P}_n(\mathbb{F})$. □

## Factorization of Polynomials over $\mathbb{C}$ (Section 4C)

**Thm 4.12** (*Fundamental Theorem of Algebra, first version*)**:**

Every non-constant polynomial with complex coefficients has a zero in $\mathbb{C}$.

**Thm 4.13** (*Fundamental Theorem of Algebra, second version:*)**:**

If $p \in \mathcal{P}(\mathbb{C})$ is a nonconstant polynomial (i.e., $\deg p \geq 1$), then $p$ has a unique factorization

$$p(z) = c \cdot (z - \lambda_1) \cdots (z - \lambda_m),$$

where $c, \lambda_1, \ldots, \lambda_m \in \mathbb{C}$. This factorization is unique except for the order of the factors.

***Proof:*** Let $p \in \mathcal{P}(\mathbb{C})$ and put $m := \deg p$. If $m = 1$, then

$$p = a_0 + a_1 z = a_1 \cdot \left(z - \frac{a_0}{a_1}\right),$$

so the desired factorization exists and is unique.

Hence, assume that $m > 1$ and that the desired factorization exists and is unique for all polynomials of degree $m-1$. By the first version of the fundamental theorem of algebra 4.12, $p$ has a zero $\lambda \in \mathbb{C}$. By 4.6, there is a polynomial $q$ of degree $m-1$ such that

$$p(z) = (z - \lambda) q(z) \quad \forall z \in \mathbb{C}.$$

Our induction hypothesis implies that $q$ has the desired factorization, which when plugged into the equation above, gives the desired factorization of $p$.

***Uniqueness:*** The number $c$ is uniquely determined as the coefficient $a_m$ of $z^m$ in $p$. So, we only need to show that except for the order, there is only one way to choose $\lambda_1, \ldots, \lambda_m$. If we have some $\tau_1, \ldots, \tau_m \in \mathbb{C}$ such that

$$\cancel{c} \cdot (z - \lambda_1) \cdots (z - \lambda_m) = \cancel{c} \cdot (z - \tau_1) \cdots (z - \tau_m) \quad \forall z \in \mathbb{C},$$

then because the left side of the equation above equals 0 when $z = \lambda_1$, one of the $\tau$'s on the right-hand side equals $\lambda_1$, because every zero of a polynomial corresponds to a degree-one factor by 4.6. Without loss of generality, we can assume by relabeling, that $\tau_1 = \lambda_1$. Now, if $z \neq \lambda_1$, we can use division to get

$$\cancel{(z-\lambda_1)}(z-\lambda_2)\cdots(z-\lambda_m) = \cancel{(z-\tau_1)}(z-\tau_2)\cdots(z-\tau_m) \qquad \forall z \in \mathbb{C}$$

$$(z-\lambda_2)\cdots(z-\lambda_m) = (z-\tau_2)\cdots(z-\tau_m) \qquad \forall z \in \mathbb{C}\backslash\{\lambda_1\} \tag{4.2}$$

$$0 = \Big((z-\lambda_2)\cdots(z-\lambda_m)\Big) - \Big((z-\tau_2)\cdots(z-\tau_m)\Big) \qquad \forall z \in \mathbb{C}\backslash\{\lambda_1\} \tag{4.3}$$

Note that the last 2 equations hold for all $z \in \mathbb{C}$ except possibly $z = \lambda_1$. Remember that our goal is to use induction on the degree of the polynomial. So, it would come in handy if the remaining polynomials (4C) were equal after cancellation and behaved like any other polynomial. But we have derived those equations by the assumption that $z \neq \lambda_1$ using division. Here comes the trick: let us assume that $(z-\lambda_2)\cdots(z-\lambda_m)$ and $(z-\tau_2)\cdots(z-\tau_m)$ are not always equal. Thus, with the standard laws of polynomial arithmetic, we can form a new polynomial $s \in \mathcal{P}_{m-1}(\mathbb{C})$ with

$$s(z) := \Big((z-\lambda_2)\cdots(z-\lambda_m)\Big) - \Big((z-\tau_2)\cdots(z-\tau_m)\Big),$$

where we assume that $s$ is not the zero polynomial. Using the last equation in (4C), we know that $s(z) = 0 \quad \forall z \in \mathbb{C}\backslash\{\lambda_1\}$. But we know that $s$ can have at most $m-1$ roots by 4.8. This is a contradiction, so it must be the case that $s$ is the zero polynomial. Hence, $\forall z \in \mathbb{C} : s(z) = 0$. The zero polynomial is allowed to have infinitely many roots. This is only the case if $\forall z \in \mathbb{C}$:

$$(z-\lambda_2)\cdots(z-\lambda_m) = (z-\tau_2)\cdots(z-\tau_m).$$

So, we get two identical polynomials after canceling out $(z-\lambda_1)$. By our induction hypothesis, the equation above implies that except for the order, the $\lambda$'s are the same as the $\tau$'s, completing the proof of uniqueness. □

## Factorization of Polynomials over $\mathbb{R}$ (Section 4D)

**Thm 4.14** (*polynomials with real coefficients have nonreal zeros in pairs*):

Suppose $p \in \mathcal{P}(\mathbb{C})$ is a polynomial with real coefficients. If $\lambda \in \mathbb{C}$ is a zero/root of $p$, then so is $\overline{\lambda}$.

***Proof:*** Let $p \in \mathcal{P}(\mathbb{C})$ be defined by

$$p(z) := a_0 + a_1 z + \ldots + a_m z^m \quad (\forall z \in \mathbb{C}),$$

where $a_0, \ldots, a_m \in \mathbb{R}$. Suppose $\lambda \in \mathbb{C}$ is a zero of $p$. Then

$$a_0 + a_1\lambda + \cdots + a_m\lambda^m = 0$$
$$\overline{a_0 + a_1\lambda + \cdots + a_m\lambda^m} = \overline{0}$$
$$\overline{a_0} + \overline{a_1\lambda} + \cdots + \overline{a_m\lambda^m} = 0$$
$$a_0 + a_1\overline{\lambda} + \cdots + a_m\overline{\lambda}^m = 0.$$

The equation above shows that $\overline{\lambda}$ is a zero of $p$ as well. □

**Thm 4.15** (*factorization of a quadratic polynomial*):

Let $p(x) = x^2 + bx + c$ $(\forall x \in \mathbb{R})$ such that $b, c \in \mathbb{R}$. Then $p$ can be written as

$$p(x) = (x-\lambda_1)(x-\lambda_2) \quad (\forall x \in \mathbb{R})$$

with $\lambda_1, \lambda_2 \in \mathbb{R} \iff b^2 \geq 4c$.

***Proof:*** Write

$$x^2 + bx + c = \left(x^2 + 2\left(\frac{b}{2}\right) + \left(\frac{b}{2}\right)^2\right) + \left(c - \frac{b^2}{4}\right)$$
$$= \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right).$$

Therefore, if $b^2 < 4c$, the first and the second term are always positive and by 4.6, no factorization is possible, because $p$ has no zeros. If $b^2 \geq 4c$, set $d^2 := \left(b^2/4 - c\right)$ and rewrite $p$ as

$$\begin{aligned} p(x) &= x^2 + bx + c \\ &= \left(x + \frac{b}{2}\right)^2 - d^2 \\ &= \left(\left(x + \frac{b}{2}\right) + d\right) \cdot \left(\left(x + \frac{b}{2}\right) - d\right) \\ &= \left(x - \left(-d - \frac{b}{2}\right)\right) \cdot \left(x - \left(d - \frac{b}{2}\right)\right) \\ &= \left(x - \lambda_1\right) \cdot \left(x - \lambda_2\right) \qquad\qquad (\forall x \in \mathbb{R}) \end{aligned}$$

for $\lambda_1 := -d - b/2$ and $\lambda_2 := d - b/2$, which is the same as the well-known midnight formula:

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4c}}{2}.$$

The formula above also shows that $b^2 \geq 4c$ is a necessary condition for real solutions (i.e., for $\lambda_{1,2}$). $\square$

***Remark:*** The next result gives a factorization of a polynomial over $\mathbb{R}$. The second version of the fundamental theorem of algebra 4.13 gives a factorization of $p$ with complex coefficients. Complex but nonreal zeros of $p$ come in pairs (see 4.14). Multiplying terms of the form $(x - \lambda)$ and $(x - \overline{\lambda})$ together yields

$$(x^2 - 2(\operatorname{Re}\lambda)x + |\lambda|^2),$$

which is a quadratic term of the required form. But 4.14 does not state that these two factors appear the same number of times, as needed to make the idea work. However, the next proof works around this point.

In the next result, either $m$ or $M$ may equal 0. The numbers $\lambda_1, \ldots, \lambda_m$ are precisely the real zeros of $p$, for these are the only real values of $x$ for which the right side of the equation in the next result equals 0.

**Thm 4.16** (*factorization of a polynomial over* $\mathbb{R}$)**:**

Suppose $p \in \mathcal{P}(\mathbb{R})$ is non-constant. Then $p$ has a unique factorization:

$$p(x) = c(x - \lambda_1)\cdots(x - \lambda_m)(x^2 + b_1 x + c_1)\cdots(x^2 + b_M x + c_M)$$

where $c, \lambda_1, \ldots, \lambda_m, b_1, \ldots, b_M, c_1, \ldots, c_M \in \mathbb{R}$, with $b_k^2 < 4c_k \quad \forall k \in \{1, \ldots, M\}$.

***Proof:*** First, we will prove that the desired factorization exists, and after that, we will prove uniqueness.

Think of $p$ as an element of $\mathcal{P}(\mathbb{C})$, not $\mathcal{P}(\mathbb{R})$, but with real coefficients like stated in the theorem. If all (complex) zeros of $p$ are real, then we have the desired factorization by the second version of the fundamental theorem of algebra 4.13. Thus, suppose $p$ has a complex zero $\lambda \in \mathbb{C}$ with $\lambda \notin \mathbb{R}$. By 4.14, $\overline{\lambda}$ is a zero of $p$ as well. Thus, we write $\forall x \in \mathbb{R}$:

$$\begin{aligned} p(x) &= (x - \lambda)(x - \overline{\lambda})q(x) \\ &= (x^2 - 2(\operatorname{Re}\lambda)x + |\lambda|^2)q(x), \end{aligned}$$

for some $q \in \mathcal{P}(\mathbb{C})$, with $\deg q = \deg p - 2$. If we can prove that $q$ has real coefficients, then using induction on the degree of such polynomial completes the proof of the existence part of this result. If we solve for $q(x)$, we get

$$q(x) = \frac{p(x)}{x^2 - 2(\operatorname{Re}\lambda)x + |\lambda|^2} \quad \forall x \in \mathbb{R}.$$

The equation above implies that $q(x) \in \mathbb{R} \quad \forall x \in \mathbb{R}$. Let $n :\equiv \deg p$. Then we can rewrite $q$ as

$$q(x) = a_0 + a_1 x + \cdots + a_{n-2} x^{n-2} \text{ where } a_0, \ldots, a_{n-2} \in \mathbb{C},$$

because we don't know if the $a$'s are all real yet. But we know that $\forall x \in \mathbb{R} : q(x) \in \mathbb{R}$. Thus, we have $\forall x \in \mathbb{R}$ :

$$0 = \operatorname{Im} q(x) = (\operatorname{Im} a_0) + (\operatorname{Im} a_1) x + \cdots + (\operatorname{Im} a_{n-2}) x^{n-2}.$$

Since according to 4.8 a complex polynomial with degree $n-2$ can have at most $n-2$ zeros, this equation cannot hold for all $x \in \mathbb{R}$. This implies that $\operatorname{Im} a_0, \ldots, \operatorname{Im} a_{n-2}$ all equal 0, because only the zero polynomial is allowed to have infinitely many zeros. Thus, all the coefficients of $q$ are real, as desired. Hence, the wanted factorization exists by induction.

***Uniqueness:*** A factor of $p$ of the form $x^2 + b_k x + c_k$ with $b_k^2 < 4c_k$ can be uniquely written as $(x - \lambda_k)(x - \overline{\lambda_k})$ with $\lambda_k \in \mathbb{C}$. Two different factorizations of $p$ as an element of $\mathcal{P}(\mathbb{R})$ would lead to two different factorizations of $p$ as an element of $\mathcal{P}(\mathbb{C})$, contradicting the second version of the fundamental theorem of algebra 4.13. $\square$

## Exercises about Polynomials 4D

(4) Suppose $m$ is a positive integer. Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) : \deg p = m\} = S$$

a subspace of $\mathcal{P}(\mathbb{F})$?

***Solution:*** The set in question is not a subspace of $\mathcal{P}(\mathbb{F})$, because it is not closed under addition. Take $m = 4$, then $p = 1 + 2x^2 + 3x^3 + 4x^4$ and $s = 4x^4$ are both in $S$, but $p - s$ is not.

(5) Is the set

$$\{0\} \cup \{p \in \mathcal{P}(\mathbb{F}) : \deg p \text{ is even}\} = S$$

a subspace of $\mathcal{P}(\mathbb{F})$?

***Solution:*** The same counter-example as in the previous question applies to this one.

(6) Suppose that $m$ and $n$ are positive integers with $m \neq n$, and suppose $\lambda_1, \dots, \lambda_m \in \mathbb{F}$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{F})$ with $\deg p = m$ such that $0 = p(\lambda_1) = \cdots = p(\lambda_m)$ and such that $p$ has no other zeros.

***Proof:*** Define $p \in \mathcal{P}_n(\mathbb{F})$ by

$$\begin{aligned} p(z) &= (z-\lambda_1)(z-\lambda_2)\cdots(z-\lambda_{m-1})\underbrace{(z-\lambda_m)(z-\lambda_m)\cdot(z-\lambda_m)}_{n-m+1 \text{ times}} \\ &= \left(\sum_{k=1}^{m-1}(z-\lambda_k)\right)(z-\lambda_m)^{n-m+1} \qquad \forall z \in \mathbb{F}. \end{aligned}$$

□

If $p$ would have another root than $\lambda_1, \dots, \lambda_m \in \mathbb{F}$, say $\lambda_{\text{fresh}}$, we could write $p$ as

$$p(z) = (z - \lambda_{\text{fresh}})q(z) \quad \forall z \in \mathbb{F},$$

where $q \in \mathcal{P}_{n-1}(\mathbb{F})$ would have one $(z - \lambda_j)$-term less than $p$. But this is impossible and contradicts the allready exising structure of $p$. Hence, $p$ has no other zeros.

# Eigenvalues and Eigenvectors (Chapter 5)

## Invariant Subspaces (Section 5A)

### Eigenvalues (Sub-Section 5A1)

**Def 5.1** (*operator*)**:**

A linear mapfrom a vector spaceto itself is called an *"operator"*.

(Suppose $T \in \mathcal{L}(V)$, then may be $T\big|_{V_k}$ is not an operator on a subspace $V_k$)

**Def 5.2** (*invariant subspace*)**:**

Let $T \in \mathcal{L}(V)$. A subspace $U$ of $V$ is called *"invariant under $T$"* if $Tu \in U$ for all $u \in U$.
Thus, $U$ is invariant under $T$ if $T\big|_U$ is an operator on $U$.

**Example 5.3:**

Let $T \in \mathcal{L}(\mathcal{P}(\mathbb{R}))$ such that $Tp = p'$. Let $U = \mathcal{P}_4(\mathbb{R}) \subseteq \mathcal{P}(\mathbb{R})$. Then $U$ is invariant under $T$ because if $p \in U$, $\deg p \leq 4$ and $\deg(p') \leq 3$.

**Example 5.4:**

Let $T \in \mathcal{L}(V)$. Then $\{0\}, V, \operatorname{null} T, \operatorname{range} T$ are all invariant.
(Sometimes, $\operatorname{null} T = \{0\}$ and $\operatorname{range} T = V$ if $T$ is invertible.)

**Invariant subspaces of dimension one:**

Take any $v \in V, v \neq 0$, and let

$$U :\equiv \{\lambda v \mid \lambda \in \mathbb{F}\} = \operatorname{span}(v),$$

then $U$ is a one-dimensional subspace of $V$.

If $U$ is invariant under an operator $T \in \mathcal{L}(V)$, then $Tv \in U \implies \exists \lambda \in \mathbb{F} : Tv = \lambda v$.
Conversely, if $Tv = \lambda v$, $\lambda \in \mathbb{F}$, then $\operatorname{span}(v)$ is a one-dimensional subspace of $V$ invariant under $T$.

**Def 5.5** (*eigenvalue*)**:**

Suppose $T \in \mathcal{L}(V)$. A scalar $\lambda \in \mathbb{F}$ is called an *"eigenvalue of $T$"* if there exists a nonzero (i.e., $v \neq 0$) $v \in V$ such that

$$Tv = \lambda v.$$

**Thm 5.7** (*equivalent conditions to be an eigenvalue*)**:**

The following are equivalent for $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$:

- **(a)** $\lambda$ is an eigenvalue of $T$.
- **(b)** $T - \lambda I$ is not injective.
- **(c)** $T - \lambda I$ is not surjective.
- **(d)** $T - \lambda I$ is not invertible.

***Proof:*** Conditions **(a)** and **(b)** are equivalent because the eigenvector $v$ is a solution to $Tv = \lambda v$, which is equivalent to $(T - \lambda I)v = 0$. Hence, $\dim \operatorname{null}(T - \lambda I) > 0$, which implies that $T - \lambda I$ is not injective. **(b)**, **(c)**, and **(d)** are therefore equivalent by 3.65. □

**Def 5.8** (*eigenvector*)**:**

Let $T \in \mathcal{L}(V)$. A vector $v \in V$ is called an *"eigenvector"* of $T$ corresponding to $\lambda$ if $v \neq 0$ and $Tv = \lambda v$.

In other words: A vector $v \in V, v \neq 0$ is an eigenvector corresponding to $\lambda \iff v \in \text{null}(T - \lambda I_V)$.

**Thm 5.11** (*linearly independent eigenvectors*):
Every list of eigenvectors of $T$ corresponding to distinct eigenvalues of $T$ is linearly independent.

***Proof:*** Suppose the desired result is false. Then there exists a smallest list of length $m$ of linearly dependent eigenvectors $v_1, \dots, v_m$ with distinct eigenvalues $\lambda_1, \dots, \lambda_m$. Since an eigenvector is unequal to the zero vector, $m$ must be $\geq 2$.
Because of the minimality of $m$ and because our list is linearly dependent:

$\exists a_1, \dots, a_m$, where not all $a$'s are equal to 0, such that $a_1 v_1 + \cdots + a_m v_m = 0$.

Now, we apply $(T - \lambda_m I)$ on both sides of the equation and get

$$(a_1\lambda_1 v_1 - a_1\lambda_m v_1) + \cdots + (a_{m-1}\lambda_{m-1} v_{m-1} - a_{m-1}\lambda_m v_{m-1}) + \underbrace{(a_m\lambda_m v_m - a_m\lambda_m v_m)}_{=0} = 0,$$

which is the same as:

$$a_1 \underbrace{(\lambda_1 - \lambda_m)}_{\neq 0} v_1 + \cdots + a_{m-1} \underbrace{(\lambda_{m-1} - \lambda_m)}_{\neq 0} v_{m-1} = 0,$$

which contradicts the minimality of $m$. Therefore, no such linearly dependent list of eigenvectors can exist. $\square$

**Thm 5.12** (*number of eigenvectors*):
Each operator $T \in \mathcal{L}(V)$ on $V$ has at most $\dim V$ distinct eigenvalues.

***Proof:*** Let $T \in \mathcal{L}(V)$. Suppose $\lambda_1, \dots, \lambda_m$ are distinct eigenvalues of $T$ with corresponding eigenvectors $v_1, \dots, v_m \in V$. Hence, 5.11 implies that the list $v_1, \dots, v_m$ is linearly independent. Thus, 2.22 implies that $m \leq \dim V$. $\square$

## Polynomials Applied to Operators (Sub-Section 5A2)

**Def 5.13** (*notation $T^m$*):
Let $T \in \mathcal{L}(V)$ and $m \in \mathbb{N}^+$:

- $T^m :\equiv \underbrace{T \cdots T}_{m \text{ times}}$ or $T^m :\equiv T^{m-1} \cdot T$ such that $T^m \in \mathcal{L}(V)$;
- $T^0 :\equiv I_V$;
- If $T$ is invertible with inverse $T^{-1}$, then $T^{-m} \in \mathcal{L}(V)$ is defined by $T^{-m} :\equiv (T^{-1})^m$.

$T^m T^n = T^{m+n}$ and $(T^m)^n = T^{mn}$ for $m, n \in \mathbb{Z}$ if $T$ is invertible. And for $m, n \in \mathbb{N}$ if $T$ is not invertible.

**Def 5.14** (*notation $p(T)$*):
For $\mathcal{P}(\mathbb{F}) \ni p(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_m z^m$ and $T \in \mathcal{L}(V)$, we define:

$$p(T) :\equiv a_0 I + a_1 T + a_2 T^2 + \cdots a_m T^m.$$

Note that $p(T) \in \mathcal{L}(V)$.

**Thm 5.17** (*multiplicative properties*):
Suppose $p, q \in \mathcal{P}(\mathbb{F})$ and $T \in \mathcal{L}(V)$. Then

$$(pq)(T) = p(T)q(T) = q(T)p(T).$$

***Proof:*** When a product of polynomials is expanded using the distributive property, it does not matter if the symbol is $z$ or $T$. Suppose $p(z) = \sum_{j=0}^{m} a_j z^j$ and $q(z) = \sum_{k=0}^{n} b_k z^k$ for all $z \in \mathbb{F}$. Then

$$(pq)(z) = \sum_{j=0}^{m}\sum_{k=0}^{n} a_j b_k z^{j+k} \quad \text{and} \quad (pq)(T) = \sum_{j=0}^{m}\sum_{k=0}^{n} a_j b_k T^{j+k} = \sum_{j=0}^{m} a_j T^j \sum_{k=0}^{n} b_k T^k,$$

which is the same as $(pq)(T) = p(T)q(T)$ as well as $(pq)(T) = q(T)p(T)$. □

**Thm 5.18** (*null space and range of $p(T)$ are invariant under $T$*)**:**
For every $T \in \mathcal{L}(V)$ and every $p \in \mathcal{P}(\mathbb{F})$,

$\operatorname{null} p(T)$ and $\operatorname{range} p(T)$ are invariant under $T$.

***Proof:*** ***Step 1:*** First, let $u \in \operatorname{null} p(T)$. Then,

$$p(T)u = 0.$$

Since $p(T)$ commutes with $T$ (by associativity and distributivity of polynomial evaluation),

$$\begin{aligned}(p(T))(Tu) &= (p(T)T)(u)\\ &= (Tp(T))(u)\\ &= T(p(T)u)\\ &= T(0) = 0.\end{aligned}$$

Hence, $Tu \in \operatorname{null} p(T)$.

***Step 2:*** Second, let $u \in \operatorname{range} p(T)$. Then, $u = p(T)v$ for some $v \in V$. Thus,

$$Tu = T(p(T)v) = p(T)(Tv),$$

so $Tu \in \operatorname{range} p(T)$. □

## Exercises about Invariant Subspaces 5A

**1** Suppose $T \in \mathcal{L}(V)$ and $U$ is a subspace of $V$. Then

**(a)** $U \subseteq \operatorname{null} T \implies U$ is invariant under $T$.

**(b)** $\operatorname{range} T \subseteq U \implies U$ is invariant under $T$.

***Proof:*** (a). If $v \in U$, then $Tv = 0 \in U$, since $U$ is a subspace and therefore $0 \in U$.

(b). Let $w \in U \subseteq V$. $\implies$ $Tw \in \operatorname{range} T$, and since $\operatorname{range} T \subseteq U$, $Tw \in U$. □

---

**2** Suppose that $T \in \mathcal{L}(V)$ and $V_1, \ldots, V_m$ are subspaces of $V$ invariant under $T$. Prove that $V_1 + \cdots + V_m$ is invariant under $T$.

***Proof:*** Let $w \in V_1 + \cdots + V_m$ such that $w = v_1 + \cdots + v_m$ where $v_1 \in V_1, \ldots, v_m \in V_m$. Then $Tw = Tv_1 + \cdots + Tv_m$ such that $Tv_1 \in V_1, \ldots, Tv_m \in V_m$ because $V_1, \ldots, V_m$ are invariant under $T$. Therefore we have $Tw \in V_1 + \cdots + V_m$. So $V_1 + \cdots + V_m$ is invariant under $T$ as well. □

---

**3** Suppose $T \in \mathcal{L}(V)$. Prove that the intersection of every collection of subspaces of $V$ invariant under $T$ is invariant under $T$.

***Proof:*** Let $w \in V_1 \cap \cdots \cap V_m \subseteq V$ where each $V_i$ is invariant under $T$. Then $w \in V_1, \ldots, w \in V_m$, and thus, $Tw \in V_1, \ldots, Tw \in V_m$. Hence, $Tw \in V_1 \cap \cdots \cap V_m$. □

---

**12** Suppose $V = U \oplus W$, where $U$ and $W$ are nonzero subspaces of $V$. Define $P \in \mathcal{L}(V)$ by $P(u+w) = u$ for each $u \in U$ and each $w \in W$. Find all eigenvalues and eigenvectors of $P$.

***Solution:*** We have an eigenvalue $\lambda_1 = 1$ for all vectors $u \in U$, because $P$ acts as the identity on $U$. We also have an eigenvalue $\lambda_0 = 0$ for all $w \in W$, because $P$ acts as the zero operator there.

---

**15** Suppose $V$ is finite-dimensional, $T \in \mathcal{L}(V)$, and $\lambda \in \mathbb{F}$. Show that $\lambda$ is an eigenvalue of $T \iff \lambda$ is an eigenvalue of the dual operator $T' \in \mathcal{L}(V')$.

***Solution:*** The definition of $T'$ states that

$$T'(\varphi) = \varphi \circ T \quad \forall \varphi \in V'.$$

Hence, being an eigenvalue $\lambda$ and an eigenvector $\varphi \in V' = \mathcal{L}(V, \mathbb{F})$ of $T'$ means:

$$T'(\varphi) = \lambda\varphi.$$

$\lambda$ is an eigenvalue of $T \iff T - \lambda I$ is not injective $\iff (T-\lambda I)'$ is not surjective $\iff T' - \lambda I$ is not invertible $\iff \lambda$ is an eigenvalue of $T'$.

---

**21** Supppose $T \in \mathcal{L}(V)$ is invertible

**(a)** Suppose $\lambda \in \mathbb{F}$ with $\lambda \neq 0$. Prove that $\lambda$ is an eigenvalue of $T \iff \frac{1}{\lambda}$ is an eigenvalue of $T^{-1}$.

**(b)** Prove that $T$ and $T^{-1}$ have the same eigenvectors.

***Solution:*** Let $v \in V$ be such that $T(v) = \lambda v$. This is the case $\iff T^{-1}T(v) = T^{-1}(\lambda v) \iff T^{-1}(v) = \frac{1}{\lambda}v$. Note that in a field $\mathbb{F}$, we can "replace" $\frac{1}{\lambda} \in \mathbb{F}$ with its multiplicative inverse $\alpha \in \mathbb{F}$. Since $\left(T^{-1}\right)^{-1} = T$, we have proven (a) and (b).

---

**22** Suppose $T \in \mathcal{L}(V)$ and there exist nonzero vectors $u, w \in V$ such that

$$Tu = 3w \quad \text{and} \quad Tw = 3u.$$

Prove that 3 or $-3$ is an eigenvalue of $T$.

***Solution:*** Compute $T(u+w) = Tu+Tw = 3w+3u = 3(u+w)$, and $T(u-w) = Tu-Tw = 3w-3u = -3(u-w)$. Since $u \neq 0$, $(u+w)+(u-w) = 2u \neq 0$, so $u+w$ and $u-w$ cannot both be zero. If $u+w \neq 0$, then $u+w$ is an eigenvector with eigenvalue 3. If $u-w \neq 0$, then $u-w$ is an eigenvector with eigenvalue $-3$. Hence, 3 or $-3$ is an eigenvalue of $T$.

---

**23** Suppose $V$ is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that $T$ has an eigenvalue $\iff$ there exists a subspace of $V$ of dimension $\dim V - 1$ that is invariant under $T$.

***Proof:*** $\Rightarrow$ ***direction:*** Suppose $T$ has an eigenvalue $\lambda$. By Exercise 15, $\lambda$ is an eigenvalue of $T' \in \mathcal{L}(V')$. Let $\varphi \in V' \setminus \{0\}$ be an eigenvector of $T'$ with eigenvalue $\lambda$, so $T'(\varphi) = \lambda\varphi$. Define $U = \operatorname{null}\varphi$. Since $\varphi \neq 0$, $\dim U = \dim V - 1$. For any $u \in U$, $\varphi(Tu) = (T'\varphi)(u) = (\lambda\varphi)(u) = \lambda \cdot 0 = 0$, so $Tu \in U$. Thus $U$ is an invariant subspace of dimension $\dim V - 1$.

$\Leftarrow$ ***direction:*** Suppose $U$ is a $T$-invariant subspace of $V$ with $\dim U = \dim V - 1$. The quotient space $V/U$ has dimension $\dim V - \dim U = 1$. The induced quotient operator $T/U \in \mathcal{L}(V/U)$ on a 1-dimensional space is multiplication by some scalar $\lambda \in \mathbb{F}$, so $T$ has an eigenvalue $\lambda$. $\square$

---

**25** Suppose $T \in \mathcal{L}(V)$ and $u, w$ are eigenvectors of $T$ such that $u + w$ is also an eigenvector of $T$. Prove that $u$ and $w$ are eigenvectors of $T$ corresponding to the same eigenvalue.

***Proof:*** Let $u, w \in V \setminus \{0\}$ such that $Tu = \lambda_u u$, $Tw = \lambda_w w$, and $T(u+w) = \lambda_{u+w}(u+w)$. Since

$$T(u+w) = Tu + Tw = \lambda_u u + \lambda_w w = \lambda_{u+w}(u+w) = \lambda_{u+w}u + \lambda_{u+w}w,$$

it follows that

$$\left(\lambda_u - \lambda_{u+w}\right)u + \left(\lambda_w - \lambda_{u+w}\right)w = 0.$$

If $u$ and $w$ are linearly independent, the coefficients must vanish, so $\lambda_u = \lambda_{u+w} = \lambda_w$. Otherwise, if $u$ and $w$ are linearly dependent, we can write $u = \alpha w$ for some $\alpha \neq 0$. Applying $T$ to $u$ in two ways gives

$$Tu = T(\alpha w) = \alpha Tw = \alpha\lambda_w w \text{ and}$$
$$Tu = \lambda_u u = \lambda_u \alpha w.$$

Hence, $\lambda_u = \lambda_w$. □

(26) Let $u, v$

(33) Suppose $T \in \mathcal{L}(V)$ and $m$ is a positive integer.

**(a)** Prove that $T$ is injective $\iff$ $T^m$ is injective.

**(b)** Prove that $T$ is surjective $\iff$ $T^m$ is surjective.

***Proof:*** We show (a) and (b) both with induction, first in the $\Rightarrow$ direction, and then in the $\Leftarrow$ direction.

**(a)** $\Rightarrow$ ***direction:*** Suppose $T$ is injective. For $m = 1$ the statement is trivial ($T^1 = T$). Fix $m \geq 1$ and suppose as induction hypothesis that $T^m$ is injective. If $T^{m+1}(v_1) = T^{m+1}(v_2)$, then

$$T(T^m(v_1)) = T(T^m(v_2)).$$

Since $T$ is injective, $T^m(v_1) = T^m(v_2)$. The induction hypothesis forces $v_1 = v_2$. Therefore, $T^{m+1}$ is injective as well. (In general, function composition is injective. If $f : X \to Y$ and $g : Y \to Z$ are injective, so is $g \circ f : X \to Z$. So, a finite composition of injective maps is injective.)

$\Leftarrow$ ***direction:*** Assume $T^m$ is injective for some fixed $m \geq 1$. If $m = 1$ we are done. If $m > 1$, take $v_1, v_2 \in V$ with $T(v_1) = T(v_2)$. Apply $T^{m-1}$ to both sides to get $T^{m-1}(T(v_1)) = T^{m-1}(T(v_2))$. Injectivity of $T^m$ forces $v_1 = v_2$. Thus, $T$ is injective.

**(b)** $\Rightarrow$ ***direction:*** Suppose $T$ is surjective. For $m = 1$ the statement is trivial. Fix $m \geq 1$ and assume that $T^m$ is surjective. Let $y \in V$ be arbitrary. Since $T$ is surjective, there exists $z \in V$ with $T(z) = y$. By the induction hypothesis, there exists $v \in V$ with $T^m(v) = z$. Thus,

$$T^{m+1}(v) = T(T^m(v)) = T(z) = y.$$

Hence, $T^{m+1}$ is surjective.

(Or: A finite composition of surjective maps (not necessarely linear) is surjective.)

$\Leftarrow$ ***direction:*** Assume $T^m$ is surjective for some fixed $m \geq 1$. Let $y \in V$ be arbitrary. Choose $v \in V$ with $T^m(v) = y$ (always possible, since $T^m$ is surjective). Put $x := T^{m-1}(v)$. Then

$$y = T(x) = T(T^{m-1}(v)) = T^m(v).$$

Hence, $T$ is surjective.

□

(40) Suppose $S, T \in \mathcal{L}(V)$, and $S$ is invertible. Suppose $p \in \mathcal{P}(\mathbb{F})$ is a polynomial. Prove that

$$p(STS^{-1}) = Sp(T)S^{-1}.$$

***Proof:*** We first want to prove that $(STS^{-1})^m = ST^mS^{-1}$ for every $m \in \mathbb{N}_{\geq 0}$. To this end, we use induction over $m$. For $m = 0$, the statement holds because

$$(STS^{-1})^0 = I = SIS^{-1} = ST^0S^{-1}.$$

If $m = 1$ the claim is immediate. Hence, assume $m \geq 1$. Then

$$\begin{aligned}(STS^{-1})^{m+1} &= STS^{-1}(STS^{-1})^m \\ &= ST\,\underbrace{S^{-1}S}_{I}\,T^mS^{-1} \\ &= ST^{m+1}S^{-1},\end{aligned}$$

This completes the induction. Let $p(z) = a_0 z^0 + a_1 z^1 + \cdots + a_m z^m$, $a_j \in \mathbb{F}$. Then

$$\begin{aligned}
p(STS^{-1}) &= a_0 I + a_1(STS^{-1}) + a_2(STS^{-1})^2 + \cdots + a_m(STS^{-1})^m && = \textstyle\sum_{j=0}^m a_j (STS^{-1})^j \\
&= a_0 I \underbrace{SS^{-1}}_{=I=(SIS^{-1})^0} + a_1(STS^{-1}) + a_2(ST^2S^{-1}) + \cdots + a_m(ST^mS^{-1}) && = \textstyle\sum_{j=0}^m a_j (ST^jS^{-1}) \\
&= S(a_0 I)S^{-1} + S(a_1 T)S^{-1} + S(a_2 T^2)S^{-1} + \cdots + S(a_m T^m)S^{-1} && = \textstyle\sum_{j=0}^m S(a_j T^j)S^{-1} \\
&= S\left(a_0 I + a_1 T + a_2 T^2 + \cdots + a_m T^m\right)S^{-1} && = S\left(\textstyle\sum_{j=0}^m a_j T^j\right)S^{-1} \\
&= Sp(T)S^{-1}.
\end{aligned}$$

Here we have been making use of the distributive property of linear maps, see 3.8. And we used that scalar multiples of the identity (i.e. $a_0 I$) commute with every operator. □

# The Minimal Polynomial (Section 5B)

## Existence of Eigenvalues on Complex Vector Spaces (Sub-Section 5B1)

**Thm 5.19** (*existence of eigenvalues*):

Every operator on a finite-dimensionalnonzero, complex vector space has an eigenvalue.

***Proof:*** Suppose $V$ is finite-dimensional with $\dim V \geq 1$ and $T \in \mathcal{L}(V)$. Let $v \in V$ be nonzero. Then the list

$$v, Tv, T^2v, \ldots, T^n v$$

of length $n+1$ cannot be linearly independent. Hence, some non-trivial linear combination of these vectors is zero, so there exists a non-constant polynomial $p \in \mathcal{P}(\mathbb{C})$ of minimal degree such that

$$p(T)v = 0.$$

By the first version of the fundamental theorem of algebra 4.12, there exists a root $\exists \lambda \in \mathbb{C}$ with

$$p(\lambda) = 0.$$

By 4.6, we know that $\exists q \in \mathcal{P}(\mathbb{C})$ such that we can factor out $(z - \lambda)$ and write

$$p(z) = (z - \lambda)q(z) \quad \forall z \in \mathbb{C}.$$

Applying this to $T$ gives

$$0 = p(T)v = (T - \lambda I)(q(T)v).$$

If $q(T)v$ were zero, then $q$ (of strictly smaller degree) would also annihilate $v$, contradicting the minimality of $p$. Thus,

$$q(T)v \neq 0$$

and so $\lambda$ is an eigenvalue of $T$ with eigenvector $q(T)v$. □

## Eigenvalues and the Minimal Polynomial (Sub-Section 5B2)

**Def 5.20** (*monic polynomials*):

A *"monic polynomial"* is a polynomial whose highest-degree coefficient equals 1.

**Example 5.21:**

$p(z) = 2 + 9z^2 + z^7, \quad \deg p = 7.$

**Thm 5.22** (*existence, uniqueness, and degree of minimal polynomial*):

Suppose $T \in \mathcal{L}(V)$. Then there exists a unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that

$$p(T) = 0.$$

Furthermore, $\deg p \leq \dim V$.

***Proof:*** Before beginning our inductive proof, we examine some base cases. We first prove the existence of a monic polynomial that annihilates $T$ and later establish its uniqueness; these two parts are independent.

***Base case:*** If $\dim V = 0$, then the identity operator $I$ is the zero-operator on $V$, and we let $p = 1$ such that

$$1I\vec{0} = 0.$$

***Another base case:*** Now, suppose that $\dim V = 1$. For any nonzero $v \in V$, the list $v, Tv$ is linearly dependent (since it has length 2), so there exists a scalar $c_0 \in \mathbb{F}$ such that

$$c_0 v + Tv = 0.$$

This follows from the linear dependence lemma 2.19. If we choose $q(z) := c_0 + z \quad \forall z \in \mathbb{C}$, then

$$q(T)v = (c_0 I + T)(v) = c_0 v + Tv = 0.$$

In the case where $T$ is the zero operator, one may instead choose $q(z) = z$. So we have $\operatorname{dim}\operatorname{null} q(T) = 1$ and therefore $\operatorname{dim}\operatorname{range} q(T) = \operatorname{dim} V - \operatorname{dim}\operatorname{range} q(T) = 1 - 1 = 0$. Thus, every vector in this one-dimensional space is a scalar multiple of $v$ and maps to zero by $q(T)$. Thus, $q(z)$ serves as a minimal polynomial for $T$.

***Yet another base case:*** Suppose $\operatorname{dim} V = 2$. For any nonzero $v \in V$, the list $v, Tv, T^2 v$ is linearly dependent (since it has length 3). According to the linear dependence lemma 2.19, we either have:

$$c_0 v + Tv = 0 \text{ or } c_0 v + c_1 Tv + T^2 v = 0,$$

for some scalars $c_0, c_1 \in \mathbb{F}$. If we define $q(z) := c_0 + z$ in the first case or $q(z) := c_0 + c_1 z + z^2$ in the second case accordingly, then $q(T)v = 0$.

In the first case, where $(-1)c_0 v = Tv$ and $q(z) = c_0 + z$, the vector $v$ by itself is linearly independent, and thus $\operatorname{dim}\operatorname{null} q(T) \geq 1$. Therefore,

$$\operatorname{dim}\operatorname{range} q(T) \leq \operatorname{dim} V - \operatorname{dim}\operatorname{null} q(T) \leq 1.$$

Since $\operatorname{range} q(T)$ is at most one-dimensional and is invariant under $T$ (by 5.18), we can restrict $T$ to this subspace. In this restricted setting $\operatorname{range} q(T)$, we know how to construct the desired polynomial; denote it by $s$, defined such that

$$s\left( T\big|_{\operatorname{range}\, q(T)} \right) = 0.$$

Just for clarity, the equation above means that $s$ is the zero-operator of $T\big|_{\operatorname{range} q(T)}$. Therefore,

$$\forall u \in V : \Big((sq)(T)\Big)(u) = s(T)(q(T)u) = 0.$$

Thus, $sq$ is a candidate for our minimal polynomial of $T$.

A similar reasoning goes for the case where $(-1)c_0 v + (-1)c_1 Tv = T^2 v$ and $q(z) = c_0 + c_1 z + z^2$. Here, we have $v$ and $Tv$ as linearly independent vectors. Note that if $k \in \mathbb{N}$, then

$$q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0. \tag{5.1}$$

Hence, the equation (5B2) above tells us that $v \in \operatorname{null} q(T)$ and $Tv \in \operatorname{null} q(T)$. Here, in this case, we conclude that $\operatorname{dim}\operatorname{null} q(T) \geq 2$ and therefore $\operatorname{dim}\operatorname{range} q(T) = 0$. Since $\operatorname{dim}\operatorname{null} q(T) = 2$, we have that

$$\operatorname{null} q(T) = V.$$

This means that every vector of $V$ gets mapped to 0 if $q(T)$ is applied. So, we have found a minimal monic polynomial!

***General case:*** As we can see a pattern now, we can use induction on $\operatorname{dim} V$ and assume $\operatorname{dim} V > 2$ and that the theorem holds for all vector spaces of smaller dimension than $\operatorname{dim} V$. Let $v \in V$ be nonzero. Then the list

$$v, Tv, \ldots, T^{\operatorname{dim} V} \tag{5.2}$$

(which has length $1 + \operatorname{dim} V$) is linearly dependent. By the linear dependence lemma (2.19), there is a smallest positive integer $m \leq \operatorname{dim} V$ and some $c_0, \ldots, c_{m-1} \in \mathbb{F}$ such that

$$c_0 v + c_1 Tv + \cdots + c_{m-1} T^{m-1} v + T^m v = 0.$$

Let us define $q \in \mathcal{P}_m(\mathbb{F})$ as

$$q(z) := c_0 + c_1 z + \cdots + c_{m-1} z^{m-1} + z^m$$

accordingly. Thus, we have

$$q(T)v = 0.$$

Note that $q(z)$ is a monic polynomial and that $\deg q = m$. Observe that if $k \in \mathbb{N}$, then

$$q(T)(T^k v) = T^k(q(T)v) = T^k(0) = 0. \tag{5.3}$$

By the linear dependence lemma (2.19) and the definition of $m$, the list

$$v, Tv, \ldots, T^{m-1}v \tag{5.4}$$

from before is linearly independent (This is not the same list as (5B2) since $m \leq \dim V$). This fact together with equation (5B2) implies that the vectors of the list (5B2) all belong to $\operatorname{null} q(T)$. Hence,

$$\dim \operatorname{null} q(T) \geq m.$$

Using the Rank-Nullity Theorem, this implies:

$$\dim \operatorname{range} q(T) = \dim V - \dim \operatorname{null} q(T) \leq \dim V - m.$$

Because $\operatorname{range} q(T)$ is invariant under $T$ by 5.18, we can apply our induction hypothesis to the operator $T\big|_{\operatorname{range} q(T)}$. Hence, there exists a monic polynomial $s \in \mathcal{P}(\mathbb{F})$ such that $\deg s \leq \dim V - m$ and

$$s\left(T\big|_{\operatorname{range} q(T)}\right) = 0.$$

In other words, $s(T)(u) = 0$ for all $u \in \operatorname{range} q(T)$. So, we can safely say that

$$s(T)\big|_{\operatorname{range} q(T)} = s\left(T\big|_{\operatorname{range} q(T)}\right).$$

Because $q(T)u \in \operatorname{range} q(T)$ for all $u \in V$, we conclude:

$$\forall u \in V : \Big((sq)(T)\Big)(u) = s(T)(q(T)u) = 0.$$

Therefore, $sq$ is a monic polynomial such that

$$\deg sq \leq (\dim V - m) + m = \dim V$$

and $(sq)(T) = 0$.

***Proof of uniqueness:*** Let $p \in \mathcal{P}(\mathbb{F})$ be a monic polynomial of smallest degree such that

$$p(T) = 0.$$

Let $r \in \mathcal{P}(\mathbb{F})$ be another monic polynomial of the same degree such that $r(T) = 0$. This implies

$$(p - r)(T) = 0.$$

Moreover, $\deg(p - r) < \deg p = \deg r$. If $p - r \neq 0$, we could divide $p - r$ by the coefficient of the highest-order term in $p - r$ to obtain a monic polynomial that, when applied to $T$, yields the zero-operator. This polynomial would have a smaller degree than $p$ or $r$, which would be a contradiction. Therefore, $p - r = 0 \iff p = r$. □

**Thm 5.23** (*Existence, uniqueness, and degree of minimal polynomial*)**:**
Suppose $T \in \mathcal{L}(V)$. Then there exists a unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$, of minimal degree, such that

$$p(T) = 0.$$

Furthermore, $\deg p \leq \dim V$.

***Proof:*** We prove existence first, and uniqueness second. The arguments are independent.

***Base case:*** If $\dim V = 0$, then $V = \{\vec{0}\}$, and any operator on $V$ is both the zero and identity operator. Let $p(z) := 1$; then $p(T) = I = 0$ on $V$, so $p(T)\vec{0} = 0$.

***Next base case:*** Suppose $\dim V = 1$. Let $v \in V$, $v \neq 0$. Then the list $(v, Tv)$ is linearly dependent (length $2 >$ dimension 1), so there exists $c_0 \in \mathbb{F}$ such that

$$Tv + c_0 v = 0.$$

Define $q(z) := z + c_0$; then $q(T)v = 0$. Since $v \neq 0$ and spans $V$, it follows that $q(T) = 0$ on all of $V$. If $T = 0$, we may instead use $q(z) := z$; in either case, $q$ is monic of degree 1 and annihilates $T$.

***Another base case:*** Suppose $\dim V = 2$. Let $v \in V$, $v \neq 0$; then the list $(v, Tv, T^2 v)$ is linearly dependent. By the linear dependence lemma, either

$$Tv + c_0 v = 0, \quad \text{or} \quad T^2 v + c_1 Tv + c_0 v = 0$$

for some $c_0, c_1 \in \mathbb{F}$. Define

$$q(z) := \begin{cases} z + c_0, & \text{in the first case,} \\ z^2 + c_1 z + c_0, & \text{in the second.} \end{cases}$$

Then $q(T)v = 0$.

If $v$ and $Tv$ are linearly dependent, then $\dim \operatorname{null} q(T) \geq 1$ and by rank-nullity,

$$\dim \operatorname{range} q(T) \leq 1.$$

Since $\operatorname{range} q(T)$ is $T$-invariant (Theorem 5.18), we restrict $T$ to this subspace, which has dimension 1, and apply the previous case to obtain a monic polynomial $s$ such that

$$s\left(T\big|_{\operatorname{range} q(T)}\right) = 0.$$

Then, for all $u \in V$,

$$(sq)(T)u = s(T)(q(T)u) = 0.$$

Thus, $sq$ is a monic annihilating polynomial of degree at most 2.

If instead $v$ and $Tv$ are linearly independent, then

$$q(T)(T^k v) = T^k(q(T)v) = 0 \quad \forall k \in \mathbb{N}.$$

Hence, both $v$ and $Tv$ lie in $\operatorname{null} q(T)$, so

$$\dim \operatorname{null} q(T) \geq 2 = \dim V \quad \Rightarrow \quad q(T) = 0.$$

In either case, we find a monic annihilating polynomial of degree $\leq 2$.

***Inductive step:*** Assume the theorem holds for vector spaces of dimension less than $n$, and let $\dim V = n > 2$.

Pick $v \in V$, $v \neq 0$. The list $(v, Tv, \ldots, T^n v)$ has length $n+1$, so it is linearly dependent. Let $m \leq n$ be minimal such that

$$T^m v + c_{m-1} T^{m-1} v + \cdots + c_0 v = 0.$$

Define the monic polynomial

$$q(z) := z^m + c_{m-1} z^{m-1} + \cdots + c_0.$$

Then $q(T)v = 0$. Since

$$q(T)(T^k v) = T^k(q(T)v) = 0 \quad \forall k \in \mathbb{N},$$

the vectors $v, Tv, \ldots, T^{m-1}v$ lie in $\operatorname{null} q(T)$. By minimality of $m$, this list is linearly independent, so

$$\dim \operatorname{null} q(T) \geq m \quad \Rightarrow \quad \dim \operatorname{range} q(T) \leq n - m.$$

As $\operatorname{range} q(T)$ is $T$-invariant, we restrict $T$ to it and apply the inductive hypothesis: there exists a monic polynomial $s$ of degree at most $n-m$ such that

$$s\left(T\big|_{\operatorname{range} q(T)}\right) = 0.$$

Since $q(T)u \in \operatorname{range} q(T)$ for all $u \in V$, it follows that

$$(sq)(T)u = s(T)(q(T)u) = 0.$$

Hence, $sq$ is a monic annihilating polynomial with

$$\deg(sq) = \deg s + \deg q \leq n.$$

***Uniqueness:*** Suppose $p$ and $r$ are two monic polynomials of minimal degree such that $p(T) = r(T) = 0$. Then

$$(p-r)(T) = 0,$$

but $\deg(p-r) < \deg p$ unless $p = r$. This contradicts the minimality of $p$ unless $p = r$. Therefore, the monic minimal polynomial is unique. □

**Def 5.24** (*minimal polynomial*):

Let $T \in \mathcal{L}(V)$. The *"minimal polynomial of $T$"* is the unique monic polynomial $p \in \mathcal{P}(\mathbb{F})$ of smallest degree such that $p(T) = 0$.

**Computation of the minimal polynomial:** Find the smallest $m \in \mathbb{N}$ such that:
$c_0 I + c_1 T + \cdots + c_{m-1} T^{m-1} = -T^m$ has a solution $c_0, \ldots, c_{m-1} \in \mathbb{F}$. Solve for $m = 1, 2, \ldots, \dim V$.
Even faster (usually), pick $v \in V$ with $v \neq 0$ and consider the equation:

$$c_0 v + c_1 T v + \cdots + c_{\dim V - 1} T^{\dim V - 1} v = -T^{\dim V} v.$$

If this equation has a unique solution, as happens most of the time,

$$c_0, c_1, c_2, \ldots, c_{\dim V - 1}, 1$$

are the coefficients of the minimal polynomial of $T$.

**Thm 5.27** (*eigenvalues are the zeros of the minimal polynomial*):

Let $T \in \mathcal{L}(V)$, which means $T$ is an operator, and let $V$ be finite-dimensional ($\dim V \neq \infty$). Then:

**(a)** The zeros of the minimal polynomial of $T$ are the eigenvalues of $T$.

**(b)** If $V$ is a complex vector space, the minimal polynomial has the form

$$(z-\lambda_1)\cdots(z-\lambda_m),$$

where $\lambda_1, \ldots, \lambda_m$ are the eigenvalues of $T$, possibly with repetitions (may be $\lambda_i = \lambda_j$ for $i \neq j$).

***Proof:*** Let $p$ be the minimal polynomial of $T$.

**(a)** $\Rightarrow$ ***direction:*** Suppose $\lambda \in \mathbb{F}$ is a zero of $p$. Hence, we can write $p$ as $p(z) = (z-\lambda)q(z)$, where $q$ is a monic polynomial with coefficients in $\mathbb{F}$ (see 4.6). As we can write

$$p(T) = 0,$$

we can also write

$$0 = (T - \lambda I)(q(T)v) \quad \forall v \in V.$$

Because $q$ is of lesser degree than $p$ (i.e., $\deg q = (\deg p) - 1$) and $p$ is the minimal polynomial of $T$, there exists at least one vector $u \in V$ such that

$$q(T)u \neq 0,$$

which makes $q(T)u$ an eigenvector with eigenvalue $\lambda$.

$\Leftarrow$ ***direction:*** Conversely, suppose $\lambda \in \mathbb{F}$ is an eigenvalue of $T$. Thus, there exists a nonzero $u \in V$ such that $Tu = \lambda u$. Repeated applications of $T$ on both sides of this equation show that

$$T^k u = \lambda^k u \quad \forall k \in \mathbb{N}.$$

This implies that

$$p(T)u = p(\lambda)u.$$

Because $p$ is the minimal polynomial of $T$, we have $p(T)u = p(\lambda)u = 0$. Since $u$ is nonzero, we have $p(\lambda) = 0$. This makes $\lambda$ a zero of $p$.

**(b)** Use (a) and the second version of the fundamental theorem of algebra (4.13). $\square$

**Thm 5.29** (*every "zero-polynomial" is a multiple of the minimal polynomial*):
Let $T \in \mathcal{L}(V)$ and $q \in \mathcal{P}(\mathbb{F})$. Then $q(T) = 0 \iff q$ is a multiple of the minimal polynomial of $T$.

Or equivalently: The minimal polynomial of $T$ divides every polynomial $q$ for which $q(T) = 0$ (sometimes written as $m_T \mid q$).

***Proof:*** Let $p$ denote the minimal polynomial of $T$.

$\Rightarrow$ ***direction:*** Suppose $q(T) = 0$. By 4.9, there exist $s, r \in \mathcal{P}(\mathbb{F})$ such that

$$q = ps + r, \quad \deg r < \deg p.$$

We have

$$0 = q(T) = p(T)s(T) + r(T) = r(T). \tag{5.5}$$

The equation above implies that $r = 0$. Otherwise, dividing $r$ by its highest-degree coefficient would produce a monic polynomial that when applied to $T$ gives 0. This is a contradiction because $\deg r < \deg p$ and $p$ is minimal. Thus, (5B2) becomes the equation $q = ps$, as desired.

$\Leftarrow$ ***direction:*** Suppose $q = ps$ for $q, p, s \in \mathcal{P}(\mathbb{F})$. We have

$$q(T) = p(T)s(T) = 0s(T) = 0,$$

as desired, which proves both directions. $\square$

**Thm 5.31** (*minimal polynomial of a restriction operator*):
If $U$ is a subspace of $V$, then the minimal polynomial of $T$ is a polynomial multiple of the minimal polynomial of $T\big|_U$.

***Proof:*** Suppose $p$ is the minimal polynomial of $T \in \mathcal{L}(V)$ and $U \subseteq V$. Then

$$p(T)v = 0 \quad \forall v \in V.$$

In particular,

$$p(T)u = 0 \quad \forall u \in U.$$

Thus, $p\left(T\big|_U\right) = 0$. Now, the previous theorem 5.29 tells us that $p$ is a polynomial multiple of the minimal polynomial of $T\big|_U$. [This means that the degree of $p$ is at least as large]. $\square$

**Thm 5.32** (*invertibility and the constant term of the minimal polynomial*):
Let $T \in \mathcal{L}(V)$. Then

$T$ is not invertible $\iff$ the constant term of the minimal polynomial of $T$ is 0.

***Proof:*** $T$ is not invertible $\iff_{(5.7)}$ 0 is an eigenvalue of $T$ $\iff_{(5.27)}$ 0 is a zero of $p$ $\iff$ the constant term of $p$ is 0. (In the first equivalence, we have actually used that 0 is an eigenvalue of $T$ if and only if $T - 0I$ is not invertible, according to 5.7.) $\square$

## Eigenvalues on Odd-Dimensional Real Vector Spaces (Sub-Section 5B3)

**Thm 5.33** (*even-dimensional null space*):
Suppose $\mathbb{F} = \mathbb{R}$ and $V$ is finite-dimensional. Suppose also that $T \in \mathcal{L}(V)$ and $b, c \in \mathbb{R}$ with $b^2 < 4c$. Then $\dim \operatorname{null}(T^2 + bT + cI)$ is an even number.

***Proof:*** Recall that $\operatorname{null}(T^2 + bT + cI)$ is invariant under $T$, by 5.18. Let

$$N := \operatorname{null}(T^2 + bT + cI) \text{ and}$$
$$S := T|_N.$$

With these definitions above, we can assume that

$$S^2 + bS + cI = 0, \tag{5.6}$$

where 0 means the zero operator. We now need to prove that $\dim N$ is even. Suppose $\lambda \in \mathbb{R}$ and $v \in N$ are such that $Sv = \lambda v$. Then

$$0 = (S^2 + bS + cI)v = (\lambda^2 + b\lambda + c)v = \underbrace{\left(\left(\lambda + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}\right)}_{\geq 0, \text{ since } b^2 \leq 4c} v.$$

The term in the large parentheses above is a positive number. Thus, the equation above implies that $v = 0$. Hence, we have shown that $S$ has no eigenvalues. Let $U$ be a subspace of $N$ with even dimension that is invariant under $S$ and has the largest dimension among all subspaces of $N$. Note that $U$ could be $\{0\}$. If $U = N$, then we are done.
Therefore, assume that $U \neq N$. By maximality, $U$ is not properly contained in any larger even-dimensional $S$-invariant subspace of $N$. Hence, there exists $w \in N$ such that $w \notin U$. Now, let

$$W := \operatorname{span}(w, Sw).$$

Clearly, $W \subseteq N$, and since $w \notin U$, $W \subsetneq U$. We can rewrite (5B3) as $S(Sw) = -bSw - cw$. Using this, we calculate for $x \in W$, written as $x = a_1 w + a_2 Sw$ for some scalars $a_1, a_2$:

$$\begin{aligned} S(x) &= a_1 Sw + a_2 S^2 w = a_1 Sw - a_2(bSw + cw) \\ &= (a_1 - a_2 b)Sw - a_2 cw \\ &\in W. \end{aligned}$$

Thus, we see that $W$ is invariant under $S$ (remember, $S = T|_N$). Furthermore, $\dim W = 2$, because otherwise $w$ would be an eigenvector of $S$, since $w$ and $Sw$ would be linearly dependent. This means that there would exist nonzero scalars $a_1$ and $a_2$, such that we have a nontrivial relation $a_1 w + a_2 Sw = 0$. In this case, $-a_1/a_2$ would be an eigenvalue of $S$ with eigenvector $w$, which is impossible. Now, using 2.43:

$$\dim(U + W) = \dim U + \underbrace{\dim W}_{=2} - \underbrace{\dim(U \cap W)}_{=0} = \dim U + 2,$$

where $U \cap W = \{0\}$, because otherwise, $U \cap W$ would be a one-dimensional $S$-invariant subspace providing an eigenvector for $S$, which is impossible. And the dimension of $U \cap W$ cannot be 2 either, because then we would have $U \cap W = W$, using 2.39. But then $W \subseteq U$, contradicting the fact that $w \notin U$. Since $U$ was assumed to be maximal among even-dimensional $S$-invariant subspaces, the existence of $U + W$, with $\dim(U + W) = \dim U + 2$, contradicts the maximality of $U$. Thus, the assumption that $U \neq N$ was incorrect. Therefore, we conclude that $U = N$. Hence, $N = \operatorname{null}(S^2 + bS + cI) = \operatorname{null}(T|_N^2 + bT|_N + cI) = \operatorname{null}(T^2 + bT + cI)$ has even dimension $\dim N$. □

**Thm 5.34** (*operators on odd-dimensional vector spaces have eigenvalues*):
Every operator on an odd-dimensional vector space has an eigenvalue.

***Proof:*** Suppose $\mathbb{F} = \mathbb{R}$ and $V$ is finite-dimensional. Let $n := \dim V$, and suppose $n$ is an odd number. Let $T \in \mathcal{L}(V)$. We will use induction on $n$ in steps of size two to show that $T$ has an eigenvalue. Note that the desired result holds if $\dim V = 1$, because then every nonzero vector in $V$ is an eigenvector of $T$.

Now, suppose $n \geq 3$ and the desired result holds for all operators on all odd-dimensional vector spaces of dimension less than $n$, i.e., with dimension $\leq n-2$. Let $p$ denote the minimal polynomial of $T$. If $p$ is a polynomial multiple of $x - \lambda$ for some $\lambda \in \mathbb{R}$, then $\lambda$ is an eigenvalue of $T$ by Theorem 5.27, and we are done. Thus, we can assume that there exist $b, c \in \mathbb{R}$ such that $b^2 < 4c$ and $p$ is a polynomial multiple of $x^2 + bx + c$ by 4.16. By looking at this theorem, we can see that there exists a monic polynomial $q \in \mathcal{P}(\mathbb{R})$ such that $p(x) = q(x)(x^2 + bx + c) \quad \forall x \in \mathbb{R}$. Now,

$$0 = p(T) = (q(T))(T^2 + bT + cI),$$

which means that $q(T)$ equals 0 on $\operatorname{range}(T^2 + bT + cI)$. In other words,

$$q(T)|_{\operatorname{range}(T^2+bT+cI)} = 0.$$

Because $\deg q < \deg p$ and $p$ is the minimal polynomial of $T$, this implies that

$$\operatorname{range}(T^2 + bT + cI) \neq V. \tag{5.7}$$

The rank-nullity theorem 3.21 tells us that

$$\dim V = \dim \operatorname{null}(T^2 + bT + cI) + \dim \operatorname{range}(T^2 + bT + cI). \tag{5.8}$$

The previous theorem 5.33 tells us that $\dim \operatorname{null}(T^2 + bT + cI)$ is even. Since $\dim V$ is odd, equation (5B3) above and equation (5B3) show that $\dim \operatorname{range}(T^2 + bT + cI)$ is odd as well and at least 2 less than $\dim V$. We know that $\operatorname{range}(T^2 + bT + cI)$ is invariant under $T$ by 5.18. Our induction hypothesis now implies that $T$ restricted to $\operatorname{range}(T^2 + bT + cI)$, which is a subspace of $V$, has an eigenvalue; this also means that $T$ has an eigenvalue. □

## Exercises about the Minimal Polynomial 5B

(1) Suppose $T \in \mathcal{L}(V)$. Prove that 9 is an eigenvalue of $T^2 \iff 3$ or $-3$ is an eigenvalue of $T$.

***Proof:*** $\Leftarrow$ ***direction:*** Suppose $-3$ is an eigenvalue of $T$. Thus, there exists $v \in V$ such that $Tv = -3v$. If we apply $T$ a second time, we get $TTv = T(-3v) = -3Tv = (-3)(-3)v = 9v$. Therefore, 9 is an eigenvalue of $T^2$. The case where we start with 3 as an eigenvalue of $T$ works analogously.

$\Rightarrow$ ***direction:*** Let $v \neq 0$ satisfy $T^2 v = 9v$. Then

$$0 = (T^2 - 9I)v = (T - 3I)(T + 3I)v.$$

Set $w := (T + 3I)v$. Then $(T - 3I)w = 0$, so $Tw = 3w$ (because $w \in \operatorname{null}(T - 3I))$^. There are two cases:

- If $w = 0$, then $(T + 3I)v = 0$. Hence, $Tv = -3v$ and $-3$ is an eigenvalue of $T$.
- If $w \neq 0$, then $w$ is an eigenvector of $T$ with eigenvalue 3.

Thus, $T$ has eigenvalue 3 or $-3$, as required.

□

(3) Suppose $n \in \mathbb{N}_{>0}$ and $T \in \mathcal{L}(\mathbb{F}^n)$ is defined by $T(x_1, \ldots, x_n) := (x_1 + \cdots + x_n, \ldots, x_1 + \cdots + x_n)$. The

matrix $\mathcal{M}(T)$ of $T$ looks like this

$$\mathcal{M}(T) = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}$$

**(a)** Find all eigenvalues and eigenvectors of $T$.

**(b)** Find the minimal polynomial of $T$.

***Solution:*** We want to find all $x \in V$ such that $Tx = \lambda x$. Therefore,

$$\begin{aligned} x_1 + \cdots + x_n &= \lambda x_1 \\ &\vdots \\ x_1 + \cdots + x_n &= \lambda x_n \end{aligned}$$

If we define $S := x_1 + \cdots + x_n$, we can rewrite the system of equations as

$$S = \lambda x_1, \ldots, S = \lambda x_n, \tag{5.9}$$

or equivalently: $S = \lambda x_1 = \lambda x_2 = \cdots = \lambda x_n$. Solving by $x_i$ for $\lambda \neq 0$ gives

$$x_1 = \frac{S}{\lambda},\ x_2 = \frac{S}{\lambda},\ \ldots,\ x_n = \frac{S}{\lambda}$$

Summing up all equations for $S$ in (**1**) gives:

$$nS = \lambda x_1 + \cdots + \lambda x_n = \lambda S$$

Hence,

$$\begin{aligned} nS - \lambda S &= 0 \\ S(n-\lambda) &= 0. \qquad \text{(which is the same as } (S-0)(n-\lambda) = 0) \end{aligned}$$

If $\lambda = n$, then $x_1 = x_2 = \cdots = x_n$. Meaning all eigenvectors corresponding to the eigenvalue $n$ are s multiples of $(1, 1, \ldots, 1) \in \mathbb{F}$.

If $\lambda \neq n$, then $S(\lambda - n) = 0$ implies that $S = 0$ and thus, $\lambda = 0$. This means that $x$ belongs to the eigenspace of 0, if $x_1 + x_2 + x_3 + \cdots + x_n = 0$.

***The minimal polynomial of T:***

$$\begin{aligned} T(e_1) &= e_1 + \cdots + e_n \\ T(e_2) &= e_1 + \cdots + e_n \\ &\vdots \\ T(e_n) &= e_1 + \cdots + e_n \end{aligned}$$

Define $v := e_1 + \cdots + e_n$. Note that $\operatorname{range} T = \operatorname{span}(v)$. Applying $T$ to $v$ gives

$$\begin{aligned} T(v) &= T(e_1 + \cdots + e_n) = T(e_1) + \cdots + T(e_n) = nv \\ T^2(v) &= nT(v) \end{aligned}$$

The last line tells us that $T^2 = nT$ or equivalently $T(T - nI) = 0$. Therefore, the minimal polynomial is

$$p(z) = z(z-n).$$

Because $\operatorname{range} T = \operatorname{span}(v)$, the polynomial is in fact minimal.

---

**4** Suppose $\mathbb{F} = \mathbb{C}$, $T \in \mathcal{L}(V)$, $p \in \mathcal{P}(\mathbb{C})$, and $\alpha \in \mathbb{C}$. Prove that $\alpha$ is an eigenvalue of $p(T) \iff \alpha = p(\lambda)$ for some eigenvalue $\lambda$ of $T$.

***Proof:*** If $\lambda$ is an eigenvalue of $T$, then $\exists v \in V$ such that

$$\begin{aligned} Tv &= \lambda v \\ T^2 v &= \lambda^2 v \\ T^k v &= \lambda^k v \quad \forall k \in \mathbb{N}_{>0}. \end{aligned}$$

$\Rightarrow$ ***direction:*** If $\alpha$ is an eigenvalue of $p(T)$, then $\exists u \in V$ such that

$$\begin{aligned} \alpha u &= p(T)(u) \\ \alpha u &= c_0 u + c_1 T u + \cdots + c_{m-1} T^{m-1} u + T^m u \\ \alpha u &= c_0 \lambda^0 u + c_1 \lambda^1 u + \cdots + c_{m-1} \lambda^{m-1} u + \lambda^m u \\ \alpha u &= (c_0 \lambda^0 + c_1 \lambda^1 + \cdots + c_{m-1} \lambda^{m-1} + \lambda^m) u \\ \alpha &= c_0 \lambda^0 + c_1 \lambda^1 + \cdots + c_{m-1} \lambda^{m-1} + \lambda^m \\ \alpha &= p(\lambda) \end{aligned}$$

But where have we been using the fact that $\mathbb{F} = \mathbb{C}$? □

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**12** Define $T \in \mathcal{L}(\mathbb{F}^n)$ by $T(x_1, x_2, x_3, \ldots, x_n) = (x_1, 2x_2, 3x_3, \ldots, nx_n)$. Find the minimal polynomial of $T$.

***Solution:*** For a standard basis $e_1, \ldots, e_n$ we have $T(e_1 + \cdots + e_n) = e_1 + 2e_2 + \cdots + ne_n$. Or equivalently:

$$\begin{aligned} T(e_1) &= e_1 \\ T(e_2) &= 2e_2 \\ &\vdots \\ T(e_n) &= ne_n. \end{aligned}$$

Hence, $e_1, \ldots, e_n$ are $\dim \mathbb{F}^n = n$ distinct eigenvectors with corresponding eigenvalues $1, \ldots, n$. This makes $Te_1, \ldots, Te_n$ linearly independent as well. Using 5.27, we conclude that the minimal polynomial of $T$ looks like this:

$$p(z) = (z-1)(z-2)\cdots(z-n) \quad \forall z \in \mathbb{F}^n. \tag{5.10}$$

We can check if $\forall v \in \mathbb{F}^n$, given as $v = a_1 e_1 + \cdots + a_n e_n$ ($a_i \in \mathbb{F}$), the following equation holds:

$$(T - I)(T - 2I)\cdots(T - nI)v = 0.$$

This is true because $a_k e_k \in \operatorname{null}(T - kI)$ for $k = 1, \ldots, n$. Therefore,

$$\begin{aligned} (T - I)(T - 2I)\cdots(T - nI)v = {} & (T - I)(T - 2I)(T - 3I)\cdots(T - nI)(a_1 e_1) \\ & + (T - I)(T - 2I)(T - 3I)\cdots(T - nI)(a_2 e_2) \\ & + (T - I)(T - 2I)(T - 3I)\cdots(T - nI)(a_3 e_3) \\ & + \cdots \\ & + (T - I)(T - 2I)(T - 3I)\cdots(T - nI)(a_n e_n) = 0. \end{aligned}$$

Thus, we have found our minimal polynomial given in equation (**1**). It is easy to conclude that there exists no polynomial of smaller degree that distributes to every basis vector of the sum $a_1 e_1 + \cdots + a_n e_n$.

---

**16** Suppose $a_0, a_1, \ldots, a_n \in \mathbb{F}$. Let $T$ be the operator on $\mathbb{F}$ whose matrix (with respect to the standard basis) is

$$\mathcal{M}(T) = \begin{array}{c c} & \begin{array}{cccccc} e_1 & e_2 & \cdots & & e_{n-1} & e_n \end{array} \\ \begin{array}{c} e_1 \\ e_2 \\ e_3 \\ \vdots \\ e_{n-1} \\ e_n \end{array} & \begin{pmatrix} 0 & & & & & -a_0 \\ 1 & 0 & & & & -a_1 \\ & 1 & \ddots & & & -a_2 \\ & & \ddots & & & \vdots \\ & & & & 0 & -a_{n-2} \\ & & & & 1 & -a_{n-1} \end{pmatrix} \end{array}.$$

Note that all the other entries of this matrix are zero. Show that the minimal polynomial of $T$ is the polynomial

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n$$

***Solution:*** Looking at the matrix $\mathcal{M}(T)$ and having a glance at 3.31, we observe that

$$\begin{aligned} Te_1 &= e_2 \\ Te_2 &= e_3 = T(Te_1) = T^2 e_1 \\ Te_3 &= e_4 = T(Te_2) = T^3 e_1 \\ Te_4 &= e_5 = T(Te_3) = T^4 e_1 \\ &\vdots \\ Te_{n-1} &= e_n = T(Te_{n-2}) = T^{n-1} e_1 \\ Te_n &= -a_0 e_1 - a_1 e_2 - a_2 e_3 - \cdots - a_{n-2} e_{n-1} - a_{n-1} e_n = T(T^{n-1} e_1) = T^n e_1. \end{aligned}$$

We can rewrite the last expression as

$$-T^n(e_1) = a_0 e_1 + a_1 T e_1 + a_2 T^2 e_1 + \cdots + a_{n-2} T^{n-2} e_1 + a_{n-1} T^{n-1} e_1$$

The list $e_1, Te_1, T^2 e_1, T^3 e_1, \cdots, T^{n-1} e_1$, which equals the list $e_1, e_2, e_3, \cdots, e_{n-1}$, is linearly independent. Therefore, no other combination of the list equals $-T^5 e_1$. Hence, the minimal polynomial of $T$ is

$$a_0 + a_1 z + \cdots + a_{n-1} z^{n-1} + z^n.$$

## Upper-Triangular Matrices (Section 5C)

**Thm 5.39** (*conditions for upper-triangular matrix*):

If $T \in \mathcal{L}(V)$ and $v_1,\dots,v_n$ is a basis of $V$, then the following are equivalent:

**(a)** The matrix $\mathcal{M}(T)$ of $T$ with respect to $v_1,\dots,v_n$ is upper triangular.

**(b)** $\mathsf{span}(v_1,\dots,v_k)$ is invariant under $T$ for every $k = 1,\dots,n$.

**(c)** $Tv_k \in \mathsf{span}(v_1,\dots,v_k)$ for every $k = 1,\dots,n$.

***Proof:*** We prove (a) $\implies$ (b) $\implies$ (c) $\implies$ (a).

***Step 1:*** Suppose (a) holds and $j,k \in \{1,\dots,n\}$ with $j \le k$. The matrix of $T$ looks like this:

$$\mathcal{M}(T) = \begin{array}{c} \\ v_1 \\ \vdots \\ v_j \\ \vdots \\ v_n \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_j & \cdots & v_n \end{array} \\ \begin{pmatrix} A_{1,1} & \cdots & A_{1,j} & \cdots & A_{1,n} \\ & \ddots & \vdots & \ddots & \vdots \\ & & A_{j,j} & \cdots & A_{j,n} \\ & & & \ddots & \vdots \\ & & & & A_{n,n} \end{pmatrix} \end{array}.$$

By definition of the matrix entries (3.31), we have that

$$Tv_j = A_{1,j}v_1 + A_{2,j}v_2 + \cdots + A_{j,j}v_j = \sum_{\ell=1}^{j} A_{\ell,j}v_\ell.$$

Hence, the upper-triangular structure of $\mathcal{M}(T)$ implies

$$Tv_j \in \mathsf{span}(v_1,\dots,v_j).$$

Because

$$\mathsf{span}(v_1,\dots,v_j) \subseteq \mathsf{span}(v_1,\dots,v_k),$$

it is immediate that

$$Tv_j \in \mathsf{span}(v_1,\dots,v_k).$$

Since we had $j \in \{1,\dots,k\}$ and so $v_j$ could be any basis vector in the list $v_1,\dots,v_k$, $\mathsf{span}(v_1,\dots,v_k)$ is invariant under $T$, completing the proof that (a) implies (b).

***Step 2:*** Now, suppose (b) holds. Hence, we assume $\mathsf{span}(v_1,\dots,v_k)$ is invariant under $T$ for each $k \in \{1,\dots,n\}$. In particular,

$$Tv_k \in \mathsf{span}(v_1,\dots,v_k) \quad \forall k \in \{1,\dots,n\}.$$

Thus, (b) implies (c).

***Step 3:*** Finally, suppose (c) holds. Hence, $Tv_k \in \mathsf{span}(v_1,\dots,v_k)$ for $k = 1,\dots,n$. This implies that there exist $a_1,\dots,a_k \in \mathbb{F}$, such that

$$Tv_k = a_1v_1 + \cdots + a_kv_k = a_1v_1 + \cdots + a_kv_k + 0v_{k+1} + \cdots + 0v_n.$$

Hence, all the entries below the diagonal of $\mathcal{M}(T)$ are 0, because the list $v_1,\dots,v_n$ is linearly independent and we cannot choose another representation for $Tv_k$ with respect to that list. Thus, $\mathcal{M}(T)$ is an upper-triangular matrix, completing the proof that (c) implies (a). After all, we have shown that (a) $\implies$ (b) $\implies$ (c) $\implies$ (a). □

**Thm 5.40** (*equation satisfied by operator with upper-triangular matrix*):

Suppose $T \in \mathcal{L}(V)$ and $V$ has a basis with respect to which $T$ has an upper-triangular matrix $\mathcal{M}(T)$ with diagonal entries $\lambda_1,\dots,\lambda_n$. Then

$$(T - \lambda_1 I)\cdots(T - \lambda_n I) = 0.$$

Note that this theorem does not mention anything about eigenvalues.

***Proof:*** Let $A :\equiv \mathcal{M}(T)$. Let $v_1,\dots,v_n$ denote a basis of $V$ with respect to which $T$ has an upper-triangular matrix with diagonal entries $\lambda_1,\dots,\lambda_n$.

$$A = \mathcal{M}(T) = \begin{pmatrix} \lambda_1 & & * \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} = \begin{pmatrix} \lambda_1 & A_{1,2} & \cdots & A_{1,n} \\ & \lambda_2 & \cdots & A_{2,n} \\ & & \ddots & \vdots \\ 0 & & & \lambda_n \end{pmatrix}$$

Then

$$Tv_1 = \lambda_1 v_1 \iff (T - \lambda_1 I)v_1 = 0, \tag{5.11}$$

which implies that

$$(T-\lambda_1 I)(T-\lambda_2 I)v_1 = 0,$$
$$(T-\lambda_1 I)(T-\lambda_2 I)(T-\lambda_3 I)v_1 = 0,$$

... or more generally speaking:

$$(T-\lambda_1 I)\cdots(T-\lambda_m I)v_1 = 0, \text{ for } m = 1,\dots,n.$$

(Using commutativity for linear maps after multiplying both sides of the equation with anything.)

Note that $(T-\lambda_2 I)v_2 \in \mathsf{span}(v_1)$ because $Tv_2 = A_{1,2}v_1 + \lambda_2 v_2$. Thus, using (5C),

$$\begin{aligned}(T-\lambda_1 I)(T-\lambda_2 I)v_2 &= (T-\lambda_1 I)A_{1,2}v_1 \\ &= A_{2,1}(T-\lambda_1 I)v_1 \\ &= 0,\end{aligned} \tag{5.12}$$

which implies that

$$(T-\lambda_1 I)(T-\lambda_2 I)(T-\lambda_3)v2 = 0,$$
$$(T-\lambda_1 I)(T-\lambda_2 I)(T-\lambda_3)(T-\lambda_4)v2 = 0,$$

... or more generally speaking:

$$(T-\lambda_1 I)\cdots(T-\lambda_m I)v_2 = 0, \text{ for } m = 2,\dots,n.$$

(Using commutativity for linear maps after multiplying both sides of the equation with anything.)

Note that $(T-\lambda_3 I)v_3 \in \mathsf{span}(v_1, v_2)$ because $Tv_3 = A_{1,3}v_1 + A_{2,3}v_2 + \lambda_3 v_3$.
Thus, using (5C) and (5C), we get

$$\begin{aligned}(T-\lambda_1 I)(T-\lambda_2 I)(T-\lambda_3 I)v_3 &= (T-\lambda_1 I)(T-\lambda_2 I)(A_{1,3}v_1 + A_{2,3}v_2) \\ &= (T-\lambda_1 I)A_{1,3}v_1(T-\lambda_2 I) + (T-\lambda_1 I)(T-\lambda_2 I)A_{2,3}v_2 \\ &= A_{1,3}(T-\lambda_1 I)v_1(T-\lambda_2 I) + A_{2,3}(T-\lambda_1 I)(T-\lambda_2 I)v_2 \\ &= 0,\end{aligned}$$

which implies that

$$(T-\lambda_1 I)(T-\lambda_2 I)\cdots(T-\lambda_m I)v_3 = 0, \text{ for } m = 3,\dots,n.$$

(Using commutativity for linear maps after multiplying both sides of the equation with anything.)

Continuing this pattern, we see that

$$(T-\lambda_1 I)\cdots(T-\lambda_n I)v_k = 0 \quad \forall k \in \{1,\dots,n\}.$$

Thus, $(T-\lambda_1 I)\cdots(T-\lambda_n I)$ is the 0 operator because it is 0 on each vector in a basis of $V$. $\square$

**EXAMPLE** (*octave, matlab*):

To test this in octave or matlab, type:

```
A=rand(4,4); I=eye(4); T=triu(A);
ZERO=(T-T(1,1)*I)*(T-T(2,2)*I)*(T-T(3,3)*I)*(T-T(4,4)*I)
```

which should yield a 4-by-4 zero-matrix.

**THM 5.41** (*determination of eigenvalues from upper-triangular matrix*):

Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of $V$. Then the eigenvalues of $T$ are precisely the entries on the diagonal of that upper-triangular matrix.

***PROOF:*** Let $v_1, \ldots, v_n$ denote a basis of $V$ with respect to which $T$ has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$.

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & \cdots & * \\ & \ddots & \vdots \\ 0 & & \lambda_n \end{pmatrix}.$$

Because $Tv_1 = \lambda_1 v_1$, we see that $\lambda_1$ is an eigenvalue of $T$. Suppose $k \in \{2, \ldots, n\}$.

Then $(T - \lambda_k I)v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$, because

$$Tv_k = \mathcal{M}(T)_{1,k}v_1 + \mathcal{M}(T)_{2,k}v_2 + \cdots + \mathcal{M}(T)_{k-1,k}v_{k-1} + \lambda_k v_k.$$

Thus, $T - \lambda_k I$ maps $\operatorname{span}(v_1, \ldots, v_k)$ into $\operatorname{span}(v_1, \ldots, v_{k-1})$.

Because

$$\dim \operatorname{span}(v_1, \ldots, v_k) = k \quad \text{and} \quad \dim \operatorname{span}(v_1, \ldots, v_{k-1}) = k - 1,$$

this implies that $T - \lambda_k I$, which is restricted to $\operatorname{span}(v_1, \ldots, v_{k-1})$, is not injective by 3.22. This theorem states that a linear map to a lower-dimensional space is not injective. Thus,

$$\exists v \in \operatorname{span}(v_1, \ldots, v_n) : v \neq 0 \text{ and } (T - \lambda_k I)v = 0.$$

Thus, $\lambda_k$ is an eigenvalue of $T$ with eigenvector $v$. Hence, we have shown that every entry of the diagonal of $\mathcal{M}(T)$ is an eigenvalue of $T$.

To prove $T$ has no other eigenvalues, let

$$q :\equiv (z - \lambda_1) \cdots (z - \lambda_2).$$

$$\implies q(T) = 0 \ \text{ by } 5.40 \ .$$

Hence, $q$ is a polynomial multiple of the minimal polynomial of $T$ by 5.29. Thus, every zero of the minimal polynomial is a zero of $q$. Because the zeros of the minimal polynomial of $T$ are the eigenvalues of $T$ by 5.27, this implies that every eigenvalue of $T$ is a zero of $q$. Hence, the eigenvalues of $T$ are all contained in the list $\lambda_1, \ldots, \lambda_n$. □

**THM 5.44** (*necessary and sufficient condition to have an upper-triangular-matrix*):

Suppose $T \in \mathcal{L}(V)$, $\dim V \neq \infty$. Then $T$ has an upper-triangular matrix $\mathcal{M}(T)$ with respect to some basis of $V$ $\iff$ the minimal polynomial of $T$ equals $(z - \lambda_1) \cdots (z - \lambda_m)$, where $\lambda_1, \ldots, \lambda_m \in \mathbb{F}$.

***PROOF:*** $\Rightarrow$ ***DIRECTION:*** Suppose $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to some basis of $V$. Let $\alpha_1, \ldots, \alpha_n$ denote the diagonal entries of that matrix. I.e.,

$$\mathcal{M}(T) = \begin{pmatrix} \alpha_1 & \cdots & * \\ & \ddots & \vdots \\ 0 & & \alpha_n \end{pmatrix}.$$

Let $q \in \mathcal{P}(\mathbb{F})$ such that

$$q(z) := (z - \alpha_1) \cdots (z - \alpha_n).$$

Then $q(T) = 0$, by 5.40. Hence, $q$ is a polynomial multiple of the minimal polynomial of $T$, by 5.29. Therefore, the minimal polynomial $p$ of $T$ equals

$$p(z) = (z - \lambda_1)\cdots(z - \lambda_m) \quad \forall z \in \mathbb{F}, \quad (\lambda_1, \dots, \lambda_m \in \mathbb{F})$$

and we have

$$\{\lambda_1, \dots, \lambda_m\} \subseteq \{\alpha_1, \dots, \alpha_n\}.$$

$\Leftarrow$ ***direction:*** Let $p \in \mathcal{P}(\mathbb{F})$ be the minimal polynomial of $T$ such that

$$p(z) = (z - \lambda_1)\cdots(z - \lambda_m) \quad \forall z \in \mathbb{F}, \quad (\lambda_1, \dots, \lambda_m \in \mathbb{F}). \tag{5.13}$$

If $m = 1$, then $z - \lambda_1$ is the minimal polynomial of $T$. Thus, $T = \lambda_1 I$, which implies that the matrix of $T$ is the identity matrix times a scalar:

$$\mathcal{M}(T) = \lambda_1 [I_{\dim V}].$$

Thus, $\mathcal{M}(T)$ is upper triangular. Therefore, we are ready to use induction on $m$. Now, suppose $m > 1$ and the result holds for all smaller integers / polynomials. Let

$$U := \mathsf{range}(T - \lambda_m I). \tag{5.14}$$

In this case, $U$ is invariant under $T$, because of 5.18, where $p(z)$ would be $z - \lambda_m$. Thus, $T\big|_U$ is an operator on $U$. If $u \in U$, then $u = (T - \lambda_m I)v$ for some $v \in V$. For this arbitrary $u$ and corresponding $v$, it holds that

$$\begin{aligned}(T - \lambda_1 I)\cdots(T - \lambda_{m-1} I)u &= (T - \lambda_1 I)\cdots(T - \lambda_{m-1} I)(T - \lambda_m I)v \\ &= 0,\end{aligned}$$

because of (5C). Hence, $(T - \lambda_1 I)\cdots(T - \lambda_{m-1} I)$ is a polynomial multiple of the minimal polynomial of $T\big|_U$, by 5.29. In other words: it is the product of at most $m-1$ terms of the form $(z - \lambda_k)$ for $k = 1, \dots, m-1$. By the induction hypothesis, there is a basis $u_1, \dots, u_M$ of $U$ with respect to which $T\big|_U$ has an upper-triangular matrix. Using **??**, we know that

$$\forall k \in \{1, \dots, M\} : Tu_k = (T\big|_U)(u_k) \in \mathsf{span}(u_1, \dots, u_k). \tag{5.15}$$

Extend $u_1, \dots, u_M$ to a basis $u_1, \dots, u_M, v_1, \dots, v_N$ of $V$. If $k \in \{1, \dots, N\}$, then

$$Tv_k = Tv_k - \lambda_m v_k + \lambda_m v_k = (T - \lambda_m I)v_k + \lambda_m v_k. \tag{5.16}$$

The definition (5C) of $U$ shows that $(T - \lambda_m I)v_k \in U = \mathsf{span}(u_1, \dots, u_M)$. Thus, equation (5C) above shows that

$$\forall k \in \{1, \dots, N\} : Tv_k \in \mathsf{span}(u_1, \dots, u_M, v_1, \dots, v_k). \tag{5.17}$$

Using Theorem **??** together with (5C) and (5C) implies that $T \in \mathcal{L}(V)$ has an upper-triangular matrix with respect to the basis $u_1, \dots, u_M, v_1, \dots, v_N$ of $V$. □

**Thm 5.47** (*sufficient condition about the field for every operator on $V$ to have an upper-triangular matrix*)**:** Let $\mathbb{F} = \mathbb{C}$. Let $T \in \mathcal{L}(V)$, $\dim V \neq \infty$. Then $T$ has an upper-triangular matrix with respect to some basis of $V$.

***Proof:*** This follows from the previous theorem 5.44 and the second version of the fundamental theorem of algebra 4.13. Because $\mathbb{F} = \mathbb{C}$, the minimal polynomial of $T$ can be expressed as $p \in \mathcal{P}(\mathbb{C})$, where

$$p(z) := (z - \lambda_1)\cdots(z - \lambda_m) \quad \forall z \in \mathbb{C}.$$

Hence, $T$ has an upper-triangular matrix $\mathcal{M}(T)$. □

## Exercises about Upper-Triangular Matrices 5C

**10** Suppose $T \in \mathcal{L}(V)$ and $v_1, \dots, v_n$ is a basis of $V$. Show that the following are equivalent.

**(a)** The matrix $\mathcal{M}(T)$ of $T$ with respect to $v_1, \dots, v_n$ is lower triangular.

**(b)** $\operatorname{span}(v_k, \dots, v_n)$ is invariant under $T$ for each $k = 1, \dots, n$.

**(c)** $Tv_k \in \operatorname{span}(v_k, \dots, v_n)$ for each $k = 1, \dots, n$.

***Proof:*** Like in 5.39, we will prove that (a) $\implies$ (b) $\implies$ (c) $\implies$ (a).

***Step 1:*** Suppose (a) holds and fix $k \in \{1, \dots, n\}$. Consider $j \in \{k, \dots, n\}$ (i.e., $k \le j$). The matrix $\mathcal{M}(T)$ of $T$ looks as follows.

$$\mathcal{M}(T) = \begin{array}{c} \\ v_1 \\ \vdots \\ v_j \\ \vdots \\ v_n \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & \cdots & v_j & \cdots & v_n \end{array} \\ \begin{pmatrix} A_{1,1} & & & & \\ \vdots & \ddots & & & \\ A_{j,1} & \cdots & A_{j,j} & & \\ \vdots & \ddots & \vdots & \ddots & \\ A_{n,1} & \cdots & A_{n,j} & \cdots & A_{n,n} \end{pmatrix} \end{array}.$$

We have

$$Tv_j = A_{j,j}v_j + A_{j+1,j}v_{j+1} + \cdots + A_{n,j}v_n = \sum_{\ell=1}^{n} A_{\ell,j}v_\ell.$$

Therefore, $Tv_j \in \operatorname{span}(v_j, \dots, v_n)$. Because

$$\operatorname{span}(v_j, \dots, v_n) \subseteq \operatorname{span}(v_k, \dots, v_n),$$

it's immediate that $Tv_j \in \operatorname{span}(v_k, \dots, v_n)$. Since $v_j$ could be any basis vector in the list $v_k, \dots, v_n$, $\operatorname{span}(v_k, \dots, v_n)$ is invariant under $T$, completing the proof that (a) implies (b).

***Step 2:*** Now, suppose (b) holds. Hence, we assume $\operatorname{span}(v_k, \dots, v_n)$ is invariant under $T$ for each $k = 1, \dots, n$. In particular,

$$Tv_j \in \operatorname{span}(v_j, \dots, v_n) \quad \forall j \in \{1, \dots, n\}.$$

Thus, (b) implies (c)

***Step 3:*** Finally, suppose (c) holds. Therefore, $Tv_k \in \operatorname{span}(v_k, \dots, v_n)$ for $k = 1, \dots, n$. This implies that for a fixed $k$, there exist scalars $a_k, \dots, a_n \in \mathbb{F}$ such that

$$Tv_k = a_kv_k + \cdots + a_nv_n = 0v_1 + 0v_2 + \cdots + 0v_{k-1} + a_kv_k + \cdots + a_nv_n.$$

This implies that all the entries above the diagonal of $\mathcal{M}(T)$ are 0, because the list $v_1, \dots, v_n$ is linearly independent and we cannot choose another representation for $Tv_k$ in respect to that list. Therefore, the scalars $a_k, \dots, a_n$ coincide with the entries of $\mathcal{M}(T)$. Thus, $\mathcal{M}(T)$ is a lower-triangular matrix, completing the proof that (c) implies (a). $\square$

# Diagonalizable Operators (Section 5D)

## Diagonal Matrices (Sub-Section 5D1)

**Def 5.48** (*diagonal matrix*):

A *"diagonal matrix"* is a square matrix that is 0 everywhere except possibly on the diagonal.

**Def 5.50** (*diagonalizable*):

An operator on $V$ is called *"diagonalizable"* if the operator has a diagonal matrix with respect to some basis of $V$.

**Def 5.52** (*eigenspace*, $E(\lambda, T)$):

Let $T \in \mathcal{L}(V)$ and $\lambda \in \mathbb{F}$. The *"eigenspace"* of $T$ corresponding to $\lambda$ is the subspace $E(\lambda, T)$ of $V$ defined by

$$E(\lambda, T) :\equiv \mathrm{null}(T - \lambda I) = \{v \in V \mid Tv = \lambda v\}.$$

**Thm 5.54** (*sum of eigenspaces is a direct sum*):

Suppose $T \in \mathcal{L}(V)$ and $\lambda_1, \ldots, \lambda_m$ are distinct eigenvalues of $T$. Then

$$E(\lambda_1, T) + \cdots + E(\lambda_m, T)$$

is a direct sum $\Big($i.e., $\sum_{i=1}^{m} E(\lambda_i, T) = \bigoplus_{i=1}^{m} E(\lambda_i, T)\Big)$.

Furthermore, if $V$ is finite-dimensional, then

$$\begin{aligned}\dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) &= \dim\big(E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)\big) \\ &\leq \dim V.\end{aligned}$$

***Proof:*** Let $\lambda_1, \ldots, \lambda_m$ be distinct eigenvalues. Suppose $v_1 + \cdots + v_m = 0$, where each $v_k \in E(\lambda_k, T)$ for $k = 1, \ldots, m$. This means $\forall k \in \{1, \ldots, m\} : v_k = 0$, because eigenvectors corresponding to distinct eigenvalues are linearly independent by 5.11. Thus, $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is a direct sum by 1.45. By 3.94, the dimensions of a direct sum add up, and by 2.37, the dimension of a subspace is always $\leq$ to the space itself (if it is finite-dimensional). Therefore,

$$\dim\left(\underbrace{E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)}_{\text{subpace of } V}\right) = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T) \leq \dim V.$$

□

## Conditions for Diagonalizability (Sub-Section 5D2)

**Thm 5.55** (*conditions equivalent to diagonalizability*):

Let $\lambda_1, \ldots, \lambda_m$ denote the distinct (i.e., $\lambda_j \neq \lambda_k$) eigenvalues of $T \in \mathcal{L}(V)$. Then the following are equivalent:

**(a)** $T$ is diagonalizable.

**(b)** $V$ has a basis consisting of eigenvectors of $T$.

**(c)** $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$.

**(d)** $\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T)$.

***Proof:*** We prove (a) $\iff$ (b), (b) $\implies$ (c), (c) $\implies$ (d), and then (d) $\implies$ (b). Let $\lambda_1, \ldots, \lambda_m$ be distinct eigenvalues of $T \in \mathcal{L}(V)$ for the whole proof.

***Step 1:*** Suppose (a) holds. The matrix of $T$ looks as follows:

$$\mathcal{M}(T) = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_m \end{pmatrix}$$

with respect to a basis $v_1, \dots, v_m$. This is the case $\iff$ $Tv_k = \lambda_k v_k$ for $k = 1, \dots, m$. Thus, (a) and (b) are equivalent, i.e., (a) $\iff$ (b).

***Step 2:*** Now, suppose (b) holds; thus, $V$ has a basis consisting of eigenvectors of $T$. Hence, every vector in $V$ is a linear combination of such basis, which implies

$$V = E(\lambda_1, T) + \cdots + E(\lambda_m, T).$$

By 5.54, the distinctness of all $\lambda_j$'s implies that this is a direct sum $\left(\text{i.e., } V = \bigoplus_{j=1}^{m} E(\lambda_j, T)\right)$. This proves that (b) $\implies$ (c).

***Step 3:*** (c) $\implies$ (d) follows immediately from 3.94, which states that the dimensions of a direct sum add up $\left(\text{i.e., } \dim\left(\bigoplus_{j=1}^{m} E(\lambda_j, T)\right) = \sum_{j=1}^{m} \dim V_j\right)$.

***Step 4:*** Finally, suppose (d) holds; thus,

$$\dim V = \dim E(\lambda_1, T) + \cdots + \dim E(\lambda_m, T).$$

Choose a basis of each $E(\lambda_m, T)$; put all these together to form a list

$$v_1, \dots, v_n \tag{5.18}$$

of eigenvectors of $T$, where $n = \dim V$. We want to show that (5D2) forms a basis of $V$. To this end, suppose for some $a_1, \dots, a_n \in \mathbb{F}$:

$$a_1 v_1 + \cdots + a_n v_n = 0. \tag{5.19}$$

Now, for every $k = 1, \dots, m$, let

$$u_k := \sum_{\substack{v_j \in E(\lambda_k, T), \text{ such that} \\ v_j \text{ is a basis vector of } E(\lambda_k, T)}} (a_j v_j)$$

denote the $k$-th sum of the terms $a_j v_j$, where the $v_j$'s belong to the basis of $E(\lambda_k, T)$. This means that $u_k \in E(\lambda_k, T)$ is an eigenvector with eigenvalue $\lambda_k$. Therefore, for every $k = 1, \dots, m$, we have

$$u_k \in E(\lambda_k, T)$$

and

$$u_1 + \cdots + u_m = 0,$$

as this is the same sum we had earlier (5D2). By 5.11, every list of eigenvectors of $T$ corresponding to distinct eigenvalues of $T$ is linearly independent. This forces

$$u_k = 0 \quad \forall k \in \{1, \dots, m\}.$$

This implies that all the $a$'s must be 0, because the $v$'s are bases of the $E$'s (eigenspaces), and their corresponding sums (the $u$'s) all equal 0. Therefore, $v_1, \dots, v_n$ is a linearly independent list, and hence, a basis of $V$, by 2.38 ($n = \dim V$). Now, we have shown that (d) $\implies$ (b), completing the proof. □

**Thm 5.58** (*enough eigenvalues implies diagonalizability*)**:**

If $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues, then $T$ is diagonalizable.

***Proof:*** Let $\lambda_1, \dots, \lambda_{\dim V}$ be the eigenvalues of the eigenvectors $v_1, \dots, v_{\dim V}$, which are linearly independent by 5.11. So, we have a basis consisting of $\dim V$ eigenvectors. Hence, $T$ is diagonalizable by 5.55. □

**Thm 5.62** (*necessary and sufficient condition for diagonalizability*):
Let $T \in \mathcal{L}(V)$, $\dim V \neq \infty$. Then $T$ is diagonalizable $\iff$ the minimal polynomial of $T$ equals

$$(z-\lambda_1)\cdots(z-\lambda_m),$$

where $\lambda_1,\ldots,\lambda_m \in \mathbb{F}$ and $\lambda_i \neq \lambda_j$ for $i,j \in \{1,\ldots,m\}$.

***Proof:*** For this proof, let $T \in \mathcal{L}(V)$ be finite-dimensional ($\dim V \neq \infty$).

$\Rightarrow$ ***direction:*** Suppose $T$ is diagonalizable. Thus, there exists a basis $v_1,\ldots,v_n$ of $V$ consisting of eigenvectors of $T$. Let $\lambda_1,\ldots,\lambda_m$ be said distinct eigenvalues of $T$. Therefore, for every $v_j$, we have $(T-\lambda_k I)v_j = 0$. Thus,

$$(T-\lambda_1)\cdots(T-\lambda_m I)v_j = 0,$$

which implies the minimal polynomial equals $(z-\lambda_1)\cdots(z-\lambda_m)$. (Why?)

$\Leftarrow$ ***direction:*** Suppose the minimal polynomial of $T$ equals $(z-\lambda_1)\cdots(z-\lambda_m)$ for some $\lambda_1,\ldots,\lambda_m \in \mathbb{F}$ with $\lambda_i \neq \lambda_j$. Write

$$(T-\lambda_1 I)\cdots(T-\lambda_{m-1}I)(T-\lambda_m I) = 0. \tag{5.20}$$

Suppose $m = 1$. Then $T-\lambda_1 I = 0 \iff T = \lambda_1 I$, so $T$ is diagonalizable. This is our base case for induction over $m \in \mathbb{N}$. Now, suppose $m > 1$ and that the desired result holds for all values smaller than $m$. Like in the proof of Theorem 5.44, we observe that $\mathrm{range}(T-\lambda_m I)$ is invariant under any polynomial of $T$ (using 5.18 with $p(z) := z-\lambda_m$). Now, define

$$U := \mathrm{range}(T-\lambda_m I).$$

This means that $T\big|_U$ is an operator on $U$. If $u \in \mathrm{range}(T-\lambda_m I)$, then $u = (T-\lambda_m I)v$ for some $v \in V$. Together with (5D2), we have that

$$\begin{aligned}(T-\lambda_1 I)\cdots(T-\lambda_{m-1}I)u &= (T-\lambda_1 I)\cdots(T-\lambda_m I)v \\ &= 0.\end{aligned} \tag{5.21}$$

Since $u \in U$ was arbitrary, $q(z) := (z-\lambda_1)\cdots(z-\lambda_{m-1})$ is a polynomial such that $q(T)(u) = 0$ for every $u \in U$. By 5.29, $q(z)$ is a polynomial multiple of the minimal polynomial of $T\big|_U$. By our induction hypothesis, there is a basis of $U$ consisting of eigenvectors of $T\big|_U$ and therefore, as well as eigenvectors of $T$. Suppose

$$x \in U \cap \mathrm{null}(T-\lambda_m I) = U \cap E(\lambda_m, T).$$

Hence, $Tx = \lambda_m x$, because $Tx - \lambda_m I x = (T-\lambda_m I)x = 0$. Since $x \in U$, equation (5D2) implies that

$$\begin{aligned}0 &= (T-\lambda_1 I)\cdots(T-\lambda_{m-1}I)x \\ &= (\lambda_m I-\lambda_1 I)\cdots(\lambda_m I-\lambda_{m-1}I)x \\ &= (\lambda_m-\lambda_1)\cdots(\lambda_m-\lambda_{m-1})x.\end{aligned}$$

Because $\forall i \neq j : \lambda_j \neq \lambda_k$, this can only happen if $x = 0$. Therefore,

$$U \cap \mathrm{null}(T-\lambda_m I) = \{0\}.$$

Thus, $U + \mathrm{null}(T-\lambda_m I)$ is a direct sum by 1.46. That is,

$$U + \mathrm{null}(T-\lambda_m I) = U \oplus \mathrm{null}(T-\lambda_m I).$$

Since $(T-\lambda_m I) \in \mathcal{L}(V)$, we can use the Rank-Nullity Theorem 3.21 to conclude that

$$\dim V = \dim U + \dim \mathrm{null}(T-\lambda_m I).$$

3.94 states that a sum is a direct sum $\iff$ the dimensions add up to the dimension of the sum. This means that we can rewrite the equation above to:

$$\dim V = \dim(U \oplus \mathrm{null}(T-\lambda_m I)).$$

According to 2.39, together with the fact that $U \oplus \mathrm{null}(T-\lambda_m I) \subseteq V$, this implies

$$V = U \oplus \mathrm{null}(T-\lambda_m I) = U \oplus E(\lambda_m, T).$$

Every vector in $\mathrm{null}(T-\lambda_m I)$ is an eigenvector of $T$ with eigenvalue $\lambda_m$. Earlier in this proof, we saw that there is a basis of $U = \mathrm{range}(T-\lambda_m I)$ consisting of eigenvectors of $T$ (using our induction hypothesis on this smaller subspace of $V$). Adjoining to that basis a basis of $E(\lambda_m, T) = \mathrm{null}(T-\lambda_m I)$ gives a basis of $V$ consisting of eigenvectors of $T$. The matrix $\mathcal{M}(T)$ of $T$ with respect to this basis is a diagonal matrix, as desired, using 5.55 (according to that theorem, $V$ has a basis consisting of eigenvectors of $T \in \mathcal{L}(V) \iff$ it is diagonalizable). As a side note, this theorem states that $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, which correlates with $V = U \oplus \mathrm{null}(T-\lambda_m I)$. □

**Thm 5.65** (*restriction of diagonalizable operator to invariant subspace*):
Let $T \in \mathcal{L}(V)$. Suppose $T$ is diagonalizable and $U$ is a subspace of $V$ that is invariant under $T$. Then $T\big|_U$ is a diagonalizable operator on $U$.

***Proof:*** Diagonalizability of $T \iff_{(5.62)}$ the minimal polynomial of $T$ equals

$$(z-\lambda_1)\cdots(z-\lambda_m) \ \text{ for } \ \lambda_j \neq \lambda_k.$$

By 5.31, the minimal polynomial of $T$ is a polynomial multiple of the minimal polynomial of $T\big|_U$. Hence, the minimal polynomial of $T\big|_U$ has the form required by 5.62, which shows that $T\big|_U$ is diagonalizable. It consists of any factor of the form $(z-\lambda_k)$. □

## Exercises about Diagonalizable Operators 5D

(2) Suppose $T \in \mathcal{L}(V)$ has a diagonal matrix $A = \mathcal{M}T$ with respect to some basis of $V$. Prove that if $\lambda \in \mathbb{F}$, then $\lambda$ appears on the diagonal of $A$ precisely $\dim E(\lambda, T)$ times.

***Proof:*** If $\lambda$ is not an eigenvalue of $T$, then

$$\dim E(\lambda, T) = \dim\{v \in V \mid Tv = \lambda v\} = \dim\{0\} = 0,$$

and therefore $\lambda$ does not appear among the diagonal entries of $A$.

Now assume $\lambda$ is an eigenvalue. Let $\mathcal{B} = \{v_1, \ldots, v_\ell\}$ be a basis of $V$ with respect to which $A = \mathcal{M}T$ is diagonal; write

$$A = \mathrm{diag}(\mu_1, \ldots, \mu_\ell),$$

so that for each $j$ we have $Tv_j = \mu_j v_j$. In particular, every $v_j$ is an eigenvector of $T$.

Define the subset of basis vectors corresponding to the eigenvalue $\lambda$ by

$$S = \{v_j \in \mathcal{B} \mid \mu_j = \lambda\}.$$

We first observe that $\mathrm{span}(S) \subseteq E(\lambda, T)$, since each vector in $S$ is by definition a $\lambda$-eigenvector and linear combinations of $\lambda$-eigenvectors remain in the $\lambda$-eigenspace.

For the reverse inclusion, let $w \in E(\lambda, T)$. Express $w$ in the basis $\mathcal{B}$:

$$w = \sum_{j=1}^{\ell} c_j v_j \qquad (c_j \in \mathbb{F}).$$

Applying $T$ to both sides and using $Tv_j = \mu_j v_j$ gives

$$\lambda \sum_{j=1}^{\ell} c_j v_j = Tw = \sum_{j=1}^{\ell} c_j \mu_j v_j.$$

By uniqueness of coordinates with respect to the basis $\mathcal{B}$, the coefficient of $v_j$ on the left must equal the coefficient of $v_j$ on the right for each $j$, hence

$$(\lambda - \mu_j)c_j = 0 \quad \text{for every } j = 1, \dots, \ell.$$

Consequently, $c_j = 0$ whenever $\mu_j \neq \lambda$, so every nonzero coefficient $c_j$ corresponds to some $v_j \in S$. Thus $w$ is a linear combination of vectors from $S$, meaning $w \in \operatorname{span}(S)$. This proves the reverse inclusion and therefore

$$\operatorname{span}(S) = E(\lambda, T).$$

Since $S$ is a subset of the basis $\mathcal{B}$, its vectors are linearly independent; hence the number of elements of $S$ equals the dimension of its span:

$$|S| = \dim \operatorname{span}(S) = \dim E(\lambda, T).$$

Finally, by definition $|S|$ is exactly the number of diagonal entries of $A$ equal to $\lambda$. We conclude that the scalar $\lambda$ appears on the diagonal of $A$ precisely $\dim E(\lambda, T)$ times. $\square$

## Commuting Operators (Section 5E)

**Def 5.66:**

Two operators or matrices $S$ and $T$ *"commute"* if $ST = TS$.

# Inner Product Spaces (Chapter 6)

## Inner Products and Norms (Section 6A)

### Inner Products (Sub-Section 6A1)

We define the norm of $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ by

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

The norm is not linear. The dot product is.

**Def 6.1** (*dot product*):

The *"dot product"* of $x, y \in \mathbb{R}^n$ is defined by

$$x \cdot y :\equiv x_1 y_1 + \cdots + x_n y_n.$$

The dot product on $\mathbb{R}^n$ has the following properties:

- $x \cdot x = \|x\|^2 \quad \forall x \in \mathbb{R}^n$;
- $x \cdot x = 0 \iff x = 0$;
- For $y \in \mathbb{R}^n$, the map from $\mathbb{R}^n$ to $\mathbb{R}$ that sends $x \in \mathbb{R}^n$ to $x \cdot y$ is linear.
  $T : \mathbb{R}^n \to \mathbb{R}, T(x) = x \cdot y$. Then $T$ is linear for a fixed $y \in \mathbb{R}^n$.
- $x \cdot y = y \cdot x$.

For $z \in \mathbb{C}$, we define the norm of $z$ by

$$\|z\| :\equiv \sqrt{|z_1|^2 + \cdots + |z_n|^2} = \sqrt{z_1 \overline{z_1} + \cdots + z_n \overline{z_n}}$$
$$\implies \|z\|^2 = z_1 \overline{z_1} + \cdots + z_n \overline{z_n}.$$

**Def 6.2** (*inner product*):

The *"inner product"* of $V$ is a function defined as follows:

- **Positivity:** $\langle v, v \rangle \geq 0 \quad \forall v \in V$;
- **Definiteness:** $\langle v, v \rangle = 0 \iff v = 0$;
- **Additivity in first slot:** $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle \quad \forall u, v, w \in V$;
- **Homogeneity in first slot:** $\langle \lambda u, v \rangle = \lambda \langle u, v \rangle \quad \forall \lambda \in \mathbb{F}$ and $\forall u, v \in V$;
- **Conjugate symmetry:** $\langle u, v \rangle = \overline{\langle v, u \rangle} \quad \forall u, v \in V$.

**Example 6.3:**

(b) $\langle w, z \rangle :\equiv c_1 w_1 \overline{z_1} + \cdots + c_n w_n \overline{z_n}$

**Def 6.4** (*inner product space*):

An *"inner product space"* is a vector space $V$ along with an inner product on $V$.

**Thm 6.6** (*basic properties of an inner product*):

$\forall u, v, w \in V$ and $\forall \lambda \in \mathbb{F}$:

**(a)** $x \in V \mapsto \langle u, x \rangle \in \mathbb{F}$ is a linear map from $V$ to $\mathbb{F}$.

**(b)** $\langle 0, v \rangle = 0$;

**(c)** $\langle v, 0 \rangle = 0$;

**(d)** $\langle u, v + w \rangle = \langle u, v \rangle + \langle u, w \rangle$;

**(e)** $\langle u, \lambda v \rangle = \overline{\lambda} \langle u, v \rangle$.

## Norms (Sub-Section 6A2)

**Def 6.7** (*norm*):

$$\|v\| :\equiv \sqrt{\langle v, v\rangle}, \quad v \in V.$$

**Example 6.8**:

For $f \in C[-1,1]$ and with inner product given as in 6.3(c), we have $\|f\| = \sqrt{\int_{-1}^{1} f^2}$.

**Thm 6.9** (*basic properties of the norm*):

Suppose $v \in V$:

**(a)** $\|v\| = 0 \iff v = 0$;

**(b)** $\|\lambda v\| = |\lambda|\,\|v\| \quad \forall \lambda \in \mathbb{F}$.

***Proof:*** Let $\lambda \in \mathbb{F}$ and $v \in V$. Then:

**(a)** $\langle v, v\rangle = 0 \iff v = 0$;

**(b)** $\|\lambda v\|^2 = \langle \lambda v, \lambda v\rangle = \lambda\overline{\lambda}\langle v, v\rangle = |\lambda|^2\|v\|^2$.

□

**Def 6.10** (*orthogonal*):

$u, v \in V$ are called *"orthogonal"* or *"perpendicular"* if

$\langle u, v\rangle = 0$. One also writes $u \perp v$.

**Thm 6.11** (*orthogonality and* 0):

**(a)** 0 is orthogonal to every vector in $V$.

**(b)** 0 is the only vector that is orthogonal to itself.

**Thm 6.12** (*Pythagorean theorem*):

If $u, v \in V$ are orthogonal, then

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2.$$

In this version of the theorem, $u$ and $v$ can equal 0.

***Proof:*** $\|u+v\|^2 = \langle u+v, u+v\rangle = \langle u, u\rangle + \underbrace{\langle u, v\rangle}_{=0} + \underbrace{\langle v, u\rangle}_{=0} + \langle v, v\rangle = \|u\|^2 + \|v\|^2$. □

**Thm 6.13** (*an orthogonal decomposition*):

Suppose $u, v \in V, v \neq 0$. Set $c :\equiv \frac{\langle u, v\rangle}{\|v\|^2}$ and $w :\equiv u - \frac{\langle u, v\rangle}{\|v\|^2}v$. Then

$$u = cv + w = \frac{\langle u, v\rangle}{\|v\|^2}v + \left(u - \frac{\langle u, v\rangle}{\|v\|^2}v\right) \quad \text{and} \quad \langle w, v\rangle = 0.$$

**Thm 6.14** (*Cauchy-Schwarz inequality*):

Suppose $u, v \in V$. Then

$$|\langle u, v\rangle| \leq \|u\|\,\|v\|.$$

Equality holds $\iff$ one of $u, v$ is a scalar multiple of the other ($u = \lambda v$).

*Proof.* If $v = 0$, both sides are 0. Thus, $v \neq 0$.

Let $u :\equiv cv + w = \frac{\langle u, v\rangle}{\|v\|^2}v + w$ such that $v \perp w$ (i.e., $\langle w, v\rangle = 0$), like above (6.13).

By Pythagoras, we have

$$\begin{aligned}
\|u\|^2 = \left\|\frac{\langle u,v\rangle}{\|v\|^2}v\right\|^2 + \|w\|^2 &= \sqrt{\left\langle \frac{\langle u,v\rangle}{\|v\|^2}v, \frac{\langle u,v\rangle}{\|v\|^2}v\right\rangle} + \|w\|^2 \\
&= \sqrt{\frac{1}{\left(\|v\|^2\right)^2}\left\langle \langle u,v\rangle v, \langle u,v\rangle v\right\rangle} + \|w\|^2 \\
&= \frac{1}{\|v\|^4}\langle u,v\rangle\overline{\langle u,v\rangle} + \|w\|^2 \\
&= |\langle u,v\rangle|\,\|v\|^2\frac{1}{\|v\|^4} + \|w\|^2 \\
&= \frac{|\langle u,v\rangle|}{\|v\|^2} + \|w\|^2 \\
&\geq \frac{|\langle u,v\rangle|}{\|v\|^2}.
\end{aligned}$$

Multiplying both sides by $\|v\|^2$ yields the Cauchy-Schwarz inequality. Equality is only given if $w = 0$, so if $u = \lambda v$. □

**Thm 6.17** (*triangle inequality*):

$u, v \in V$ :

$$\|u+v\| \leq \|u\| + \|v\|.$$

This inequality is an equality $\iff$ $u = rv$ for some $r \in \mathbb{R}\setminus\{0\}$.

***Proof:*** Let $u, v \in V$, where both might equal 0. Thus,

$$\begin{aligned}
\|u+v\|^2 &= \langle u+v, u+v\rangle \\
&= \langle u, u+v\rangle + \langle v, u+v\rangle \\
&= \langle u,u\rangle + \langle u,v\rangle + \langle v,u\rangle + \langle v,v\rangle \\
&= \langle u,u\rangle + \langle v,v\rangle + \langle u,v\rangle + \overline{\langle u,v\rangle} && \text{conjugate symmetry} \\
&= \|u\|^2 + \|v\|^2 + 2\,\mathrm{Re}\langle u,v\rangle && z+\overline{z} = 2\,\mathrm{Re}\,z \\
&\leq \|u\|^2 + \|v\|^2 + 2|\langle u,v\rangle| && z = \mathrm{Re}\,z + (\mathrm{Im}\,z)i \\
&\leq \|u\|^2 + \|v\|^2 + 2\,\|u\|\,\|v\| && \text{Cauchy-Schwarz} \\
&= (\|u\| + \|v\|)^2.
\end{aligned}$$

We have equality in the triangle inequality $\iff$ $\langle u,v\rangle = \|u\|\cdot\|v\|$ $\iff$ $u = \lambda v$. □

# Orthonormal Bases (Section 6B)

## Orthonormal Lists and the Gram-Schmidt Procedure (Sub-Section 6B1)

**Def 6.22** (*orthonormal*):

A list of vectors is called *"orthonormal"* if each vector in the list has norm 1 and is orthogonal to all the other vectors in the list. Since the square root of 1 is 1, we can use the following definition as well:

A list $e_1, \ldots, e_m$ of vectors in $V$ is *"orthonormal"* if, for all $j, k \in \{1, \ldots, m\}$,

$$\left\langle e_j, e_k\right\rangle = \begin{cases} 1, & \text{if } j = k \\ 0, & \text{if } j \neq k. \end{cases}$$

Recall that $\|v\| \equiv \sqrt{\langle v,v\rangle}$.

**Thm 6.24** (*norm of an orthonormal linear combination*)**:**

If $e_1, \dots, e_m$ is an orthonormal list of vectors in $V$, then

$$\left\|a_1 e_1 + \cdots + a_m e_m\right\|^2 = |a_1|^2 + \cdots + |a_m|^2 \quad \forall a_1, \dots, a_m \in \mathbb{F}.$$

***Proof:*** Because $\forall k : \left\|e_k\right\| = 1$, this follows from repeated applications of the Pythagorean theorem 6.12:

$$\begin{aligned}
\left\|a_1 e_1 + \cdots + a_m e_m\right\|^2 &= \left\|(a_1 e_1 + \cdots + a_{m-1} e_{m-1}) + a_m e_m\right\|^2 \\
&= \left\|(a_1 e_1 + \cdots + a_{m-1} e_{m-1})\right\|^2 + \left\|a_m e_m\right\|^2 \\
&= \left\|(a_1 e_1 + \cdots + a_{m-2} e_{m-2}) + a_{m-1} e_{m-1}\right\|^2 + |a_m|^2 \\
&= \left\|(a_1 e_1 + \cdots + a_{m-2} e_{m-2})\right\|^2 + \left\|a_{m-1} e_{m-1}\right\|^2 + |a_m|^2 \\
&= \left\|(a_1 e_1 + \cdots + a_{m-3} e_{m-3}) + a_{m-2} e_{m-2}\right\|^2 + |a_{m-1}|^2 + |a_m|^2 \\
&= \cdots \\
&= |a_1|^2 + |a_2|^2 + \cdots + |a_{m-1}|^2 + |a_m|^2.
\end{aligned}$$

□

**Thm 6.25** (*linear independence*)**:**

Every orthonormal list of vectors is linearly independent.

***Proof:*** Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in $V$ and $a_1, \dots, a_m \in \mathbb{F}$ are such that

$$a_1 e_1 + \cdots + a_m e_m = 0.$$

This implies that $|a_1|^2 + \cdots + |a_m|^2 = 0$ by 6.24, which means that all the $a_k$'s are 0. This is all that is needed to show. □

**Thm 6.26** (*Bessel's inequality*)**:**

Suppose $e_1, \dots, e_m$ is an orthonormal list of vectors in $V$. If $v \in V$, then

$$|\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2 \le \|v\|^2.$$

***Proof:*** Suppose $v \in V$. Then

$$v = \underbrace{\langle v, e_1\rangle e_1 + \cdots + \langle v, e_m\rangle e_m}_{u} + \underbrace{v - \langle v, e_1\rangle e_1 + \cdots + \langle v, e_m\rangle e_m}_{w}.$$

Let $u$ and $w$ be defined as in the equation above. If $k \in \{1, \dots, m\}$, then

$$\begin{aligned}
\langle w, e_k\rangle &= \left\langle v - \underbrace{\langle v, e_1\rangle}_{\in \mathbb{F}} e_1 + \cdots + \underbrace{\langle v, e_m\rangle}_{\in \mathbb{F}} e_m, e_k\right\rangle \\
&= \langle v, e_k\rangle - \langle v, e_1\rangle\langle e_1, e_k\rangle - \langle v, e_2\rangle\langle e_2, e_k\rangle - \cdots - \langle v, e_m\rangle\langle e_m, e_k\rangle \\
&= \langle v, e_k\rangle - \langle v, e_k\rangle\langle e_k, e_k\rangle \\
&= 0. \text{ Hence,} \\
\langle w, u\rangle &= \left\langle w, \underbrace{\langle v, e_1\rangle}_{\in \mathbb{F}} e_1 + \cdots + \underbrace{\langle v, e_m\rangle}_{\in \mathbb{F}} e_m\right\rangle \\
&= \left\langle w, \langle v, e_1\rangle e_1\right\rangle + \cdots + \left\langle w, \langle v, e_m\rangle e_m\right\rangle \\
&= \overline{\langle v, e_1\rangle}\langle w, e_1\rangle + \cdots + \overline{\langle v, e_m\rangle}\langle w, e_m\rangle \\
&= 0.
\end{aligned}$$

Since this means that $u$ and $w$ are orthogonal, we can use the well-known Pythagorean theorem

6.12 to conclude that

$$\begin{aligned}\|v\|^2 &= \|u\|^2 + \|w\|^2 \\ &\geq \|u\|^2 \\ &= |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_m\rangle|^2,\end{aligned}$$

where the last line comes from 6.24. If you look at the definition of $u = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_m\rangle e_m$, the inner products are the coefficients in regards to the orthonormal basis $e_1, \ldots, e_m$. □

**Def 6.27** (*orthonormal basis*):

An *"orthonormal"* basis of $V$ is an orthonormal list of vectors in $V$ that is also a basis in $V$.

**Def 6.28** (*orthonormal lists of the right length are orthonormal bases*):

Suppose $\dim V \neq \infty$.

**Thm 6.30** (*writing a vector as a linear combination of an orthonormal basis*):

Suppose $e_1, \ldots, e_n$ is an orthonormal basis of $V$ and $u, v \in V$. Then:

**(a)** $v = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n$;

**(b)** $\|v\|^2 = |\langle v, e_1\rangle|^2 + \cdots + |\langle v, e_n\rangle|^2$;

**(c)** $\langle u, v\rangle = \langle u, e_1\rangle \overline{\langle v, e_1\rangle} + \cdots + \langle u, e_n\rangle \overline{\langle v, e_n\rangle}$.

***Proof:*** Let $v = a_1 e_1 + \cdots + a_n e_n$. Then, $\forall k \in \{1, \ldots, n\}$, we have

$$\begin{aligned}\langle v, e_k\rangle &= \langle a_1 e_1 + \cdots + a_n e_n, e_k\rangle \\ &= \langle a_1 e_1, e_k\rangle + \cdots + \langle a_n e_n, e_k\rangle \\ &= \langle a_k e_k, e_k\rangle \\ &= a_k \langle e_k, e_k\rangle \\ &= a_k.\end{aligned}$$

Thus, **(a)** holds.

Now, **(b)** follows immediately from **(a)** and 6.24.

Using conjugate linearity,

$$\begin{aligned}\langle u, v\rangle &= \left\langle u, \langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n\right\rangle \\ &= \left\langle u, \langle v, e_1\rangle e_1\right\rangle + \cdots + \left\langle u, \langle v, e_n\rangle e_n\right\rangle \\ &= \overline{\langle v, e_1\rangle}\langle u, e_1\rangle + \cdots + \overline{\langle v, e_n\rangle}\langle u, e_n\rangle \\ &= \langle u, e_1\rangle \overline{\langle v, e_1\rangle} + \cdots + \langle u, e_n\rangle \overline{\langle v, e_n\rangle}.\end{aligned}$$

Therefore, **(c)** holds. □

## Linear Functionals on Inner Product Spaces (Sub-Section 6B2)

## Orthogonal Complements and Minimization Problems (Section 6C)

### Orthogonal Complements (Sub-Section 6C1)

**Def 6.46** (*orthogonal complement, $U^\perp$*):

If $U \subset V$, then the *"orthogonal complement"* of $U$, denoted by $U^\perp$, is defined as follows:

$U^\perp :\equiv \{v \in V \mid \langle u, v\rangle = 0 \quad \forall u \in U\}$.

**Thm 6.48** (*properties of the orthogonal complement*):

**(a)** If $U \subseteq V$, then $U^\perp$ is a subspace of $V$.

**(b)** $\{0\}^\perp = V$.

**(c)** $V^\perp = \{0\}$.

**(d)** If $U$ is a subset of $V$, then $U \cap U^\perp \subseteq \{0\}$.

**(e)** If $G$ and $H$ are subsets of $V$ and $G \subseteq H$, then

$H^\perp \subseteq G^\perp$. This is not a typo.

***Proof:***

**(a)** We have to check for additive identity, closedness under addition, and closedness under scalar multiplication:

- $\langle u, 0\rangle = 0 \quad \forall u \in U \implies 0 \in U^\perp$.
- Let $v, w \in U^\perp, u \in U$. $\langle u, v + w\rangle = \langle u, v\rangle + \langle v, w\rangle = 0 \implies v + w \in U^\perp$.
- Let $\lambda \in \mathbb{F}, w \in U^\perp, u \in U : \langle u, \lambda w\rangle = \overline{\lambda}\langle u, w\rangle = \overline{\lambda}0 = 0 \implies \lambda w \in U^\perp$.

Therefore, $U^\perp$ is a subspace of $V$.

**(b)** Let $v \in V \implies \langle 0, v\rangle = 0 \implies v \in \{0\}^\perp$. Since $v$ was arbitrary, $\implies \{0\}^\perp = V$.

**(c)** Let $v \in V^\perp \implies \langle v, v\rangle = 0 \implies v = 0 \implies V^\perp = \{0\}$.

**(d)** Suppose $U \subseteq V$ and $u \in U \cap U^\perp \implies \langle u, u\rangle = 0 \implies u = 0 \implies U \cap U^\perp \subseteq \{0\}$.

**(e)** Suppose $G \subseteq V$ and $H \subseteq V$ with $G \subseteq H$. [As usual, we misuse $\subseteq$ to denote subspaces.]

$$\text{Let } v \in H^\perp. \implies \langle u, v\rangle = 0 \quad \forall u \in H.$$
$$\implies \langle u, v\rangle = 0 \quad \forall u \in G.$$
$$\implies v \in G^\perp. \text{ Thus, } H^\perp \subseteq G^\perp.$$

□

**Thm 6.49** (*direct sum of a subspace and its orthogonal complement*):

Suppose $U$ is a finite dimensional subspace of $V$. Then

$$V = U \oplus U^\perp \quad (v \in V \text{ can be uniquely written as } v = u + w, \text{ where } u \in U, w \in U^\perp).$$

***Proof:*** Let $v \in V$ and $e_1, \dots, e_m$ be an orthonormal basis of $U$. Like in 6.26, we have that

$$v = \underbrace{\left\langle v, e_1\right\rangle e_1 + \cdots + \left\langle v, e_m\right\rangle e_m}_{u} + \underbrace{v - \left\langle v, e_1\right\rangle e_1 + \cdots + \left\langle v, e_m\right\rangle e_m}_{w},$$

and we let $u$ and $w$ be defined as above.

$$\forall k \in \{1, \dots, m\} : e_k \in U \implies u \in U, \quad \text{because it is a linear combination of } e_1, \dots, e_m.$$

Again, like in 6.26, $\forall k \in \{1, \dots, m\}$:

$$\begin{aligned}\left\langle w, e_k\right\rangle &= \left\langle v - \underbrace{\left\langle v, e_1\right\rangle}_{\in \mathbb{F}} e_1 + \cdots + \underbrace{\left\langle v, e_m\right\rangle}_{\in \mathbb{F}} e_m, e_k\right\rangle \\ &= \left\langle v, e_k\right\rangle - \left\langle v, e_1\right\rangle\left\langle e_1, e_k\right\rangle - \left\langle v, e_2\right\rangle\left\langle e_2, e_k\right\rangle - \cdots - \left\langle v, e_m\right\rangle\left\langle e_m, e_k\right\rangle \\ &= \left\langle v, e_k\right\rangle - \left\langle v, e_k\right\rangle\left\langle e_k, e_k\right\rangle \\ &= 0.\end{aligned}$$

$\implies w \perp x$, whenever $x \in \operatorname{span}(e_1, \dots, e_m)$. This is very easy to verify using linearity of the inner product in the second slot. Since we had $\forall k \in \{1, \dots, m\} : e_k \in U$,

$$\implies w \in U^\perp.$$

$$\implies v = u + w, \text{ where } u \in U \text{ and } w \in U^\perp.$$

□

**Thm 6.51** (*dimension of orthogonal complement*):

Suppose $V$ is finite-dimensional and $U$ is a subspace of $V$. Then

$$\dim U^\perp = \dim V - \dim U.$$

***Proof:*** Because $V$ is a subspace of itself, this follows immediately from 6.49 and 3.94. □

**Thm 6.52** (*orthogonal complement of the orthogonal complement*):

Suppose $U$ is a finite-dimensional subspace of $V$. Then

$$U = (U^\perp)^\perp.$$

***Proof:*** We prove this theorem by showing that $U \subseteq (U^\perp)^\perp$ and $(U^\perp)^\perp \subseteq U$.

***Step 1:*** Let $u \in U \implies \langle u, w\rangle = 0 \ \forall w \in U^\perp$. Therefore, for the same $w$'s using conjugate symmetry:

$$\langle u, w\rangle = \overline{\langle w, u\rangle} = 0 \implies \overline{\overline{\langle w, u\rangle}} = \overline{0} \implies \langle w, u\rangle = 0.$$

Because $u$ is orthogonal to every vector $w$ in $U^\perp$, we have $u \in (U^\perp)^\perp$. Hence, $U \subseteq (U^\perp)^\perp$.

***Step 2:*** Let $v \in (U^\perp)^\perp \subseteq V$. By 6.49, we can write $v$ as a sum like this:

$$v = u + w, \quad \text{where} \quad u \in U \quad \text{and} \quad w \in U^\perp.$$

Hence, $v - u = w \in U^\perp$. Because $v \in (U^\perp)^\perp$ and $u \in U \subseteq (U^\perp)^\perp$ from step 1, we have

$$v - u = w \in U^\perp \cap (U^\perp)^\perp \subseteq \{0\}.$$

$\implies v = u \implies v \in U \implies (U^\perp)^\perp \subseteq U$, because $v$ was arbitrary. □

**Thm 6.54**:

Suppose $U$ is a finite-dimensional subspace of $V$. Then

$$U^\perp = \{0\} \iff U = V.$$

***Proof:*** *"$\Rightarrow$-direction:"* $U^\perp = \{0\}$. Then by first using 6.52 and then using 6.48(c), $U = (U^\perp)^\perp = \{0\}^\perp = V$.

*"$\Leftarrow$-direction:"* Only using 6.48(c): if $U = V$, then $U^\perp = V^\perp = \{0\}$. □

**Def 6.55** (*orthogonal projection, $P_u$*):

Suppose $U \subseteq V$, $\dim U \neq \infty$. The *"orthogonal projection"* of $V$ onto $U$ is the operator $P_U \in \mathcal{L}(V)$ defined as follows:

$\forall v \in V$, we can write $v = u + w$, where $u \in U$ and $w \in U^\perp$ (Because $V = U \oplus U^\perp$). Then let:

$$P_U v :\equiv u.$$

**Example 6.56** (*orthogonal projection onto one-dimensional subspace*):

Suppose $u \in V, u \neq 0$, and $U :\equiv \operatorname{span}(u)$ is a one-dimensional subspace of $V$. If $v \in V$, then

$$v = \underbrace{\frac{\langle v, u\rangle}{\|u\|^2}u}_{\in \operatorname{span}(u) = U} + \underbrace{\left(v - \frac{\langle v, u\rangle}{\|u\|^2}u\right)}_{\in U^\perp}.$$

Therefore,

$$P_U v = \frac{\langle v, u\rangle}{\|u\|^2}u \quad \forall v \in V.$$

Notice that $P_U u = u$ and $P_U v = 0 \iff v \in \{u\}^\perp$. Recall that $\langle u, u\rangle = \|u\|^2$.

**Thm 6.57** (*properties orthogonal projection $P_U$*):

Suppose $U \subseteq V$, $\dim U \neq \infty$. Then:

**(a)** $P_U \in \mathcal{L}(V)$;

**(b)** $P_U u = u \quad \forall u \in U$;

**(c)** $P_U w = 0 \quad \forall w \in U^\perp$;
**(d)** $\operatorname{range} P_U = U$;
**(e)** $\operatorname{null} P_U = U^\perp$;
**(f)** $v - P_U v \in U^\perp \quad \forall v \in V$;
**(g)** $P_U^2 = P_U$;
**(h)** $\left\|P_U v\right\| \leq \|v\| \quad \forall v \in V$;
**(i)** If $e_1, \ldots, e_m$ is an orthonormal basis of $U$ and $v \in V$, then
$$P_U v = \left\langle v, e_1 \right\rangle e_1 + \cdots + \left\langle v, e_m \right\rangle e_m.$$

## Minimization Problems (Sub-Section 6C2)

Given a subspace $U \subseteq V$ and $v \in V$, find a $u \in U$ such that $\|v - u\|$ is as small as possible.

**Thm 6.58** (*minimizing distance to a subspace*)**:**
Suppose $U$ is a finite-dimensional subspace of $V$, $v \in V$, and $u \in U$. Then
$$\left\|v - P_U v\right\| \leq \|v - u\|.$$
Furthermore, the inequality above is an equality $\iff u = P_U v$.

***Proof:*** For this proof, visualize that $v - P_U v$ is pointing from $P_U v$ to $v$ and $P_U v - u$ is pointing from $u$ to the projection of $v$, namely $P_U v$, which lies on the vector space $U$.

We also have the properties that $v - P_U v \in U^\perp$ and $P_U v - u \in U$. Therefore, we can use the Pythagorean theorem 6.12 after adding an always-positive term to the left-hand side of the inequality:
$$\begin{aligned}\left\|v - P_U v\right\|^2 &\leq \left\|v - P_U v\right\|^2 + \underbrace{\left\|P_U v - u\right\|^2}_{\geq 0} \\ &= \left\|(v - P_U v) + (P_U v - u)\right\|^2 \qquad \text{(the triangle inequality does not apply here)} \\ &= \|v - u\|^2.\end{aligned}$$
The inequality proved above is an equality $\iff \left\|P_U v - u\right\|^2 = 0 \iff u = P_U v$. Note that in this version of the Pythagorean theorem 6.12, one or both vectors are allowed to be 0. $\square$

## Pseudoinverse (Sub-Section 6C3)

If $T \in \mathcal{L}(V, W)$ is not invertible, instead of solving the equation $Tv = w$, which is the same as $Tv - w = 0$, we can try to find $v \in V$ such that $\left\|Tv - w\right\|$ is as small as possible.

**Thm 6.59** (*restriction of a linear map to obtain a one-to-one and onto map*)**:**
Suppose $\dim V \neq 0$ and $T \in \mathcal{L}(V, W)$. Then $T\big|_{(\operatorname{null} T)^\perp}$ is an injective map of $(\operatorname{null} T)^\perp$ onto $\operatorname{range} T$.

***Proof:*** We will first prove injectivity and then surjectivity. Suppose $v \in (\operatorname{null} T)^\perp$ such that $T\big|_{(\operatorname{null} T)^\perp} v = 0$. Hence, $Tv = 0$ and $v \in (\operatorname{null} T) \cap (\operatorname{null} T)^\perp$, which implies that $v = 0$ by 6.48, because $(\operatorname{null} T) \cap (\operatorname{null} T)^\perp \subseteq \{0\}$.
$$\implies \operatorname{null} T\big|_{(\operatorname{null} T)^\perp} = \{0\},$$
which by 3.15 implies that $T\big|_{(\operatorname{null} T)^\perp}$ is injective.

To prove that range $T\big|_{(\operatorname{null} T)^\perp}$ is onto $\operatorname{range} T$, we will show that range $T\big|_{(\operatorname{null} T)^\perp} \subseteq \operatorname{range} T$ and then $\operatorname{range} T \subseteq$ range $T\big|_{(\operatorname{null} T)^\perp}$. Because this will imply that $\operatorname{range} T =$ range $T\big|_{(\operatorname{null} T)^\perp}$ and therefore surjectivity.

***Proof of*** range $T\big|_{(\text{null}\,T)^\perp} \subseteq$ range $T$**:** The fact that range $T\big|_{(\text{null}\,T)^\perp} \subseteq$ range $T$ is actually obvious, because range $T\big|_{(\text{null}\,T)^\perp}$ is just $T$ on a restricted domain $(\text{null}\,T)^\perp$.

***Proof of*** range $T \subseteq$ range $T\big|_{(\text{null}\,T)^\perp}$ : Now, suppose $w \in \text{range}\,T \implies \exists v \in V$ such that $w = Tv$. By 6.49, there exist $u \in \text{null}\,T$ and $x \in (\text{null}\,T)^\perp$ such that

$$v = u + x \text{ or } x = v - u. \text{ Now,}$$

$$\implies T\big|_{(\text{null}\,T)^\perp} x = T(x) = T(v-u) = T(v) - T(u) = T(v) - 0 = w,$$

which shows that $w \in \text{range}\,T\big|_{(\text{null}\,T)^\perp}$ and thus range $T \subseteq$ range $T\big|_{(\text{null}\,T)^\perp}$.

We conclude: range $T$ = range $T\big|_{(\text{null}\,T)^\perp}$. □

**<u>Def 6.68</u>:**

Suppose $\dim V \neq \infty$ and $T \in \mathcal{L}(V,W)$. The *"pseudoinverse"* $T^\dagger \in \mathcal{L}(V,W)$ of $T$ is defined by

$$T^\dagger w :\equiv \left(T\big|_{(\text{null}\,T)^\perp}\right)^{-1} P_{\text{range}\,T}\, w \quad \forall w \in W.$$

Theorem 6.59 ensures that $\left(T\big|_{(\text{null}\,T)^\perp}\right)^{-1} \in \mathcal{L}\Big(W, (\text{null}\,T)^\perp\Big)$ is well defined.

Recall that:

- By 6.57(b): $P_{\text{range}\,T}\, w = w$ if $w \in \text{range}\,T$;
- By 6.57(c): $P_{\text{range}\,T}\, w = 0$ if $w \in (\text{range}\,T)^\perp$.

Since inverse maps are always injective and onto, we have that:

- $w \in \text{range}\,T \implies T^\dagger w$ is the unique element of $(\text{null}\,T)^\perp$ such that $T(T^\dagger w) = w$;
- $w \in (\text{range}\,T)^\perp \implies T^\dagger w = 0$.

**<u>Thm 6.69 (*algebraic properties of the pseudoinverse*)</u>:**

Suppose $\dim V \neq \infty$ and $T \in \mathcal{L}(V,W)$.

(a) If $T$ is invertible, then $T^\dagger = T^{-1}$.

(b) $TT^\dagger = P_{\text{range}\,T}$ = the orthogonal projection of $W$ onto range $T$.

(c) $T^\dagger T = P_{(\text{null}\,T)^\perp}$ = the orthogonal projection of $V$ onto $(\text{null}\,T)^\perp$.

***<u>Proof:</u>*** Suppose $T \in \mathcal{L}(V,W)$, $\dim V \neq \infty$, is invertible. This means $\text{null}\,T = \{0\}$. Hence, by 6.48(b), we have that

$$(\text{null}\,T)^\perp = \{0\}^\perp = V \implies T\big|_{(\text{null}\,T)^\perp} = T\big|_V = T. \tag{6.1}$$

Invertibility also implies that range $T = W$. Hence,

$$P_{\text{range}\,T} = I_W. \tag{6.2}$$

Equations (6C3) and (6C3) imply that $\forall w \in W$:

$$T^\dagger w = \left(T\big|_{(\text{null}\,T)^\perp}\right)^{-1} P_{\text{range}\,T}\, w = T^{-1} I_W w = T^{-1} w.$$

Therefore, we have shown that $T^\dagger = T^{-1}$, which proves (a).

***Proof of (b):*** By 6.49, we have that $W = \text{range}\,T \oplus (\text{range}\,T)^\perp$.

So, suppose $w \in \text{range}\,T$. Then $P_{\text{range}\,T}\, w = w$. Hence,

$$TT^\dagger w = T\left(T\big|_{(\text{null}\,T)^\perp}\right)^{-1} w = P_{\text{range}\,T}\, w.$$

Now, suppose $w \in (\text{range}\,T)^\perp$. We know from before that $T^\dagger w = 0$. Now,

$$TT^\dagger w = 0 = P_{\text{range}\,T}\, w.$$

Thus, $TT^\dagger$ and $P_{\text{range}\,T}$ both agree on range $T$ and on $(\text{range}\,T)^\perp$. Hence, these two linear maps are equal. So, we have proven (b).

***Proof of (c):*** By 6.49, it is true that $V = \text{null}\,T \oplus (\text{null}\,T)^\perp$.

So, let $v \in (\operatorname{null} T)^\perp$. Because $Tv \in \operatorname{range} T$, it is the case that $P_{\operatorname{range} T}(Tv) = Tv$ and therefore

$$T^\dagger(Tv) = \left(T\big|_{(\operatorname{null} T)^\perp}\right)^{-1} P_{\operatorname{range} T}(Tv) = \left(T\big|_{(\operatorname{null} T)^\perp}\right)^{-1} (Tv) = v = P_{(\operatorname{null} T)^\perp} v.$$

Now, let $v \in \operatorname{null} T$. Then

$$T^\dagger T v = 0 = P_{(\operatorname{null} T)^\perp} v,$$

because of 6.57(c). Thus, $T^\dagger T$ and $P_{(\operatorname{null} T)^\perp}$ agree on $(\operatorname{null} T)^\perp$ and $\operatorname{null} T$. Hence, these two linear maps are equal. So, we have proven (c). □

**Thm 6.70** (*pseudoinverse provides best approximate solution or best solution*)**:**

Suppose $\dim V \neq \infty$, $T \in \mathcal{L}(V, W)$, and $w \in W$.

**(a)** If $v \in V$, then

$$\left\|T(T^\dagger w) - w\right\| \leq \left\|Tv - w\right\|,$$

with equality $\iff v \in T^\dagger w + \operatorname{null} T$.

**(b)** If $v \in T^\dagger w + \operatorname{null} T$, then

$$\left\|T^\dagger w\right\| \leq \|v\|,$$

with equality $\iff v = T^\dagger w$.

***Proof:*** Suppose $v \in V$. Then, with some algebra,

$$Tv - w = \underbrace{(Tv - TT^\dagger w)}_{\in \operatorname{range} T} + \underbrace{(TT^\dagger w - w)}_{\in (\operatorname{range} T)^\perp}. \tag{6.3}$$

This is because from 6.69(b), we know that

$$TT^\dagger = P_{\operatorname{range} T}.$$

Thus, $Tv - TT^\dagger w \in \operatorname{range} T$. From 6.57(f), we know that

$$w - TT^\dagger w = w - P_{\operatorname{range} T} w \in (\operatorname{range} T)^\perp \quad \forall w \in W.$$

Of course, because of closedness, the same goes for its negative: $TT^\dagger w - w \in (\operatorname{range} T)^\perp \quad \forall w \in W$.

This means we can use the Pythagorean theorem 6.12 and (6C3) becomes

$$\begin{aligned}\left\|Tv - w\right\|^2 &= \left\|(Tv - TT^\dagger w) + (TT^\dagger w - w)\right\|^2 \\ &= \underbrace{\left\|(Tv - TT^\dagger w)\right\|^2}_{\geq 0} + \left\|(TT^\dagger w - w)\right\|^2 \\ &\geq \left\|(TT^\dagger w - w)\right\|^2,\end{aligned}$$

with equality only if $Tv - TT^\dagger w = 0 \iff v - T^\dagger w \in \operatorname{null} T$, which is equivalent to $v \in T^\dagger w + \operatorname{null} T$, completing the proof of (a).

***Proof of (b):*** Suppose $v \in T^\dagger w + \operatorname{null} T$. This is equivalent to $v - T^\dagger w \in \operatorname{null} T$. Now,

$$v = (v - T^\dagger w) + T^\dagger w.$$

The definition of $T^\dagger$ implies that $T^\dagger w \in (\operatorname{null} T)^\perp$. This is because $\left(T\big|_{(\operatorname{null} T)^\perp}\right)^{-1}$ is a map from $W$ to $(\operatorname{null} T)^\perp \subseteq V$. Thus, the Pythagorean theorem implies that $\left\|T^\dagger w\right\| \leq \|v\|$, with equality $\iff v = T^\dagger w$. □

# Operators on Inner Product Spaces (Chapter 7)

$\mathbb{F}$ denotes $\mathbb{R}$ or $\mathbb{C}$. $V$ and $W$ are nonzero finite-dimensional inner product spaces over $\mathbb{F}$.

## Self-Adjoint and Normal Operators (Section 7A)

### Adjoints (Sub-Section 7A1)

**Def 7.1** (*adjoint, $T^*$*):

Suppose $T \in \mathcal{L}(V,W)$. The *"adjoint"* of $T$ is the function $T^* : W \to V$ such that

$$\underbrace{\langle Tv, w\rangle}_{\text{takes place in } W} = \underbrace{\left\langle v, T^*w\right\rangle}_{\text{takes place in } V} \qquad \forall v \in V \ \text{ and } \ \forall w \in W.$$

**Recall: Riesz Representation Theorem:** For a linear functional $\varphi \in \mathcal{L}(V,\mathbb{F})$, it holds that

$$\exists!\, v \in V : \varphi(u) = \langle u, v\rangle \quad \forall u \in V.$$

**Example 7.2** (*adjoint linear map from $\mathbb{R}^3$ to $\mathbb{R}$*):

Let $T : \mathbb{R}^3 \to \mathbb{R}^2$, $T(x_1,x_2,x_3) :\equiv (x_2 + 3x_3, 2x_1)$.

**Compute:**
$$\begin{aligned}\left\langle T(x_1,x_2,x_3),(y_1,y_2)\right\rangle &= \left\langle (x_2+3x_3,2x_1),(y_1,y_2)\right\rangle \\ &= x_2y_1 + 3x_3y_1 + 2x_1y_2 \\ &= \left\langle (x_1,x_2,x_3),(2y_2,y_1,3y_1)\right\rangle.\end{aligned}$$

$\Longrightarrow T^* = (2y_2, y_1, 3y_1)$.

**Example 7.3** (*adjoint of a linear map with range of dimension at most 1*):

Fix $u \in V$ and $x \in W$. Define $T \in \mathcal{L}(V,W)$ by

$$Tv :\equiv \langle v, u\rangle x \quad \forall v \in V.$$

Let $\widetilde{v} \in V$ and $\widetilde{w} \in W$. Then:

**Compute:**
$$\begin{aligned}\left\langle T\widetilde{v}, \widetilde{w}\right\rangle &= \left\langle \left\langle \widetilde{v}, u\right\rangle x, \widetilde{w}\right\rangle = \left\langle \widetilde{v}, u\right\rangle \left\langle x, \widetilde{w}\right\rangle \\ &= \left\langle x, \widetilde{w}\right\rangle \left\langle \widetilde{v}, u\right\rangle = \overline{\left\langle \widetilde{w}, x\right\rangle} \left\langle \widetilde{v}, u\right\rangle \\ &= \left\langle \widetilde{v}, \overline{\overline{\left\langle \widetilde{w}, x\right\rangle}} u\right\rangle = \left\langle \widetilde{v}, \left\langle \widetilde{w}, x\right\rangle u\right\rangle.\end{aligned}$$

$\Longrightarrow T^*\widetilde{w} = \left\langle \widetilde{w}, x\right\rangle u$.

**Thm 7.4** (*adjoint of a linear map is a linear map*):

If $T \in \mathcal{L}(V,W)$, then $T^* \in \mathcal{L}(W,V)$.

***Proof:*** Let $T \in \mathcal{L}(V,W)$. If $v \in V$ and $w_1, w_2 \in W$, then:

$$\begin{aligned}\left\langle v, T^*(w_1+w_2)\right\rangle &= \left\langle Tv, w_1 + w_2\right\rangle \\ &= \left\langle Tv, w_1\right\rangle + \left\langle Tv, w_2\right\rangle \\ &= \left\langle v, T^*w_1\right\rangle + \left\langle v, T^*w_2\right\rangle \\ &= \left\langle v, T^*(w_1) + T^*(w_2)\right\rangle.\end{aligned}$$

$\Longrightarrow T^*(w_1+w_2) = T^*(w_1) + T^*(w_2)$.

If $v \in V$, $\lambda \in \mathbb{F}$, and $w \in W$, then:

$$\begin{aligned}\Big\langle v, T^*(\lambda w)\Big\rangle = \langle Tv, \lambda w\rangle &= \overline{\lambda}\langle Tv, w\rangle \\ &= \overline{\lambda}\Big\langle v, T^*w\Big\rangle \\ &= \Big\langle v, \lambda T^*w\Big\rangle.\end{aligned}$$

$\implies T^*(\lambda w) = \lambda T^*w.$

Thus, $T^*$ is a linear map, as desired. □

**Thm 7.5** (*properties of the adjoint*)**:**

Suppose $T \in \mathcal{L}(V, W)$. Then:

- **(a)** $(S+T)^* = S^* + T^* \quad \forall S \in \mathcal{L}(V, W)$;
- **(b)** $(\lambda T)^* = \overline{\lambda} T^* \quad \forall \lambda \in \mathbb{F}$;
- **(c)** $(T^*)^* = T$;
- **(d)** $(ST)^* = T^*S^* \quad \forall S \in \mathcal{L}(W, U)$ [here $U$, with $\dim U \neq \infty$, is an inner product space over $\mathbb{F}$];
- **(e)** $I_V^* = I_V$;
- **(f)** If $T$ is invertible, then $T^*$ is invertible and $(T^*)^{-1} = (T^{-1})^*$.

***Proof:*** Let $v \in V$, $w \in W$, $T \in \mathcal{L}(V, W)$, and $\lambda \in \mathbb{F}$. [Where $V$ and $W$ are inner product spaces.]

**(a)** $$\begin{aligned}\Big\langle v, (S+T)^*w\Big\rangle &= \Big\langle (S+T)v, w\Big\rangle = \langle Sv + Tv, w\rangle \\ &= \langle Sv, w\rangle + \langle Tv, w\rangle = \langle v, S^*w\rangle + \langle v, S^*w\rangle = \langle v, S^*w + T^*w\rangle\end{aligned}$$

Thus, $(S+T)^*w = S^*w + T^*w \implies (S+T)^* = S^* + T^*$, as desired.

**(b)** $$\begin{aligned}\Big\langle v, (\lambda T)^*w\Big\rangle &= \Big\langle (\lambda T)v, w\Big\rangle = \lambda\langle Tv, w\rangle \\ &= \lambda\langle v, T^*w\rangle = \Big\langle v, \overline{\lambda}T^*w\Big\rangle\end{aligned}$$

Thus, $(\lambda T)^* = \overline{\lambda}T^*$.

**(c)** $\Big\langle w, (T^*)^*\Big\rangle = \langle T^*w, v\rangle = \overline{\langle w, T^*w\rangle} = \overline{\langle Tv, w\rangle} = \langle w, Tw\rangle$. Thus, $(T^*)^* = T$.

**(d)** Suppose $S \in \mathcal{L}(W, U)$ and $u \in U$ [where $U$ is an inner product space]. Then:

$$\Big\langle v, (ST)^*u\Big\rangle = \Big\langle (ST)v, u\Big\rangle = \Big\langle S(Tv), u\Big\rangle = \Big\langle Tv, S^*u\Big\rangle = \Big\langle v, T^*(S^*u)\Big\rangle = \Big\langle v, (T^*S^*)u\Big\rangle.$$

Therefore, $(ST)^* = T^*S^*$.

**(e)** Let $v_1, v_2 \in V$. Then $\Big\langle v_1, I_V^*v_2\Big\rangle = \Big\langle I_Vv_1, v_2\Big\rangle = \Big\langle v_1, v_2\Big\rangle = \Big\langle v_1, I_Vv_2\Big\rangle$. Thus, $I^* = I$.

**(f)** $$\begin{aligned}&T^{-1}T = I \iff (T^{-1}T)^* = I^* \overset{\textbf{(d)}}{\implies} T^*(T^{-1})^* = I \\ &TT^{-1} = I \iff (TT^{-1})^* = I^* \overset{\textbf{(d)}}{\implies} (T^{-1})^*T^* = I\end{aligned}$$

$\implies (T^{-1})^*$ is the inverse of $T^*$, as desired.

□

**Thm 7.6** (*null space and range of $T^*$*)**:**

If $T \in \mathcal{L}(V, W)$, then:

- **(a)** $\operatorname{null} T^* = (\operatorname{range} T)^\perp$;
- **(b)** $\operatorname{range} T^* = (\operatorname{null} T)^\perp$;
- **(c)** $\operatorname{null} T = (\operatorname{range} T^*)^\perp$;
- **(d)** $\operatorname{range} T = (\operatorname{null} T^*)^\perp$.

***Proof:*** We begin by proving (a). Let $w \in W$. Then:

$$\begin{aligned}w \in \operatorname{null} T^* &\iff T^*w = 0 \\ &\iff \Big\langle v, T^*w\Big\rangle = 0 \quad \forall v \in V \\ &\iff \langle Tv, w\rangle = 0 \quad \forall v \in V \\ &\iff w \in (\operatorname{range} T)^\perp.\end{aligned}$$

Thus, $\operatorname{null} T^* = (\operatorname{range} T)^\perp$, proving (a).

If we use 6.52 where $U = (U^\perp)^\perp$ and we take orthogonal complements on both sides in (a), we get:

$$(\operatorname{null} T^*)^\perp = \left((\operatorname{range} T)^\perp\right)^\perp$$

$$(\operatorname{null} T^*)^\perp = \operatorname{range} T,$$

which proves (d).

Using 7.5(c) where $(T^*)^* = T$, replacing $T$ with $T^*$ in (a) will get us:

$$\operatorname{null}\left((T^*)^*\right) = (\operatorname{range} T^*)^\perp$$

$$\operatorname{null} T = (\operatorname{range} T^*)^\perp,$$

which gives (c).

Finally, replacing $T$ with $T^*$ in (d) gives (b):

$$\operatorname{range} T^* = (\operatorname{null}(T^*)^*)^\perp \iff \operatorname{range} T^* = (\operatorname{null} T)^\perp.$$

□

**Def 7.7** (*conjugate transpose, $A^*$*)**:**

For $A \in \mathbb{F}^{m,n}$, we define its *"conjugate transpose"* $A^* \in \mathbb{F}^{n,m}$ as follows:

If $j \in \{1, \dots, n\}$ and $k \in \{1, \dots, m\}$, then

$$(A^*)_{j,k} :\equiv \overline{A_{k,j}}.$$

If a matrix has only real entries, then

$$A^* = A^t.$$

**Example 7.8:**

$$\begin{pmatrix} 2 & 3+4i & 7 \\ 6 & 5 & 8i \end{pmatrix}^* = \begin{pmatrix} 2 & 6 \\ 3-4i & 5 \\ 7 & -8i \end{pmatrix}$$

**Thm 7.9** (*matrix of $T^*$ equals conjugate transpose matrix of $T$*)**:**

Let $T \in \mathcal{L}(V, W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of $V$ and $f_1, \dots, f_m$ is an orthonormal basis of $W$. Then

$$\mathcal{M}(T^*, (f_1, \dots, f_m), (e_1, \dots, e_m)) = (\mathcal{M}(T, (e_1, \dots, e_n), (f_1, \dots, f_m)))^*.$$

Or the short version:

$$\mathcal{M}(T^*) = (\mathcal{M}(T))^*.$$

***Proof:*** From 6.30(c), we know that for any $v \in V$: $v = \left\langle v, e_1\right\rangle e_1 + \cdots + \left\langle v, e_m\right\rangle e_m$. This means that for all $k \in \{1, \dots, n\}$:

$$Te_k = \left\langle Te_k, f_1\right\rangle f_1 + \cdots + \left\langle Te_k, f_m\right\rangle f_m.$$

So $\mathcal{M}(T)$ looks as follows:

$$\mathcal{M}(T) = \begin{array}{c} \\ f_1 \\ \vdots \\ f_j \\ \vdots \\ f_m \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & \cdots & e_k & \cdots & e_n \end{array} \\ \begin{pmatrix} & & \left\langle Te_k, f_1\right\rangle & & \\ & & \vdots & & \\ & & \left\langle Te_k, f_j\right\rangle & & \\ & & \vdots & & \\ & & \left\langle Te_k, f_m\right\rangle & & \end{pmatrix} \end{array}$$

Similarly, for all $k \in \{1, \dots, m\}$:

$$T^* f_k = \left\langle T^* f_k, e_1 \right\rangle e_1 + \cdots + \left\langle T^* f_k, e_n \right\rangle e_n.$$

Since for all $k \in \{1, \dots, m\}$ and for all $j \in \{1, \dots, n\}$ we have

$$\left\langle T^* f_k, e_j \right\rangle = \left\langle f_k, T e_j \right\rangle = \overline{\left\langle T e_j, f_k \right\rangle},$$

now $\mathcal{M}(T^*)$ looks as follows:

$$\mathcal{M}(T^*) = \begin{array}{c} \\ e_1 \\ \vdots \\ e_j \\ \vdots \\ e_n \end{array} \begin{array}{c} \begin{array}{ccccc} f_1 & \cdots & f_k & \cdots & f_m \end{array} \\ \begin{pmatrix} & & \left\langle T^* f_k, e_1 \right\rangle & & \\ & & \vdots & & \\ & & \left\langle T^* f_k, e_j \right\rangle & & \\ & & \vdots & & \\ & & \left\langle T^* f_k, e_n \right\rangle & & \end{pmatrix} \end{array} = \begin{array}{c} \\ e_1 \\ \vdots \\ e_j \\ \vdots \\ e_n \end{array} \begin{array}{c} \begin{array}{ccccc} f_1 & \cdots & f_k & \cdots & f_m \end{array} \\ \begin{pmatrix} & & \overline{\left\langle T e_1, f_k \right\rangle} & & \\ & & \vdots & & \\ & & \overline{\left\langle T e_j, f_k \right\rangle} & & \\ & & \vdots & & \\ & & \overline{\left\langle T e_n, f_k \right\rangle} & & \end{pmatrix} \end{array}$$

Thus, the entry in row $j$, column $k$, of $\mathcal{M}(T)$ equals the complex conjugate of the entry in row $k$, column $j$, of $\mathcal{M}(T)$. Thus, $\mathcal{M}(T^*) = (\mathcal{M}(T))^*$. □

## Self-Adjoint Operators (Sub-Section 7A2)

**Def 7.10** (*self-adjoint*)**:**

An operator $T \in \mathcal{L}(V)$ is called *"self-adjoint"* if

$$T = T^* \iff \langle Tv, w \rangle = \langle v, Tw \rangle \quad \forall v, w \in V.$$

If $\mathbb{F} = \mathbb{R}$, then every eigenvalue is real, so the next result is interesting only when $\mathbb{F} = \mathbb{C}$.

**Thm 7.12** (*eigenvalues of self-adjoint operators*)**:**

Every eigenvalue of a self-adjoint operator is real.

***Proof:*** Suppose $T$ is a self-adjoint operator on $V$. Let $\lambda$ be an eigenvalue of $T$, and let $v \in V, v \neq 0$, such that $Tv = \lambda v$. Then:

$$\lambda \|v\|^2 = \langle \lambda v, v \rangle = \langle Tv, v \rangle = \langle v, Tv \rangle = \langle v, \lambda v \rangle = \overline{\lambda} \|v\|^2.$$

Thus, $\lambda = \overline{\lambda}$, which means that $\lambda$ is real, as desired. □

**Thm 7.13** (*$Tv$ orthogonality to $v$*)**:**

Suppose $V$ is a complex inner product space [$\mathbb{F} = \mathbb{C}$] and $T \in \mathcal{L}(V)$. Then

$$\langle Tv, v \rangle = 0 \quad \forall v \in V \iff T = 0.$$

***Proof:*** $\Rightarrow$ ***direction:*** If $u, w \in V$, then:

- $\left\langle T(u+w), u+w \right\rangle = \langle Tu, u \rangle + \langle Tw, u \rangle + \langle Tu, w \rangle + \langle Tw, w \rangle;$
- $\left\langle T(u-w), u-w \right\rangle = \langle Tu, u \rangle - \langle Tw, u \rangle - \langle Tu, w \rangle + \langle Tw, w \rangle;$
- $\left\langle T(u+iw), u+iw \right\rangle = \langle Tu, u \rangle + i \langle Tw, u \rangle + \overline{i} \langle Tu, w \rangle + i\overline{i} \langle Tw, w \rangle$
  $= \langle Tu, u \rangle + i \langle Tw, u \rangle - i \langle Tu, w \rangle + \langle Tw, w \rangle;$
- $\left\langle T(u-iw), u-iw \right\rangle = \langle Tu, u \rangle - i \langle Tw, u \rangle - \overline{i} \langle Tu, w \rangle + i\overline{i} \langle Tw, w \rangle$
  $= \langle Tu, u \rangle - i \langle Tw, u \rangle + i \langle Tu, w \rangle + \langle Tw, w \rangle.$

That means that:

- $\left\langle T(u+w), u+w \right\rangle - \left\langle T(u-w), u-w \right\rangle = 2 \langle Tw, u \rangle + 2 \langle Tu, w \rangle;$
- $\left\langle T(u+iw), u+iw \right\rangle - \left\langle T(u-iw), u-iw \right\rangle = 2i \langle Tw, u \rangle - 2i \langle Tu, w \rangle.$

Therefore, it holds for all $u, w \in V$:

$$\begin{aligned}\frac{\Big\langle T(u+w), u+w\Big\rangle - \Big\langle T(u-w), u-w\Big\rangle}{4} + \frac{\Big\langle T(u+iw), u+iw\Big\rangle - \Big\langle T(u-iw), u-iw\Big\rangle}{4} \cdot i \\ = \frac{1}{2}\langle Tw, u\rangle + \frac{1}{2}\langle Tu, w\rangle + \left(\frac{1}{2}i\langle Tw, u\rangle - \frac{1}{2}i\langle Tu, w\rangle\right)\cdot i \\ = \frac{1}{2}\langle Tw, u\rangle + \frac{1}{2}\langle Tu, w\rangle - \frac{1}{2}\langle Tw, u\rangle + \frac{1}{2}\langle Tu, w\rangle \\ = \langle Tu, w\rangle.\end{aligned}$$

Note that each term in the numerator of the two big expressions which have the denominator 4 are of the form $\langle Tv, v\rangle$ for appropriate $v \in V$, for example $\Big\langle T(u-iw), u-iw\Big\rangle$, where $u - iw \in V$. Note that these two fractions added together are equal to $\langle Tu, w\rangle \quad \forall u, w \in V$ at the end.

Now, suppose $\langle Tv, v\rangle = 0 \quad \forall v \in V$. Then the (big) equation above implies that $\langle Tu, w\rangle = 0 \quad \forall u, w \in V$. We can further conclude that $Tu = 0 \quad \forall u \in V$, because only $\vec{0}$ is always orthogonal to every vector (take $w = Tu$. Only if $w = Tu = \vec{0}$, the expression $\langle Tu, Tu\rangle$ can always be 0). Hence, $T = 0$, as desired. Here, the last 0 means the zero operator.

$\Leftarrow$ ***direction:*** If $T = 0$, then trivially $\langle Tv, v\rangle = 0 \quad \forall v \in V$. ☐

**Thm 7.14** (*condition for $T$ such that $\langle Tv, v\rangle$ is always real*):

Suppose $V$ is a complex inner product space and $T \in \mathcal{L}(V)$. Then

$$T \text{ is self-adjoint } \iff \langle Tv, v\rangle \in \mathbb{R} \quad \forall v \in V.$$

***Proof:*** If $v \in V$, then:

$$\Big\langle T^*v, v\Big\rangle = \overline{\langle v, T^*v\rangle} = \overline{\langle Tv, v\rangle}. \tag{7.1}$$

Now,

$$\begin{aligned} T \text{ is self-adjoint } &\iff T - T^* = 0 \\ &\iff \Big\langle (T-T^*)v, v\Big\rangle = 0 \quad \forall v \in V \\ &\iff \langle Tv, v\rangle - \Big\langle T^*v, v\Big\rangle = 0 \quad \forall v \in V \\ &\iff \langle Tv, v\rangle - \overline{\langle Tv, v\rangle} = 0 \quad \forall v \in V \quad (\text{which means } \langle Tv, v\rangle = \overline{\langle Tv, v\rangle}) \\ &\iff \langle Tv, v\rangle \in \mathbb{R} \quad \forall v \in V,\end{aligned}$$

where we used 7.13 for the second equivalence as applied to $T - T^*$, and the fourth equivalence follows from (7A2). ☐

**Thm 7.16:**

Suppose $T$ is a self-adjoint operator on $V$. Then

$$\langle Tv, v\rangle = 0 \quad \forall v \in V \iff T = 0.$$

***Proof:*** We have already proved this (without the hypothesis that $T$ is self-adjoint) when $V$ is a complex inner product space. Thus, we can assume that $V$ is a real inner product space. If $w, u \in V$, then:

$$\langle Tw, u\rangle = \langle w, Tu\rangle = \overline{\langle Tu, w\rangle} = \langle Tu, w\rangle.$$

Thus, we can compute:

$$\begin{aligned}\frac{\Big\langle T(u+w), u+w\Big\rangle - \Big\langle T(u-w), u-w\Big\rangle}{4} &= \frac{1}{2}\langle Tw, u\rangle + \frac{1}{2}\langle Tu, w\rangle \\ &= \frac{1}{2}\langle Tu, w\rangle + \frac{1}{2}\langle Tu, w\rangle \\ &= \langle Tu, w\rangle.\end{aligned} \tag{7.2}$$

Now, suppose $\langle Tv, v\rangle = 0 \quad \forall v \in V$. Because each term in the numerator in the first expression of (7A2) is of the form $\langle Tv, v\rangle$ for appropriate $v$, this implies $\langle Tu, w\rangle = 0 \quad \forall u, w \in V$. Because only the zero vector is orthogonal to every vector, we have $Tu = 0 \quad \forall u \in V$ [take $w = Tu$]. Hence, $T = 0$, as desired. □

This theorem means that for a self-adjoint operator $T$, it cannot happen that $\langle Tv, v\rangle = 0 \quad \forall v \in V$ other than the zero operator. But there are other operators for which this is true, for example a counterclockwise rotation of 90° around the origin; thus $T(x,y) = (-y,x)$. Then we have $\Big\langle T(x,y),(x,y)\Big\rangle = \Big\langle(-y,x),(x,y)\Big\rangle = -yx + xy = 0$.

## Normal Operators (Sub-Section 7A3)

**Def 7.18** (*normal*):
- An operator on an inner product space is called *"normal"* if it commutes with its adjoint.
- In other words, $T \in \mathcal{L}(V)$ is normal if $TT^* = T^*T$.

Every self-adjoint operator is normal, because if $T^* = T$, then $T$ commutes with $T^*$.

**Example 7.19** (*an operator that is normal but not self-adjoint*):

Let $T$ be the operator on $\mathbb{F}^2$ whose matrix with respect to the standard basis is

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}$$

Thus, $T(x_1, x_2) = (2x_1 - 3x_2, 3x_1 + 2x_2)$.

This operator $T$ is not self-adjoint because $\mathcal{M}(T)_{2,1} = 3$, which is not equal to the complex conjugate of the entry in $\mathcal{M}(T)_{1,2} = -3$.

Nonetheless, for the matrix of $TT^*$, we have:

$$\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}$$

Similarly, for the matrix of $T^*T$, we have:

$$\begin{pmatrix} 2 & 3 \\ -3 & 2 \end{pmatrix}\begin{pmatrix} 2 & -3 \\ 3 & 2 \end{pmatrix} = \begin{pmatrix} 13 & 0 \\ 0 & 13 \end{pmatrix}$$

Because $TT^*$ and $T^*T$ have the same matrix, we see that $TT^* = T^*T$. Thus, $T$ is normal.

**Thm 7.20** (*condition for $T$ to be normal*):

Let $T \in \mathcal{L}(V)$. Then

$$T \text{ is normal} \iff \|Tv\| = \|T^*v\| \quad \forall v \in V.$$

***Proof:*** 
$$\begin{aligned} T \text{ is normal} &\iff T^*T - TT^* = 0 \\ &\iff \Big\langle (T^*T - TT^*)v, v\Big\rangle = 0 \quad \forall v \in V \\ &\iff \langle T^*Tv, v\rangle = \langle TT^*v, v\rangle \\ &\iff \langle Tv, Tv\rangle = \langle T^*v, T^*v\rangle \quad \forall v \in V \\ &\iff \|Tv\|^2 = \|Tv\|^2 \quad \forall v \in V \\ &\iff \|Tv\| = \|T^*v\| \quad \forall v \in V, \end{aligned}$$

where the second equivalence follows from 7.16. The operator $T^*T - TT^*$ is self-adjoint for every operator $T \in \mathcal{L}(V)$. This is because, for all $v \in V$:

$$\begin{aligned}\left\langle (T^*T - TT^*)v, v\right\rangle &= \left\langle T^*Tv, v\right\rangle - \left\langle TT^*v, v\right\rangle \\ &= \left\langle v, (T^*T)^*v\right\rangle - \left\langle v, (T^*T)^*v\right\rangle \\ &= \left\langle v, T^*(T^*)^*\right\rangle - \left\langle v, (T^*)^*T^*v\right\rangle \\ &= \left\langle v, T^*Tv\right\rangle - \left\langle v, TT^*v\right\rangle \\ &= \left\langle v, (T^*T - TT^*)v\right\rangle.\end{aligned}$$

One can also just calculate $(T^*T - TT^*)^*$ directly using 7.5 and verify that $T^*T - TT^* = (T^*T - TT^*)^*$. □

**Thm 7.21** (*range, null space, and eigenvectors of a normal operator*):

Suppose $T \in \mathcal{L}(V)$ is normal. Then:

**(a)** $\operatorname{null} T = \operatorname{null} T^*$;

**(b)** $\operatorname{range} T = \operatorname{range} T^*$;

**(c)** $V = \operatorname{null} T \oplus \operatorname{range} T$;

**(d)** $T - \lambda I$ is normal $\quad \forall \lambda \in \mathbb{F}$;

**(e)** If $v \in V$ and $\lambda \in \mathbb{F}$, then $Tv = \lambda v \iff T^*v = \overline{\lambda} v$.

***Proof:*** Suppose $v \in V$, $\lambda \in \mathbb{F}$, and let $T \in \mathcal{L}(V)$ be a normal operator. Then:

**(a)** $v \in \operatorname{null} T \iff \left\|Tv\right\| = 0 \overset{7.20}{\iff} \left\|T^*v\right\| = 0 \iff v \in \operatorname{null} T^*$.

**(b)** $\operatorname{range} T \overset{7.6\,(\mathbf{d})}{=} (\operatorname{null} T^*)^\perp \overset{(\mathbf{a})}{=} (\operatorname{null} T)^\perp \overset{7.6\,(\mathbf{b})}{=} \operatorname{range} T^*$.

**(c)** $V \overset{6.49}{=} (\operatorname{null} T) \oplus (\operatorname{null} T)^\perp \overset{7.6}{=} \operatorname{null} T \oplus \operatorname{range} T^* \overset{(\mathbf{b})}{=} \operatorname{null} T \oplus \operatorname{range} T$.

**(d)** $$\begin{aligned}(T-\lambda)(T-\lambda I) &= (T - \lambda I)(T^* - \overline{\lambda} I) \\ &= TT^* - \overline{\lambda} T - \lambda T^* + |\lambda|^2 I \\ &= T^*T - \overline{\lambda} T - \lambda T^* + |\lambda|^2 I \\ &= (T^* - \overline{\lambda} I)(T - \lambda I) \\ &= (T - \lambda I)^*(T - \lambda I).\end{aligned}$$

Thus, $T - \lambda I$ commutes with its adjoint. Hence, $T - \lambda I$ is normal.

**(e)** Part **(d)** tells us that $T - \lambda I$ is normal, and we can use 7.20 to yield that

$$\left\|(T-\lambda I)v\right\| = \left\|(T-\lambda I)^*v\right\| = \left\|(T^* - \overline{\lambda} I)v\right\|.$$

Thus, $\left\|(T-\lambda I)v\right\| = 0 \iff \left\|(T^* - \overline{\lambda} I)v\right\| = 0$. Hence, $Tv = \lambda v \iff T^*v = \overline{\lambda} v$. □

**Thm 7.22** (*orthogonal eigenvectors for normal operators*):

Suppose $T \in \mathcal{L}(V)$ is normal. Then eigenvectors of $T$ corresponding to distinct eigenvalues are orthogonal.

***Proof:*** Suppose $\alpha, \beta$ are distinct eigenvalues of $T$, with corresponding eigenvectors $u, v \in V$. Thus, $Tu = \alpha u$ and $Tv = \beta v$. From 7.21(e), we have $T^*v = \overline{\beta} v$. Now,

$$\begin{aligned}(\alpha - \beta)\langle u, v\rangle &= \langle \alpha u, v\rangle - \left\langle u, \overline{\beta} v\right\rangle \\ &= \langle Tu, v\rangle - \left\langle u, T^*v\right\rangle \\ &= 0.\end{aligned}$$

Because $\alpha \neq \beta \implies \alpha - \beta \neq 0$, the equation above implies that $\langle u, v\rangle = 0$. Thus, $u, v$ are orthogonal, as desired. □

**Thm 7.23** ($T$ *is normal* $\iff$ *the real and imaginary parts of* $T$ *commute*):

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then $T$ is normal $\iff$ there exist commuting self-adjoint operators

$A$ and $B$ such that $T = A + iB$.

***Proof:*** $\Rightarrow$ ***direction:*** Let $T$ be normal. Let:

$$A :\equiv \frac{T + T^*}{2} \quad \text{and} \quad B :\equiv \frac{T - T^*}{2i}.$$

So, $T = A + iB$. For $A$, we have $A^* = \left(\frac{T+T^*}{2}\right)^* = \frac{1}{2}(T^* + T) = A$. For $B$, we get the following computation:

$$\begin{aligned} B^* &= \overline{\left(\frac{1}{2i}\right)}(T - T^*)^* = \overline{\left(-\frac{i}{2}\right)}(T^* - T) \\ &= \left(\frac{i}{2}\right)(T^* - T) = \left(-\frac{1}{2i}\right)(T^* - T) \\ &= \frac{T - T^*}{2i} = B. \end{aligned}$$

So $A = A^*$ and $B = B^*$; therefore, $A$ and $B$ are self-adjoint. A quick computation shows that

$$\begin{aligned} AB - BA &= \frac{TT - TT^* + T^*T - T^*T^*}{4i} - \frac{TT + TT^* - T^*T - T^*T^*}{4i} \\ &= \frac{-2TT^* + 2T^*T}{4i} \\ &= \frac{T^*T - TT^*}{2i} \\ &= \frac{T^*T - T^*T}{2i} \qquad \text{because } T \text{ is normal} \\ &= 0, \quad \text{and therefore } AB = BA. \end{aligned}$$

Thus, the operators $A$ and $B$ commute, as desired.

$\Leftarrow$ ***direction:*** Now, suppose there exist commuting self-adjoint operators $A$ and $B$ $[AB = BA]$ such that

$$T = A + iB \implies T^* = A - iB.$$

Now, we can compute $A$, because $T + T^* = 2A$, and therefore we get the same equation for $A$ as before, namely

$$A = \frac{T + T^*}{2}.$$

Therefore, if we subtract the equation for $T^*$ from the equation for $A$, we get the same equation for $B$:

$$\begin{aligned} A - A + iB &= \frac{T + T^*}{2} - T^* \\ B &= \frac{T - T^*}{2i}. \end{aligned}$$

Since $A$ and $B$ commute, the expression for $AB - BA$ from the first part, which evaluates to $\frac{T^*T - TT^*}{2i}$, is equal to 0, because we assumed $AB = BA$. Therefore, $T^*T = TT^*$, which means that $T$ is normal, as desired. [We have not used the fact that $A$ and $B$ are self-adjoint in this direction of the proof.] $\square$

## Spectral Theorem (Section 7B)

### Real Spectral Theorem (Sub-Section 7B1)

Let $x, b, c \in \mathbb{R}$ and $b^2 < 4c$. Then

$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right) > 0.$$

**Thm 7.24** (*invertible quadratic expressions*):

Suppose $T \in \mathcal{L}(V)$ is self-adjoint and $b, c \in \mathbb{R}$ are such that $b^2 < 4c$. Then

$$T^2 + bT + cI$$

is an invertible operator.

***Proof:*** Let $v \in V, v \neq 0, T \in \mathcal{L}(V)$, and let $b, c \in \mathbb{R}$ such that $b^2 < 4c$. Then:

$$\begin{aligned}
\left\langle (T^2 + bT + cI)v, v \right\rangle &= \left\langle T^2 v, v \right\rangle + b\langle Tv, v\rangle + c\langle v, v\rangle \\
&= \langle Tv, Tv\rangle + b\langle Tv, v\rangle + c\|v\|^2 \\
&\geq \left\|Tv\right\|^2 - |b|\left\|Tv\right\|\|v\| + c\|v\|^2 && \text{[Cauchy-Schwarz inequality 6.14]} \\
&= \left(\left\|Tv\right\| - \frac{|b|\|v\|}{2}\right)^2 + \left(c - \frac{b^2}{4}\right)\|v\|^2 \\
&> 0.
\end{aligned}$$

The last inequality implies that $(T^2 + bT + cI)v \neq 0$. Since $v \neq 0$, it follows that $T^2 + bT + cI$ is injective by 3.15, because only $(T^2 + bT + cI)0 = 0$. This also implies that it is invertible by 3.65. □

**Thm 7.27** (*minimal polynomial of self-adjoint operator*):

Suppose $T \in \mathcal{L}(V)$ is self-adjoint. Then the minimal polynomial of $T$ equals $(z - \lambda_1)\cdots(z - \lambda_m)$ for some $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$.

***Proof:*** First, suppose $\mathbb{F} = \mathbb{C}$. The zeros of the minimal polynomial of $T$ are the eigenvalues of $T$ by 5.27(**a**). Because $T$ is self-adjoint, all eigenvalues of $T$ are real by 7.12. Thus, the second version of the fundamental theorem of algebra 4.13 tells us that the minimal polynomial of $T$ has the desired form.

Now, suppose $\mathbb{F} = \mathbb{R}$. By 4.16, there exist $\lambda_1, \ldots, \lambda_m \in \mathbb{R}$ and $b_1, \ldots, b_N, c_1, \ldots, c_N \in \mathbb{R}$ with $b_k^2 < 4c_k \;\; \forall k \in \{1, \ldots, N\}$ such that the minimal polynomial of $T$ equals

$$(z - \lambda_1)\cdots(z - \lambda_m) \cdot (z^2 + b_1 z + c_1)\cdots(z^2 + b_N z + c_N). \tag{7.3}$$

Note that it might be the case that $m = 0$ or $N = 0$, meaning that there are no terms of the corresponding form. If $N = 0$, we are done with this proof. Now,

$$(T - \lambda_1 I)\cdots(T - \lambda_m I) \cdot (T^2 + b_1 T + c_1 I)\cdots(T^2 + b_N T + c_N I) = 0.$$

If $N > 0$, then we could multiply both sides of the equation above on the right by the inverse $(T^2 + b_N T + c_N I)^{-1}$ of $T^2 + b_N T + c_N I$, which is an invertible operator by 7.24. We would then obtain a polynomial expression of $T$ that equals 0. The corresponding polynomial would have a degree 2 less than the degree of (7B1), violating the minimality of the minimal polynomial. Thus, we must have $N = 0$, which means that the minimal polynomial in (7B1) has the form $(z - \lambda_1)\cdots(z - \lambda_m)$, as desired. □

The previous result along with 5.27 implies that every self-adjoint operator has an eigenvalue. The next result will show that self-adjoint operators have enough eigenvalues to form a basis. It gives a complete description of self-adjoint operators on a real inner product space and is one of the major theorems in linear algebra.

**Thm 7.28** (*real spectral theorem*):

Suppose $\mathbb{F} = \mathbb{R}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent:

**(a)** $T$ is self-adjoint;

**(b)** $T$ has a diagonal matrix with respect to some orthonormal basis of $V$;

**(c)** $V$ has an orthonormal basis consisting of eigenvectors of $T$.

***Proof:*** ***Step 1:*** First, suppose (a) holds, so $T$ is self-adjoint. Our results on minimal polynomials, specifically:

- Every orthonormal list extends to an orthonormal basis (Todo: 6.37)
- minimal polynomial of self-adjoint operator

imply that $T$ has an upper-triangular matrix with respect to some orthonormal basis of $V$. With respect to this basis, the matrix $\mathcal{M}(T^*)$ of $T^*$ is the transpose of the matrix $\mathcal{M}(T)$ of $T$:

$$\mathcal{M}(T^*)^t = \mathcal{M}(T).$$

However, $T^* = T$. Thus, $\mathcal{M}(T)^t = \mathcal{M}(T)$. Because $\mathcal{M}(T)$ is upper-triangular, this means that all entries of the matrix above and below the diagonal are 0. Hence, $\mathcal{M}(T)$ is a diagonal matrix with respect to the orthonormal basis. Thus, (a) implies (b).

***Step 2:*** Second, suppose (b) holds, so $T$ has a diagonal matrix $\mathcal{M}(T)$ with respect to some orthonormal basis of $V$. That diagonal matrix $\mathcal{M}(T)$ equals its transpose (i.e., $\mathcal{M}(T) = \mathcal{M}(T)^t$). Thus, with respect to that basis, $T^* = T$, proving that (b) implies (a).

***Step 3:*** The equivalence of (b) and (c) follows from the definitions. Suppose (b) has a diagonal matrix whose entries are $\lambda_1, \dots, \lambda_m$ with an orthonormal basis $e_1, \dots, e_m$. Hence, $Te_k = \lambda_k e_k$ (for $k = 1, \dots, m$). Therefore, $e_1, \dots, e_m$ are the eigenvectors of $T$. □

## Complex Spectral Theorem (Sub-Section 7B2)

**Thm 7.29** (*complex spectral theorem*)**:**

Suppose $\mathbb{F} = \mathbb{C}$ and $T \in \mathcal{L}(V)$. Then the following are equivalent:

**(a)** $T$ is normal ($TT^* = T^*T$);

**(b)** $T$ has a diagonal matrix with respect to some orthonormal basis of $V$;

**(c)** $V$ has an orthonormal basis consisting of eigenvectors of $T$.

***Proof:*** Since $\mathbb{F} = \mathbb{C}$, by Schur's theorem, there is an orthonormal basis $e_1, \dots, e_n$ of $V$ with respect to which $T$ has an upper-triangular matrix. Thus, we can write:

$$\mathcal{M}(T) = \mathcal{M}(T, (e_1, \dots, e_n)) = \begin{pmatrix} a_{1,1} & \cdots & a_{1,n} \\ & \ddots & \vdots \\ 0 & & a_{n,n} \end{pmatrix}. \tag{7.4}$$

We see that:

$$\left\|Te_1\right\|^2 = \sqrt{\left\langle Te_1, Te_1 \right\rangle}^2 = \left\langle a_{1,1} \cdot e_1, a_{1,1} \cdot e_1 \right\rangle = a_{1,1}\overline{a_{1,1}} \cdot \left\langle e_1, e_1 \right\rangle = |a_{1,1}|^2 \left\|e_1\right\|^2 = |a_{1,1}|^2,$$

as well as

$$\begin{aligned}\left\|T^*\right\| = \sqrt{\left\langle T^*e_1\,,T^*e_1\right\rangle}^{\,2} &= \left\langle (a_{1,1}\cdot e_1+\cdots a_{1,n}e_n)\,,(a_{1,1}\cdot e_1+\cdots a_{1,n}e_n)\right\rangle\\ &= \left\langle (a_{1,1}\cdot e_1)\,,(a_{1,1}\cdot e_1+\cdots+a_{1,n}e_n)\right\rangle+\cdots+\left\langle (a_{1,n}\cdot e_n)\,,(a_{1,1}\cdot e_1+\cdots+a_{1,n}e_n)\right\rangle\\ &= \Big(\left\langle a_{1,1}\cdot e_1\,,a_{1,1}\cdot e_1\right\rangle+\underbrace{\left\langle a_{1,1}\cdot e_1\,,a_{1,2}\cdot e_2\right\rangle}_{=0}+\cdots+\underbrace{\left\langle a_{1,1}\cdot e_1\,,a_{1,n}\cdot e_n\right\rangle}_{=0}\Big)\\ &\quad+\Big(\underbrace{\left\langle a_{1,2}\cdot e_2\,,a_{1,1}\cdot e_1\right\rangle}_{=0}+\left\langle a_{1,2}\cdot e_2\,,a_{1,2}\cdot e_2\right\rangle+\cdots+\underbrace{\left\langle a_{1,2}\cdot e_2\,,a_{1,n}\cdot e_n\right\rangle}_{=0}\Big)\\ &\quad+\ \ldots\ +\Big(\underbrace{\left\langle a_{1,n}\cdot e_n\,,a_{1,1}\cdot e_1\right\rangle}_{=0}+\underbrace{\left\langle a_{1,n}\cdot e_n\,,a_{1,2}\cdot e_2\right\rangle}_{=0}+\cdots+\left\langle a_{1,n}\cdot e_n\,,a_{1,n}\cdot e_n\right\rangle\Big)\\ &= a_{1,1}\overline{a_{1,1}}\cdot\left\langle e_1\,,e_1\right\rangle+a_{1,2}\overline{a_{1,2}}\cdot\left\langle e_2\,,e_2\right\rangle+\cdots+a_{1,n}\overline{a_{1,n}}\cdot\left\langle e_n\,,e_n\right\rangle\\ &= |a_{1,1}|^2+\cdots+|a_{1,n}|^2.\end{aligned}$$

***Step 1:*** Now, suppose (a) holds. Because $T$ is normal, $\left\|Te_1\right\|^2 = \left\|T^*e_1\right\|^2$; see 7.20. Thus,

$$|a_{1,1}|^2 = |a_{1,1}|^2+\cdots+|a_{1,n}|^2$$

implies that all entries in the first row of the matrix $\mathcal{M}(T)$, except possibly the first entry $a_{1,1}$, equal 0.

Now, (7B2) implies that

$$\left\|Te_2\right\|^2 = |a_{2,1}|^2+|a_{2,2}|^2 = |a_{2,2}|^2 \quad \text{and}$$
$$\left\|T^*e_2\right\|^2 = |a_{2,2}|^2+|a_{2,3}|^2+\cdots+|a_{2,n}|^2.$$

Because $T$ is normal, $\left\|Te_2\right\| = \left\|T^*e_2\right\|$, and therefore

$$|a_{2,2}|^2 = |a_{2,2}|^2+|a_{2,3}|^2+\cdots+|a_{2,n}|^2.$$

Thus, the equations above imply that all entries in the second row of the matrix in (7B2), except possibly for the diagonal entry $a_{2,2}$, equal 0. Continuing in this fashion, we see that all nondiagonal entries in the matrix (7B2) equal 0. Thus, (b) holds, completing the proof that $(a) \implies (b)$.

***Step 2:*** Now, suppose (b) holds, so $T$ has a diagonal matrix $\mathcal{M}(T)$ with respect to some orthonormal basis of $V$. The matrix $T^*$ (with respect to some basis) is obtained by taking the conjugate transpose of the matrix of $T$; hence, $T^*$ also has a diagonal matrix. Any two diagonal matrices commute; thus, $T$ commutes with $T^*$, which means that $T$ is normal. In other words, (a) holds, completing the proof that $(b) \implies (a)$.

***Step 3:*** The equivalence of (b) and (c) follows from the definitions; see 5.55. □

## Positive (Semidefinite) Operators (Section 7C)

**Def 7.34** (*positive (semidefinite) operators*):

An operator $T \in \mathcal{L}(V)$ is called *"positive (semidefinite)"* if $T$ is self-adjoint and

$$\langle Tv\,,v\rangle \geq 0 \quad \forall v \in V.$$

One can either use the term *"positive operator"* or *"positive semidefinite operator"*, which means the same.

**Def 7.36** (*square root*):

An operator $R$ is called a *"square root"* of an operator $T$ if $R^2 = T$.

**Thm 7.38** (*characterizations of positive operators*)**:**

Let $T \in \mathcal{L}(V)$. Then the following are equivalent:

**(a)** $T$ is a positive operator;

**(b)** $T$ is self-adjoint and all eigenvalues of $T$ are nonnegative;

**(c)** With respect to some orthonormal basis of $V$, the matrix $\mathcal{M}(T)$ of $T$ is a diagonal matrix with only nonnegative numbers on the diagonal;

**(d)** $T$ has a positive square root;

**(e)** $T$ has a self-adjoint square root;

**(f)** $T = R^*R$ for some $R \in \mathcal{L}(V)$.

***Proof:*** We will prove $(a) \implies (b) \implies (c) \implies (d) \implies (e) \implies (f) \implies (a)$.

***Step 1:*** If (a) holds, that means that $T$ is also self-adjoint. Then we have for an eigenvalue $\lambda \in \mathbb{F}$:

$$0 \le \langle Tv, v\rangle = \langle \lambda v, v\rangle = \lambda \langle v, v\rangle \implies 0 \le \lambda.$$

***Step 2:*** Suppose (b) holds. By the spectral theorem, there is an orthonormal basis $e_1, \dots, e_n$ of $V$ consisting of eigenvectors of $T$. Let $\lambda_1, \dots, \lambda_n$ be the eigenvalues of $T$ corresponding to $e_1, \dots, e_n$. By (a), we know that they are all nonnegative. So, $(b) \implies (c)$.

***Step 3:*** Now, suppose (c) holds. Then there is an orthonormal basis $e_1, \dots, e_n$ of $V$ such that $\mathcal{M}(T) = \mathsf{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_i \ge 0$. The linear map lemma 3.4 implies that $\exists R \in \mathcal{L}(V)$:

$$Re_k = \sqrt{\lambda_k} e_k \in V \quad \forall k \in \{1, \dots, n\}.$$

$$R^2 e_k = RRe_k = R\sqrt{\lambda_k} e_k = \sqrt{\lambda_k} Re_k = \sqrt{\lambda_k}\sqrt{\lambda_k} e_k = \lambda_k e_k = Te_k \quad \forall k \in \{1, \dots, n\},$$

which implies that $R^2 = T$. Thus, $R$ is a positive (semidefinite) square root of $T$. Hence, (d) holds, which shows $(c) \implies (d)$.

***Step 4:*** Every positive (semidefinite) operator is self-adjoint by the definition of a positive (semidefinite) operator. Thus, $(d) \implies (e)$.

***Step 5:*** Now, suppose (e) holds, meaning that there exists a self-adjoint operator $R = R^* \in \mathcal{L}(V)$ on $V$ such that $T = R^2$. Then $T = R^*R$. Hence, $(e) \implies (f)$.

***Step 6:*** Finally, suppose (f) holds. Let $R \in \mathcal{L}(V)$ be such that $T = R^*R$. Then $T^* = (R^*R)^* = R^*(R^*)^* = R^*R = T$. Hence, $T$ is self-adjoint. Note that $\forall v \in V$:

$$\langle Tv, v\rangle = \left\langle R^*Rv, v\right\rangle = \left\langle Rv, (R^*)^*v\right\rangle = \langle Rv, Rv\rangle \ge 0.$$

Thus, $T$ is positive (semidefinite), showing that $(f) \implies (a)$. $\square$

**Thm 7.39** (*uniqueness of square root*)**:**

Each positive operator on $V$ has a unique positive square root.

**Def 7.40** (*notation:* $\sqrt{T}$)**:**

For $T$ a positive operator, $\sqrt{T}$ denotes the unique positive square root of $T$.

**Thm 7.41:**

Suppose $T$ is a positive operator on $V$ and $v \in V$ such that $\langle Tv, v\rangle = 0$. Then:

$$Tv = 0.$$

***Proof:*** We have $0 = \langle Tv, v\rangle = \left\langle \sqrt{T}\sqrt{T}v, v\right\rangle = \left\langle \sqrt{T}v, \sqrt{T}v\right\rangle = \left\|\sqrt{T}v\right\|^2 \implies \sqrt{T}v = 0$. Thus, $Tv = \sqrt{T}\left(\sqrt{T}v\right) = 0$, as desired. $\square$

# Isometries, Unitary Operators, and Matrix Factorization (Section 7D)

## Isometries (Sub-Section 7D1)

**Def 7.42** (*isometry*):

$S \in \mathcal{L}(V,W)$ is called an *"isometry"* if

$$\|Sv\| = \|v\| \quad \forall v \in V.$$

In other words, a linear map is an isometry if it preserves norms.

If $S \in \mathcal{L}(V,W)$ is an isometry and $v \in V$ is such that $Sv = 0$, then

$$\|v\| = \|Sv\| = \|0\| = 0,$$

which implies that $v = 0$. Thus, every isometry is injective.

(The Greek word **isos** means *"equal"*; the Greek word **metron** means *"measure"*. Thus **isometry** literally means *"equal measure"*.)

**Thm 7.49** (*characterizations of isometries*):

Suppose $S \in \mathcal{L}(V,W)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of $V$ and $f_1, \dots, f_m$ is an orthonormal basis of $W$. Then the following are equivalent:

- **(a)** $S$ is an isometry;
- **(b)** $S^*S = I$;
- **(c)** $\langle Su, Sv \rangle = \langle u, v \rangle \quad \forall u, v \in V$;
- **(d)** $Se_1, \dots, Se_n$ is an orthonormal list in $W$;
- **(e)** The columns of $\mathcal{M}(S, (e_1, \dots, e_n), (f_1, \dots, f_m))$ form an orthonormal list in $\mathbb{F}^m$ with respect to the Euclidean inner product.

***Proof:*** We will prove $(a) \implies (b) \implies (c) \implies (d) \implies (e) \implies (a)$.

*"First step:"* First, suppose (a) holds, so $S$ is an isometry. If $v \in V$, then:

$$\left\langle (I - S^*S)v, v \right\rangle = \langle v, v \rangle - \left\langle S^*Sv, v \right\rangle = \|v\|^2 - \langle Sv, Sv \rangle = \|v\|^2 - \|Sv\|^2 = 0.$$

Since $I - S^*S$ is a self-adjoint operator because $(I - S^*S)^* = I - S^*S$, we can use 7.16 to conclude that $I - S^*S$ equals the zero operator. Thus, $S^*S = I$, proving $(a) \implies (b)$.

*"Second step:"* Now, suppose (b) holds, so $S^*S = I$. If $u, v \in V$, then:

$$\langle Su, Sv \rangle = \left\langle S^*Su, v \right\rangle = \langle Iu, v \rangle = \langle u, v \rangle,$$

proving $(b) \implies (c)$.

*"Third step:"* Now, suppose that (c) holds, so $\langle Su, Sv \rangle = \langle u, v \rangle \quad \forall u, v \in V$. Thus, if $j, k \in \{1, \dots, n\}$, then:

$$\left\langle Se_j, Se_k \right\rangle = \left\langle e_j, e_k \right\rangle.$$

Hence, $Se_1, \dots, Se_n$ is an orthonormal list in $W$, proving that $(c) \implies (d)$. □

## Unitary Operators (Sub-Section 7D2)

**Def 7.51** (*unitary operator*):

An operator $S \in \mathcal{L}(V)$ is called *"unitary"* if $S$ is an invertible isometry.

Note: Every isometry is injective. Every injective operator on a finite-dimensional vector space is invertible. So *"unitary"* and *"isometry"* mean the same thing for operators on finite-dimensional inner product spaces. But remember: A unitary operator maps a vector space to itself, while an isometry maps a vector space to another (possibly different) vector space.

**Example 7.52** (*rotation of* $\mathbb{R}^2$)**:**
Suppose $\theta \in \mathbb{R}$ and $S$ is the operator on $\mathbb{F}^2$ whose matrix $\mathcal{M}(S)$ with respect to the standard basis of $\mathbb{F}^2$ is

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \mathcal{M}(S).$$

The two columns of this matrix form an orthonormal list in $\mathbb{F}^2$. Therefore, $S$ is an isometry by the equivalence of (a) and (e) in 7.49. Thus, $S$ is a unitary operator.

If $\mathbb{F} = \mathbb{R}$, then $S$ is the counterclockwise rotation by $\theta$ radians around the origin of $\mathbb{R}^2$. So it preserves norms.

**Thm 7.53** (*characterizations of unitary operators*)**:**
Suppose $S \in \mathcal{L}(V)$. Suppose $e_1, \dots, e_n$ is an orthonormal basis of $V$. Then:

- **(a)** $S$ is a unitary operator;
- **(b)** $S^*S = SS^* = I$;
- **(c)** $S$ is invertible and $S^{-1} = S^*$;
- **(d)** $Se_1, \dots, Se_n$ is an orthonormal basis of $V$;
- **(e)** The rows of $\mathcal{M}(S, (e_1, \dots, e_n))$ form an orthonormal basis of $\mathbb{F}^n$ with respect to the Euclidean inner product;
- **(f)** $S^*$ is a unitary operator.

**Thm 7.54** (*eigenvalues of unitary operators*)**:**
Suppose $\lambda$ is an eigenvalue of a unitary operator. Then $|\lambda| = 1$.

**Thm 7.55** (*description of unitary operators on complex inner product spaces*)**:**
Suppose $\mathbb{F} = \mathbb{C}$ and $S \in \mathcal{L}(V)$. Then the following are equivalent:

- **(a)** $S$ is a unitary operator;
- **(b)** There is an orthonormal basis of $V$ consisting of eigenvectors of $S$ whose corresponding eigenvalues all have absolute value 1.

## QR Factorization (Sub-Section 7D3)

**Def 7.56** (*unitary matrix*)**:**
An $n$-by-$n$ matrix is called *"unitary"* if its columns form an orthonormal list in $\mathbb{F}^n$.

**Thm 7.57** (*characterizations of unitary matrices*)**:**
Let $Q \in \mathbb{F}^{n,n}$. Then the following are equivalent:

- **(a)** $Q$ is a unitary matrix;
- **(b)** The rows of $Q$ form an orthonormal list in $\mathbb{F}^n$;
- **(c)** $\|Qv\| = \|v\| \quad \forall v \in \mathbb{F}^n$;
- **(d)** $Q^*Q = QQ^* = I$, the $n$-by-$n$ matrix with 1's on the diagonal and 0's elsewhere.

**Thm 7.58** (*QR factorization*)**:**
Suppose $A$ is a square matrix with linearly independent columns. Then there exist unique matrices $Q$ and $R$ such that $Q$ is unitary, $R$ is upper triangular with only positive numbers on its diagonal, and

$$A = QR.$$

## Cholesky Factorization (Sub-Section 7D4)

**Thm 7.61** (*positive invertible operator*):
A self-adjoint operator $T \in \mathcal{L}(V)$ is a positive invertible operator $\iff \langle Tv, v\rangle > 0 \quad \forall v \in V, v \neq 0.$

**Def 7.62** (*positive definite*):
A matrix $B \in \mathbb{F}^{n,n}$ is called *"positive definite"* if $B^* = B$ and

$$\langle Bx, x\rangle > 0 \quad \forall x \in \mathbb{F}^n, x \neq 0.$$

**Thm 7.63** (*Cholesky factorization*):
Suppose $B$ is a positive definite matrix. Then there exists a unique upper triangular matrix $R$ with only positive numbers on its diagonal such that

$$B = R^*R.$$

# Singular Value Decomposition (Section 7E)

## Singular Values (Sub-Section 7E1)

**Thm 7.64** (*properties of $T^*T$*):
Suppose $T \in \mathcal{L}(V, W)$. Then:

- **(a)** $T^*T$ is a positive operator on $V$;
- **(b)** $\operatorname{null} T^*T = \operatorname{null} T$;
- **(c)** $\operatorname{range} T^*T = \operatorname{range} T^*$;
- **(d)** $\dim \operatorname{range} T = \dim \operatorname{range} T^* = \dim \operatorname{range} T^*T$.

**Def 7.65** (*singular values*):
Suppose $T \in \mathcal{L}(V, W)$. The *"singular values"* of $T$ are the nonnegative square roots of the eigenvalues of $T^*T$, listed in decreasing order, each included as many times as the dimension of the corresponding eigenspace of $T^*T$.

**Thm 7.68** (*role of positive singular values*):
Suppose that $T \in \mathcal{L}(V, W)$. Then:

- **(a)** $T$ is injective $\iff$ 0 is not a singular value of $T$;
- **(b)** the number of positive singular values of $T$ equals $\dim \operatorname{range} T$;
- **(c)** $T$ is surjective $\iff$ the number of positive singular values of $T$ equals $\dim W$.

The table below compares eigenvalues with singular values.

| List of eigenvalues | List of singular values |
|---|---|
| context: vector spaces | context: inner product spaces |
| defined only for linear maps from a vector space to itself | defined for linear maps from an inner product space to a possibly different inner product space |
| can be arbitrary real numbers (if $\mathbb{F} = \mathbb{R}$) or complex numbers (if $\mathbb{F} = \mathbb{C}$) | are nonnegative numbers |
| can be the empty list if $\mathbb{F} = \mathbb{R}$ | length of list equals dimension of domain |
| includes 0 $\iff$ operator is not invertible | includes 0 $\iff$ linear map is not injective |
| no standard order, especially if $\mathbb{F} = \mathbb{C}$ | always listed in decreasing order |

**Thm 7.69** (*isometries characterized through singular values*):

Suppose $S \in \mathcal{L}(V, W)$. Then

$S$ is an isometry $\iff$ all singular values of $S$ equal 1.

## SVD for Linear Maps and for Matrices (Sub-Section 7E2)

**Thm 7.70** (*singular value decomposition*):

Let $T \in \mathcal{L}(V, W)$, and let the positive singular values of $T$ be $s_1, \ldots, s_m$. Then there exist two orthonormal lists $e_1, \ldots, e_m$ in $V$ and $f_1, \ldots, f_m$ in $W$ such that

$$Tv = s_1 \left\langle v, e_1 \right\rangle f_1 + \cdots + s_m \left\langle v, e_m \right\rangle f_m \quad \forall v \in V.$$

**Def 7.74** (*diagonal matrix*):

An $M$-by-$N$ matrix $A$ is called a *"diagonal matrix"* if all entries of the matrix are 0 except possibly $A_{k,k}$ for $k \in \{1, \ldots, \min\{M, N\}\}$.

The table below compares the spectral theorem with the singular value decomposition:

| Spectral theorem | Singular value decomposition |
|---|---|
| describes only self-adjoint operators (when $\mathbb{F} = \mathbb{R}$) or normal operators (when $\mathbb{F} = \mathbb{C}$) | describes arbitrary linear maps from an inner product space to a possibly different inner product space |
| produces a single orthonormal basis | produces two orthonormal lists, one for domain space and one for range space, that are not necessarily the same even when range space equals domain space |
| different proofs depending on whether $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ | same proof works regardless of whether $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$ |

**Thm 7.75** (*singular value decomposition of adjoint and pseudoinverse*):

Suppose $T \in \mathcal{L}(V, W)$ and the positive singular values of $T$ are $s_1, \ldots, s_m$. Suppose $e_1, \ldots, e_m$ and $f_1, \ldots, f_m$ are orthonormal lists in $V$ and $W$ such that we have the following SVD:

$$Tv = s_1 \left\langle v, e_1 \right\rangle f_1 + \cdots + s_m \left\langle v, e_m \right\rangle f_m \quad \forall v \in V.$$

Then we have for all $w \in W$:

- $T^* w = s_1 \left\langle w, f_1 \right\rangle e_1 + \cdots + s_m \left\langle w, f_m \right\rangle e_m$;
- $T^\dagger w = \dfrac{\left\langle w, f_1 \right\rangle}{s_1} e_1 + \cdots + \dfrac{\left\langle w, f_m \right\rangle}{s_m} e_m$.

**Thm 7.76** (*matrix version of SVD*):

Suppose $A$ is a $p$-by-$n$ matrix of rank $m \geq 1$. Then there exist a $p$-by-$n$ matrix $B$ with orthonormal columns, an $m$-by-$m$ diagonal matrix $D$ with positive numbers on the diagonal, and an $n$-by-$m$ matrix $C$ with orthonormal columns such that

$$A = BDC^*.$$

# Consequences of Singular Value Decomposition (Section 7F)

## Norms of Linear Maps (Sub-Section 7F1)

**Thm 7.82** (*upper bound for $\|Tv\|$*):

Suppose $T \in \mathcal{L}(V,W)$. Let $s_1$ be the largest singular value of $T$. Then

$$\|Tv\| \le s_1 \|v\| \quad \forall v \in V.$$

**Def 7.86** (*norm of a linear map*):

Suppose $T \in \mathcal{L}(V,W)$. Then the *"norm"* of $T$, denoted by $\|T\|$, is defined by

$$\|T\| :\equiv \max\left\{\|Tv\| \;\middle|\; v \in V \text{ and } \|v\| \le 1\right\}.$$

**Thm 7.87** (*basic properties of norms of linear maps*):

Let $T \in \mathcal{L}(V,W)$. Then:

- **(a)** $\|T\| \ge 0$;
- **(b)** $\|T\| = 0 \iff T = 0$;
- **(c)** $\|\lambda T\| = |\lambda|\|T\| \quad \forall \lambda \in \mathbb{F}$;
- **(d)** $\|S+T\| \le \|S\| + \|T\| \quad \forall S \in \mathcal{L}(V,W)$.

**Thm 7.88** (*alternative formulas for $\|T\|$*):

Let $T \in \mathcal{L}(V,W)$. Then:

- **(a)** $\|T\| =$ the largest singular value of $T$;
- **(b)** $\|T\| = \max\{\|Tv\| \mid v \in V \text{ and } \|v\| = 1\}$;
- **(c)** $\|T\| =$ the smallest number $c$ such that $\|Tv\| \le c\|v\| \quad \forall v \in V$.

**Thm 7.89** (*norm of the adjoint*):

Suppose $T \in \mathcal{L}(V,W)$. Then $\|T^*\| = \|T\|$.

## Approximation by Linear Maps with Lower-Dimensional Range (Sub-Section 7F2)

**Thm 7.92** (*best approximation by linear map whose range has dimension $\le k$*):

Let $T \in \mathcal{L}(V,W)$, and let $s_1 \ge \cdots \ge s_m$ be the positive singular values of $T$. Suppose $1 \le k < m$. Then

$$\min\left\{\|T-S\| \;\middle|\; S \in \mathcal{L}(V,W) \text{ and } \dim\operatorname{range} S \le k\right\} = s_{k+1}.$$

Furthermore, if

$$Tv = s_1\left\langle v, e_1\right\rangle f_1 + \cdots + s_m\left\langle v, e_m\right\rangle f_m$$

is a singular value decomposition of $T$, and $T_k \in \mathcal{L}(V,W)$ is defined by

$$T_k v = s_1\left\langle v, e_1\right\rangle f_1 + \cdots + s_k\left\langle v, e_k\right\rangle f_k \quad \forall v \in V,$$

then $\dim\operatorname{range} T_k = k$ and $\|T - T_k\| = s_{k+1}$.

**Thm 7.93** (*polar decomposition*):

Suppose $T \in \mathcal{L}(V)$. Then there exists a unique unitary operator $S \in \mathcal{L}(V)$ such that

$$T = S\sqrt{T^*T}.$$

## Operators Applied to Ellipsoids and Parallelepipeds (Sub-Section 7F3)

**Def 7.95** (*ball*, $B$)**:**

The *"ball"* in $V$ of radius 1 centered at 0, denoted by $B$, is defined by

$$B := \left\{ v \in V \;\middle|\; \|v\| < 1 \right\}.$$